

APPENDIX B

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UTILITY PATENT APPLICATION TRANSMITTAL

(Only for new nonprovisional applications under 37 CFR 1.53(b))

Attorney Docket No.	0202-1231-1
First Inventor	Gihsung CHUNG
Title	METHOD FOR DETECTING BIOMETRIC...
Express Mail Label No.	

APPLICATION ELEMENTS

See MPEP chapter 600 concerning utility patent application contents.

1. ☐ **Fee Transmittal Form.**
(PTO/SB/17 or equivalent)
2. ☐ **Applicant claims small entity status.**
See 37 CFR 1.27.
3. ☒ **Specification.** [Total Pages 80]
Both the claims and abstract must start on a new page
(For information on the preferred arrangement, see MPEP § 608.01(a))
4. ☒ **Drawing(s).** (35 U.S.C. 113) [Total Sheets 23]
5. **Inventor's Oath or Declaration.** [Total Sheets 9]
(including substitute statements under 37 CFR 1.64 and assignments serving as an oath or declaration under 37 CFR 1.63(e))
 - a. ☐ Newly executed (original or copy)
 - b. ☒ A copy from a prior application (37 CFR 1.63(d))
6. ☒ **Application Data Sheet.** *See Note below.
See 37 CFR 1.76 (PTO/AIA/14 or equivalent)
7. ☐ **CD-ROM or CD-R.**
in duplicate, large table or Computer Program (Appendix)
☐ Landscape Table on CD
8. **Nucleotide and/or Amino Acid Sequence Submission.**
(if applicable, items a. – c. are required)
 - a. ☐ Computer Readable Form (CRF)
 - b. ☐ Specification Sequence Listing on:
 - i. ☐ CD-ROM or CD-R (2 copies); or
 - ii. ☐ Paper
 - c. ☐ Statements verifying identity of above copies

ADDRESS TO: Commissioner for Patents
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Alexandria VA 22313-1450

ACCOMPANYING APPLICATION PARTS

9. ☐ **Assignment Papers.**
(cover sheet & document(s))
Name of Assignee _____
10. ☐ **37 CFR 3.73(c) Statement.** ☒ **Power of Attorney.**
(when there is an assignee)
11. ☐ **English Translation Document.**
(if applicable)
12. ☒ **Information Disclosure Statement.**
(PTO/SB/08 or PTO-1449)
☐ Copies of citations attached
13. ☐ **Preliminary Amendment.**
14. ☐ **Return Receipt Postcard.**
(MPEP § 503) (Should be specifically itemized)
15. ☐ **Certified Copy of Priority Document(s).**
(if foreign priority is claimed)
16. ☐ **Nonpublication Request.**
Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent.
17. ☐ **Other:** _____

***Note:** (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 **must** be included in an Application Data Sheet (ADS).
(2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).

18. CORRESPONDENCE ADDRESS

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This collection of information is required by 37 CFR 1.53(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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**METHOD FOR DETECTING BIOMETRIC INFORMATION AND
ELECTRONIC DEVICE USING SAME**

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation application of prior Application number 15/173,099, filed on June 3, 2016, which claimed priority under 35 U.S.C § 119(a) of a Korean patent application number 10-2015-0113972, filed on August 12, 2015, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an electronic device. More particularly, the present disclosure relates to a device and method for detecting biometric information.

BACKGROUND

[0003] Electronic devices have become smaller and slimmer, and have evolved to be easily carried while performing the same or more diverse functions. Although compact electronic devices are generally carried while being received in a user's pocket, they may be worn on the wrist, the head, or the arm of a human body.

[0004] In addition, electronic devices may be equipped with biometric sensors (e.g., health care sensors) in order to measure biometric information (e.g., health data) that individuals manage. For example, the biometric information may include blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen, and the like.

[0005] The above information is presented as background information only to assist with an understanding of the present disclosure. No determination has been

made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the present disclosure.

SUMMARY

[0006] Electronic devices may operate biometric sensors (e.g., heart rate sensors) in order to measure biometric information (e.g., a heart rate). The operation of the biometric sensors may consume the battery power of the electronic devices. Accordingly, the electronic devices may have difficulty in measuring biometric information for a long period of time (e.g., while users wear the electronic devices).

[0007] In another example, when electronic devices measure users' biometric information, the measurement of biometric information is impossible, or the accuracy of the measured biometric information may be deteriorated according to the motion of the electronic devices.

[0008] Aspects of the present disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the present disclosure is to provide an electronic device and a method that can selectively operate a biometric sensor based on a selected (or specified) condition.

[0009] Another aspect of the present disclosure is to provide an electronic device and a method that can continually (e.g., periodically (e.g., in 1-minute periods)) operate a biometric sensor according to a selected (or specified) period.

[0010] Another aspect of the present disclosure is to provide an electronic device and a method that can measure biometric information based on the motion of the electronic device.

[0011] In accordance with an aspect of the present disclosure, an electronic device is provided. The electronic device includes a housing, a fastener coupled to a part of the housing and detachably attached to a part of a user's body, at least one first sensor configured to detect a motion of the electronic device, at least one second sensor configured to detect a change in the user's body, a memory, and a processor

electrically connected to the first sensor, the second sensor, and the memory, wherein the memory may store instructions that configure the processor, when being executed, to determine whether the electronic device is worn on a part of the user's body using the at least one first sensor, acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time, which is shorter than the first period of time, shortly before and/or immediately after a first period of time, which is selected (or specified), passes from a time when the electronic device is worn on a part of the user's body, and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

[0012] In accordance with another aspect of the present disclosure, a method of measuring biometric information by an electronic device is provided. The method includes determining whether the electronic device is worn on a part of a user's body using at least one first sensor, activating (operating or triggering) at least one second sensor based on at least a part of the a motion of the electronic device from a time when the electronic device is worn on a part of the user's body to the time when a first period of time selected (or specified) passes, and measuring biometric information based on the at least one second sensor.

[0013] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 is a diagram illustrating an electronic device in a network environment according to various embodiments of the present disclosure;

[0016] FIG. 2 is a block diagram of an electronic device according to various embodiments of the present disclosure;

[0017] FIG. 3 is a block diagram of a program module according to various embodiments of the present disclosure;

[0018] FIGS. 4A, 4B, and 4C are views illustrating an electronic device according to various embodiments of the present disclosure;

[0019] FIG. 5A is a diagram illustrating the configuration of a processor according to various embodiments of the present disclosure;

[0020] FIG. 5B is a view illustrating an operation of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

[0021] FIG. 6 is a flowchart illustrating a biometric information measurement operation of an electronic device according to various embodiments of the present disclosure;

[0022] FIG. 7 is a flowchart illustrating a performance procedure of a method of determining the state of an electronic device according to various embodiments of the present disclosure;

[0023] FIG. 8 is a flowchart illustrating the performance procedure of a method of operating a sensor by an electronic device according to various embodiments of the present disclosure;

[0024] FIG. 9 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

[0025] FIG. 10 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

[0026] FIG. 11 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0027] FIG. 12 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0028] FIG. 13 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0029] FIG. 14 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0030] FIG. 15 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0031] FIG. 16 is a flowchart illustrating the performance procedure of a method of changing a measurement trigger according to various embodiments of the present disclosure;

[0032] FIG. 17 is a view illustrating an operation of changing a measurement trigger by an electronic device according to various embodiments of the present disclosure;

[0033] FIG. 18 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

[0034] FIG. 19 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure; and

[0035] FIG. 20 is a view illustrating a screen configuration of an electronic device according to various embodiments of the present disclosure.

[0036] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

[0037] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the present disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the present disclosure. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

[0038] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the present disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the present disclosure is provided for illustration purpose only and not for the purpose of limiting the present disclosure as defined by the appended claims and their equivalents.

[0039] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0040] The terms “have,” “may have,” “include,” or “may include” used in the various embodiments of the present disclosure indicate the presence of disclosed corresponding functions, operations, elements, and the like, and do not limit additional one or more functions, operations, elements, and the like. In addition, it should be understood that the terms “include” or “have” used in the various embodiments of the present disclosure are to indicate the presence of features, numbers, operations, elements, parts, or a combination thereof described in the specifications, and do not

preclude the presence or addition of one or more other features, numbers, operations, elements, parts, or a combination thereof.

[0041] The terms “A or B”, “at least one of A or/and B” or “one or more of A or/and B” used in the various embodiments of the present disclosure include any and all combinations of words enumerated with it. For example, “A or B”, “at least one of A and B” or “at least one of A or B” means (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0042] Although the term such as “first” and “second” used in various embodiments of the present disclosure may modify various elements of various embodiments, these terms do not limit the corresponding elements. For example, these terms do not limit an order and/or importance of the corresponding elements. These terms may be used for the purpose of distinguishing one element from another element. For example, a first user device and a second user device all indicate user devices and may indicate different user devices. For example, a first element may be named a second element without departing from the scope of right of various embodiments of the present disclosure, and similarly, a second element may be named a first element.

[0043] It will be understood that when an element (e.g., first element) is “connected to” or “(operatively or communicatively) coupled with/to” to another element (e.g., second element), the element may be directly connected or coupled to another element, and there may be an intervening element (e.g., third element) between the element and another element. To the contrary, it will be understood that when an element (e.g., first element) is “directly connected” or “directly coupled” to another element (e.g., second element), there is no intervening element (e.g., third element) between the element and another element.

[0044] The expression “configured to (or set to)” used in various embodiments of the present disclosure may be replaced with “suitable for”, “having the capacity to”, “designed to”, “adapted to”, “made to”, or “capable of” according to a situation. The term “configured to (set to)” does not necessarily mean “specifically designed to” in a hardware level. Instead, the expression “apparatus configured to...” may mean that

the apparatus is “capable of...” along with other devices or parts in a certain situation. For example, “a processor configured to (set to) perform A, B, and C” may be a dedicated processor, e.g., an embedded processor, for performing a corresponding operation, or a generic-purpose processor, e.g., a central processing unit (CPU) or an application processor (AP), capable of performing a corresponding operation by executing one or more software programs stored in a memory device.

[0045] Further, all the terms used herein, including technical and scientific terms, should be interpreted to have the same meanings as commonly understood by those skilled in the art to which the present disclosure pertains, and should not be interpreted to have ideal or excessively formal meanings unless explicitly defined in various embodiments of the present disclosure.

[0046] An electronic device according to various embodiments of the present disclosure may be a device. For example, the electronic device according to various embodiments of the present disclosure may include at least one of: a smart phone; a tablet personal computer (PC); a mobile phone; a video phone; an e-book reader; a desktop PC; a laptop PC; a netbook computer; a workstation, a server, a personal digital assistant (PDA); a portable multimedia player (PMP); a Moving Picture Experts Group phase 1 or phase 2 (MPEG-1 or MPEG-2) audio layer 3 (MP3) player; a mobile medical device; a camera; or a wearable device (e.g., a head-mount-device (HMD), an electronic glasses, an electronic clothing, an electronic bracelet, an electronic necklace, an electronic appcessory, an electronic tattoo, a smart mirror, or a smart watch).

[0047] In other embodiments, an electronic device may be a smart home appliance. For example, of such appliances may include at least one of: a television (TV); a digital versatile disc (DVD) player; an audio component; a refrigerator; an air conditioner; a vacuum cleaner; an oven; a microwave oven; a washing machine; an air cleaner; a set-top box; a home automation control panel; a security control panel; a TV box (e.g., Samsung HomeSync™, Apple TV™, or Google TV™); a game console(e.g., Xbox™ PlayStation™); an electronic dictionary; an electronic key; a camcorder; or an electronic frame.

[0048] In other embodiments, an electronic device may include at least one of: a medical equipment (e.g., a mobile medical device (e.g., a blood glucose monitoring device, a heart rate monitor, a blood pressure monitoring device or a temperature meter), a magnetic resonance angiography (MRA) machine, a magnetic resonance imaging (MRI) machine, a computed tomography (CT) scanner, or an ultrasound machine); a navigation device; a global positioning system (GPS) receiver; an event data recorder (EDR); a flight data recorder (FDR); an in-vehicle infotainment device; an electronic equipment for a ship (e.g., ship navigation equipment and/or a gyrocompass); an avionics equipment; a security equipment; a head unit for vehicle; an industrial or home robot; an automatic teller's machine (ATM) of a financial institution, point of sale (POS) device at a retail store, or an internet of things device (e.g., a lightbulb, various sensors, an electronic meter, a gas meter, a sprinkler, a fire alarm, a thermostat, a streetlamp, a toaster, a sporting equipment, a hot-water tank, a heater, or a boiler and the like).

[0049] In certain embodiments, an electronic device may include at least one of: a piece of furniture or a building/structure; an electronic board; an electronic signature receiving device; a projector; and various measuring instruments (e.g., a water meter, an electricity meter, a gas meter, or a wave meter).

[0050] An electronic device according to various embodiments of the present disclosure may also include a combination of one or more of the above-mentioned devices. Further, it will be apparent to those skilled in the art that an electronic device according to various embodiments of the present disclosure is not limited to the above-mentioned devices.

[0051] FIG. 1 is a view illustrating a network environment 100 including an electronic device 101 according to various embodiments of the present disclosure.

[0052] Referring to FIG. 1, the electronic device 101 may include a bus 110, a processor 120, a memory 130, an input/output (I/O) interface 150, a display 160, and a communication interface 170.

[0053] The bus 110 may be a circuit for connecting the above-described elements (e.g., the processor 120, the memory 130, the I/O interface 150, the display

160 or the communication interface 170, etc.) with each other, and transferring communication (e.g., a control message) between the above-described elements.

[0054] The processor 120 may include a CPU, a communication processor (CP), a graphics processing unit (GPU).

[0055] The processor 120 may receive, for example, an instruction from the above-described other elements (e.g., the memory 130, the I/O interface 150, the display 160, or the communication interface 170, etc.) via the bus 110, decipher the received instruction, and execute an operation or a data process corresponding to the deciphered instruction.

[0056] The memory 130 may include any suitable type of volatile or non-volatile memory. The memory 130 may store an instruction or data received from the processor 120 or other elements (e.g., the I/O interface 150, the display 160, or the communication interface 170, etc.), or generated by the processor 120 or other elements. The memory 130 may include, for example, programming modules 140 such as a kernel 141, a middleware 143, an application programming interface (API) 145, or an application 147. The each of the programming modules may be configured using a software, a firmware, a hardware, or a combination of two or more of these.

[0057] The kernel 141 may control or manage system resources (e.g., the bus 110, the processor 120, or the memory 130, etc.) used for executing an operation or a function implemented in the rest of the programming modules, for example, the middleware 143, the API 145, or the application 147. Also, the kernel 141 may provide an interface for allowing the middleware 143, the API 145, or the application 147 to access an individual element of the electronic device 101 and control or manage the same.

[0058] The middleware 143 may perform a mediation role so that the API 145 or the application 147 may communicate with the kernel 141 to give and take data. Also, in connection with task requests received from the application 147, the middleware 143 may perform a control (e.g., scheduling or load balancing) for a task request using, for example, a method of assigning priority that may use a system

resource (e.g., the bus 110, the processor 120, or the memory 130, etc.) of the electronic device 101 to at least one application 134.

[0059] The API 145 is an interface for allowing the application 147 to control a function provided by the kernel 141 or the middleware 143, and may include at least one interface or function (e.g., an instruction) for file control, window control, image processing, or character control, etc.

[0060] The I/O interface 150 may transfer an instruction or data input from a user via an I/O unit (e.g., a sensor, a keyboard, or a touchscreen) to the processor 120, the memory 130, or the communication interface 170 via the bus 110, for example. For example, the I/O interface 150 may provide data regarding a user's touch input via the touchscreen to the processor 120. Also, the I/O interface 150 may, for example, output an instruction or data received via the bus 110 from the processor 120, the memory 130, or the communication interface 170 via the I/O unit (e.g., a speaker or a display). For example, the I/O interface 150 may output voice data processed by the processor 120 to a user via a speaker.

[0061] The display 160 may include, for example, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic LED (OLED) display, a micro electro mechanical system (MEMS) display, or an electronic paper display. The display 160 may display various types of contents (for example, text, images, videos, icons, or symbols) for users. The display 160 may include a touch screen, and may receive, for example, a touch, gesture, proximity, or hovering input by using an electronic pen or a part of the user's body.

[0062] The communication interface 170 may connect communication (e.g., communication 164) between the electronic device 101 and an external device (for example, the electronic device 102, electronic device 104, or the server 106). For example, the communication interface 170 may be connected to a network 162 through wireless communication or wired communication, and may communicate with an external device.

[0063] The wireless communication may use at least one of, for example, long term evolution (LTE), LTE-advanced (LTE-A), code division multiple access

(CDMA), wideband CDMA (WCDMA), universal mobile telecommunications system (UMTS), wireless broadband (WiBro), and global system for mobile communications (GSM) as a cellular communication protocol.

[0064] The wired communication may include, for example, at least one of universal serial bus (USB), high definition multimedia interface (HDMI), recommended standard 232 (RS-232), and plain old telephone service (POTS).

[0065] The network 162 may include at least one of communication networks such as a computer network (for example, a local area network (LAN) or a wide area network (WAN)), the internet, and a telephone network.

[0066] The electronic devices 102 and 104 may be devices of the same type as that the electronic device 101 or devices of different types from that of the electronic device 101. According to an embodiment, the server 106 may include a group of one or more servers. According to various embodiments, all or some of the operations executed in the electronic device 101 may be carried out in another electronic device or a plurality of electronic devices (for example, the electronic device 102 or 104 and the server 106). According to an embodiment, when the electronic device 101 should perform some functions or services automatically or by a request, the electronic device 101 may make a request for performing at least some functions related to the functions or services to another device (for example, the electronic device 102 or 104, or the server 106) instead of performing the functions or services by itself or additionally. The electronic device (for example, the electronic device 102 or 104, or the server 106) may carry out the functions requested by the electronic device 101 or additional functions and provide results thereof to the electronic device 101. The electronic device 101 may provide the requested functions or services to another electronic device based on the received results or after additionally processing the received results. To this end, for example, cloud computing, distributed computing, or client-server computing technology may be used.

[0067] FIG. 2 is a block diagram 200 illustrating an electronic device 201 according to various embodiments of the present disclosure. The electronic device

201 may configure, for example, all or a portion of the electronic device 101 illustrated in FIG. 1.

[0068] Referring to FIG. 2, the electronic device 201 may include one or more APs 210, a communication module 220, a subscriber identification module (SIM) card 224, a memory 230, a sensor module 240, an input unit 250, a display 260, an interface 270, an audio module 280, a camera module 291, a power management module 295, a battery 296, an indicator 297, or a motor 298.

[0069] The AP 210 may drive an OS or an application to control a plurality of hardware or software elements connected to the AP 210, and perform various data processes including multimedia data and operations. The AP 210 may be implemented, for example, as a system on chip (SoC). According to an embodiment, the AP 210 may further include at least one of a GPU or image signal processor (ISP). According to an embodiment, the AP 210 may be implemented to include at least a portion (e.g., the cellular module 221) of the above-described elements. Also, the AP 210 may store data received from at least one of other elements or generated by at least one of other elements in a non-volatile memory.

[0070] The communication module 220 (e.g., the communication interface 170) may perform data transmission/reception in communication between the electronic device 201 (e.g., the electronic device 101) and other electronic devices (e.g., the electronic device 104 or the server 106) connected via a network. According to an embodiment, the communication module 220 may include a cellular module 221, a Wi-Fi module 223, a Bluetooth (BT) module 225, a GPS module 227, a near field communication (NFC) module 228, and a radio frequency (RF) module 229.

[0071] The cellular module 221 may provide voice communication, image communication, a short message service, or an internet service, etc. via a communication network (e.g., LTE, LTE-A, CDMA, WCDMA, UMTS, WiBro, or GSM, etc.). Also, the cellular module 221 may perform discrimination and authentication of an electronic device within a communication network using, for example, a subscriber identify module (e.g., a SIM card 224). According to an embodiment, the cellular module 221 may perform at least a portion of functions that

may be provided by the AP 210. According to an embodiment, the cellular module 221 may include a CP. Also, the cellular module 221 may be, for example, implemented as an SoC. Though elements such as the cellular module 221 (e.g., a CP), the memory 230, or the power management module 295, etc. are illustrated as elements separated from the AP 210 in FIG. 2, according to an embodiment, the AP 210 may be implemented to include at least a portion (e.g., the cellular module 221) of the above-described elements.

[0072] Each of the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may include, for example, a processor for processing data transmitted/received via a relevant module. Though the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 are illustrated as separate blocks in FIG. 2, according to an embodiment, at least a portion (e.g., two or more elements) of the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may be included in one integrated circuit (IC) or an IC package. For example, at least a portion (e.g., a CP corresponding to the cellular module 221 and a Wi-Fi processor corresponding to the Wi-Fi module 223) of processors corresponding to each of the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may be implemented as one SoC.

[0073] The RF module 229 may perform transmission/reception of data, for example, transmission/reception of an RF signal. The RF module 229 may include, for example, a transceiver, a power amp module (PAM), a frequency filter, or a low noise amplifier (LNA), etc., though not shown. Also, the RF module 229 may further include a part for transmitting/receiving an electromagnetic wave on a free space in wireless communication, for example, a conductor or a conducting line, etc. Though FIG. 2 illustrates the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, and the NFC module 228 share one RF module 229, according to an embodiment, at least one of the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may perform transmission/reception of an RF signal via a separate RF module.

[0074] The SIM card 224 may be a card including a subscriber identify module, and may be inserted into a slot formed in a specific position of the electronic device. The SIM card 224 may include unique identify information (e.g., integrated circuit card identifier (ICCID)) or subscriber information (e.g., international mobile subscriber identity (IMSI)).

[0075] The memory 230 (e.g., the memory 20) may include an internal memory 232 or an external memory 234. The internal 232 may include, for example, at least one of a volatile memory (e.g., dynamic random access memory (DRAM), static RAM (SRAM), synchronous dynamic RAM (SDRAM)) and a non-volatile memory (e.g., one time programmable read only memory (OTPROM), programmable ROM (PROM), erasable and programmable ROM (EPROM), electrically erasable and programmable ROM (EEPROM), mask ROM, flash ROM, NAND flash memory, NOR flash memory, etc.).

[0076] According to an embodiment, the internal memory 232 may be a solid state drive (SSD). The external memory 234 may further include a flash drive, for example, compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), or a memory stick. The external memory 234 may be functionally connected with the electronic device 201 via various interfaces. According to an embodiment, the electronic device 201 may further include a storage device (or a storage medium) such as a hard drive.

[0077] The sensor module 240 may measure a physical quantity or detect an operation state of the electronic device 201, and convert the measured or detected information to an electric signal. The sensor module 240 may include, for example, at least one of a gesture sensor 240A, a gyro sensor 240B, an atmospheric pressure sensor 240C, a magnetic sensor 240D, an acceleration sensor 240E, a grip sensor 240F, a proximity sensor 240G, a color sensor 240H (e.g., red, green, blue (RGB) sensor), a biometric sensor 240I, a temperature/humidity sensor 240J, an illuminance sensor 240K, or an ultra violet (UV) sensor 240M. Additionally or alternatively, the sensor module 240 may include, for example, an E-nose sensor (not shown), an electromyography (EMG) sensor (not shown), an electroencephalogram (EEG) sensor

(not shown), an electrocardiogram (ECG) sensor (not shown), an infrared (IR) sensor (not shown), an iris sensor (not shown), or a fingerprint sensor (not shown), etc. The sensor module 240 may further include a control circuit for controlling at least one sensor belonging thereto.

[0078] The input unit 250 may include a touch panel 252, a (digital) pen sensor 254, a key 256, or an ultrasonic input unit 258. The touch panel 252 may recognize a touch input using at least one of capacitive, resistive, infrared, or ultrasonic methods. Also, the touch panel 252 may further include a control circuit. A capacitive touch panel may perform detection by a physical contact or proximity recognition. The touch panel 252 may further include a tactile layer. In this case, the touch panel 252 may provide a tactile reaction to a user.

[0079] The (digital) pen sensor 254 may be implemented using, for example, a method which is the same as or similar to receiving a user's touch input, or using a separate sheet for detection. The key 256 may include, for example, a physical button, an optical key or keypad. The ultrasonic input unit 258 is a unit for recognizing data by detecting a sound wave using a microphone (e.g., a microphone 288) in the electronic device 201 via an input tool generating an ultrasonic signal, and enables wireless recognition. According to an embodiment, the electronic device 201 may receive a user input from an external device (e.g., a computer or a server) connected to the communication module 220 using the communication module 220.

[0080] The display 260 (e.g., the display 160) may include a panel 262, a hologram device 264, or a projector 266. The panel 262 may be, for example, an LCD, or an active-matrix OLED (AM-OLED), etc. The panel 262 may be implemented, for example, such that it is flexible, transparent, or wearable. The panel 262 may be configured as one module together with the touch panel 252. The hologram device 264 may show a three-dimensional image in the air using interferences of light. The projector 266 may project light onto a screen to display an image. The screen may be positioned, for example, inside or outside the electronic device 201. According to an embodiment, the display 260 may further include a

control circuit for controlling the panel 262, the hologram device 264, or the projector 266.

[0081] The interface 270 may include, for example, a high-definition multimedia interface (HDMI) 272, a USB 274, an optical interface 276, or a D-subminiature (D-sub) 278. The interface 270 may be included, for example, in the communication interface 170 illustrated in FIG. 1. Additionally or alternatively, the interface 270 may include, for example, a mobile high-definition link (MHL) interface, an SD card/multi-media card (MMC) interface, or an infrared data association (IrDA) standard interface.

[0082] The audio module 280 may convert a sound and an electric signal in dual directions. At least a partial element of the audio module 280 may be included, for example, in the I/O interface 150 illustrated in FIG. 1. The audio module 280 may process sound information input or output via, for example, a speaker 282, a receiver 284, an earphone 286, or a microphone 288, etc.

[0083] The camera module 291 is a device that may shoot a still image and a moving picture. According to an embodiment, the camera module 291 may include one or more image sensors (e.g., a front sensor or a rear sensor), a lens (not shown), an ISP (not shown), or a flash (not shown) (e.g., an LED or xenon lamp).

[0084] The power management module 295 may manage power of the electronic device 201. Though not shown, the power management module 295 may include, for example, a power management IC (PMIC), a charger IC, or a battery or a battery or fuel gauge.

[0085] The PMIC may be mounted, for example, inside an IC or a SoC semiconductor. A charging method may be classified into a wired charging method and a wireless charging method. The charging IC may charge a battery and prevent introduction of an overvoltage or an overcurrent from a charger. According to an embodiment, the charging IC may include a charging IC for at least one of the wired charging method and the wireless charging method. The wireless charging method may be, for example, a magnetic resonance method, a magnetic induction method, or an electromagnetic wave method, etc., and may additionally include an additional

circuit for wireless charging, for example, a circuit such as a coil loop, a resonance circuit, or a rectifier, etc.

[0086] The battery gauge may measure, for example, a remnant of the battery 296, a voltage, a current, or a temperature while charging. The battery 296 may store or generate electricity, and supply power to the electronic device 201 using the stored or generated electricity. The battery 296 may include, for example, a rechargeable battery or a solar battery.

[0087] The indicator 297 may display a specific state of the electronic device 201 or a portion thereof (e.g., the AP 210), for example, a booting state, a message state, or a charging state, etc. The motor 298 may convert an electric signal to mechanical vibration. Though not shown, the electronic device 201 may include a processor (e.g., a GPU) for supporting a mobile TV. The processor for supporting the mobile TV may process media data corresponding to standards, for example, such as digital multimedia broadcasting (DMB), digital video broadcasting (DVB), or a media flow, etc.

[0088] The aforementioned elements of the electronic device according to various embodiments of the present disclosure may be constituted by one or more components, and the name of the corresponding element may vary with a type of electronic device. The electronic device according to various embodiments of the present disclosure may include at least one of the aforementioned elements. Some elements may be omitted or other additional elements may be further included in the electronic device. Further, some of the components of the electronic device according to the various embodiments of the present disclosure may be combined to form a single entity, and thus, may equivalently execute functions of the corresponding elements prior to the combination.

[0089] FIG. 3 is a block diagram 300 of a program module 310 according to various embodiments of the present disclosure.

[0090] Referring to FIG. 3, the program module 310 (for example, the programming modules 140) may include an operating system (OS) for controlling resources related to the electronic device (for example, the electronic device 101)

and/or various applications (for example, the application 147) executed in the operating system. The OS may be, for example, Android, iOS, Windows, Symbian, Tizen, Bada, or the like.

[0091] The program module 310 may include a kernel 320, middleware 330, an API 360, and/or applications 370. At least some of the program module 310 may be preloaded in the electronic device or downloaded from the server.

[0092] The kernel 320 (for example, the kernel 141 of FIG. 1) may include, for example, a system resource manager 321 or a device driver 323. The system resource manager 321 may control, allocate, or collect the system resources. According to an embodiment, the system resource manager 321 may include a process management unit, a memory management unit, or a file system management unit. The device driver 323 may include, for example, a display driver, a camera driver, a BT driver, a shared-memory driver, a USB driver, a keypad driver, a Wi-Fi driver, an audio driver, or an inter-process communication (IPC) driver.

[0093] The middleware 330 may provide a function used by the applications 370 in common or provide various functions to the applications 370 through the API 360 so that the applications 370 can efficiently use limited system resources within the electronic device. According to an embodiment, the middleware 330 (for example, the middleware 143) may include, for example, at least one of a runtime library 335, an application manager 341, a window manager 342, a multimedia manager 343, a resource manager 344, a power manager 345, a database manager 346, a package manager 347, a connectivity manager 348, a notification manager 349, a location manager 350, a graphic manager 351, and a security manager 352.

[0094] The runtime library 335 may include, for example, a library module that a compiler uses to add new functions through a programming language while the application 370 is executed. The runtime library 335 may perform I/O management, memory management, or a function for an arithmetic function.

[0095] The application manager 341 may manage, for example, a life cycle of at least one of the applications 370. The window manager 342 may manage graphical user interface (GUI) resources used by a screen. The multimedia manager 343 may

grasp formats used for the reproduction of various media files, and may perform an encoding or decoding of the media file by using a codec suitable for the corresponding format. The resource manager 344 may manage resources such as a source code, a memory, and a storage space of at least one of the applications 370.

[0096] The power manager 345 may operate together with a basic input/output system (BIOS) to manage a battery or power and may provide power information used for the operation of the electronic device. The database manager 346 may generate, search for, or change a database to be used by at least one of the applications 370. The package manager 347 may manage the installation or the updating of applications distributed in the form of package file.

[0097] The connectivity manager 348 may manage wireless connection of, for example, Wi-Fi or BT. The notification manager 349 can display or notify of an event such as an arrival message, promise, proximity notification, and the like in such a way that does not disturb a user. The location manager 350 may manage location information of the electronic device. The graphic manager 351 may manage graphic effects to be provided to a user and user interfaces related to the graphic effects. The security manager 352 may provide one or more security functions used for system security or user authentication. According to an embodiment, when the electronic device (for example, electronic device 101) has a call function, the middleware 330 may further include a telephony manager for managing a voice call function or a video call function of the electronic device.

[0098] The middleware 330 may include a middleware module for forming a combination of various functions of the aforementioned components. The middleware 330 may provide modules specialized according to types of operating systems in order to provide differentiated functions. Further, the middleware 330 may dynamically remove some of the existing components or add new components.

[0099] The API 360 (for example, the API 145) is, for example, a set of API programming functions, and a different configuration thereof may be provided according to an operating system. For example, Android or iOS may provide one API set per platform, and Tizen may provide two or more API sets per platform.

[00100] The applications 370 (for example, the application 147) may include, for example, one or more applications which can provide functions such as home 371, dialer 372, short message service (SMS)/multimedia message service (MMS) 373, instant message (IM) 374, browser 375, camera 376, alarm 377, contacts 378, voice dialer 379, email 380, calendar 381, media player 382, album 383, clock 384, health care (for example, measure exercise quantity or blood sugar), or environment information (for example, atmospheric pressure, humidity, or temperature information).

[00101] According to an embodiment, the applications 370 may include an application (hereinafter, referred to as an “information exchange application” for convenience of the description) supporting information exchange between the electronic device (for example, the electronic device 101) and an external electronic device. The information exchange application may include, for example, a notification relay application for transferring predetermined information to an external electronic device or a device management application for managing an external electronic device.

[00102] For example, the notification relay application may include a function of transferring, to the external electronic device, notification information generated from other applications of the electronic device 101 (for example, an SMS/MMS application, an e-mail application, a health management application, or an environmental information application). Further, the notification relay application may receive notification information from, for example, a control device and provide the received notification information to the user. The device management application may manage (for example, install, delete, or update), for example, a function for at least a part of the external electronic device communicating with the electronic device (for example, turning on/off the external electronic device itself (or some elements thereof) or adjusting brightness (or resolution) of a display), applications executed in the external electronic device, or services provided from the external electronic device (for example, a telephone call service or a message service).

[00103] According to an embodiment, the applications 370 may include an application (for example, health management application) designated according to attributes of the external electronic device (for example, attributes of the electronic device such as the type of electronic device which corresponds to a mobile medical device). According to an embodiment, the applications 370 may include an application received from the external electronic devices (for example, the server or the electronic device). According to an embodiment, the applications 370 may include a preloaded application or a third party application which can be downloaded from the server. The names of the components of the program module 310 according to the embodiment illustrated in FIG. 3 may vary according to the type of operating system.

[00104] According to various embodiments, at least some of the program module 310 may be implemented by software, firmware, hardware, or a combination of two or more thereof. At least some of the program module 310 may be implemented (for example, executed) by, for example, the processor (for example, the application program). At least some of the program module 310 may include, for example, a module, program, routine, sets of instructions, or process for performing one or more functions.

[00105] FIGS. 4A, 4B, and 4C are views illustrating an electronic device according to various embodiments of the present disclosure.

[00106] A three-dimensional X-Y-Z Cartesian coordinate system is illustrated in FIGS. 4A, 4B, and 4C, where the 'Z-axis' may denote the vertical direction (thickness direction) of the main body of the electronic device; the 'X-axis' may denote the lateral direction of the main body; and the 'Y-axis' may denote the longitudinal direction of the main body.

[00107] FIG. 4A is a perspective view of the electronic device according to various embodiments of the present disclosure.

[00108] Referring to FIG. 4A, the electronic device 400 (e.g., the electronic device 101 of FIG. 1) may be a wrist-mounted wearable electronic device that can be worn on a user's wrist. However, the various embodiments of the present disclosure

are not limited thereto, and the electronic device may also be one of a bracelet, a strip, a band, an attachment type (band-aid type) band, a belt, ear-mounted earphones, headphones, a clothing type electronic device, a shoe type electronic device, an HMD, a hat type electronic device, a glove type electronic device, a finger-tip type electronic device, a clip type electronic device, an arm band type electronic device, a contact lens device, digital clothing, and a remote controller.

[00109] According to various embodiments, the electronic device 400 may include the main body 410 and connection parts 420 (e.g., straps) that are equipped to the main body 410.

[00110] According to various embodiments, the electronic device 400 may have a battery (e.g., a rechargeable battery, etc.) therein as a power supply means. For example, the electronic device 400 may be implemented to be selectively mounted on a portable charging cradle in order to recharge the battery.

[00111] According to various embodiments, the main body 410 may include a display 415, at least one key button 430 (e.g., a side key K, a touch sensor, or a pressure sensor), a sensor module (e.g., a biometric sensor, etc.), and/or the like. According to various embodiments, the display 415 may include a touch screen to receive a touch input. According to an embodiment, the main body 410 may include the front surface F and the rear surface R which makes contact with the user's body while the main body 410 is worn on the user's body. For example, the display 415 may be disposed on the front surface F of the main body 410, and the sensor module may be disposed on the rear surface R of the main body 410. However, the various embodiments of the present disclosure are not limited thereto, and the display 415 or the sensor module may be disposed in various regions of the electronic device 400. For example, the sensor module may also be disposed on the connection parts 420.

[00112] According to various embodiments, the main body 410 may have a bar shape and may at least partially have a curvature that corresponds to the user's body. For example, the main body 410 may have a curved rectangular shape extending in the longitudinal direction (Y-axis direction). According to an embodiment, the main

body 410 may have fastening recesses formed on opposite sides thereof with which the connection parts 420 are engaged.

[00113] According to various embodiments, the connection parts 420 may be configured to be coupled to, or separated from, the main body 410. According to various embodiments, the connection parts 420 may be formed of a resilient material. Accordingly, it is possible to bring the main body 410 close to the skin of the user's body. According to an embodiment, the connection parts 420 may have a changeable structure and may be implemented in various designs or colors. Accordingly, the user may change the connection parts 420 according to his/her preference.

[00114] According to various embodiments, the connection parts 420 may be coupled to the main body 410 and may include a fastener 422 that enables the connection parts 420 to be detachably attached to a part of the user's body. According to an embodiment, the electronic device 400 may be worn on the wrist by a method of tying and fastening the connection parts 420 together while the electronic device 400 is placed on the wrist. For example, one of the connection parts 420 may have a plurality of openings for wrist adjustment that are formed therein at a constant interval, and the other connection part 420 may include the fastener 422 that is coupled through one of the openings for wrist adjustment.

[00115] According to various embodiments, the main body 410 may have a rotary body (not illustrated) (e.g., a rotary input module) mounted thereon for adjusting various user interface environments that are displayed on the display 415. According to various embodiments, the rotary body may have a protrusion shape similar to a crown and may be rotatably disposed on the side surface of the main body 410. According to various embodiments, in a case where the main body 410 has a circular shape, the rotary body may be rotatably mounted on the top, bottom, or side surface of the main body 410. For example, the rotary body may be disposed on a bezel that surrounds the edge of the display 415 that is disposed on the top of the main body 410.

[00116] FIG. 4B is a perspective view of the main body of the electronic device 400 according to various embodiments of the present disclosure, and FIG. 4C is a

perspective view of the main body of the electronic device 400 according to various embodiments of the present disclosure, when viewed in a different direction.

[00117] Referring to FIGS. 4B and 4C, the main body may include a main body housing 440 and a display 450 mounted on the main body housing 440. According to various embodiments, the main body housing 440 may have the front surface F, the rear surface R, and the side surface that connects the front surface F and the rear surface R. However, the various embodiments of the present disclosure are not limited thereto, and the main body housing 440 may be constituted by the front surface F and the rear surface R. Furthermore, the front surface F, the rear surface R, and the side surface of the main body housing 440 may be integrally formed with each other. According to an embodiment, the front surface F and the rear surface R may be configured to have different curvatures.

[00118] According to various embodiments, the main body housing 440 may further include a key button (e.g., the key button 430, the side key K, a touch sensor, or a pressure sensor) for inputting various types of information. According to an embodiment, the front surface F may be a surface on which the display 450 is mounted, and the rear surface R may be a wearing surface that makes contact with the user's body. However, the various embodiments of the present disclosure are not limited thereto, and the display 415 or the sensor module may be disposed in various regions of the electronic device 400.

[00119] According to various embodiments, the display 450 may be disposed on the front surface F of the main body housing 440. Accordingly, it may be easy for the user to view a screen displayed through the electronic device. The display 450 is illustrated in a shape that reflects the curved surface of the user's body, but may be constituted by a flat panel display (e.g., an LCD, an OLED, etc.), a curved display, or a flexible display. For example, in the specific embodiments of the present disclosure, the main body 410 is illustrated as including a curved display, but the main body 410 may include a flat panel display or a flexible display.

[00120] Referring to FIG. 4C, a sensor module 470 (e.g., a biometric sensor) may be disposed on the rear surface R of the main body housing 440 according to an

embodiment. For example, the sensor module may have a shape that is brought close to the user's wrist. Accordingly, it is possible to enhance the comfort that the user experiences when wearing the electronic device.

[00121] The sensor module 470 (e.g., the sensor module 240 or the biometric sensor) provided in the main body housing 440 may include at least one of an acceleration sensor, a gyro sensor, a terrestrial magnetism sensor, a heart rate sensor, a proximity sensor, an optical sensor, a galvanic skin response (GSR) sensor, an ECG sensor, an EMG sensor, a blood glucose sensor, a blood pressure sensor, a pressure sensor, and a temperature sensor. In addition, the sensor module may also include other sensors that detect a sensing value for determining whether the electronic device 400 is worn. According to various embodiments, the sensor module 470 may be provided on the rear surface R of the main body housing 440 in a single module shape that includes: a sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor) for measuring the motion of the electronic device 400; a sensor (e.g., a proximity sensor or a pressure sensor) that is used to determine whether the electronic device 400 is worn on a body; and a biometric sensor (e.g., a heart rate sensor, a blood pressure sensor, or a body temperature sensor) for measuring a biometric signal.

[00122] According to various embodiments, connection members 460 (e.g., a charging terminal, the interface 270, etc.) may be disposed on the rear surface R of the main body housing 440.

[00123] FIG. 5A is a diagram illustrating the configuration of a processor 500 according to various embodiments of the present disclosure. According to various embodiments, the processor may be a processor (e.g., the processor 120) of an electronic device (e.g., the electronic device 101 or 400).

[00124] Referring to FIG. 5A, the processor 500 may include a device state determination module 502, a parameter change module 504, a user state determination module 505, a measurement module 506, and a storage module 508.

[00125] According to various embodiments, the device state determination module 502 may determine the state of the electronic device (e.g., a wearing or non-wearing state, a proximity or non-proximity state, or an in-use (holding) or not-in-use

(non-holding) state). According to various embodiments, the device state determination module 502 may determine the state of the electronic device based on at least one sensor. According to an embodiment, the device state determination module 502 may determine the state of the electronic device based on information acquired through at least one of an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, an atmospheric pressure sensor (or a pressure sensor), and a temperature sensor.

[00126] For example, when the motion of the electronic device is detected through, for example, an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor, the device state determination module 502 may determine that the electronic device has been worn on a user's body. Furthermore, when the motion of the electronic device is not detected through, for example, an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor, the device state determination module 502 may determine that the electronic device has not been worn on the user's body.

[00127] For example, when an object (e.g., a user) is detected within a selected (or specified) distance (e.g., about 10 cm) from the electronic device through, for example, a proximity sensor, the device state determination module 502 may determine that the electronic device is close to the object (e.g., the electronic device is used or held by the user). Further, when an object (e.g., a user) is not detected within a selected (or specified) distance from the electronic device through, for example, a proximity sensor, the device state determination module 502 may determine that the electronic device is not close to the object (e.g., the electronic device is not used or held by the user).

[00128] For example, when a selected (or specified) pressure or a selected (or specified) temperature (e.g., about 37 degrees Celsius) is detected through, for example, a pressure sensor or a temperature sensor, the device state determination module 502 may determine that the electronic device is used or held by a user. Further, when a selected pressure or a selected temperature is not detected through a pressure sensor or a temperature sensor, the device state determination module 502 may determine that the electronic device is not used or held by the user.

[00129] According to various embodiments, the device state determination module 502 may determine the state of the electronic device based on whether the electronic device is electrically or mechanically connected or not. For example, in a case where the electronic device is a pair of smart eyeglasses, a switch circuit included in the smart eyeglass temples may be connected when the smart eyeglasses are worn on a user's head. Accordingly, the device state determination module 502 may determine the state of the electronic device to be a wearing state. Further, in a case where the smart eyeglasses are not worn on the user's head, the switch circuit included in the smart eyeglass temples may not be connected. Accordingly, the device state determination module 502 may determine the state of the electronic device to be a non-wearing state.

[00130] For example, in a case where the electronic device is a smart watch, the device state determination module 502 may determine the state of the electronic device to be a wearing state when the buckle of the smart watch is connected. Further, the device state determination module 502 may determine the state of the electronic device to be a non-wearing state when the buckle of the smart watch is not connected.

[00131] According to various embodiments, the user state determination module 505 may determine a user's state (e.g., a sleep state). According to an embodiment, the user state determination module 505 may determine the user's state based on the user's pattern. For example, the user's pattern may include the user's motion pattern, a life pattern relating to a specified schedule, etc.

[00132] According to various embodiments, the user state determination module 505 may determine, for example, the user's motion, the strength of the motion, etc. when biometric information is measured. For example, the user state determination module 505 may determine the user's state to be a sleep state when the user's motion strength is lower than a selected (or specified) strength. Further, the user state determination module 505 may determine the user's state to be an active state (e.g., a state in which the user does not sleep) when the user's motion strength is higher than, or equal to, the selected (or specified) strength. According to an embodiment, the user state determination module 505 may provide information on the user's motion to the

measurement module 506 or the parameter change module 504 in order to change a biometric information measurement method.

[00133] According to various embodiments, the user state determination module 505 may determine the user's state based on schedule information stored in the electronic device. For example, a first time (e.g., from about 11:00 P.M. to about 8:00 A.M.) may be specified as a 'sleeping time,' and a second time may be specified as an 'active time' (e.g., from about 8:00 A.M. to about 11:00 P.M.) in the schedule stored in the electronic device. In this case, the user state determination module 505 may determine the user's state at the first time to be a 'sleep state' based on the schedule. Further, the user state determination module 505 may determine the user's state at the second time to be an 'active state' based on the schedule.

[00134] According to various embodiments, the measurement module 506 may measure a user's biometric information. According to various embodiments, the measurement module 506 may measure biometric information based on at least one sensor. According to an embodiment, the measurement module 506 may measure at least one of a blood pressure, a heart rate, an ECG, skin moisture, saturation of peripheral oxygen, a body temperature, blood glucose, an EEG, an EMG, and the like based on a sensor that is brought into contact with at least a part of a body. According to various embodiments, the measurement module 506 may be activated when the state of the electronic device is determined to be a selected (or specified) state (e.g., a wearing state) through the device state determination module 502.

[00135] According to various embodiments, the measurement module 506 may perform a measurement operation when a biometric information measurement period (e.g., about 1 minute) arrives. According to various embodiments, the measurement module 506 may perform the measurement operation based on the motion of the electronic device or a user when the biometric information measurement period arrives. According to an embodiment, when the measurement period arrives, the measurement module 506 may identify the motion of the user or the electronic device that was measured before the period arrives, and when the motion satisfies a selected (or specified) condition (e.g., when the strength of the motion is lower than a selected

(or specified) strength), the measurement module 506 may measure biometric information. According to another embodiment, when the motion of the electronic device or the user does not satisfy the selected (or specified) condition (e.g., when the strength of the motion is higher than the selected (or specified) strength), the measurement module 506 may monitor a change in the motion for a predetermined period of time (e.g., about 10 seconds) with respect to the time when the period arrives. According to an embodiment, when the change in the motion for the selected (or specified) period of time satisfies the selected (or specified) condition (e.g., when the average of motion data is smaller than a selected (or specified) value), the measurement module 506 may measure biometric information.

[00136] According to various embodiments, the parameter change module 504 may change a measurement parameter used for measuring biometric information. According to an embodiment, the parameter change module 504 may change the light-emitting time, the biometric information measurement period, the sampling rate, the biometric information measurement trigger, the motion determination condition, etc. of at least one sensor. Accordingly, it is possible to reduce power consumption that is generated when the electronic device measures biometric information, or to enhance the accuracy of measurement of biometric information. Further, according to an embodiment, when biometric information is measured, the parameter change module 504 may change a measurement parameter based on situation information that is associated with the motion strength of the electronic device or a body, the physical characteristic or state of an object to be measured, a measurement quality, a measurement result, or the like while the biometric information is being measured.

[00137] According to various embodiments, the storage module 508 may store at least one program for the operation of the electronic device. The storage module 508 may store information for the operation of at least one of the device state determination module 502, the parameter change module 504, the user state determination module 505, and the measurement module 506. According to various embodiments, the device state determination module 502, the parameter change

module 504, the user state determination module 505, the measurement module 506, or the storage module 508 may be at least one software element.

[00138] According to various embodiments, the processor 500 may include at least one of the device state determination module 502, the parameter change module 504, the user state determination module 505, the measurement module 506, and the storage module 508 in order to perform the above-described measurement operations.

[00139] Although not illustrated, according to various embodiments, the device state determination module 502, the parameter change module 504, the user state determination module 505, the measurement module 506, and the storage module 508 may be included in a plurality of processors (or chips). For example, the device state determination module 502 and the user state determination module 505 may be included in a first processor (or a first chip). The parameter change module 504 may be included in a second processor (or a second chip). Further, the measurement module 506 may be included in a third processor (or a third chip).

[00140] According to various embodiments, the electronic device may perform the above-described operation of measuring biometric information through a separate module that is different from the processor 500.

[00141] FIG. 5B is a view illustrating an operation of measuring biometric information by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the electronic device 400, or the processor 500), according to various embodiments of the present disclosure.

[00142] Referring to FIG. 5B, the electronic device (e.g., the processor 500) may attempt to measure biometric information according to a period (e.g., about 1 minute) having a selected (or specified) period of time. According to an embodiment, the electronic device (e.g., the processor 500) may operate a first timer that represents a selected (or specified) period of time in order to determine when to attempt to measure biometric information. For example, the electronic device (e.g., the processor 500) may attempt to measure biometric information at every 1 minute by operating the first timer.

[00143] According to various embodiments, the electronic device (e.g., the processor 500) may measure the motion of the electronic device in at least one time interval 522 before the operation of the first timer ends (520) (e.g., for about 10 seconds before the operation of the first timer ends), and when the operation of the first timer ends, the electronic device (e.g., the processor 500) may determine whether at least a part of the measured motion satisfies a selected (or specified) reference. For example, the electronic device (e.g., the processor 120 or 500) may measure the average of the motion that is acquired in at least one time interval 522 before the first timer ends, and may determine whether the measured average of the motion is less than, or equal to, a selected (or specified) threshold value.

[00144] According to various embodiments, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the measured motion of the electronic device satisfies a selected (or specified) reference.

[00145] According to various embodiments, when the measured motion of the electronic device does not satisfy the selected (or specified) reference, the electronic device (e.g., the processor 500) may determine whether the motion of the electronic device is changed to satisfy the selected (or specified) reference for the time 524 (e.g., about 30 seconds) during which a second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may not measure biometric information when the previously measured motion (e.g., the average of the motion) is maintained at a threshold value or more. According to another embodiment, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the previously measured motion (e.g., the average of the motion) is changed to the threshold value or less. According to yet another embodiment, the electronic device (e.g., the processor 500) may additionally measure the motion of the electronic device for the time 524 during which the second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may additionally measure the motion of a selected (or specified) time interval (e.g., a time interval that has a

magnitude that is the same as, or similar to, that of the time interval 522 in which the average of the motion has been measured) for the time 524 during which the second timer operates, and may determine whether the additionally measured motion satisfies a selected (or specified) reference. For example, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the additionally measured motion satisfies the selected (or specified) reference.

[00146] According to various embodiments, the electronic device (e.g., the processor 500) may change the time 524 during which the second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may increase the time 524 during which the average of the previously measured motion is identified, and may determine whether the measurement of biometric information is possible for the increased time. According to an embodiment, based on the motion (e.g., the strength of the motion) of the electronic device, the electronic device (e.g., the processor 500) may change the time 524 during which the second timer operates. For example, the electronic device (e.g., the processor 500) may increase or decrease the operating time 524 of the second timer based on the previously measure motion (e.g., the average of the motion). For example, when the motion (e.g., the average of the motion) of the electronic device decreases or closely approaches a threshold value, the electronic device (e.g., the processor 500) may increase the operating time 524 of the second timer. For example, when the motion (e.g., the average of the motion) of the electronic device increases or does not closely approach the threshold value, the electronic device (e.g., the processor 500) may decrease the operating time 524 of the second timer.

[00147] An electronic device, according to various embodiments, may include: a housing; a fastener coupled to a part of the housing and detachably attached to a part of a user's body; at least one first sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, or a combination thereof) configured to detect the motion of the electronic device; at least one second sensor (e.g., a heart rate sensor) configured to detect a change in the user's body; a memory;

and a processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first sensor; acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

[00148] According to various embodiments, the determining of whether the electronic device is worn on the user's body part may include determining liveness of the body. According to an embodiment, the determining of the liveness of the body may be performed using at least one of a heart rate sensor, a blood pressure sensor, a body temperature sensor, a skin moisture sensor, an oxygen saturation sensor, a blood glucose sensor, an ECG sensor, an EEG sensor, and an EMG sensor. According to an embodiment, even though the determining of whether the electronic device is worn on (or close to) the user's body part is performed using the at least one first sensor, when it is determined that a signal received from a heart rate sensor is not a signal relating to the body, the electronic device may be determined to not be worn on the user's body part. According to an embodiment, even though the determining of whether the electronic device is worn on (or close to) the user's body part is performed using the at least one first sensor, when it is determined that a signal received from a body temperature sensor is not within a range suitable for the body, the electronic device may be determined to not be worn on the user's body part.

[00149] According to various embodiments, the second period of time may be shorter than the first period of time.

[00150] According to various embodiments, the at least one first sensor may include a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor.

[00151] According to various embodiments, the at least one second sensor may include a biometric sensor. According to an embodiment, the biometric sensor may include at least one of a heart rate sensor, a blood pressure sensor, an ECG sensor, a skin moisture sensor, and an oxygen saturation sensor.

[00152] According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

[00153] According to various embodiments, at least one of the second period of time and the third period of time may be set to include the time point when the first period of time passes.

[00154] According to various embodiments, the second period of time may be set to include the time point when the first period of time passes.

[00155] According to various embodiments, when data is not received from the activated (operated or triggered) second sensor within a selected (or specified) time, the instructions may be set to configure the processor to activate (operate or trigger) the second sensor again after the selected (or specified) time passes.

[00156] According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a period of time that is substantially the same as the second period of time before the selected (or specified) time passes; and activate (operate or trigger) the at least one second sensor again based on at least a part of the additional information.

[00157] According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change the first period of time and/or the

second period of time and/or the third period of time based on at least a part of the acquired data.

[00158] According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00159] According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change at least one of a period of time and a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00160] According to various embodiments, an electronic device may include a memory and a processor electrically connected to the memory. According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of a user's body using at least one first sensor; activate (operate or trigger) at least one second sensor based on at least a part of the motion of the electronic device within a selected (specified) time range at the time when a selected (or specified) period of time passes in a case where the electronic device is worn on (or close to) a part of the user's body; and measure the user's biometric information through the at least one second sensor.

[00161] According to various embodiments, the at least one first sensor may include at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, and an acceleration sensor.

[00162] According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a selected (or specified) time from the time when the period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional

information, without activating (operating or triggering) the second sensor when at least a part of the motion does not satisfy a selected (or specified) condition.

[00163] According to various embodiments, the instructions may be set to configure the processor to change the selected (or specified) time for acquiring the additional information based on data acquired from at least one of the at least one first sensor and the at least one second sensor.

[00164] According to various embodiments, the instructions may be set to configure the processor to: accumulate the number of failures when the measurement of the biometric information fails; and deactivate the second sensor when the cumulative number of failures satisfies a selected (or specified) condition.

[00165] According to various embodiments, the instructions may be set to configure the processor to acquire data from at least one of the at least one first sensor and the at least one second sensor and activate (operate or trigger) the at least one second sensor based on at least a part of the data.

[00166] According to various embodiments, the instructions may be set to configure the processor to acquire data from at least one of the at least one first sensor and the at least one second sensor and change at least one of a period of time or a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00167] According to various embodiments, an electronic device may include a memory and a processor electrically connected to the memory. According to an embodiment, the memory may be set to configure the processor to: determine whether the electronic device is close to a user's body; acquire information on the motion of the electronic device using at least one first sensor within a selected (or specified) time range from the time when a selected (or specified) period passes in a case where the electronic device is close to the user's body part; and acquire the user's biometric information through at least one second sensor based on at least a part of the information on the motion.

[00168] FIG. 6 is a flowchart illustrating a biometric information measurement operation of an electronic device (e.g., the electronic device 101, the processor 120 of

the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00169] Referring to FIG. 6, in operation 601 the electronic device (e.g., the processor 120 or 500 or the device state determination module 502) may determine the state (e.g., wearing state, proximity state, in-use state, or holding state) of the electronic device based on a first sensor. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may identify the state of the electronic device relating to wearing (or proximity) based on at least one first sensor. According to an embodiment, the state of the electronic device relating to wearing may be a state in which the electronic device makes contact with at least a part of a body, a state in which the electronic device accesses at least a part of a body, etc. According to various embodiments, the first sensor may be constituted by at least one of an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, an atmospheric pressure sensor, and a temperature sensor.

[00170] In operation 603, the electronic device (e.g., the processor 120 or 500) may identify the state (e.g., wearing state or non-wearing state) of the electronic device relating to wearing through a state determination operation.

[00171] When it is identified in operation 603 that the state of the electronic device relating to wearing does not correspond to a selected (or specified) state (e.g., when the state of the electronic device corresponds to a non-wearing state), the electronic device (e.g., the processor 120 or 500) may perform the operation of determining the state of the electronic device in operation 601.

[00172] When it is identified in operation 603 that the state of the electronic device relating to wearing corresponds to the selected (or specified) state (e.g., when the state of the electronic device corresponds to a wearing state), the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor (e.g., the sensor module 470 or the biometric sensor 240I) in operation 605. According to various embodiments, at least one second sensor may be a sensor for measuring biometric information, and may measure at least one of a blood pressure, a heart rate, an ECG, skin moisture, saturation of peripheral oxygen, a body temperature, blood

glucose, an EEG, an EMG, and the like. For example, the second sensor may be disposed on one surface of the electronic device so as to be brought into contact with a body.

[00173] According to various embodiments, the electronic device may operate at least one second sensor based on at least one of when a selected (or specified) period of time has elapsed, when the motion of the electronic device that is acquired using at least a part of at least one first sensor satisfies a selected (or specified) reference, and when a selected (or specified) time arrives. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may operate a first timer that represents a selected (or specified) period of time and may operate at least one second sensor when the first timer ends. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may identify the motion of the electronic device and may operate at least one second sensor when the motion satisfies a selected (or specified) reference. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may identify whether a selected (or specified) time (e.g., about 6:11 in a case where a wearing determination time is about 6:10) arrives and may operate at least one second sensor.

[00174] In operation 607, the electronic device (e.g., the processor 120 or 500 or the measurement module 506) may measure biometric information based on the second sensor. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may output the biometric information measured based on the second sensor on a screen thereof. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may provide a health care function (e.g., providing a heart rate or a work rate) based on the measured biometric information.

[00175] FIG. 7 is a flowchart illustrating a performance procedure of a method of determining the state of an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00176] Referring to FIG. 7, the performance procedure of the state determination method may be a detailed operation for the operation 601 illustrated in FIG. 6.

[00177] In operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of an external object (e.g., a user) to the electronic device is detected based on at least one first sensor (e.g., proximity sensor). According to various embodiments, the proximity may be associated with a state in which the electronic device is close to a body, a state in which the electronic device makes contact with a body, etc. For example, the electronic device (e.g., the processor 120 or 500) may determine whether the electronic device is close to a body based on information acquired through the proximity sensor, which is the first sensor.

[00178] When the proximity of the user (e.g., body) to the electronic device is detected in operation 701, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor) in operation 703. According to various embodiments, the motion may be associated with the motion of the electronic device or user. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the motion based on information acquired through at least one of an acceleration sensor, a terrestrial magnetism sensor, and a gyro sensor in response to the detection of the proximity.

[00179] In operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether a motion that satisfies a selected (or specified) condition is detected. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may identify whether a continuous motion is detected for a selected (or specified) period of time while the electronic device is close to a body. For example, when the continuous motion is detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine that the motion that satisfies the condition is detected.

[00180] When the motion that satisfies the selected (or specified) condition is detected in operation 705, the electronic device (e.g., the processor 120 or 500) may

determine the state of the electronic device to be a state relating to wearing in operation 707.

[00181] When the motion that satisfies the selected (or specified) condition is not detected in operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of a body to the electronic device is detected based on at least one first sensor (e.g., a proximity sensor) in operation 709. For example, the electronic device may determine whether the electronic device is close to the body based on information acquired through the proximity sensor, which is the first sensor.

[00182] When the proximity of the body to the electronic device is detected in operation 709, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing (e.g., wearing state) in operation 707. In one embodiment, when the electronic device makes contact with, or is close to, a body in a motionless state (e.g., sleep state), the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing.

[00183] When the proximity of a body to the electronic device is not detected in operation 709, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713.

[00184] When the proximity of an external object to the electronic device is not detected in operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected in operation 711. According to various embodiments, in the state in which no proximity is detected, the electronic device (e.g., the processor 120 or 500) may determine whether to perform the operation of determining the detection of proximity again based on the motion of the electronic device. For example, although no proximity is detected, the electronic device (e.g., the processor 120 or 500) may determine that proximity is likely to be detected when the motion of the electronic device is detected.

[00185] When the motion of the electronic device is detected in operation 711, the electronic device (e.g., the processor 120 or 500) may determine whether the electronic device is close to a body based on at least one first sensor in operation 701.

[00186] When the motion of the electronic device is not detected in operation 711, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713. For example, the electronic device (e.g., the processor 120 or 500) may determine that the electronic device is unlikely to access a body and may determine the state of the electronic device to be a state not relating to wearing.

[00187] According to various embodiments, at least one of the operations 701 to 713 may be omitted. For example, in operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of a user to the electronic device is detected. When the proximity of the user to the electronic device is detected, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing (e.g., wearing state) in operation 707. Furthermore, when the proximity of the user to the electronic device is not detected, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713.

[00188] According to various embodiments, in operation 703, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device. In operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether the motion (e.g., continuous motion) is detected for a specified (or selected) period of time. When the motion (continuous motion) is detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state (e.g., wearing state) relating to wearing (or a state in which the user holds or uses the electronic device) in operation 707. When the continuous motion is not detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state (non-wearing state) not

relating to wearing (or a state in which the user does not hold or use the electronic device) in operation 713.

[00189] FIG. 8 is a flowchart illustrating the performance procedure of a method of operating a second sensor by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00190] Referring to FIG. 8, the performance procedure of the method of operating the second sensor may be a detailed operation for the operation 605 illustrated in FIG. 6.

[00191] In operation 801, the electronic device (e.g., the processor 120 or 500) may operate a first timer that operates for a selected (or specified) period of time. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information based on the operation of the first timer. According to an embodiment, the first timer may be time on the basis of which when to attempt to measure biometric information is determined. For example, the first timer may operate in about 1-minute periods.

[00192] In operation 803, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device in response to the operation of the timer. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may detect the motion of the electronic device based on at least one first sensor (e.g., an acceleration sensor or a gyro sensor). According to various embodiments, based on the operation of the first timer, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device until the first timer ends.

[00193] In operation 805, the electronic device (e.g., the processor 120 or 500) may identify whether the first timer ends. The time when the first timer ends may be, for example, the time when whether to perform an operation of measuring biometric information is determined. The operation of measuring the biometric information may be, for example, an operation of driving at least one second sensor for measuring biometric information.

[00194] When the first timer does not end in operation 805, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device in operation 803. For example, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device until the first timer ends.

[00195] When the first timer ends in operation 805, the electronic device (e.g., the processor 120 or 500) may determine whether the motion of the electronic device, which is measured during the operation of the first timer, satisfies a selected (or specified) reference in operation 807. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine whether a motion by which biometric information can be accurately measured is generated. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may determine whether at least a part of the motion measured during the operation of the first timer satisfies a selected (or specified) reference. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the average of the motion measured during the operation of the first timer and may determine whether the measured average of the motion satisfies a selected (or specified) reference. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may measure the average of the motion acquired in at least one time interval before the first timer ends and may determine whether the measured average of the motion satisfies a selected (or specified) reference. According to various embodiments, the situation in which the selected (or specified) condition is satisfied may be a situation in which the motion of the electronic device that is less than, or equal to, a selected (or specified) threshold value is measured. For example, the situation in which the motion of the electronic device that is less than, or equal to, the selected (or specified) threshold value is measured may be a state in which the motion of the electronic device is not generated, or a state in which the motion of the electronic device is so small that it is recognized as not being generated.

[00196] When the motion of the electronic device, which is measured during the operation of the first timer, satisfies the selected (or specified) reference in operation

807, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor for measuring biometric information in operation 809.

[00197] When the motion of the electronic device, which is measured during the operation of the first timer, does not satisfy the selected (or specified) reference in operation 807, the electronic device (e.g., the processor 120 or 500) may operate a second timer that operates for a selected (or specified) period of time in operation 811. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate the second timer in order to determine whether the measured motion of the electronic device that does not satisfy the reference is changed to satisfy the selected (or specified) reference. The operating time of the second timer may be shorter than that of the first timer. For example, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device changes until the next period for the measurement of biometric information arrives.

[00198] In operation 813, the electronic device (e.g., the processor 120 or 500) may measure a change in the motion of the electronic device.

[00199] In operation 815, the electronic device (e.g., the processor 120 or 500) may identify whether the measured change in the motion satisfies a selected (or specified) reference. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine whether the measured motion (e.g., the motion of the electronic device that does not satisfy the reference) is changed into a motion by which biometric information can be accurately measured. For example, the electronic device (e.g., the processor 120 or 500) may identify whether the average value of the motion that is measured when the first timer ends is changed to a selected (or specified) threshold value or less.

[00200] When the change in the motion satisfies the selected (or specified) reference in operation 815, the electronic device (e.g., the processor 120 or 500) may measure biometric information. For example, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor for measuring biometric information in operation 809.

[00201] When the change in the motion does not satisfy the selected (or specified) reference in operation 815, the electronic device (e.g., the processor 120 or 500) may identify whether the second time ends in operation 817.

[00202] When the second timer does not end in operation 817, the electronic device (e.g., the processor 120 or 500) may identify whether the change in the motion of the electronic device (e.g., the processor 120 or 500) satisfies the selected (or specified) reference again while the second timer operates. For example, the electronic device (e.g., the processor 120 or 500) may measure a change in the motion of the electronic device in operation 813.

[00203] When the second timer ends in operation 817, the electronic device (e.g., the processor 120 or 500) may determine when to operate the first timer in operation 819. For example, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device relating to wearing. In another example, as the electronic device (e.g., the processor 120 or 500) determines the state (e.g., wearing state) of the electronic device relating to wearing, the electronic device may omit the operation of determining the state of the electronic device and may operate the first timer.

[00204] According to various embodiments, at least one of the operations 801 to 819 may be omitted. When the motion of the electronic device 101, which is measured during the operation of the first timer, satisfies the selected (or specified) reference in operation 807, the electronic device 101 (e.g., the processor 120 or 500) may operate at least one second sensor for measuring biometric information in operation 809. When the motion of the electronic device 101, which is measured during the operation of the first timer, does not satisfy the selected (or specified) reference in operation 807, the electronic device (e.g., the processor 120 or 500) may determine when to operate the first timer in operation 819. For example, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device relating to wearing. In another example, as the electronic device (e.g., the processor 120 or 500) determines the state (e.g., wearing state) of the

electronic device relating to wearing, the electronic device may omit the operation of determining the state of the electronic device and may operate the first timer.

[00205] FIG. 9 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00206] Referring to FIG. 9, the performance procedure of the method of measuring biometric information may be a detailed operation for the operation 607 illustrated in FIG. 6.

[00207] In operation 901, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information. According to various embodiments, the measurement of biometric information may be performed by operating at least one second sensor. According to various embodiments, the second sensor may include a light emitting part that emits light and a light receiving part that receives the light of the light emitting part, and the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a signal output from the second sensor. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information (e.g., a heart rate) based on the amount of light output through at least one second sensor that makes contact with at least a part of a body. For example, slightly dark light may be received for a systolic period of the heart cycle due to a large amount of blood in a body part, and bright light may be received for a diastolic period of the heart cycle due to a relatively small amount of blood in the body part. The electronic device (e.g., the processor 120 or 500) may measure heart rate information based on the amount of light.

[00208] In operation 903, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information is successfully measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained based on the time devoted to measuring the biometric

information. For example, when a measurement result is not obtained within a selected (or specified) time after the measurement of the biometric information starts, the electronic device (e.g., the processor 120 or 500) may determine that the measurement of the biometric information has failed. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained based on the motion of the electronic device. For example, when a motion that does not satisfy a selected (or specified) condition is detected after the measurement of the biometric information starts, the electronic device (e.g., the processor 120 or 500) may determine that the measurement of the biometric information has failed.

[00209] When the biometric information is successfully measured in operation 903, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information in operation 905. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may display the measurement result on a display thereof (e.g., a screen) based on the second sensor. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may provide a health care function (e.g., providing whether a heart rate is normal or not) based on the measurement result.

[00210] When no biometric information is measured in operation 903, the electronic device (e.g., the processor 120 or 500) may accumulate the number of failures in operation 907. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may accumulate the number of failures every time a normal result is not obtained after the electronic device attempts to measure biometric information. For example, when a normal measurement result on biometric information is not obtained once, the electronic device (e.g., the processor 120 or 500) may determine the number of failures to be one in operation 907. Furthermore, when a normal measurement result on biometric information is not obtained three times, the electronic device (e.g., the processor 120 or 500) may determine the number of failures to be three in operation 907.

[00211] In operation 909, the electronic device (e.g., the processor 120 or 500) may identify the cumulative number of failures. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the cumulative number of failures is greater than, or equal to, a threshold value (e.g., about three times). According to various embodiments, when the cumulative number of failures is greater than, or equal to, the threshold value, the electronic device may determine that the measurement has continually failed so that it is impossible to measure biometric information.

[00212] When it is identified in operation 909 that the cumulative number of failures is smaller than the threshold value, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information again. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may perform an operation associated with operation 901 again. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the motion of the electronic device. For example, the electronic device (e.g., the processor 120 or 500) may perform at least one of the operations 801 to 819.

[00213] When it is identified in operation 909 that the cumulative number of failures is greater than, or equal to, the threshold value, the electronic device (e.g., the processor 120 or 500) may deactivate at least one second sensor. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is impossible to measure biometric information, and may stop the operation of the second sensor accordingly in operation 911.

[00214] According to various embodiments, at least one of the operations 901 to 911 may be omitted. For example, the operations 907 to 911 may be omitted. Accordingly, in a case where the measurement of biometric information fails in operation 903, the electronic device (e.g., the processor 120 or 500) may perform operation 901 again.

[00215] FIG. 10 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor

500, or the electronic device 400) according to various embodiments of the present disclosure.

[00216] Referring to FIG. 10, the performance procedure of the method of measuring biometric information may be a detailed operation for the operation 607 illustrated in FIG. 6.

[00217] In operation 1001, the electronic device (e.g., the processor 120 or 500) may acquire situation information associated with a measurement. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire the situation information based on at least one first sensor and/or at least one second sensor. According to an embodiment, the situation information may include the motion of the electronic device or a body while biometric information is being measured, the physical characteristic of an object to be measured, the state of the object to be measured, a measurement quality, a measurement result, or the like.

[00218] In operation 1003, the electronic device (e.g., the processor 120 or 500) may change a measurement parameter based on the situation information. For example, the measurement parameter that is changed based on the situation information may be defined as listed in Table 1 below. However, various embodiments of the present disclosure are not limited to Table 1. For example, in a case where the situation information is 'measurement quality,' the parameter to be changed may be 'timer period.' Furthermore, in a case where the situation information is 'physical characteristic,' the parameter to be changed may be 'sampling rate.'

Table 1

Situation information	Parameter to be changed
Measurement quality	Measurement time (light-emitting time)
Motion	Sampling rate
Measurement result	Measurement trigger
Physical condition	Timer period
User's state	Condition for determining motion

[00219] According to an embodiment, the electronic device (e.g., the processor 120 or 500) may reduce power consumption that is generated when biometric information is measured by changing the measurement parameter based on the situation information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may increase the accuracy of measurement when measuring biometric information by changing the measurement parameter based on the situation information.

[00220] For example, the electronic device (e.g., the processor 120 or 500) may determine that it is useless to measure biometric information in a state in which a measurement quality is not good. Accordingly, the electronic device (e.g., the processor 120 or 500) may decrease the light-emitting time of the second sensor in order to reduce power consumption. In another example, when the motion strength of the electronic device that cannot be normally measured is detected while biometric information is being measured, the electronic device (e.g., the processor 120 or 500) may stop the measurement operation in order to reduce power consumption. In yet another example, when the electronic device (e.g., the processor 120 or 500) continually fails to measure biometric information due to the motion thereof, the electronic device may omit the measurement operation until the motion is not generated in order to reduce power consumption. In yet another example, when it is determined that the strength of a motion while biometric information is being measured can be corrected, the electronic device (e.g., the processor 120 or 500) may increase a measurement sampling rate during the measurement operation in order to enhance the accuracy of measurement. In yet another example, when the electronic device (e.g., the processor 120 or 500) detects a physical characteristic (e.g., a dark skin color) of an object to be measured that uses an amount of light more than a reference value, which is emitted from the second sensor, the electronic device may change the measurement period in order to decrease the number of times that biometric information is measured, thereby reducing power consumption. In yet another example, in the case of a user in a motionless state (e.g., a sleep state), the

electronic device (e.g., the processor 120 or 500) may omit an operation of determining the motion of the electronic device in order to reduce power consumption.

[00221] In operation 1005, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed parameter.

[00222] In operation 1007, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information is completely measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information measurement result is successfully obtained.

[00223] When the measurement of the biometric information is not completed in operation 1007, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the biometric information based on the changed parameter in operation 1005.

[00224] When the measurement of biometric information is completed in operation 1007, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information in operation 1009. For example, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information through a display or a speaker.

[00225] According to various embodiments, at least one of the operations 1001 to 1009 may be omitted. For example, operation 1009 may be omitted.

[00226] FIG. 11 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00227] Referring to FIG. 11, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00228] According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is useless to measure biometric information in a state in which a measurement quality is not good and may decrease the light-emitting time of a second sensor in order to reduce power consumption.

[00229] In operation 1101, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00230] In operation 1103, the electronic device (e.g., the processor 120 or 500) may identify the measurement quality. According to an embodiment, the measurement quality may be the quality of a signal output from the second sensor. For example, the electronic device (e.g., the processor 120 or 500) may acquire a signal to noise ratio for the signal output from the second sensor as the measurement quality.

[00231] In operation 1105, the electronic device (e.g., the processor 120 or 500) may identify whether the measurement quality satisfies a selected (or specified) reference. According to an embodiment, the measurement quality may not satisfy the selected (or specified) reference, which may mean that a signal used for measuring biometric information is not acquired. For example, the measurement quality that fails to satisfy the selected (or specified) reference may indicate that a normal measurement result is not obtained. In another example, the measurement quality that fails to satisfy the selected (or specified) reference may indicate that a measurement result is not obtained for a selected (or specified) measurement time.

[00232] When the measurement quality satisfies the selected (or specified) reference in operation 1105, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a selected (or specified) measurement time in operation 1107. According to various embodiments, the measurement time may be associated with the light-emitting time of the second sensor. According to an embodiment, when the measurement quality satisfies the selected (or specified)

reference, the electronic device (e.g., the processor 120 or 500) may acquire biometric information by emitting light through the second sensor for the selected (or specified) time (e.g., about 10 seconds) every time a biometric information measurement period (e.g., about 1-minute period) arrives.

[00233] When the measurement quality does not satisfy the selected (or specified) reference in operation 1105, the electronic device (e.g., the processor 120 or 500) may change the measurement time selected (or specified) based on the quality in operation 1109.

[00234] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement time in order to reduce power consumption. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may shorten the measurement time in order to reduce power consumption when the measurement result on the biometric information is not obtained for the selected (or specified) measurement time.

[00235] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement time in order to enhance the accuracy of the measurement result. According to an embodiment, when the measurement result on the biometric information is not obtained for the selected (or specified) measurement time, the electronic device (e.g., the processor 120 or 500) may increase the measurement time in order to sufficiently ensure the information used for measuring the biometric information.

[00236] In operation 1111, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement time.

[00237] According to various embodiments, at least one of the operations 1101 to 1111 may be omitted.

[00238] FIG. 12 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00239] Referring to FIG. 12, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00240] In operation 1201, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00241] In operation 1203, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the biometric information is being measured. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

[00242] When the motion of the electronic device is not detected in operation 1203, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1205. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire information output from the second sensor based on a selected (or specified) sampling rate.

[00243] When the motion of the electronic device is detected in operation 1203, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference value in operation 1207. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a low motion strength that does not affect the measurement result is detected. For example, the electronic device (e.g., the processor 120 or 500) may determine the motion strength that does not affect the measurement result to be a motion that satisfies the reference value. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a motion strength that may affect the measurement result, but can be corrected, is detected. For example, the electronic device (e.g., the processor 120 or 500) may determine a value larger than

the motion strength, which does not affect the measurement result, by a predetermined value or more to be a reference value.

[00244] When the motion is identified to satisfy the reference value in operation 1207, the electronic device (e.g., the processor 120 or 500) may change the sampling rate of a signal output from the second sensor in operation 1209. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may increase a signal output from the second sensor by increasing the sampling rate in order to correct the noise according to the motion thereof.

[00245] When it is identified in operation 1207 that the motion does not satisfy the reference value, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1211. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is impossible to correct the noise according to the motion, and may accordingly stop measuring the biometric information.

[00246] According to various embodiments, at least one of the operations 1201 to 1211 may be omitted. For example, the operations 1207 and 1211 may be omitted. In this case, when the motion of the electronic device is detected in operation 1203, the electronic device may change the sampling rate in operation 1209. The electronic device may measure biometric information according to the changed sampling rate in operation 1205.

[00247] FIG. 13 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00248] Referring to FIG. 13, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00249] In operation 1301, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic

device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00250] In operation 1303, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the biometric information is being measured. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

[00251] In operation 1305, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference. According to an embodiment, the measured motion may satisfy the selected (or specified) reference, which may indicate that a motion that does not affect the acquisition of biometric information is generated.

[00252] When the measured motion satisfies the selected (or specified) reference in operation 1305, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1307. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire information output from the second sensor based on a selected (or specified) measurement time.

[00253] When the measured motion does not satisfy the selected (or specified) reference in operation 1305, the electronic device (e.g., the processor 120 or 500) may determine whether information for obtaining a result is collected in operation 1309.

[00254] When the information for obtaining a result is collected in operation 1309, the electronic device (e.g., the processor 120 or 500) may perform an operation of determining the motion thereof in operation 1311. According to an embodiment, in order to prevent the measurement operation from being stopped at the time when the measurement is nearly completed, the electronic device (e.g., the processor 120 or 500) may not deactivate the second sensor even though the motion is generated and may perform the operation of determining the motion until the motion satisfies the selected (or specified) reference.

[00255] When the information for obtaining a result is not collected in operation 1309, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1313. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that the motion that does not satisfy the selected (or specified) reference is generated in a state in which there is a predetermined period of time before the measurement is completed, and may accordingly stop measuring the biometric information.

[00256] According to various embodiments, at least one of the operations 1301 to 1313 may be omitted. For example, the operations 1309 and 1311 may be omitted. In this case, when it is determined in operation 1305 that the measured motion does not satisfy the selected (or specified) reference, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1313. Furthermore, when it is determined in operation 1305 that the measured motion satisfies the selected (or specified) reference, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1307.

[00257] FIG. 14 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00258] Referring to FIG. 14, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00259] In operation 1401, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00260] In operation 1403, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the

biometric information is being measured. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

[00261] In operation 1405, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference. According to an embodiment, the measured motion may satisfy the selected (or specified) reference, which may indicate that a motion that does not affect the acquisition of biometric information is generated.

[00262] When the measured motion satisfies the selected (or specified) reference in operation 1405, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the operation of a timer in operation 1407. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time.

[00263] When the measured motion does not satisfy the selected (or specified) reference in operation 1405, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the motion in operation 1409. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information with a period having a selected (or specified) period of time as a trigger, but may not perform the measurement operation even though the measurement period arrives in a situation in which the motion that satisfies the selected (or specified) reference is not measured. For example, the electronic device (e.g., the processor 120 or 500) may identify whether the motion that satisfies the selected (or specified) reference is measured without performing the biometric information measurement operation, and when the motion that satisfies the selected (or specified) reference is detected, the electronic device may perform the biometric information measurement operation.

[00264] In operation 1411, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement trigger.

According to various embodiments, at least one of the operations 1401 to 1411 may be omitted.

[00265] FIG. 15 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00266] Referring to FIG. 15, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00267] In operation 1501, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00268] In operation 1503, the electronic device (e.g., the processor 120 or 500) may identify the previous measurement result while measuring the biometric information. The previous measurement result may be a result based on data acquired by at least one first sensor or at least one second sensor of the electronic device (e.g., the processor 120 or 500).

[00269] In operation 1505, the electronic device (e.g., the processor 120 or 500) may identify whether a measurement failure is continuously generated a selected (or specified) number of times or more based on the previous measurement result.

[00270] When it is identified in operation 1505 that the measurement failure is not continuously generated a selected (or specified) number of times or more, the electronic device (e.g., the processor 120 or 500) may perform the biometric information measurement operation based on the operation of a timer in operation 1507. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time.

[00271] When it is identified in operation 1505 that the measurement failure is continuously generated a selected (or specified) number of times or more, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the motion in operation 1509. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the motion that satisfies the selected (or specified) reference is measured and may measure biometric information when the motion that satisfies the selected (or specified) reference is detected, instead of measuring biometric information with a period having a selected (or specified) period of time as a trigger.

[00272] In operation 1511, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement trigger. According to various embodiments, at least one of the operations 1501 to 1511 may be omitted.

[00273] FIG. 16 is a flowchart illustrating the performance procedure of a method of changing a measurement trigger by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00274] Referring to FIG. 16, the performance procedure of the method of changing a measurement trigger may be a detailed operation for the operation 1409 illustrated in FIG. 14 or the operation 1509 illustrated in FIG. 15.

[00275] In operation 1601, the electronic device (e.g., the processor 120 or 500) may identify the motion of the electronic device. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device when the measurement trigger is changed based on the motion. For example, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

[00276] In operation 1603, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device satisfies a condition. For example, the electronic device (e.g., the processor 120 or 500) may determine whether the motion that does not affect the measurement of biometric information is generated.

[00277] When the motion that satisfies the condition is not detected in operation 1603, the electronic device (e.g., the processor 120 or 500) may stand by without measuring biometric information in operation 1609. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may stand by without performing the measurement operation until the motion that does not affect the measurement is detected. For example, when the measurement trigger is changed based on the motion, the electronic device (e.g., the processor 120 or 500) may not perform the measurement operation even though a measurement period arrives.

[00278] When the motion that satisfies the condition is detected in operation 1603, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1605.

[00279] In operation 1607, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on a timer after measuring the biometric information based on the motion that satisfies the condition. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that the motion that does not affect the measurement is detected, and may accordingly change the measurement trigger to measure biometric information according to a period having a selected (or specified) period of time. According to various embodiments, at least one of the operations 1601 to 1609 may be omitted.

[00280] FIG. 17 is a view illustrating an operation of changing a measurement trigger by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00281] Referring to FIG. 17, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time (e.g., about 1 minute). For example, the electronic device (e.g., the processor 120 or 500) may measure biometric information with a selected (or specified) period as a measurement trigger.

[00282] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on at least one second sensor when the selected (or specified) period arrives.

[00283] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger when the motion of the electronic device does not satisfy a selected (or specified) condition (e.g., when the motion by which biometric information cannot be accurately measured is detected) at the time points t_1 and t_2 when the selected (or specified) period arrives. For example, the electronic device (e.g., the processor 120 or 500) may not perform the biometric information measurement operation until the motion by which biometric information can be accurately measured is detected. In other words, the electronic device (e.g., the processor 120 or 500) may measure biometric information with the motion satisfying the condition (e.g., the motion by which biometric information can be accurately measured) as the measurement trigger.

[00284] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger when a measurement failure according to the motion is continuously generated (1701) a selected (or specified) number of times (e.g., about two times) or more.

[00285] For example, the electronic device (e.g., the processor 120 or 500) may monitor the motion by changing the measurement trigger based on the detection of the motion after the measurement failure is continuously generated a selected (or specified) number of times or more, and may perform the measurement operation at time t_3 when the motion satisfies the selected (or specified) condition without performing the measurement operation (1705) based on the timer for the time interval 1703 for which the motion is detected. The electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the timer when the electronic device succeeds in acquiring biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger to measure biometric information according to a period 1707 having a selected (or specified) period of time $t_3 - t_4$ when the electronic device succeeds in acquiring

biometric information by changing the measurement trigger based on the detection of the motion.

[00286] FIG. 18 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00287] Referring to FIG. 18, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00288] In operation 1801, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00289] In operation 1803, the electronic device (e.g., the processor 120 or 500) may determine a characteristic of a body with which the second sensor makes contact. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine a physical characteristic of an object for which biometric information is measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine the skin color of the body with which the second sensor makes contact as the physical characteristic.

[00290] In operation 1805, the electronic device (e.g., the processor 120 or 500) may determine whether the physical characteristic uses an amount of emitted light more than a threshold value. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure a melanin index based on the amount of reflected light that is received by a light-receiving part of the second sensor, and may identify the skin color and the amount of emitted light of a light-emitting part that is used during measurement based on the measured melanin index. For example, the

amounts of emitted light according to melanin indices may be defined as listed in Table 2 below.

Table 2

Melanin Index	Type of skin color	Current intensity
5-10	Very light	31~40%
11-20	Light	41~50%
21-30	Intermediate	51~60%
31-40	Tan	61~70%
41-70	Brown	71~80%
71-99	black	81~100%

[00291] For example, a higher melanin index represents a darker skin color, and the electronic device (e.g., the processor 120 or 500) may determine that a larger amount of power is consumed for a darker skin color than for a lighter skin color.

[00292] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may store a user's physical characteristic, or receive the same from an external device.

[00293] When it is determined in operation 1805 that the physical characteristic does not require an amount of emitted light more than the threshold value, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a selected (or specified) measurement period in operation 1811.

[00294] When it is determined in operation 1805 that the physical characteristic uses an amount of emitted light more than the threshold value, the electronic device (e.g., the processor 120 or 500) may change the measurement period based on the physical characteristic in operation 1807. For example, the electronic device (e.g., the processor 120 or 500) may increase the measurement period to reduce the number of measurements when an amount of emitted light more than the threshold value is used when biometric information is measured.

[00295] When the measurement period is changed in operation 1807, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement period in operation 1809. According to various embodiments, at least one of the operations 1801 to 1811 may be omitted.

[00296] FIG. 19 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

[00297] Referring to FIG. 19, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

[00298] In operation 1901, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

[00299] In operation 1903, the electronic device (e.g., the processor 120 or 500) may identify a user's state. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify the user's state, such as a sleep state, an active state, etc.

[00300] In operation 1905, the electronic device (e.g., the processor 120 or 500) may determine whether a motion condition needs to be changed. According to various embodiments, when the user is in a sleep state, the motion condition may not be required since it may be determined that there is no motion, or the strength of a motion is low. When the motion condition is not required, it may be determined that the motion condition needs to be changed.

[00301] When it is determined in operation 1905 that the motion condition does not need to be changed, the electronic device (e.g., the processor 120 or 500) may

measure biometric information based on a selected (or specified) motion condition in operation 1911.

[00302] When it is determined in operation 1905 that the motion condition needs to be changed, the electronic device (e.g., the processor 120 or 500) may change the motion condition based on the user's state in operation 1907. For example, the electronic device (e.g., the processor 120 or 500) may omit the condition for determining a motion, or may decrease a desired motion reference value.

[00303] When the motion condition is changed in operation 1907, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed motion condition in operation 1909. According to various embodiments, at least one of the operations 1901 to 1911 may be omitted.

[00304] FIG. 20 is a view illustrating a screen configuration of an external electronic device (e.g., a smart phone) that is connected to an electronic device (e.g., the electronic device 101 or 400) through wireless or wired communication according to various embodiments of the present disclosure.

[00305] Referring to FIG. 20, the electronic device (e.g., the processor 120 or 500) may store biometric information (e.g., heart rate information), which is measured at every selected (or specified) period, in a memory and may display the measurement result 2000 on the screen thereof based on an input.

[00306] According to various embodiments, the electronic device (e.g., the processor 120 or 500) may display the measured heart rate information in units of hours 2001, days 2003, and months 2005. According to an embodiment, when displaying the measurement result in units of hours, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information measured at every hour in units of hours. According to another embodiment, when displaying the measurement result in units of days, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information measured at every day in units of days. According to yet another embodiment, when displaying the measurement result in units of months, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information

measured at every month in units of months. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may also display the highest and lowest values 2011 and 2013 of the heart rate information that is displayed in units of hours, as well as the average value of the heart rate information.

[00307] According to various embodiments, when displaying (2007) the result of the heart rate information, the electronic device (e.g., the processor 120 or 500) may display an indicator together for the time point at which the average value is higher or lower than a predetermined reference. For example, the electronic device (e.g., the processor 120 or 500) may display the indicator using a selected (or specified) color, a selected (or specified) text, a selected (or specified) icon, or the like. In the drawing, an exclamation mark 2009 displayed in the time interval between 3 hour and 4 hour may be an indicator that represents that heart rate information more than the average value is measured.

[00308] According to various embodiments, when an input for the indicator is detected, the electronic device (e.g., the processor 120 or 500) may display detailed information relating to the corresponding time interval. For example, the electronic device (e.g., the processor 120 or 500) may display the heart rate information 2021 that corresponds to the time interval relevant to the indicator for which the input is detected and the state information 2023 (e.g., calorie consumption, etc.) of the electronic device or a user for the corresponding time interval together. The drawing illustrates that the user practiced running from 3:15 to 3:20 so that the user's heart rate increased.

[00309] A method of operating an electronic device, according to various embodiments, may include: determining whether the electronic device is worn on a part of the user's body using at least one first sensor (e.g., at least one of a proximity sensor, an acceleration sensor, a gyro sensor, and a terrestrial magnetism sensor); acquiring information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on a part of the user's body; and activating (operating or

triggering) at least one second sensor (e.g., a heart rate sensor) based on at least a part of the information on the motion.

[00310] According to various embodiments, the second period of time may be shorter than the first period of time.

[00311] According to various embodiments, the at least one second sensor may include a biometric sensor.

[00312] According to various embodiments, the biometric sensor may include at least one of a heart rate sensor, a blood pressure sensor, an ECG sensor, a skin moisture sensor, and an oxygen saturation sensor.

[00313] According to various embodiments, the activating (operating or triggering) may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating (operating or triggering) the at least one second sensor based on at least a part of the additional information in a case where the information on the motion does not satisfy a selected condition.

[00314] According to various embodiments, at least one of the second period of time and the third period of time may include the time point when the first period of time passes.

[00315] According to various embodiments, the second period of time may include the time point when the first period of time passes.

[00316] According to various embodiments, the activating (operating or triggering) may include: when data is not received from the activated (operated or triggered) second sensor within a selected (or specified) time, activating (operating or triggering) the second sensor again after the selected (specified) time passes.

[00317] According to various embodiments, the activating (operating or triggering) may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a period of time that is substantially the same as the second period of time before the selected (or

specified) time passes; and activating (operating or triggering) the at least one second sensor again based on at least a part of the additional information.

[00318] According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing the first period of time and/or the second period of time and/or the third period of time based on at least a part of the acquired data.

[00319] According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00320] According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one of a period of time and a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00321] A method of operating an electronic device, according to various embodiments, includes: determining whether the electronic device is worn on a part of a user's body using at least one first sensor; activating (operating or triggering) at least one second sensor based on at least a part of the measured motion of the electronic device within a selected (or specified) time range when a selected (or specified) period of time passes from the time when the electronic device is worn on the user's body part; and measuring biometric information based on the at least one second sensor.

[00322] According to various embodiments, the at least one first sensor may include at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, and an acceleration sensor.

[00323] According to various embodiments, the method may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a selected (or specified) time from the time when the

period of time passes; and activating (operating or triggering) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when at least a part of the motion does not satisfy a selected (or specified) condition.

[00324] According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: accumulating the number of failures when the measurement of the biometric information fails; and deactivating the second sensor when the cumulative number of failures satisfies a selected (or specified) condition.

[00325] According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00326] According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one of a period of time or a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00327] According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data about the user's physical characteristic (e.g., skin color) or state (e.g., a sleep state or an active state); and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

[00328] According to various embodiments, the acquiring of the additional information on the motion may include changing the selected (or specified) time for acquiring the additional information based on data acquired from at least one of the at least one first sensor and the at least one second sensor.

[00329] According to an embodiment, a method of operating an electronic device may include: determining whether the electronic device is close to a user's

body; acquiring information on the motion of the electronic device using at least one first sensor within a selected (or specified) time range from the time when a selected (or specified) period passes in a case where the electronic device is close to the user's body part; and acquiring the user's biometric information through at least one second sensor based on at least a part of the information on the motion.

[00330] The electronic device and the method, according to the various embodiments of the present disclosure, can selectively operate a biometric sensor based on a selected (or specified) condition, thereby measuring biometric information while reducing power consumption.

[00331] Further, the electronic device and the method, according to the various embodiments of the present disclosure, can operate a biometric sensor according to a selected (or specified) period, thereby continually measuring biometric information.

[00332] In addition, the electronic device and method, according to the various embodiments of the present disclosure, can measure the motion of the electronic device and can selectively measure a user's biometric information based on the measured motion, thereby improving the accuracy of the measured biometric information.

[00333] A terminology “module” used for the present disclosure may mean, for example, a unit including a combination of one or two or more among a hardware, a software, or a firmware. A “module” may be interchangeably used with a terminology such as a unit, a logic, a logical block, a component, or a circuit, etc. A “module” may be a minimum unit of an integrally configured part or a portion thereof. A “module” may be a minimum unit performing one or more functions or a portion thereof. A “module” may be mechanically or electronically implemented. For example, a “module” according to the present disclosure may include at least one of an application-specific IC (ASIC) chip, a field-programmable gate arrays (FPGAs), or a programmable-logic device which are known, or to be developed in the future, and performing certain operations.

[00334] According to various embodiments, at least a portion of an apparatus (e.g., modules or functions thereof) or a method (e.g., operations) according to the

present disclosure may be implemented as an instruction stored in a non-transitory computer-readable storage media, for example, in the form of a programming module. An instruction, when executed by one or more processors (e.g., the processor 120), may configure the one or more processors to perform a function corresponding to the instruction. The non-transitory computer-readable storage media may be, for example, the memory 130. At least a portion of a programming module may be implemented (e.g., executed) by, for example, the processor 120. At least a portion of the programming module may include, for example, a module, a program, a routine, sets of instructions, or a process, etc. for performing one or more functions.

[00335] The non-transitory computer-readable storage media may include a hard disk, a magnetic media such as a floppy disk and a magnetic tape, compact disc ROM (CD-ROM), optical media such as DVD, magneto-optical media such as a floptical disk, and a hardware device specially configured for storing and performing a program instruction (e.g., a programming module) such as ROM, RAM, a flash memory, etc. Also, the program instruction may include not only a machine language code generated by a compiler but also a high-level language code executable by a computer using an interpreter, etc. The above-described hardware device may be configured to operate as one or more software modules in order to perform an operation of the present disclosure, and vice versa.

[00336] A module or a programming module according to the present disclosure may include at least one of the above-described elements, omit a portion thereof, or further include additional other elements. Operations performed by a module, a programming module, or other elements according to the present disclosure may be executed in a sequential, parallel, or heuristic method. Also, a portion of the operations may be executed in a different sequence, omitted, or other operations may be added.

[00337] While the present disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing

from the spirit and scope of the present disclosure as defined by the appended claims and their equivalents.

WHAT IS CLAIMED IS:

1. A wearable electronic device comprising:
a motion sensor;
one or more optical sensors includes a light emitting unit and a light receiving unit; and
at least one processor configured to:
 obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,
 emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,
 obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,
 obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and
 refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.
2. The wearable electronic device of claim 1, wherein the at least one processor is further configured to:
 determine whether the wearable electronic device is attached to the user using the one or more optical sensors; and
 perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.

3. The wearable electronic device of claim 2, wherein the at least one processor is further configured to:

perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.

4. The wearable electronic device of claim 2, wherein the at least one processor is further configured to:

refrain from the emitting of the light further based at least in part on a determination that the wearable electronic device is not attached to the user.

5. The wearable electronic device of claim 1, wherein the at least one processor is further configured to:

perform the emitting of the light based at least in part on a specified time period.

6. The wearable electronic device of claim 5, wherein the at least one processor is further configured to:

adjust the specified time period based at least in part on context information corresponding to the biological signal.

7. The wearable electronic device of claim 1, wherein the at least one processor is further configured to:

generate a heart rate information corresponding to the user using the biological signal.

8. The wearable electronic device of claim 7, wherein the at least one processor is further configured to:

present a notification corresponding to the heart rate information based at least in part on a determination that the heart rate information is higher than a threshold.

9. A wearable electronic device comprising:

a first sensor;

a second sensor; and

at least one processor configured to:

obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,

activate the second sensor based at least in part of the movement,

obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and

refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.

10. The wearable electronic device of claim 9, wherein the at least one processor is further configured to:

control the second sensor to inactivate at least a portion thereof, as at least part of the refraining.

11. The wearable electronic device of claim 10, wherein the at least one processor is further configured to:

control the second sensor to activate the at least a portion thereof, based at least in part on a determination that the first information satisfies another specified range lower than the specified range after the refraining.

12. The wearable electronic device of claim 10, wherein the at least one processor is further configured to:

control the second sensor to activate the at least a portion thereof, further based at least in part on a specified time period.

13. The wearable electronic device of claim 12, wherein the at least one processor is further configured to:

adjust the specified time period based at least in part on context information generated using the first information and the second information.

14. The wearable electronic device of claim 9, wherein the at least one processor is further configured to:

generate a heart rate information corresponding to the user using the second information.

15. The wearable electronic device of claim 14, wherein the at least one processor is further configured to:

present a notification corresponding to the heart rate information based at least in part on a determination that the heart rate information is higher than a threshold.

16. A wearable electronic device comprising:

a first sensor;

a second sensor; and

at least one processor configured to:

obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,

obtain, using the second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and

refrain from the obtaining of the second information based at least in part on a determination that the first information satisfies a specified condition.

17. The wearable electronic device of claim 16, wherein the at least one processor is further configured to:

activate the second sensor prior to obtaining of the second information, in response to the movement of the wearable electronic device.

18. The wearable electronic device of claim 17, wherein the at least one processor is further configured to:

perform the activating based at least in part on a determination that the first information does not satisfy the specified condition.

19. The wearable electronic device of claim 16, wherein the at least one processor is further configured to:

control the second sensor to inactivate at least a portion thereof, as at least part of the refraining.

20. The wearable electronic device of claim 16,
wherein the second sensor includes a light emitting unit, and
wherein the processor is further configured to:

emit light, using the emitting unit, as at least part of the obtaining of the second information, and

stop the emitting of the light, using the emitting unit, as at least part of the refraining.

ABSTRACT

A wearable electronic device for detecting biometric information is provided. The device includes a motion sensor, one or more optical sensors including a light emitting unit and a light receiving unit, and at least one processor configured to obtain a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained, emit light using the light emitting unit based on a determination that the first motion signal satisfies a specified condition, obtain the biological signal corresponding to a user using the light receiving unit based on reflected light which the light is reflected on the user, obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and refrain from the emitting of the light, based on a determination that the second motion signal does not satisfy the specified condition.

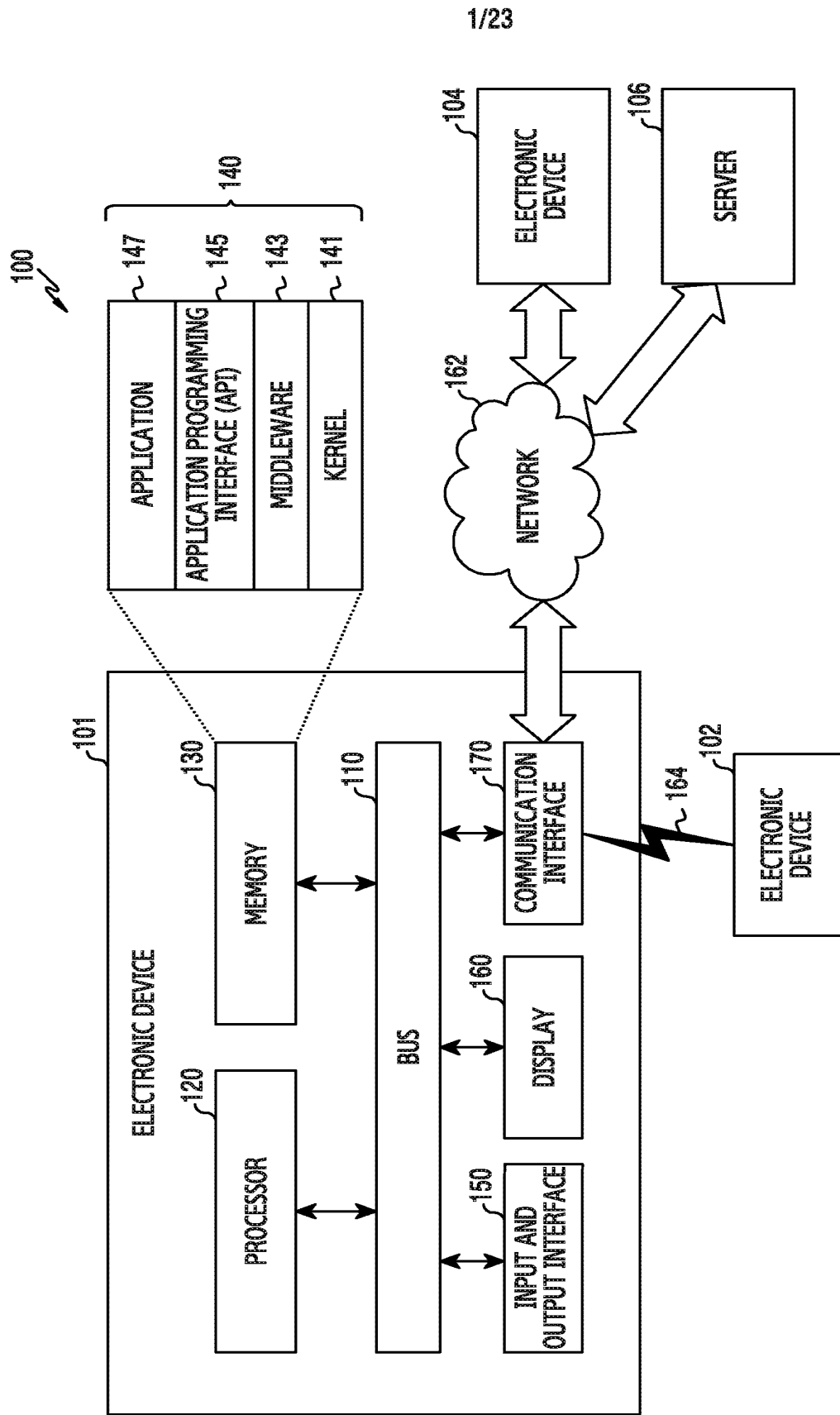


FIG.1

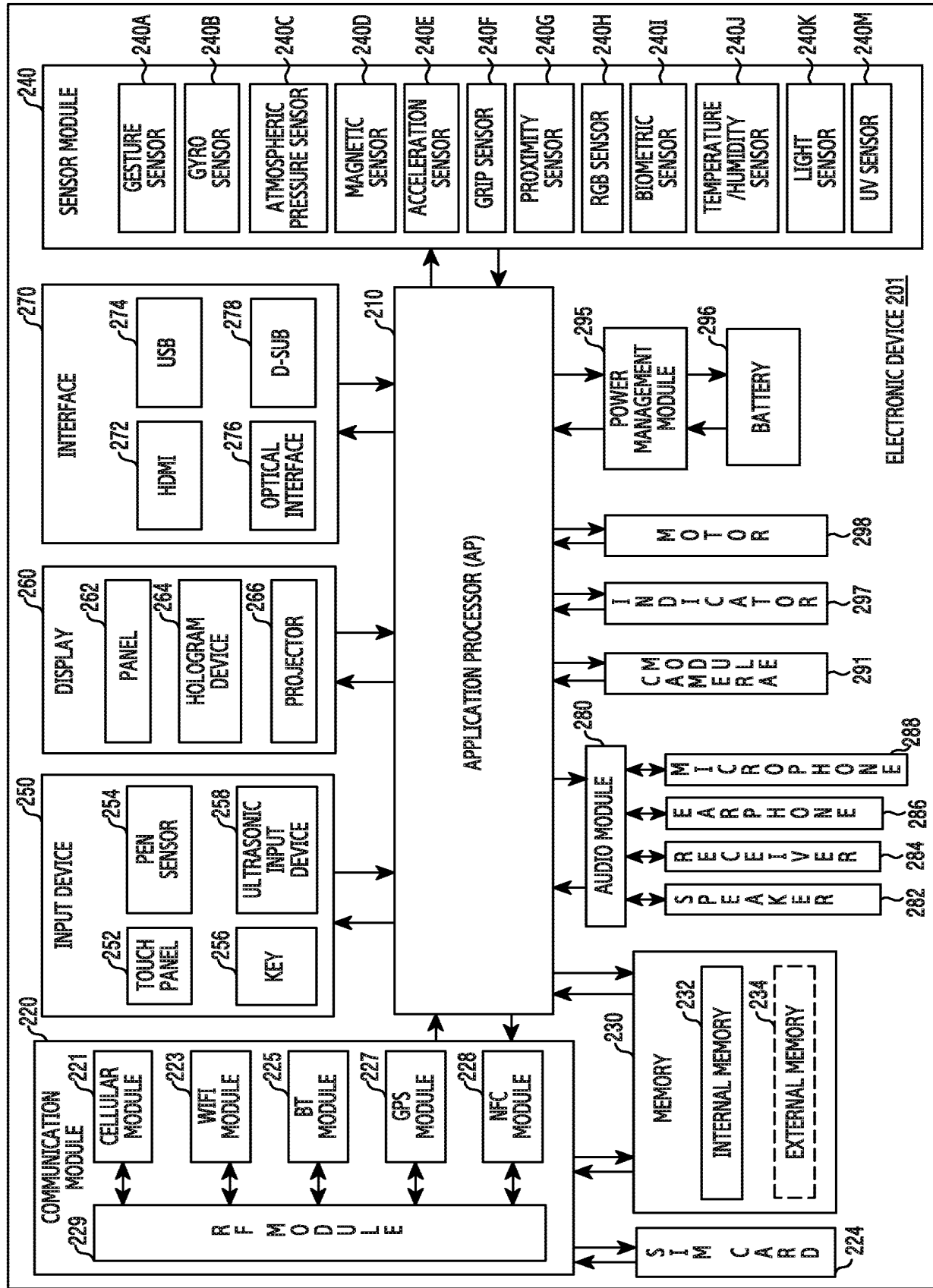


FIG.2

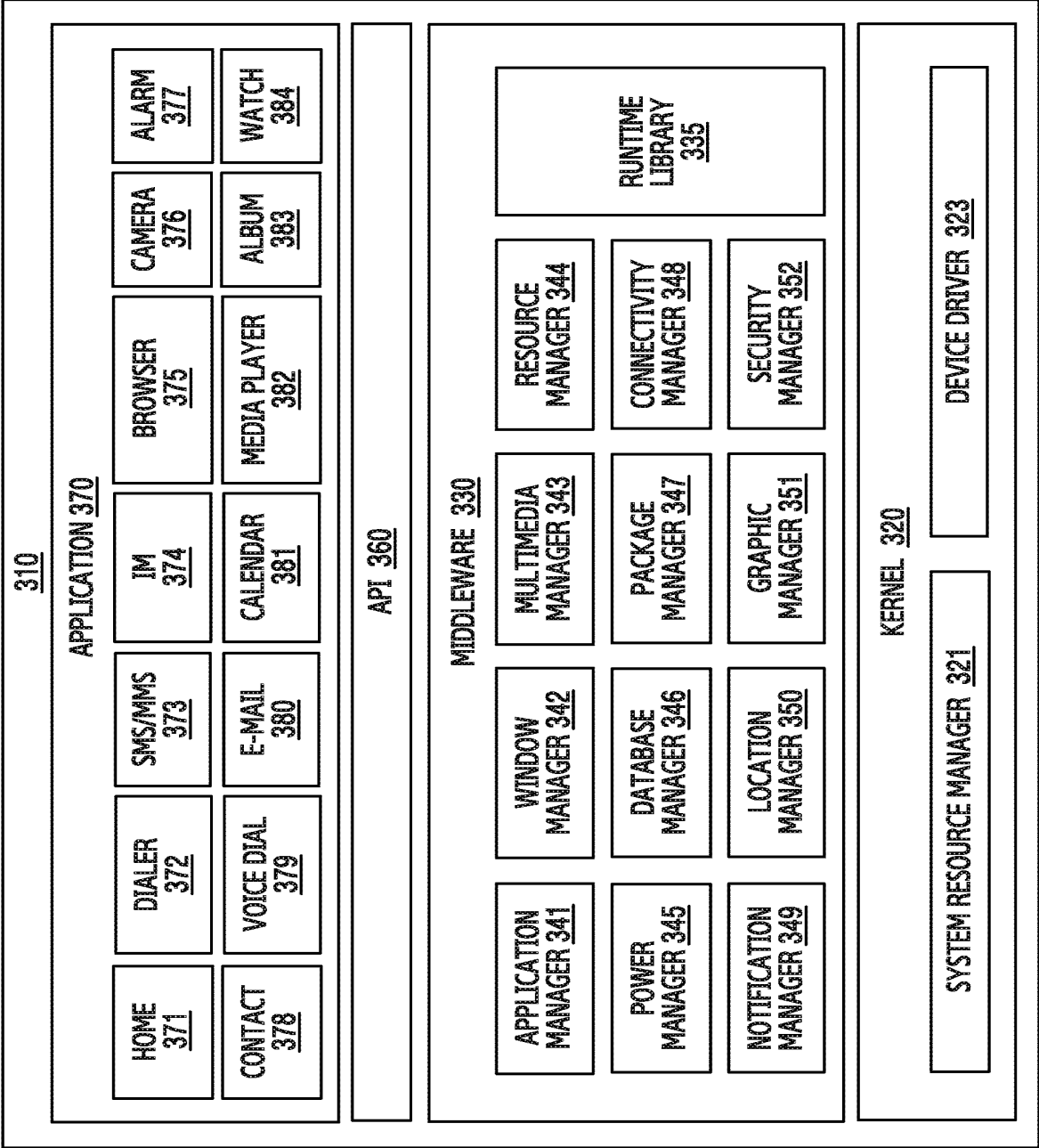


FIG.3

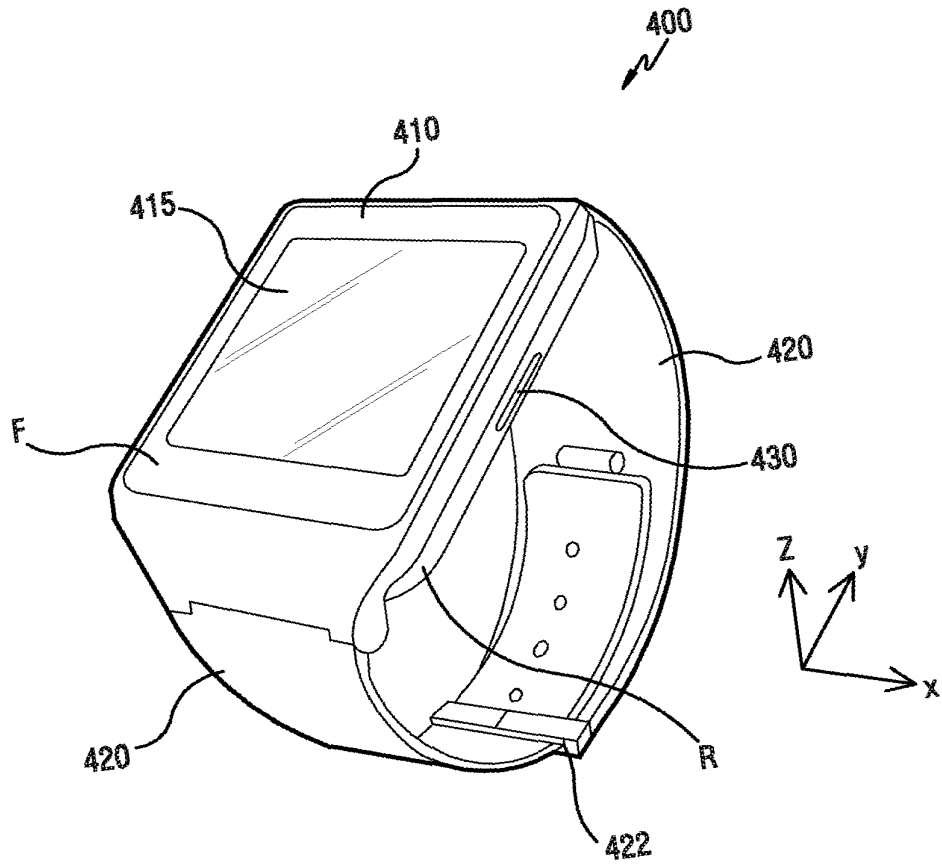


FIG. 4A

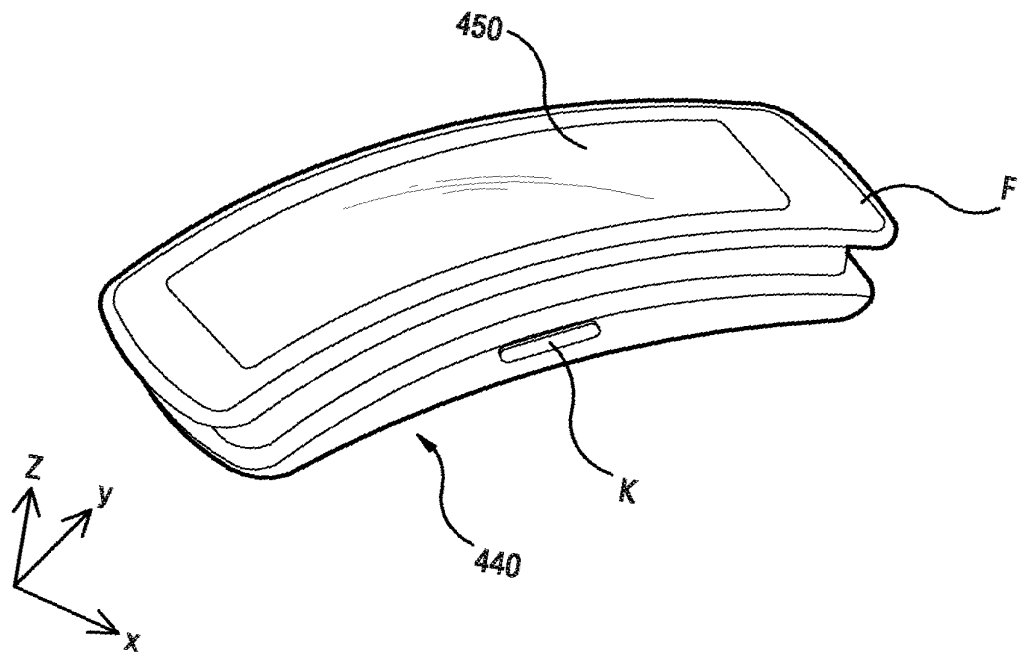


FIG. 4B

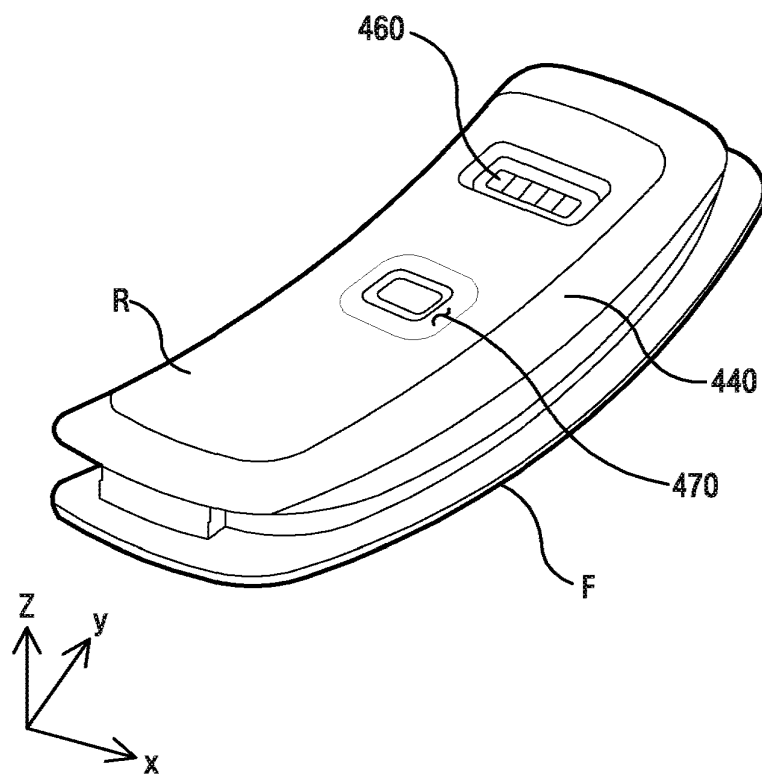


FIG. 4C

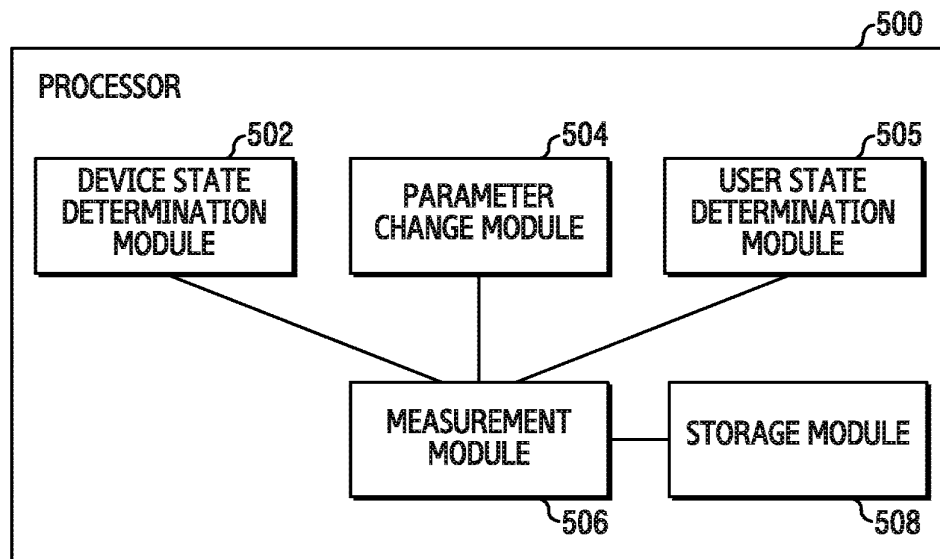


FIG.5A

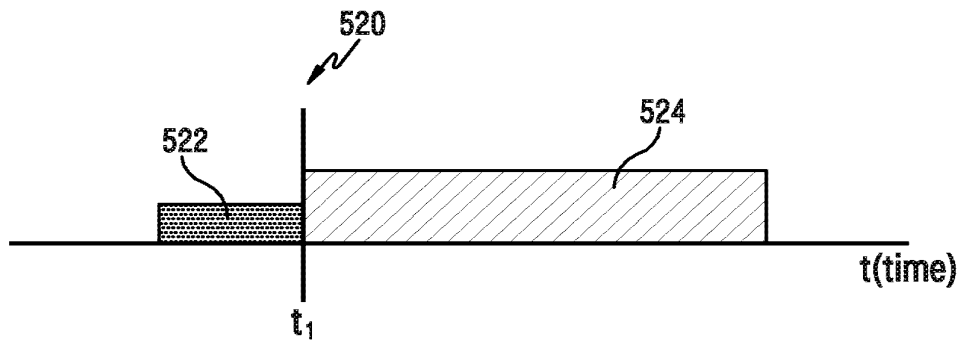


FIG.5B

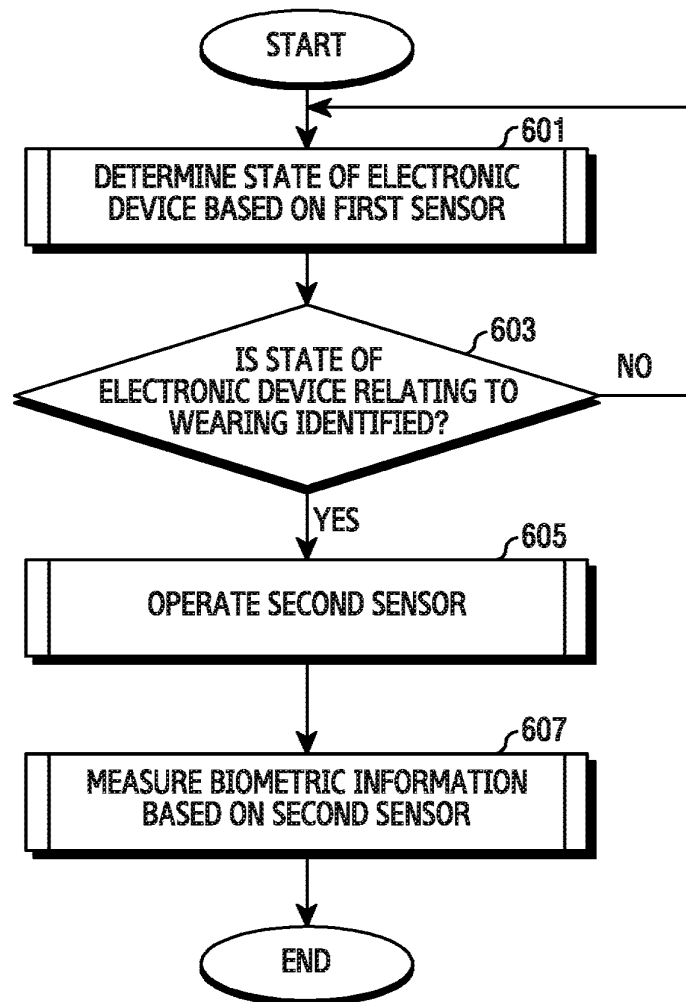


FIG.6

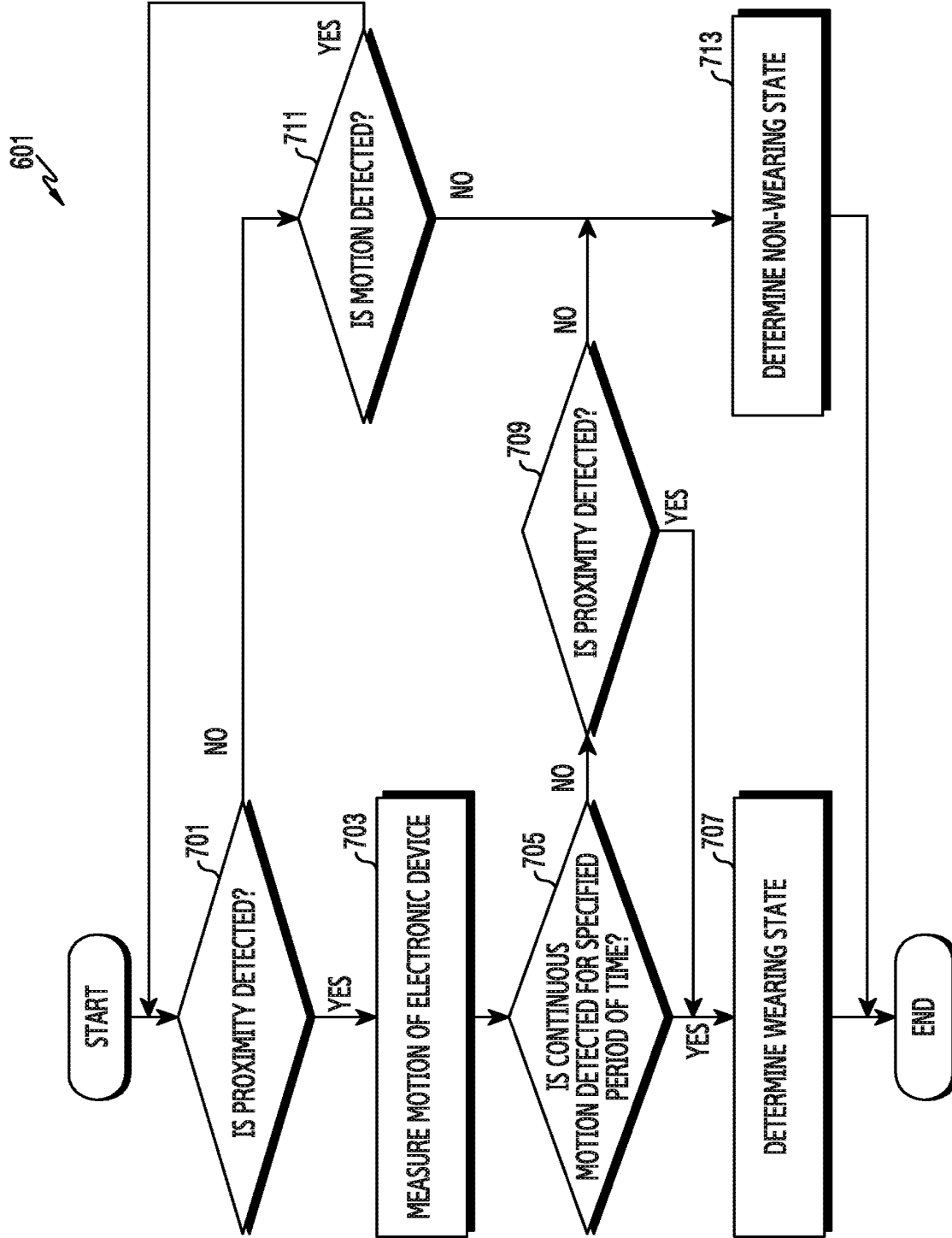


FIG. 7

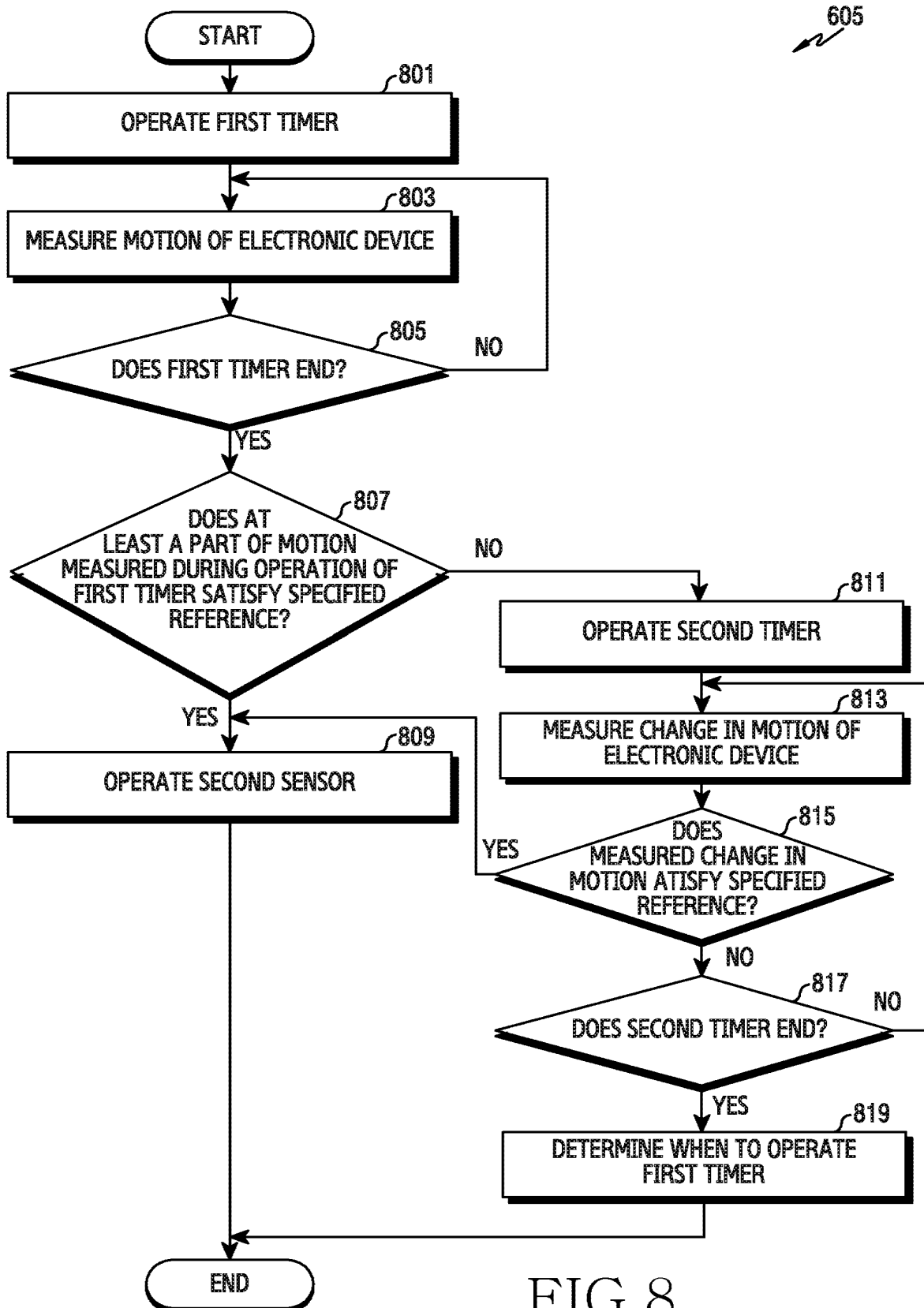


FIG.8

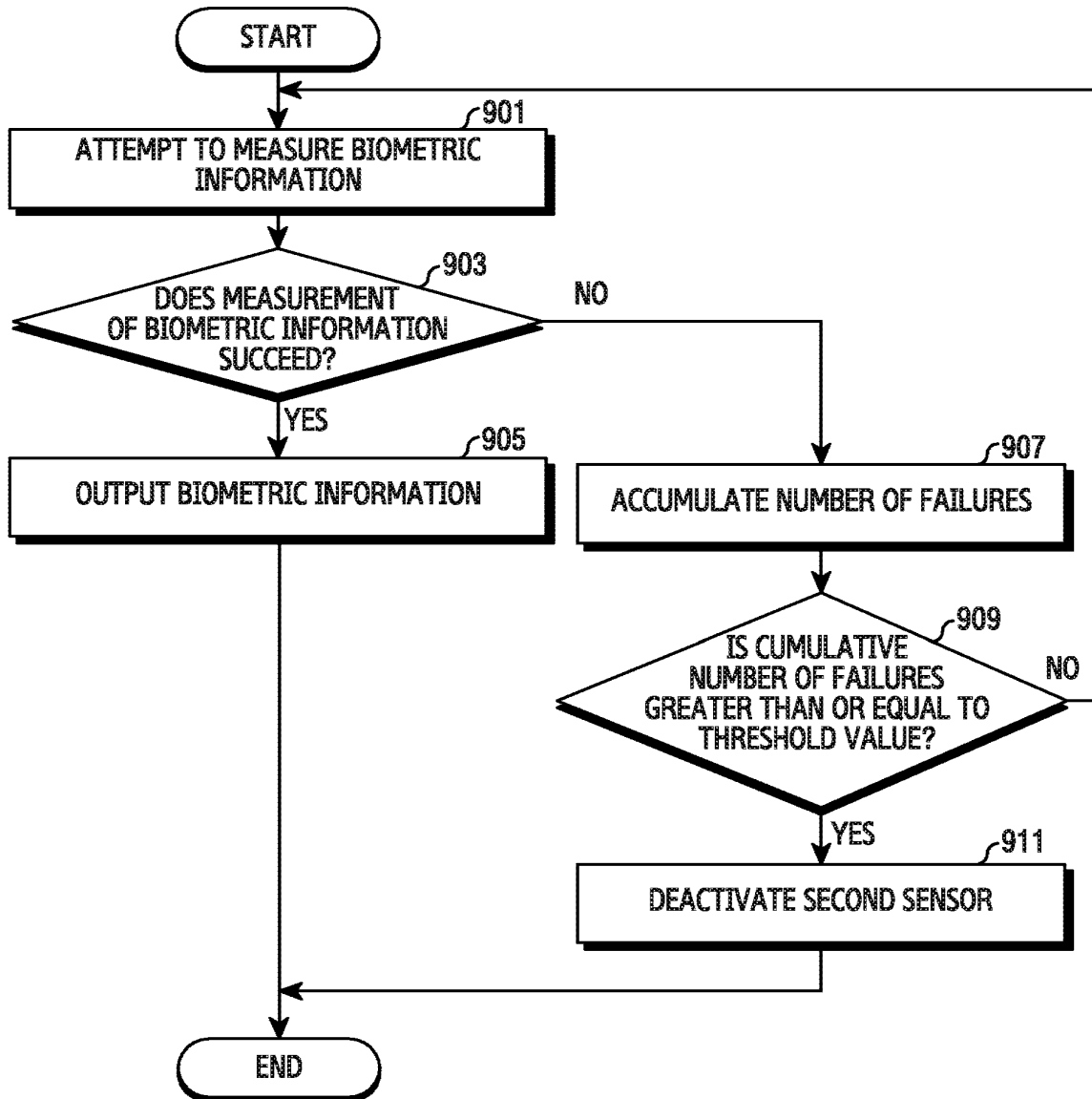


FIG.9

607

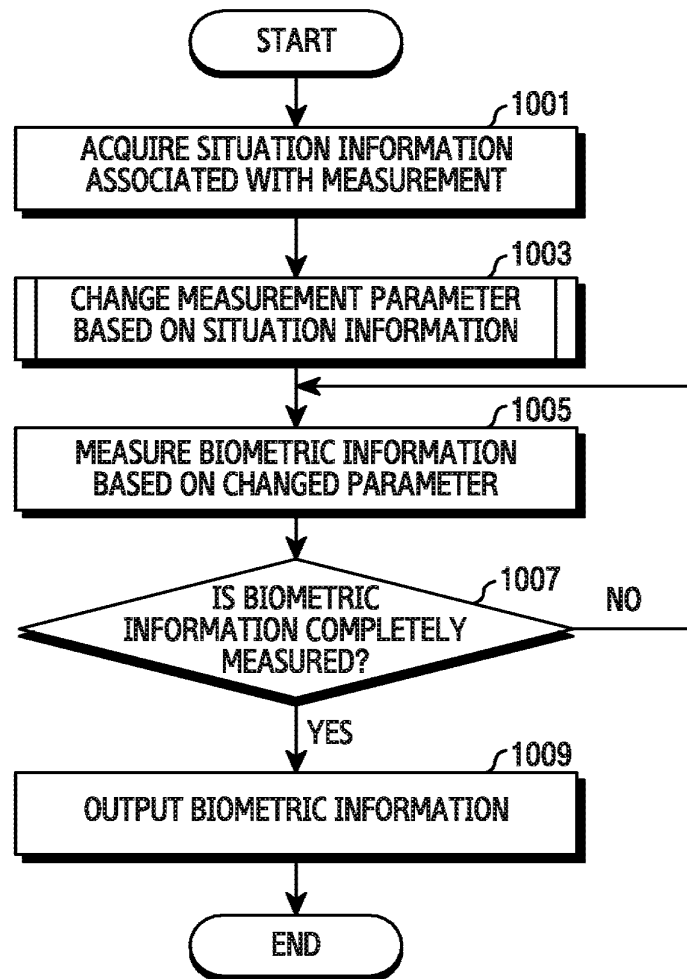


FIG. 10

1003

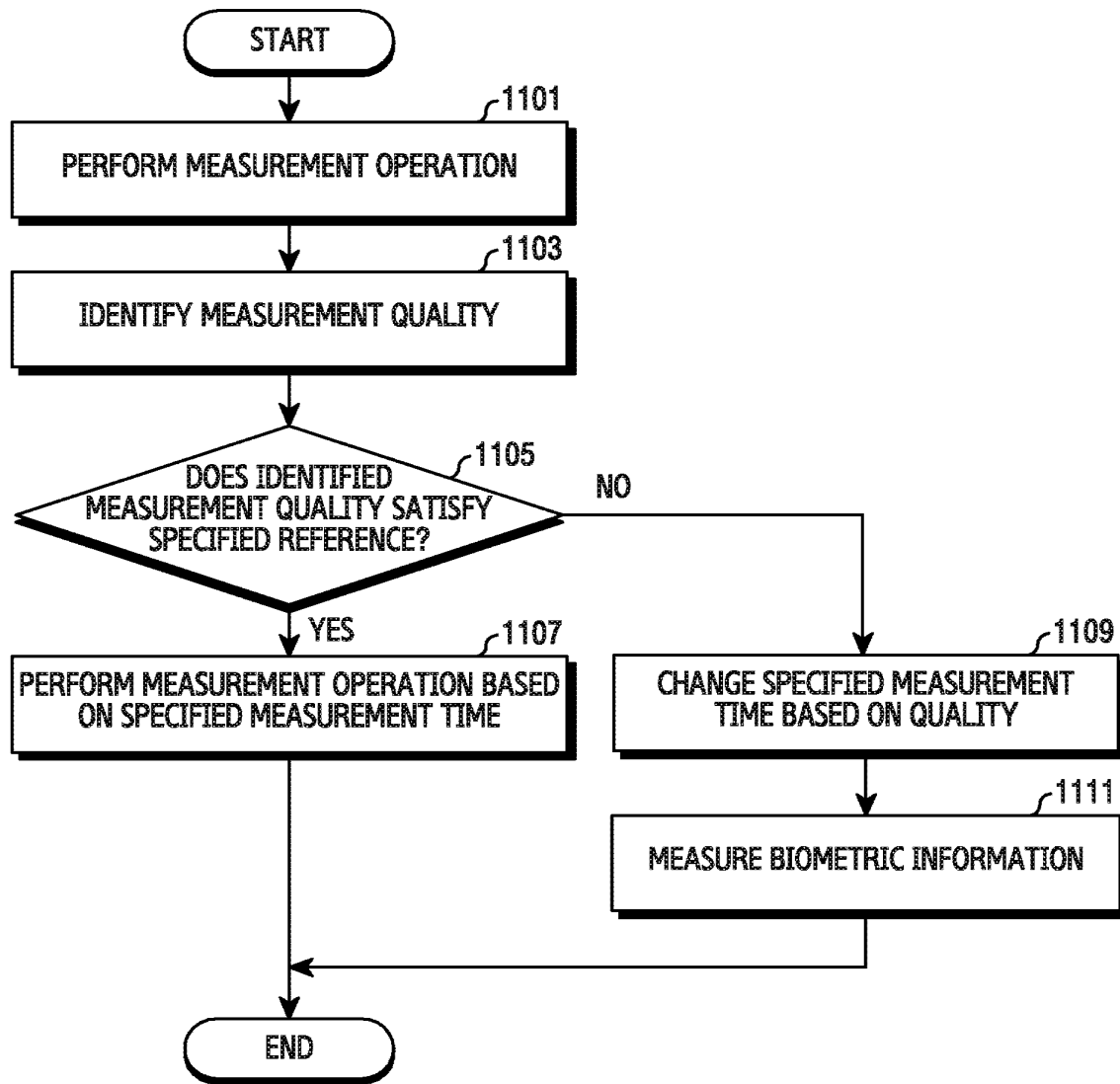


FIG.11

1003

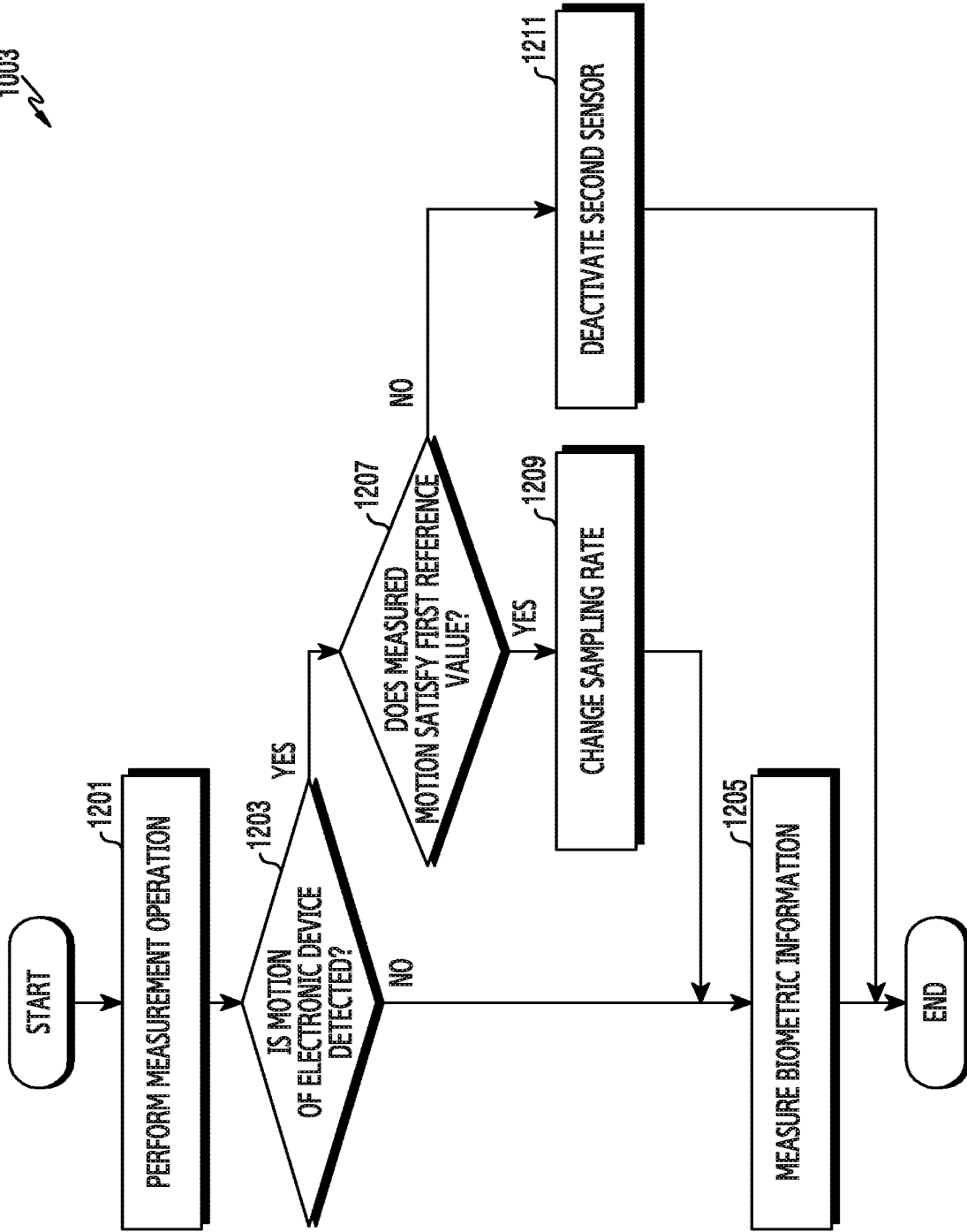


FIG.12

1003

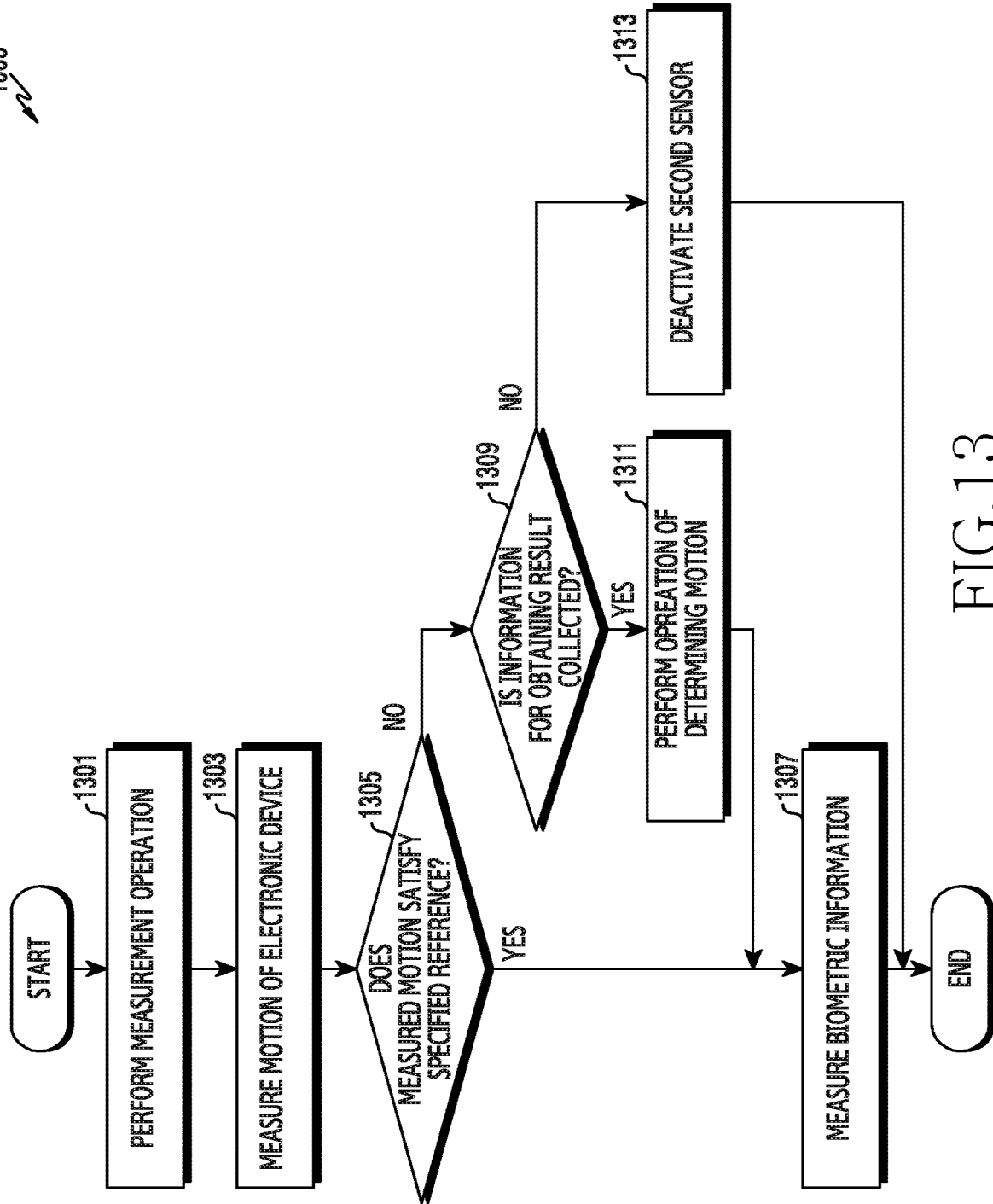


FIG.13

1003

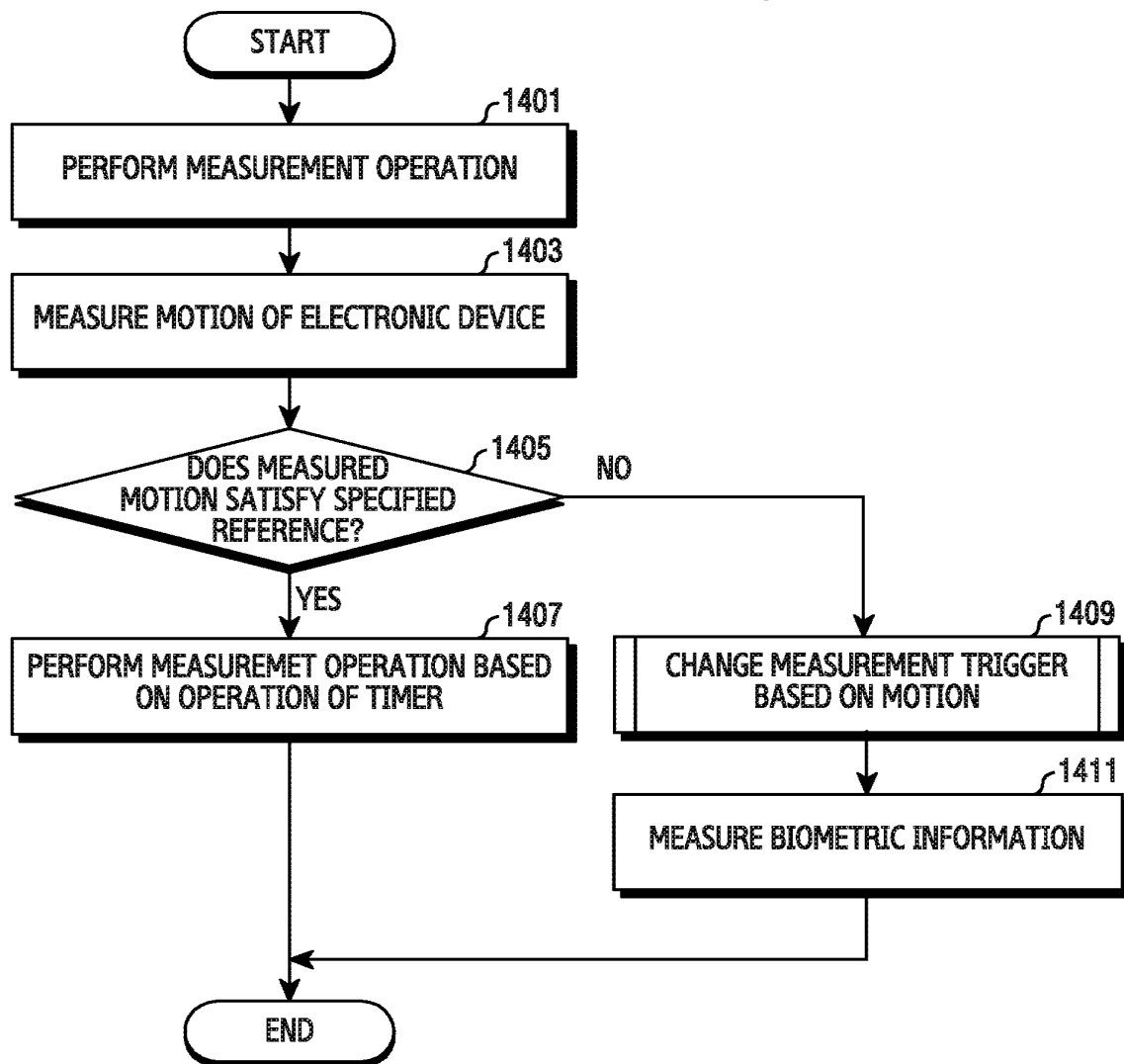


FIG.14

1003

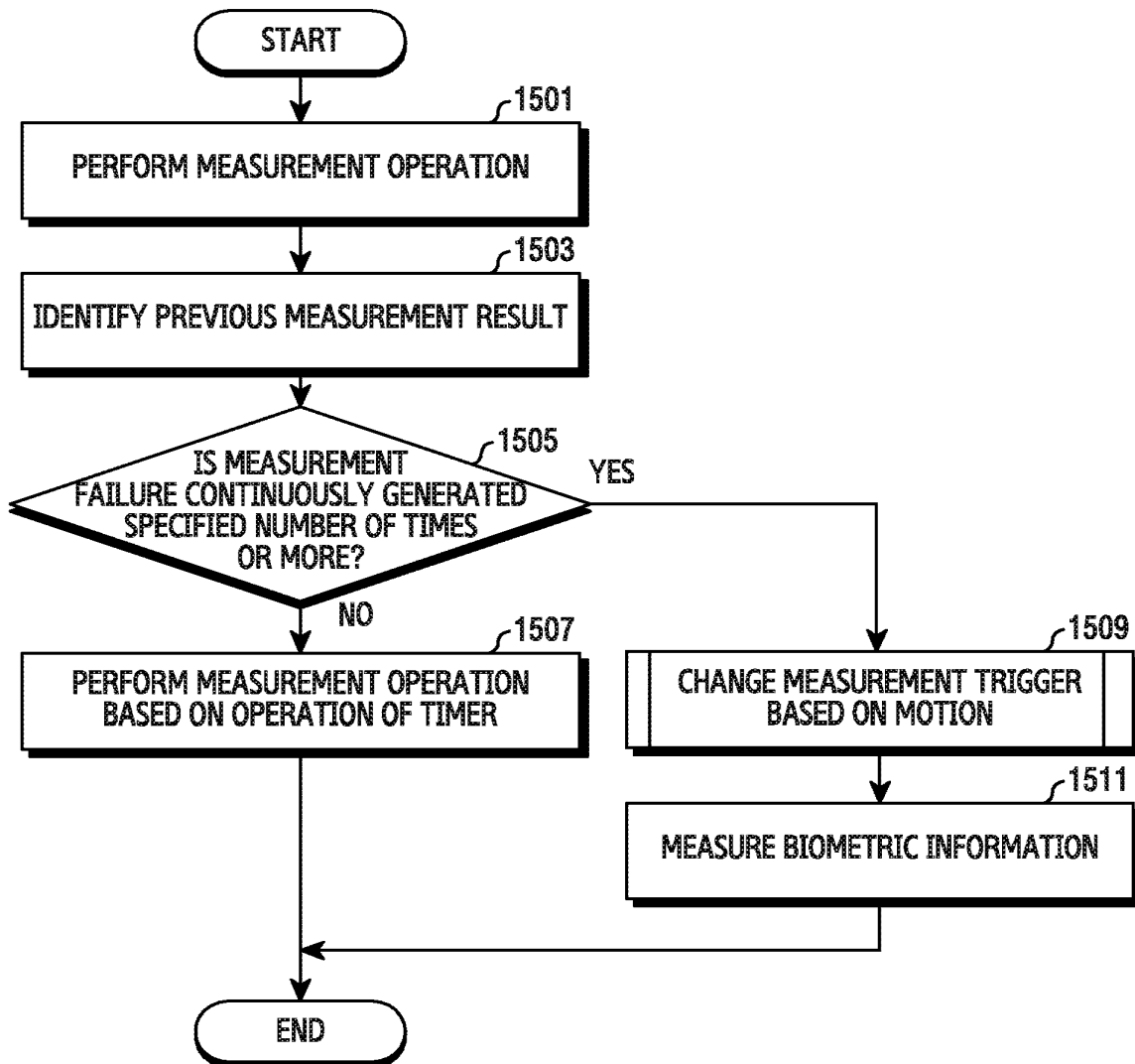


FIG.15

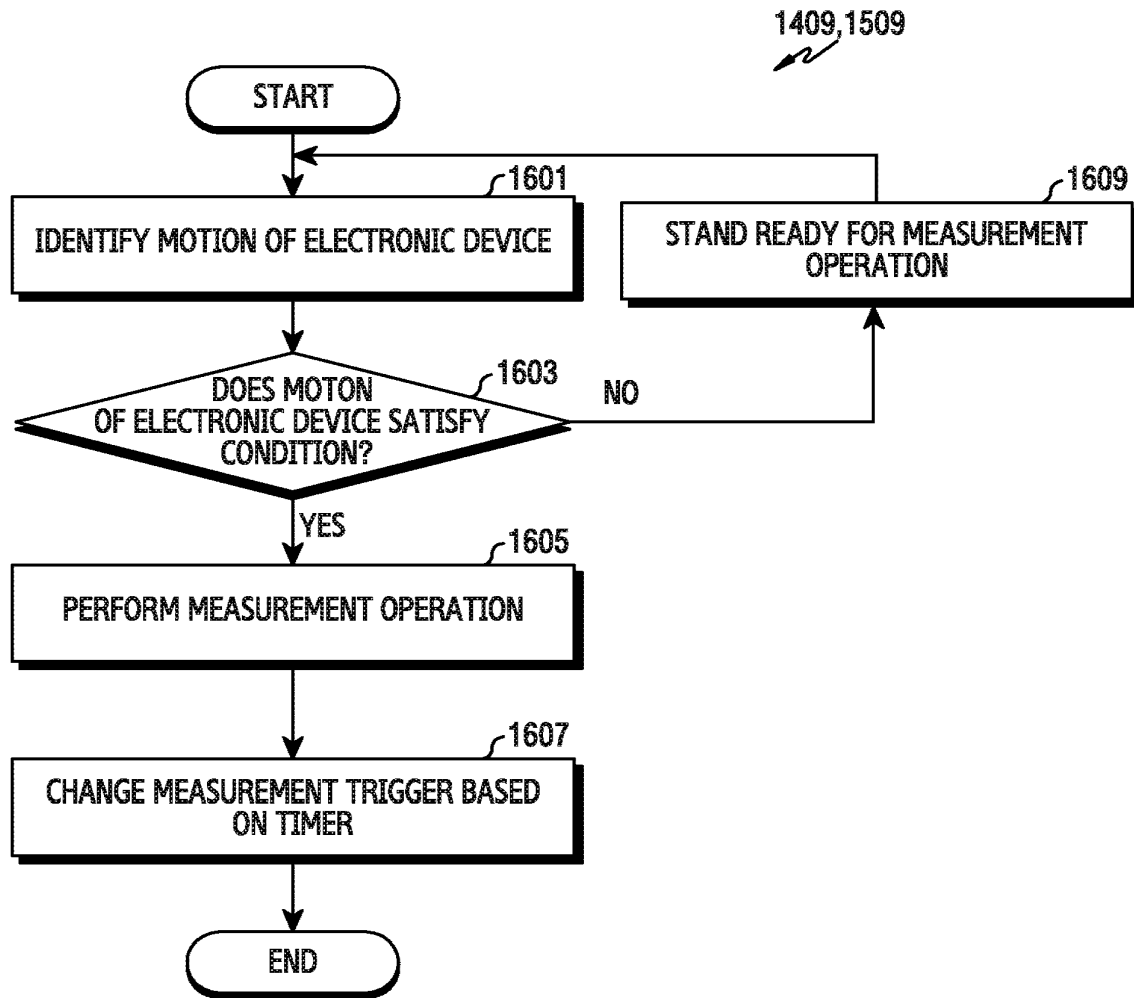


FIG.16

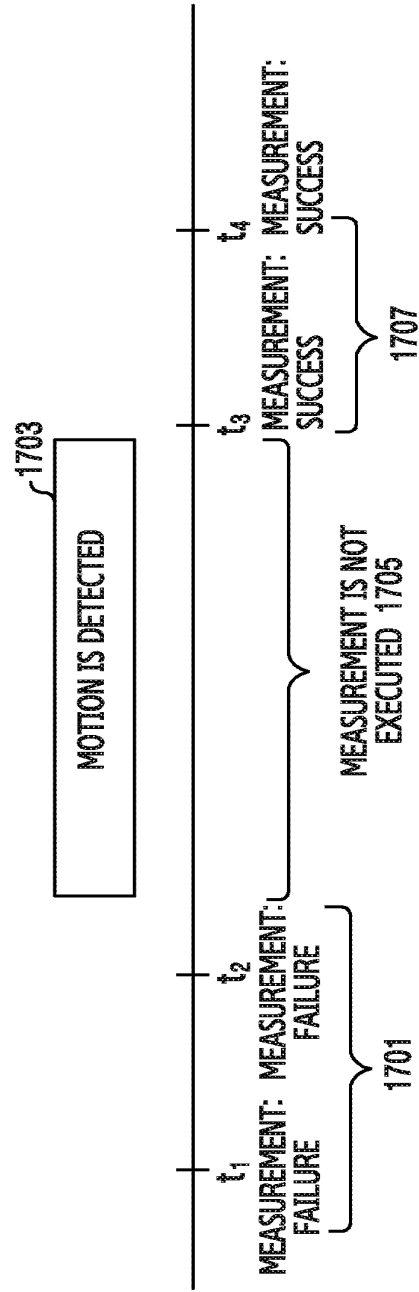


FIG.17

1003

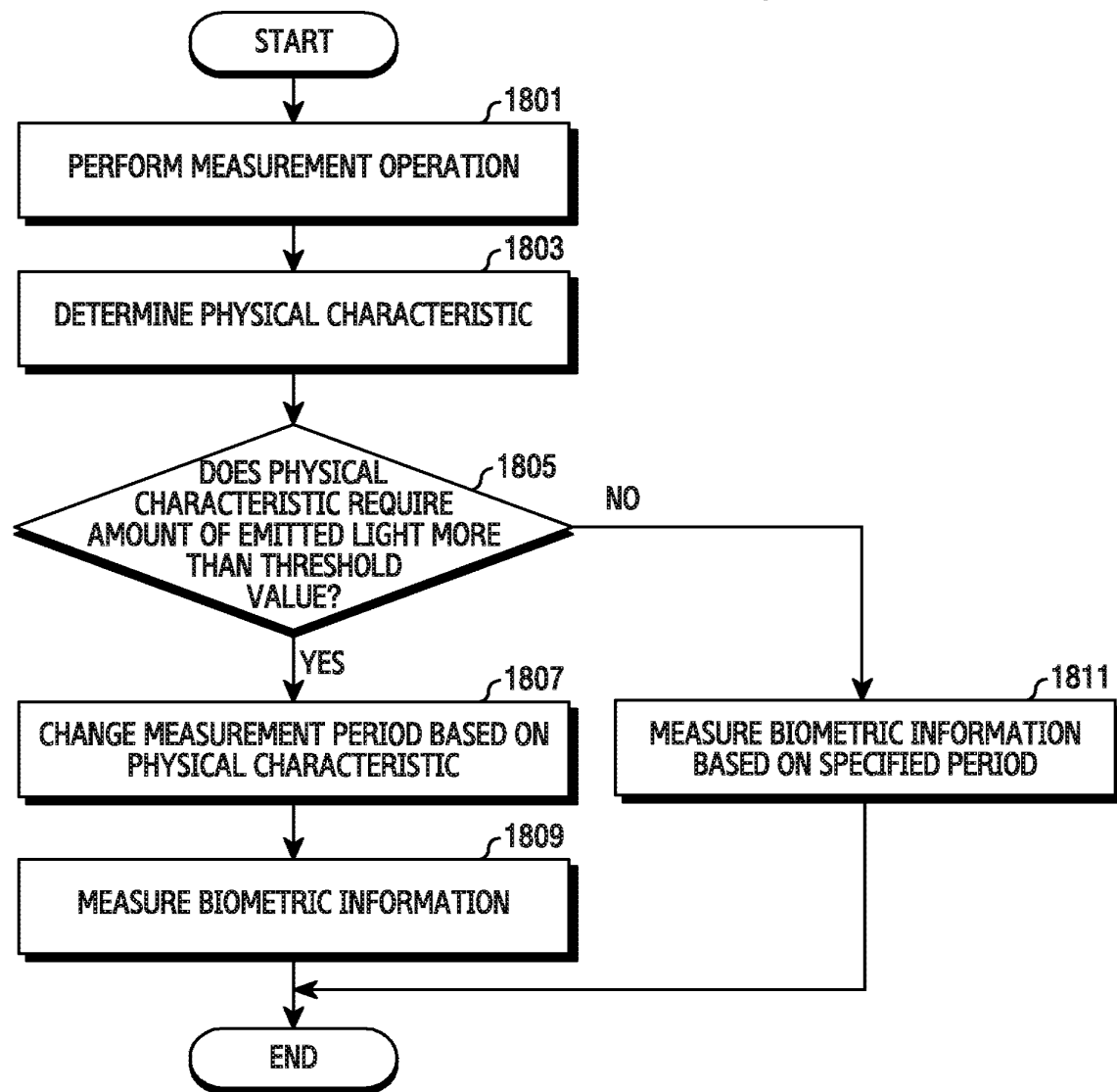


FIG.18

1003

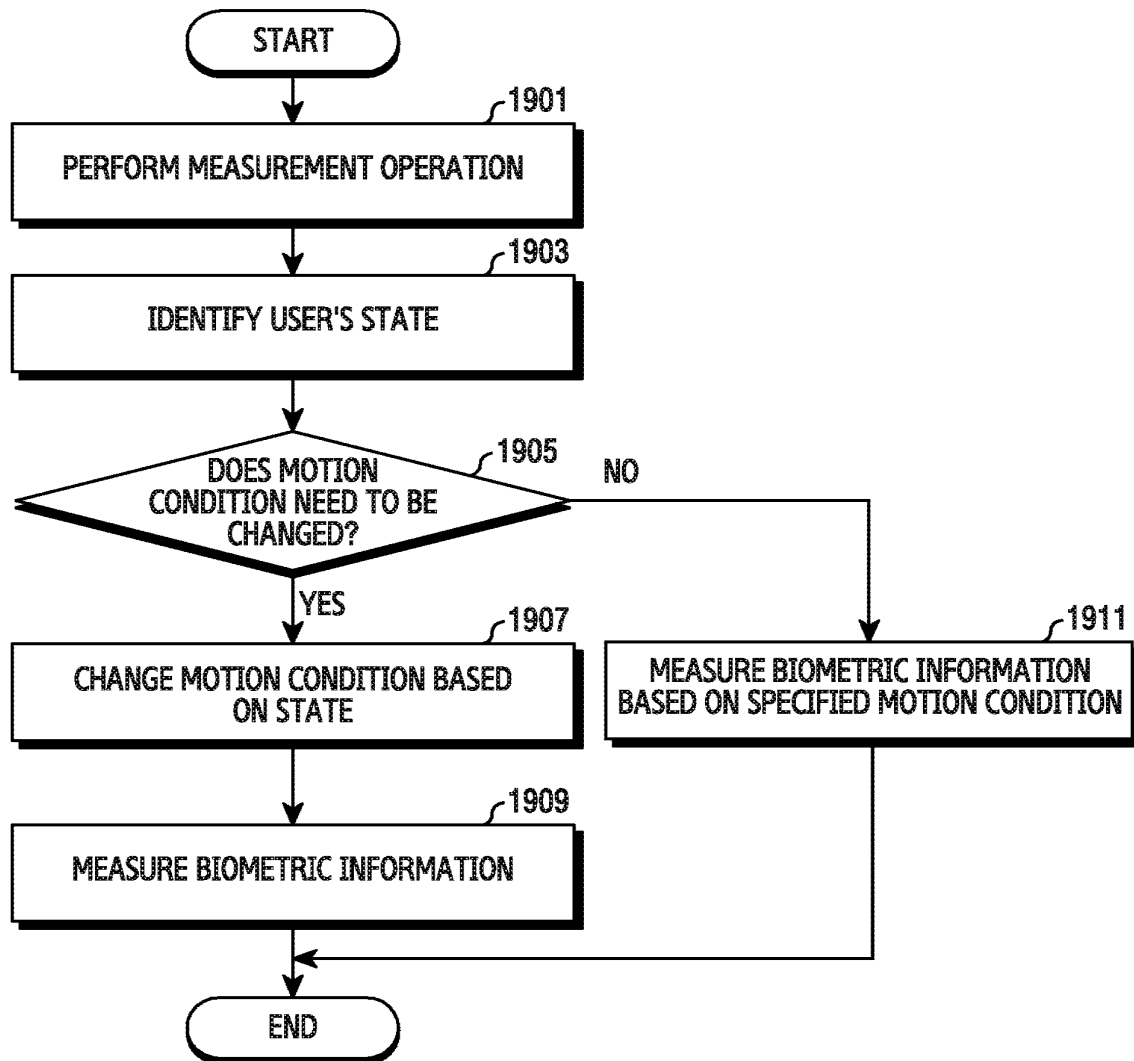


FIG.19

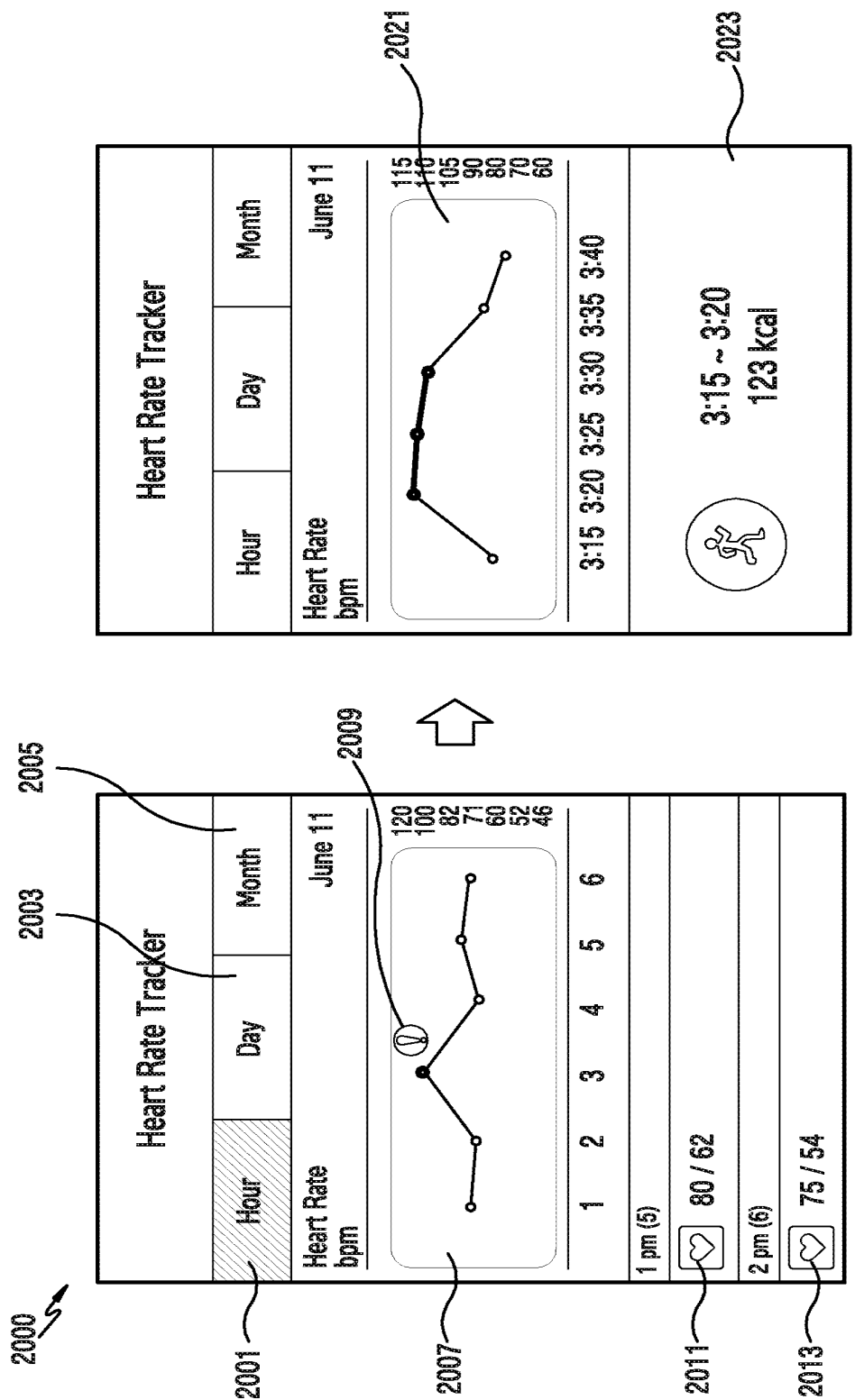


FIG.20

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)**

**TITLE: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC
DEVICE USING SAME**

As the below named inventor, I hereby declare that:

This declaration is directed to:

- ☒ The attached application, or
- ☐ United States application or PCT international application number
_____ filed on _____.

The above-identified application was made or authorized to be made by me.

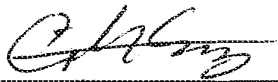
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor: Gihsung CHUNG

Date: May, 30, 2016.

Signature: 

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)**

**TITLE: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC
DEVICE USING SAME**

As the below named inventor, I hereby declare that:

This declaration is directed to:

- ☒ The attached application, or
- ☐ United States application or PCT international application number
_____ filed on _____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor: Jooman HAN

Date: May 30, 2016

Signature: Jooman Han

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)**

**TITLE: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC
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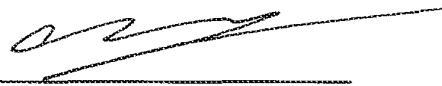
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor: So Hyun CHUNG

Date: May 30, 2016

Signature: 

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
UTILITY OR DESIGN APPLICATION
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LEGAL NAME OF INVENTOR

Inventor: Dae-Yong LEE

Date: May 30, 2016

Signature: 

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
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LEGAL NAME OF INVENTOR

Inventor: Hongsig KIM

Date: May 30, 2016

Signature: Hongsig KIM

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
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LEGAL NAME OF INVENTOR

Inventor: Hyunwoo JANG

Date: May. 30. 2011

Signature: _____



DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
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_____ filed on _____.

The above-identified application was made or authorized to be made by me.

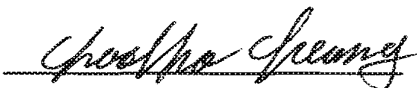
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTOR

Inventor: Cheolho CHEONG

Date: May. 26. 2016

Signature: 

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
UTILITY OR DESIGN APPLICATION
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**TITLE: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC
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This declaration is directed to:

- ☒ The attached application, or
- ☐ United States application or PCT international application number
_____ filed on _____.

The above-identified application was made or authorized to be made by me.

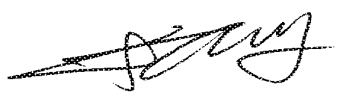
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor: Jae-Woong CHUN

Date: May 26, 2016

Signature: 

DOCKET NO.: 0202-1231
CLIENT REF.: SP16061-US_DMC
SAMSUNG REF.: GM-201507-140-1-US0

**DECLARATION (37 CFR 1.63) FOR
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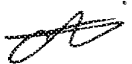
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor: Won Suk CHOI

Date: May 26, 2016

Signature: 

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

**TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE
REGISTERED PRACTITIONERS**

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B or equivalent) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5. If the Power of Attorney by Applicant form is not accompanied by this transmittal form or an equivalent, the Power of Attorney will not be recognized in the application.

Application Number	
Filing Date	January 24, 2019
First Named Inventor	Gihsung CHUNG
Title	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

SIGNATURE of Applicant or Patent Practitioner

Signature	/Wonki PARK/	Date	January 24, 2019
Name	Wonki Park	Telephone	202-293-0804
Registration Number	38,991		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY BY APPLICANT

REGISTERED PRACTITIONERS

I hereby revoke all previous powers of attorney given in the application identified in the attached transmittal letter:

- ☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

68103

OR

- ☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

Name	Registration Number	Name	Registration Number

Please recognize or change the correspondence address for the application identified in the attached transmittal letter to:

- ☒ The address associated with the above-mentioned Customer Number.

OR

- ☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document)

SIGNATURE of Applicant for Patent

Signature

Name

Heungmo Lee

Date

2012.10.11

Telephone

+82-31-279-4761

Title and Company

Vice President, Samsung Electronics Co., Ltd.

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms for more than one signature, see below.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Gihung		CHUNG		
Residence Information (Select One) US Residency ● Non US Residency Active US Military Service					
City	Suwon-si		Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	#2-403, 1035, Dongtanwoncheon-ro, Yeongtong-gu				
Address 2					
City	Suwon-si, Gyeonggi-do		State/Province		
Postal Code		Country ⁱ	KR		

Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jooman		HAN		
Residence Information (Select One) US Residency ● Non US Residency Active US Military Service					
City	Seongnam-si		Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	#208-1101, 179, Buljeong-ro, Bundang-gu				
Address 2					
City	Seongnam-si, Gyeonggi-do		State/Province		
Postal Code		Country ⁱ	KR		

Inventor	3				Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

Prefix	Given Name	Middle Name	Family Name	Suffix
	So Hyun		CHUNG	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Seoul	Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	#122-1504, 1218, Yangjae-daero, Songpa-gu			
Address 2				
City	Seoul	State/Province		
Postal Code		Country ⁱ	KR	
Inventor	4			Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Dae-Yong		EE	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Suwon-si	Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	#202, 84-16, Maetan-ro 140beon-gil, Yeongtong-gu			
Address 2				
City	Suwon-si, Gyeonggi-do	State/Province		
Postal Code		Country ⁱ	KR	
Inventor	5			Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Hongsig		KIM	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Seongnam-si	Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	#109-1403, 184, Migeum-ro, Bundang-gu			
Address 2				
City	Seongnam-si, Gyeonggi-do	State/Province		
Postal Code		Country ⁱ	KR	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

Inventor	6				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyunwoo		JANG		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Suwon-si	Country of Residence ⁱ		KR	

Mailing Address of Inventor:

Address 1	#506, 17, Yeongtong-ro 331beon-gil, Yeongtong-gu				
Address 2					
City	Suwon-si, Gyeonggi-do	State/Province			
Postal Code		Country ⁱ	KR		

Inventor	7				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Cheolho		CHEONG		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Seoul	Country of Residence ⁱ		KR	

Mailing Address of Inventor:

Address 1	#301, 14, Dongmak-ro 32an-gil, Mapo-gu				
Address 2					
City	Seoul	State/Province			
Postal Code		Country ⁱ	KR		

Inventor	8				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jae-Woong		CHUN		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Suwon-si	Country of Residence ⁱ		KR	

Mailing Address of Inventor:

Address 1	#6103-2104, 76, Central town-ro, Yeongtong-gu				
Address 2					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

City	Suwon-si, Gyeonggi-do	State/Province	
Postal Code		Country i	KR
Inventor	9	Remove	
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Won Suk		CHOI
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service			
City	Seoul	Country of Residence i	KR

Mailing Address of Inventor:

Address 1	#8-602, 391, Hyoryeong-ro, Seocho-gu		
Address 2			
City	Seoul	State/Province	
Postal Code		Country i	KR
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add			

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	68103		
Email Address	usdocketing@jeffersonip.com	Add Email	Remove Email

Application Information:

Title of the Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		
Attorney Docket Number	0202-1231-1	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	23	Suggested Figure for Publication (if any)	

Filing By Reference :

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	68103		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the application number blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	15173099	2016-06-03

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

[Add](#)

Foreign Priority Information:

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)	Remove
10-2015-0113972	KR	2015-08-12	F066	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.				Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization to Permit Access:

☒ Authorization to Permit Access to the Instant Application by the Participating Offices
--

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.		
Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.		
Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		

Name of the Deceased or Legally Incapacitated Inventor :			
If the Applicant is an Organization check here. ☒			
Organization Name	Samsung Electronics Co., Ltd.		
Mailing Address Information For Applicant:			
Address 1	129, Samsung-ro, Yeongtong-gu		
Address 2			
City	Suwon-si, Gyeonggi-do	State/Province	
Country	KR	Postal Code	16677
Phone Number		Fax Number	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button.			Add

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee	1			
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
				Remove
If the Assignee or Non-Applicant Assignee is an Organization check here.				☐
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications.				
Signature	Wonki PARK/		Date (YYYY-MM-DD)	2019-01-24
First Name	Wonki	Last Name	Park	Registration Number
				38991
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0202-1231-1
		Application Number	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		2019-01-24
	First Named Inventor	Gihung CHUNG	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	0202-1231-1	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4129125	A	1978-12-12	LESTER et al.	
	2	8942796	B2	2015-01-27	KASAMA	

If you wish to add additional U.S. Patent citation information please click the Add button.

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20150074797	A1	2015-03-12	CHOI et al.	
	2	20110092780	A1	2011-04-21	ZHANG et al.	
	3	20150057967	A1	2015-02-26	ALBINALI	
	4	20150223355	A1	2015-08-06	FLECK et al.	

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FOREIGN PATENT DOCUMENTS						Remove
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2019-01-24
First Named Inventor	Gihsung CHUNG	
Art Unit		
Examiner Name		
Attorney Docket Number	0202-1231-1	

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	2007/209430	JP	A	2007-08-23	TAKASHI et al.		
	2	10-2015-0046976	KR	A	2015-05-04	SONG		
	3	2989973	EP	A1	2016-03-02	CHOI et al.		
	4	2009140360	WO	A1	2009-11-19	KATERAAS et al.		
	5	2013043747	WO	A1	2013-03-28	KAHN et al.		
	6	2851001	EP	A2	2015-03-25	KOMAROMI et al.		
	7	2014120832	WO	A1	2014-08-07	LEE et al.		

If you wish to add additional Foreign Patent Document citation information please click the Add button

NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1		

If you wish to add additional non-patent literature document citation information please click the Add button

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2019-01-24
Name/Print	Wonki Park	Registration Number	38,991

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No.	:	To be determined	Confirmation No.	:	To be determined
First Named Inventor	:	Gihsung CHUNG	TC/A.U.	:	To be determined
Filed	:	January 24, 2019	Examiner	:	To be determined
Title	:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME	Docket No.	:	0202-1231-1

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with 37 C.F.R. §§ 1.56 and 1.97-1.98, Applicants submit the attached Form PTO/SB/08, which lists references that may be deemed material to the examination of the Applicants' above-identified application.

Notably, all references cited on the attached form PTO/SB/08 were either previously cited by or submitted to the Office in prior application 15/173,099, which prior application is relied upon for an earlier filing date under 35 U.S.C. §120. Thus, as allowed by 37 C.F.R. §1.98(d)(1) and (2), and as explained in MPEP §609.04(a)(II), Applicants are not providing copies of any cited references, as these references are available to the Office.

The Examiner is requested to consider the attached documents in connection with the Applicants' above-identified application and to return a copy of the Form PTO/SB/08 to the Applicants with the Examiner's initials in the spaces provided.

Submission of this Information Disclosure Statement does not constitute an admission by the Applicants as to the materiality of the attached documents to the Applicants' application, nor do the Applicants waive any right to challenge the validity of the documents as prior art should such action be deemed appropriate.

Respectfully submitted,

/Wonki PARK/
Wonki Park
Reg. No. 38,991

Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, D.C. 20036
(202) 293-0804

Dated: January 24, 2019

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC
DEVICE USING SAME

First Named Inventor/Applicant Name:

Gihsung CHUNG

Filer:

Maeng-Ho Shin/christine peoples

Attorney Docket Number:

0202-1231-1

Filed as Large Entity

Filing Fees for Utility under 35 USC 111(a)

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	300	300
UTILITY SEARCH FEE	1111	1	660	660
UTILITY EXAMINATION FEE	1311	1	760	760

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1720

Electronic Acknowledgement Receipt

EFS ID:	34950036
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/christine peoples
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	24-JAN-2019
Filing Date:	
Time Stamp:	15:43:12
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1720
RAM confirmation Number	012519INTEFSW15435300
Deposit Account	504100
Authorized User	christine peoples

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		0202-1231-1ApplicationAsFiled.pdf	3074289	yes	115
			7a463ffb844119710aa9fd605af7f5854a7c4254		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal of New Application		1	1	
	Specification		2	75	
	Claims		76	80	
	Abstract		81	81	
	Drawings-only black and white line drawings		82	104	
	Oath or Declaration filed		105	113	
	Power of Attorney		114	115	
Warnings:					
Information:					
2	Application Data Sheet	0202-1231-1ApplicationDataSheetAsFiled.pdf	1819821	no	10
			e073443bae24ab24137604bfe687dfd95e07683		
Warnings:					
Information:					
3	Information Disclosure Statement (IDS) Form (SB08)	0202-1231-1IDSAsFiled01-24-2019.pdf	612664	no	5
			f10ec7c27544ce23e6f3e5a4e1d6faf27aa5671e		
Warnings:					
Information:					

4	Transmittal Letter	0202-1231-1IDSLetterAsFiled.pdf	80560	no	2
			bf4c11fda1fddf61c9a740624c5900894533f440		

Warnings:

Information:

5	Fee Worksheet (SB06)	fee-info.pdf	34759	no	2
			4af6314583b3930b5cc02ea8b5910e6c58edf92d		

Warnings:

Information:

Total Files Size (in bytes):			5622093
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/256,497	01/24/2019		1720	0202-1231-1	20	3

CONFIRMATION NO. 6284

FILING RECEIPT

68103

Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036



0000000105828138

Date Mailed: 02/13/2019

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Gihsung CHUNG, Suwon-si, KOREA, REPUBLIC OF;
Jooman HAN, Seongnam-si, KOREA, REPUBLIC OF;
So Hyun CHUNG, Seoul, KOREA, REPUBLIC OF;
Dae-Yong LEE, Suwon-si, KOREA, REPUBLIC OF;
Hongsig KIM, Seongnam-si, KOREA, REPUBLIC OF;
Hyunwoo JANG, Suwon-si, KOREA, REPUBLIC OF;
Cheolho CHEONG, Seoul, KOREA, REPUBLIC OF;
Jae-Woong CHUN, Suwon-si, KOREA, REPUBLIC OF;
Won Suk CHOI, Seoul, KOREA, REPUBLIC OF;

Applicant(s)

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 68103

Domestic Priority data as claimed by applicant

This application is a CON of 15/173,099 06/03/2016

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2015-0113972 08/12/2015 Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

Request to Retrieve - This application either claims priority to one or more applications filed in an intellectual property Office that participates in the Priority Document Exchange (PDX) program or contains a proper **Request to Retrieve Electronic Priority Application(s)** (PTO/SB/38 or its equivalent). Consequently, the USPTO will attempt to electronically retrieve these priority documents.

If Required, Foreign Filing License Granted: 02/11/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/256,497**

Projected Publication Date: 05/23/2019

Non-Publication Request: No

Early Publication Request: No

Title

METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING
SAME

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/256,497

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	300
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	660
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	760
TOTAL CLAIMS (37 CFR 1.16(j))	20 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*				x 460 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1720

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

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Feb 13, 2019 03:41:02 AM

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Application	Document	Mailroom Date	Attorney Docket No.
16256497	APP.FILE.REC	02/13/2019	0202-1231-1

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16256497
	Filing Date		2019-01-24
	First Named Inventor	Gihung CHUNG	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		0202-1231-1

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²ⁱ	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16256497
	Filing Date		2019-01-24
	First Named Inventor	Gihsung CHUNG	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		0202-1231-1

1	European Office Action dated February 13, 2019; Application #: 16 183 812.3 - 1132; Ref. #P6061727EP
---	--

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2019-05-07
Name/Print	Wonki Park	Registration Number	38991

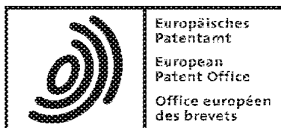
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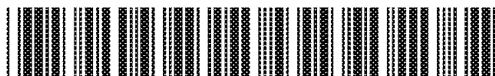
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Substantive Examiner
Name: Pohjamo, Terhi
Tel: +49 30 25901 - 577

Application No. 16 183 812.3 - 1132	Ref. P6061727EP	Date 13.02.2019
Applicant Samsung Electronics Co., Ltd.		

Communication pursuant to Article 94(3) EPC

The examination of the above-identified application has revealed that it does not meet the requirements of the European Patent Convention for the reasons enclosed herewith. If the deficiencies indicated are not rectified the application may be refused pursuant to Article 97(2) EPC.

You are invited to file your observations and insofar as the deficiencies are such as to be rectifiable, to correct the indicated deficiencies within a period

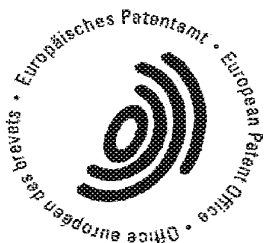
of 4 months

from the notification of this communication, this period being computed in accordance with Rules 126(2) and 131(2) and (4) EPC. One set of amendments to the description, claims and drawings is to be filed within the said period on separate sheets (R. 50(1) EPC).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the Examining Division requesting that you correct this deficiency (R. 137(4) EPC).

In case you withdraw the application within the above-mentioned period, the examination fee will be refunded at a rate of 50% (Art. 11 (b) RFees).

Failure to comply with this invitation in due time will result in the application being deemed to be withdrawn (Art. 94(4) EPC).



Pohjamo, Terhi
Primary Examiner
For the Examining Division

Enclosure(s): 2 page/s reasons (Form 2906)

The examination is being carried out on the **following application documents**

Description, Pages

1-66 as originally filed

Claims, Numbers

1-15 filed in electronic form on 13-06-2017

Drawings, Sheets

1/23-23/23 as originally filed

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 EP 2 989 973 A1 (SAMSUNG ELECTRONICS CO LTD [KR]) 2 March 2016 (2016-03-02)

D2 WO 2009/140360 A1 (ESPENUDA HOLDING LLC [US]; KATERAAS ESPEN D [US]; MEDELIUS PEDRO J [US]) 19 November 2009 (2009-11-19)

D3 WO 2013/043747 A1 (DP TECHNOLOGIES INC [US]; KAHN PHILLIPPE [US]; KINSOLVING ARTHUR [US]) 28 March 2013 (2013-03-28)

D4 EP 2 851 001 A2 (SENSIRION AG [CH]) 25 March 2015 (2015-03-25)

D5 US 2015/057967 A1 (ALBINALI FAHD KHALAF [US]) 26 February 2015 (2015-02-26)

D6 WO 2014/120832 A1 (SALUTRON INC [US]) 7 August 2014 (2014-08-07)

2 In reply to the invitation to correct deficiencies pursuant to Rule 70a EPC, the applicant has failed to suitably address and/or rectify the deficiencies of the European search opinion (ESOP). The objections in the ESOP are upheld. The arguments on clarity, and on novelty over D1-D3 are not well substantiated, and not convincing.

Additionally to those objections, claims 1-15 also do not fulfill the requirements of Art. 123(2) EPC as there is no basis directly and unambiguously derivable from the originally filed application for the amendments. See also Guidelines (eg. H-V.3.1.) In addition, also the requirements of Rule 137(4) EPC are not fulfilled as there is no basis whatsoever provided for the amendments. See also Guidelines H-III.2.1.

- 3 If the applicant fails to overcome the objections, the next procedural step will be the summoning of oral proceedings, as requested by the applicant in his letter of 13.6.2017.

Electronic Acknowledgement Receipt

EFS ID:	35937750
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Jorge Ivan Negron-Garcia
Filer Authorized By:	
Attorney Docket Number:	0202-1231-1
Receipt Date:	07-MAY-2019
Filing Date:	24-JAN-2019
Time Stamp:	13:07:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	0202-1231-1IDSAsFiled05-07-2019.pdf	1053160 798dc46b488bc9367af33d1bb275d1ae71015e16	no	4

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/256,497	01/24/2019	Gihsung CHUNG	0202-1231-1

CONFIRMATION NO. 6284

68103
Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036

PUBLICATION NOTICE



OC000000108413468

Title:METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

Publication No.US-2019-0150854-A1

Publication Date:05/23/2019

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Application	Document	Mailroom Date	Attorney Docket No.
16256497	NTC.PUB	05/23/2019	0202-1231-1

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Application number: 10-2015-0113972

Date of availability of document: 24 Jan 2019 (24.01.2019)

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출 원 번 호 : 10-2015-0113972

Application Number

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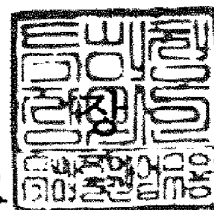
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Applicant(s) SAMSUNG ELECTRONICS CO., LTD.

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특 허 청

COMMISSIONER



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【대리인번호】	9-1998-000115-1
【포괄위임등록번호】	2002-060519-2
【대리인】	
【성명】	이정순
【대리인번호】	9-1998-000404-2
【포괄위임등록번호】	2005-005297-6
【발명의 국문명칭】	생체 정보 검출 방법 및 그 방법을 처리하는 전자 장치
【발명의 영문명칭】	METHOD FOR DETECTING BIOMETRIC INFORMATION AND AN ELECTRONIC DEVICE THEREOF
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【주민등록번호】 770527-1XXXXXX

제출 일자 : 2015-08-12

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【성명의 영문표기】 CHOI Won-Suk

【주민등록번호】 800922-1XXXXXX

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【국적】 KR

【출원언어】 국어

【취지】 위와 같이 특허청장에게 제출합니다.

대리인 권혁록 (서명 또는 인)

대리인 이정순 (서명 또는 인)

【수수료】

【출원료】 0 면 46,000 원

【가산출원료】 121 면 0 원

【우선권주장료】 0 건 0 원

【심사청구료】 0 항 0 원

【합계】 46,000 원

【발명의 설명】

【발명의 명칭】

생체 정보 검출 방법 및 그 방법을 처리하는 전자 장치(METHOD FOR DETECTING BIOMETRIC INFORMATION AND AN ELECTRONIC DEVICE THEREOF)

【기술분야】

【0001】 본 발명의 다양한 실시예는 전자 장치에 관한 것으로서, 예를 들면, 생체 정보 검출을 위한 장치 및 방법에 관한 것이다.

【발명의 배경이 되는 기술】

【0003】 전자 장치는 동일하거나 좀더 다양한 기능을 수행하면서 점차 소형 화되고 슬림화되어 휴대하기 용이한 방향으로 발전하고 있다. 이러한 소형 전자 장치들은 일반적으로 사용자의 주머니 등에 수납되어 휴대하기도 하지만, 손목에 착용하거나, 인체의 두부(head portion) 또는 팔에 착용하기도 한다.

【0004】 더하여, 전자 장치는 개인이 관리하는 생체 정보(예: 건강 데이터)를 측정하기 위하여 생체 센서(예: 건강 감지 센서(health care sensor))들을 탑재할 수 있다. 예를 들어, 생체 정보는 혈압(blood pressure), 심박(heart rate), 심전도(electrocardiogram), 피부수분도(skin moisture), 또는 산소포화도(saturation of peripheral oxygen) 등을 포함할 수 있다.

【발명의 내용】

【해결하고자 하는 과제】

【0006】 전자 장치는 생체 정보(예: 심박수)를 측정하기 위해 생체 센서(예: 심박 센서)를 작동시킬 수 있다. 생체 센서의 구동은 전자 장치의 배터리를 소모시키는 요소가 될 수 있다. 이에 따라, 전자 장치에서 장시간 동안(예: 사용자가 전자 장치를 착용하고 있는 시간 동안) 생체 정보를 측정하기 어려울 수 있다.

【0007】 다른 예로, 전자 장치가 사용자의 생체 정보를 측정하는 경우, 전자 장치의 움직임에 따라서 생체 정보의 측정이 불가하거나 측정된 생체 정보의 정확도가 떨어질 수 있다.

【0008】 본 발명의 다양한 실시예에 따른 전자 장치 및 방법은 선택된(또는, 지정된) 조건에 기반하여 생체 센서를 선택적으로 동작시킬 수 있다.

【0009】 본 발명의 다양한 실시예에 따른 전자 장치 및 방법은 선택된(또는, 지정된) 주기에 따라서 생체 센서를 지속적(예: 주기적(예: 약 1분 간격))으로 동작시킬 수 있다.

【0010】 본 발명의 다양한 실시예에 따른 전자 장치 및 방법은 전자 장치의 움직임에 기반하여 생체 정보를 측정할 수 있다.

【과제의 해결 수단】

【0012】 다양한 실시예에 따른 전자 장치는 하우징, 상기 하우징의 일부에 결합되고, 사용자의 신체의 일부에 착탈가능하게 부착되는 결합구, 상기 전자 장치의 움직임에 감지할 수 있도록 구성된 적어도 하나의 제1 센서, 상기 사용자의 신체의 변화를 감지할 수 있도록 구성된 적어도 하나의 제2 센서, 메모리 및 상기 제1 센서, 상기 제2 센서, 및 상기 메모리와 전기적으로 연결된 프로세서를 포함하고, 상기 메모리는, 실행 시에, 상기 프로세서가, 상기 적어도 하나의 제1 센서를 이용하여 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지 여부를 판단하고, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된(또는, 지정된) 제1 시간 주기(a first period of time)를 경과하는 시점 바로 전(immediately before) 및/또는 바로 후(immediately after)의, 상기 제1 시간 주기보다 짧은 제2 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하고, 상기 움직임에 관한 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 인스트럭션들을 저장할 수 있다.

【0013】 다양한 실시예에 따른 전자 장치의 생체 정보 측정 방법은 적어도 하나의 제1 센서를 이용하여 전자 장치가 사용자의 신체의 일부에 착용되었는지 여부를 판단하는 동작, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된(또는, 지정된) 제1 시간 주기(a first period of time)를 경과하는 시점까지 측정된 전자 장치의 움직임의 적어도 일부에 기반하여 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하는 동작 및 상기 적어도 하나의 제2

센서에 기반하여 생체 정보를 측정하는 동작을 포함할 수 있다.

【발명의 효과】

【0015】 본 발명의 다양한 실시예에 따르는 전자 장치 및 방법은, 예를 들면, 선택된(또는, 지정된) 조건에 기반하여 생체 센서를 선택적으로 동작시킬 수 있는 바, 배터리 소모를 줄이면서 생체 정보를 측정할 수 있다.

【0016】 본 발명의 다양한 실시예에 따르는 전자 장치 및 방법은, 예를 들면, 선택된(또는, 지정된) 주기에 따라서 생체 센서를 동작시킬 수 있는 바, 생체 정보를 지속적으로 측정할 수 있다.

【0017】 또한, 본 발명의 다양한 실시예에 따르는 전자 장치 및 방법은, 예를 들면, 전자 장치의 움직임에 측정하고, 측정된 움직임에 기반하여 선택적으로 사용자의 생체 정보를 측정할 수 있는 바, 측정된 생체 정보의 정확도를 개선할 수 있다.

【도면의 간단한 설명】

【0019】 도 1은 다양한 실시예에 따른 네트워크 환경 내의 전자 장치를 도시한 도면이다.

도 2는 다양한 실시예에 따른 전자 장치의 블록도이다.

도 3은 다양한 실시예에 따른 프로그램 모듈의 블록도이다.

도 4a 내지 도 4c는 다양한 실시예들에 따른 전자 장치를 도시한 도면이다.

도 5a는 다양한 실시예들에 따른 프로세서의 구성을 도시한 도면이다.

도 5b는 다양한 실시예들에 따른 전자 장치의 생체 정보의 측정 동작을 도시한 도면이다.

도 6은 본 발명의 다양한 실시예에 따른 전자 장치의 생체 정보 측정 동작을 도시한 흐름도이다.

도 7은 본 발명의 다양한 실시예에 따른 전자 장치의 상태 판단 방법의 수행 절차를 도시한 흐름도이다.

도 8은 본 발명의 다양한 실시예에 따른 전자 장치의 센서 동작 방법의 수행 절차를 도시한 흐름도이다.

도 9는 다양한 실시예에 따른 전자 장치의 생체 정보 측정 방법의 수행 절차를 도시한 흐름도이다.

도 10은 다양한 실시예에 따른 전자 장치의 생체 정보 측정 방법의 수행 절차를 도시한 흐름도이다.

도 11은 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 12는 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 13은 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행

절차를 도시한 흐름도이다.

도 14는 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 15는 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 16은 다양한 실시예에 따른 측정 트리거 변경 방법의 수행 절차를 도시한 흐름도이다.

도 17은 다양한 실시예에 따른 전자 장치의 측정 트리거 변경 동작을 도시한 도면이다.

도 18은 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 19는 다양한 실시예에 따른 전자 장치의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

도 20은 다양한 실시예에 따른 전자 장치의 화면 구성을 도시한 도면이다.

【발명을 실시하기 위한 구체적인 내용】

【0020】 이하, 본 개시의 다양한 실시예가 첨부된 도면을 참조하여 기재된다. 그러나, 이는 본 개시를 특정한 실시 형태에 대해 한정하려는 것이 아니며, 본 개시의 실시예의 다양한 변경(modification), 균등물(equivalent), 및/또는 대체물(alternative)을 포함하는 것으로 이해되어야 한다. 도면의 설명과

관련하여, 유사한 구성요소에 대해서는 유사한 참조 부호가 사용될 수 있다.

【0021】 본 문서에서, “가진다”, “가질 수 있다”, “포함한다” 또는 “포함할 수 있다” 등의 표현은 해당 특징(예: 수치, 기능, 동작, 또는 부품 등의 구성요소)의 존재를 가리키며, 추가적인 특징의 존재를 배제하지 않는다.

【0022】 본 문서에서, “A 또는 B”, “A 또는/및 B 중 적어도 하나” 또는 “A 또는/및 B 중 하나 또는 그 이상” 등의 표현은 함께 나열된 항목들의 모든 가능한 조합을 포함할 수 있다. 예를 들면, “A 또는 B”, “A 및 B 중 적어도 하나” 또는 “A 또는 B 중 적어도 하나”는, (1) 적어도 하나의 A를 포함, (2) 적어도 하나의 B를 포함, 또는 (3) 적어도 하나의 A 및 적어도 하나의 B 모두를 포함하는 경우를 모두 지칭할 수 있다.

【0023】 다양한 실시예에서 사용된 “제 1”, “제 2”, “첫째” 또는 “둘째” 등의 표현들은 다양한 구성요소들을, 순서 및/또는 중요도에 상관없이 수식할 수 있고, 해당 구성요소들을 한정하지 않는다. 상기 표현들은 한 구성요소를 다른 구성요소와 구분하기 위해 사용될 수 있다. 예를 들면, 제 1 사용자 기기와 제 2 사용자 기기는, 순서 또는 중요도와 무관하게, 서로 다른 사용자 기기를 나타낼 수 있다. 예를 들면, 본 개시의 권리 범위를 벗어나지 않으면서 제 1 구성요소는 제 2 구성요소로 명명될 수 있고, 유사하게 제 2 구성요소도 제 1 구성요소로 바꾸어 명명될 수 있다.

【0024】 어떤 구성요소(예: 제 1 구성요소)가 다른 구성요소(예: 제 2 구성요소)에 "(기능적으로 또는 통신적으로) 연결되어((operatively or communicatively) coupled with/to)" 있다거나 "접속되어(connected to)" 있다고 언급된 때에는, 상기 어떤 구성요소가 상기 다른 구성요소에 직접적으로 연결되거나, 다른 구성요소(예: 제 3 구성요소)를 통하여 연결될 수 있다고 이해되어야 할 것이다. 반면에, 어떤 구성요소(예: 제 1 구성요소)가 다른 구성요소(예: 제 2 구성요소)에 "직접 연결되어" 있다거나 "직접 접속되어" 있다고 언급된 때에는, 상기 어떤 구성요소와 상기 다른 구성요소 사이에 다른 구성요소(예: 제 3 구성요소)가 존재하지 않는 것으로 이해될 수 있다.

【0025】 본 문서에서 사용된 표현 “~하도록 구성된(또는 설정된)(configured to)”은 상황에 따라, 예를 들면, “~에 적합한(suitable for)”, “~하는 능력을 가지는(having the capacity to)”, “~하도록 설계된(designed to)”, “~하도록 변경된(adapted to)”, “~하도록 만들어진(made to)” 또는 “~할 수 있는(capable of)”과 바꾸어 사용될 수 있다. 용어 “~하도록 구성(또는 설정)된”은 하드웨어적으로 “특별히 설계된(specifically designed to)” 것만을 반드시 의미하지 않을 수 있다. 대신, 어떤 상황에서는, “~하도록 구성된 장치”라는 표현은, 그 장치가 다른 장치 또는 부품들과 함께 “~할 수 있는” 것을 의미할 수 있다. 예를 들면, 문구 “A, B, 및 C를 수행하도록 구성(또는 설정)된 프로세서”는 해당 동작을 수행하기 위한 전용 프로세서(예: 임베디드 프로세서), 또는 메모리 장치에 저장된 하나 이상의 소프트웨어 프로그램들을 실행함으로써, 해당

동작들을 수행할 수 있는 범용 프로세서(generic-purpose processor)(예: CPU 또는 application processor)를 의미할 수 있다.

【0026】 본 문서에서 사용된 용어들은 단지 특정한 실시예를 설명하기 위해 사용된 것으로, 다른 실시예의 범위를 한정하려는 의도가 아닐 수 있다. 단수의 표현은 문맥상 명백하게 다르게 뜻하지 않는 한, 복수의 표현을 포함할 수 있다. 기술적이거나 과학적인 용어를 포함해서 여기서 사용되는 모든 용어들은 본 개시의 기술 분야에서 통상의 지식을 가진 자에 의해 일반적으로 이해되는 것과 동일한 의미를 가질 수 있다. 일반적으로 사용되는 사전에 정의된 용어들은 관련 기술의 문맥 상 가지는 의미와 동일 또는 유사한 의미를 가지는 것으로 해석될 수 있으며, 본 문서에서 명백하게 정의되지 않는 한, 이상적이거나 과도하게 형식적인 의미로 해석되지 않는다. 경우에 따라서, 본 문서에서 정의된 용어일지라도 본 개시의 실시예들을 배제하도록 해석될 수 없다.

【0027】 본 개시의 다양한 실시예들에 따른 전자 장치는, 예를 들면, 전자 장치는 스마트폰(smartphone), 태블릿 PC(tablet personal computer), 이동 전화기(mobile phone), 화상 전화기, 전자북 리더기(e-book reader), 데스크탑 PC(desktop personal computer), 랩탑 PC(laptop personal computer), 넷북 컴퓨터(netbook computer), 워크스테이션(workstation), 서버, PDA(personal digital assistant), PMP(portable multimedia player), MP3 플레이어, 모바일 의료기기, 카메라(camera), 또는 웨어러블 장치(wearable device)(예: 스마트 안경, 머리 착용형 장치(head-mounted-device(HMD))), 전자 의복, 전자 팔찌, 전자 목걸이, 전자

앱세서리(appcessory), 전자 문신, 스마트 미러, 또는 스마트 워치(smart watch)) 중 적어도 하나를 포함할 수 있다.

【0028】 어떤 실시예들에서, 전자 장치는 스마트 가전 제품(smart home appliance)일 수 있다. 스마트 가전 제품은, 예를 들면, 텔레비전, DVD(digital versatile disk) 플레이어, 오디오, 냉장고, 에어컨, 청소기, 오븐, 전자레인지, 세탁기, 공기 청정기, 셋톱 박스(set-top box), 홈 오토메이션 컨트롤 패널(home automation control panel), 보안 컨트롤 패널(security control panel), TV 박스(예: 삼성 HomeSync™, 애플TV™, 또는 구글 TV™), 게임 콘솔(예: Xbox™, PlayStation™), 전자 사전, 전자 키, 캠코더(camcorder), 또는 전자 액자 중 적어도 하나를 포함할 수 있다.

【0029】 다른 실시예에서, 전자 장치는, 각종 의료기기(예: 각종 휴대용 의료측정기기(혈당 측정기, 심박 측정기, 혈압 측정기, 또는 체온 측정기 등), MRA(magnetic resonance angiography), MRI(magnetic resonance imaging), CT(computed tomography), 촬영기, 또는 초음파기 등), 네비게이션(navigation) 장치, GPS 수신기(global positioning system receiver), EDR(event data recorder), FDR(flight data recorder), 자동차 인포테인먼트(infotainment) 장치, 선박용 전자 장비(예: 선박용 항법 장치, 자이로 콤팩스 등), 항공 전자기기(avionics), 보안 기기, 차량용 헤드 유닛(head unit), 산업용 또는 가정용 로봇, 금융 기관의 ATM(automatic teller's machine), 상점의 POS(point of sales), 또는 사물 인터넷 장치(internet of things)(예: 전구, 각종 센서, 전기 또는 가스 미터기, 스포

링클러 장치, 화재경보기, 온도조절기(thermostat), 가로등, 토스터(toaster), 운동기구, 온수탱크, 히터, 보일러 등) 중 적어도 하나를 포함할 수 있다.

【0030】 어떤 실시예에 따르면, 전자 장치는 가구(furniture) 또는 건물/구조물의 일부, 전자 보드(electronic board), 전자 사인 수신 장치(electronic signature receiving device), 프로젝터(projector), 또는 각종 계측 기기(예: 수도, 전기, 가스, 또는 전파 계측 기기 등) 중 적어도 하나를 포함할 수 있다. 다양한 실시예에서, 전자 장치는 전술한 다양한 장치들 중 하나 또는 그 이상의 조합일 수 있다. 어떤 실시예에 따른 전자 장치는 플렉서블 전자 장치일 수 있다. 또한, 본 개시의 실시예에 따른 전자 장치는 전술한 기기들에 한정되지 않으며, 기술 발전에 따른 새로운 전자 장치를 포함할 수 있다.

【0031】 이하, 첨부 도면을 참조하여, 다양한 실시예에 따른 전자 장치가 설명된다. 본 문서에서, 사용자라는 용어는 전자 장치를 사용하는 사람 또는 전자 장치를 사용하는 장치(예: 인공지능 전자 장치)를 지칭할 수 있다.

【0033】 도 1을 참조하여, 다양한 실시예에서의, 네트워크 환경(100) 내의 전자 장치(101)가 기재된다. 전자 장치(101)는 버스(110), 프로세서(120), 메모리(130), 입출력 인터페이스(150), 디스플레이(160) 및/또는 통신 인터페이스(170)를 포함할 수 있다. 어떤 실시예에서는, 전자 장치(101)는 구성요소들 중 적어도 하나를 생략하거나 다른 구성 요소를 추가적으로 구비할 수 있다.

【0034】 버스(110)는 예를 들면, 구성요소들(120 내지 170)을 서로 연결하고, 구성요소들 간의 통신(예: 제어 메시지 및/또는 데이터)을 전달하는 회로를 포함할 수 있다.

【0035】 프로세서(120)는 중앙처리장치(central processing unit(CPU)), 어플리케이션 프로세서(application processor(AP)), 또는 커뮤니케이션 프로세서(communication processor(CP)) 중 하나 또는 그 이상을 포함할 수 있다. 프로세서(120)는 예를 들면, 전자 장치(101)의 적어도 하나의 다른 구성요소들의 제어 및/또는 통신에 관한 연산이나 데이터 처리를 실행할 수 있다.

【0036】 메모리(130)는 휘발성 및/또는 비휘발성 메모리를 포함할 수 있다. 메모리(130)는 예를 들면, 전자 장치(101)의 적어도 하나의 다른 구성요소에 관계된 명령 또는 데이터를 저장할 수 있다. 한 실시예에 따르면, 메모리(130)는 소프트웨어 및/또는 프로그램(140)을 저장할 수 있다. 프로그램(140)은, 예를 들면, 커널(141), 미들웨어(143), 어플리케이션 프로그래밍 인터페이스(application programming interface(API))(145) 및/또는 어플리케이션 프로그램(또는 “어플리케이션”)(147) 등을 포함할 수 있다. 커널(141), 미들웨어(143), 또는 API(145)의 적어도 일부는, 운영 시스템(operating system(OS))라 불릴 수 있다.

【0037】 커널(141)은, 예를 들면, 다른 프로그램들(예: 미들웨어(143), API(145), 또는 어플리케이션 프로그램(147))에 구현된 동작 또는 기능을 실행하는데 사용되는 시스템 리소스들(예: 버스(110), 프로세서(120), 또는 메모리(130) 등)을 제어 또는 관리할 수 있다. 또한, 커널(141)은 미들웨어(143), API(145), 또

는 어플리케이션 프로그램(147)에서 전자 장치(101)의 개별 구성요소에 접근함으로써, 시스템 리소스들을 제어 또는 관리할 수 있는 인터페이스를 제공할 수 있다.

【0038】 미들웨어(143)는 예를 들면, API(145) 또는 어플리케이션 프로그램(147)이 커널(141)과 통신하여 데이터를 주고받을 수 있도록 중개 역할을 수행할 수 있다. 또한, 미들웨어(143)는 어플리케이션 프로그램(147)으로부터 수신된 작업 요청들과 관련하여, 예를 들면, 어플리케이션 프로그램(147) 중 적어도 하나의 어플리케이션에 전자 장치(101)의 시스템 리소스(예: 버스(110), 프로세서(120), 또는 메모리(130) 등)를 사용할 수 있는 우선 순위를 배정하는 등의 방법을 이용하여 작업 요청에 대한 제어(예: 스케줄링 또는 로드 밸런싱)을 수행할 수 있다.

【0039】 API(145)는 예를 들면, 어플리케이션(147)이 커널(141) 또는 미들웨어(143)에서 제공되는 기능을 제어하기 위한 인터페이스로, 예를 들면, 파일 제어, 창 제어, 화상 처리, 또는 문자 제어 등을 위한 적어도 하나의 인터페이스 또는 함수(예: 명령어)를 포함할 수 있다.

【0040】 입출력 인터페이스(150)는 예를 들면, 사용자 또는 다른 외부 기기로부터 입력된 명령 또는 데이터를 전자 장치(101)의 다른 구성요소(들)에 전달할 수 있는 인터페이스의 역할을 할 수 있다. 또한, 입출력 인터페이스(150)는 전자 장치(101)의 다른 구성요소(들)로부터 수신된 명령 또는 데이터를 사용자 또는 다른 외부 기기로 출력할 수 있다.

【0041】 디스플레이(160)는 예를 들면, 액정 디스플레이(LCD), 발광 다이오드(LED) 디스플레이, 유기 발광 다이오드(OLED) 디스플레이, 또는 마이크로 전자기계 시스템(microelectromechanical systems(MEMS)) 디스플레이, 또는 전자종이(electronic paper) 디스플레이를 포함할 수 있다. 디스플레이(160)는 예를 들면, 사용자에게 각종 콘텐츠(예: 텍스트, 이미지, 비디오, 아이콘, 또는 심볼 등)를 표시할 수 있다. 디스플레이(160)는 터치 스크린을 포함할 수 있으며, 예를 들면, 전자 펜 또는 사용자의 신체의 일부를 이용한 터치, 제스처, 근접, 또는 호버링 입력을 수신할 수 있다.

【0042】 통신 인터페이스(170)는 예를 들면, 전자 장치(101)와 외부 장치(예: 제 1 외부 전자 장치(102), 제 2 외부 전자 장치(104), 또는 서버(106)) 간의 통신을 설정할 수 있다. 예를 들면, 통신 인터페이스(170)는 무선 통신 또는 유선 통신을 통해서 네트워크(162)에 연결되어 외부 장치(예: 제 2 외부 전자 장치(104) 또는 서버(106))와 통신할 수 있다.

【0043】 무선 통신은, 예를 들면, 셀룰러 통신 프로토콜로서, 예를 들면, LTE, LTE-A, CDMA, WCDMA, UMTS, WiBro, 또는 GSM 등 중 적어도 하나를 사용할 수 있다. 유선 통신은, 예를 들면, USB(universal serial bus), HDMI(high definition multimedia interface), RS-232(recommended standard 232), 또는 POTS(plain old telephone service) 등 중 적어도 하나를 포함할 수 있다. 네트워크(162)는 통신 네트워크(telecommunications network), 예를 들면, 컴퓨터 네트워크(computer network)(예: LAN 또는 WAN), 인터넷, 또는 전화 망(telephone network) 중 적어도

하나를 포함할 수 있다.

【0044】 제 1 및 제 2 외부 전자 장치(102), (104) 각각은 전자 장치(101)와 동일한 또는 다른 종류의 장치일 수 있다. 한 실시예에 따르면, 서버(106)는 하나 또는 그 이상의 서버들의 그룹을 포함할 수 있다.

【0045】 다양한 실시예에 따르면, 전자 장치(101)에서 실행되는 동작들의 전부 또는 일부는 다른 하나 또는 복수의 전자 장치(예: 전자 장치(102), (104), 또는 서버(106))에서 실행될 수 있다. 한 실시예에 따르면, 전자 장치(101)가 어떤 기능이나 서비스를 자동으로 또는 요청에 의하여 수행해야 할 경우에, 전자 장치(101)는 기능 또는 서비스를 자체적으로 실행시키는 대신에 또는 추가적으로, 그와 연관된 적어도 일부 기능을 다른 장치(예: 전자 장치(102), (104), 또는 서버(106))에게 요청할 수 있다. 다른 전자 장치(예: 전자 장치(102), (104), 또는 서버(106))는 요청된 기능 또는 추가 기능을 실행하고, 그 결과를 전자 장치(101)로 전달할 수 있다. 전자 장치(101)는 수신된 결과를 그대로 또는 추가적으로 처리하여 요청된 기능이나 서비스를 제공할 수 있다. 이를 위하여, 예를 들면, 클라우드 컴퓨팅, 분산 컴퓨팅, 또는 클라이언트-서버 컴퓨팅 기술이 이용될 수 있다.

【0047】 도 2는 다양한 실시예에 따른 전자 장치(201)의 블록도(200)이다. 전자 장치(201)는 예를 들면, 도 1에 도시된 전자 장치(101)의 전체 또는 일부를 포함할 수 있다. 전자 장치(201)는 하나 이상의 어플리케이션 프로세서(AP: application processor)(210), 통신 모듈(220), SIM(subscriber identification

module) 카드(224), 메모리(230), 센서 모듈(240), 입력 장치(250), 디스플레이(260), 인터페이스(270), 오디오 모듈(280), 카메라 모듈(291), 전력 관리 모듈(295), 배터리(296), 인디케이터(297) 및 모터(298)를 포함할 수 있다.

【0048】 AP(210)는 예를 들면, 운영 체제 또는 응용 프로그램을 구동하여 AP(210)에 연결된 다수의 하드웨어 또는 소프트웨어 구성요소들을 제어할 수 있고, 각종 데이터 처리 및 연산을 수행할 수 있다. AP(210)는 예를 들면, SoC(system on chip)로 구현될 수 있다. 한 실시예에 따르면, AP(210)는 GPU(graphic processing unit) 및/또는 이미지 신호 프로세서(image signal processor)를 더 포함할 수 있다. AP(210)는 도 2에 도시된 구성요소들 중 적어도 일부(예: 셀룰러 모듈(221))를 포함할 수도 있다. AP(210)는 다른 구성요소들(예: 비휘발성 메모리) 중 적어도 하나로부터 수신된 명령 또는 데이터를 휘발성 메모리에 로드(load)하여 처리하고, 다양한 데이터를 비휘발성 메모리에 저장(store)할 수 있다.

【0049】 통신 모듈(220)은, 도 1의 통신 인터페이스(170)와 동일 또는 유사한 구성을 가질 수 있다. 통신 모듈(220)은, 예를 들면, 셀룰러 모듈(221), WIFI 모듈(223), BT 모듈(225), GPS 모듈(227), NFC 모듈(228) 및 RF(radio frequency) 모듈(229)을 포함할 수 있다.

【0050】 셀룰러 모듈(221)은, 예를 들면, 통신망을 통해서 음성 통화, 영상 통화, 문자 서비스, 또는 인터넷 서비스 등을 제공할 수 있다. 한 실시예에 따르면, 셀룰러 모듈(221)은 가입자 식별 모듈(예: SIM 카드(224))을 이용하여 통신 네트워크 내에서 전자 장치(201)의 구별 및 인증을 수행할 수 있다. 한 실시예

에 따르면, 셀룰러 모듈(221)은 AP(210)가 제공할 수 있는 기능 중 적어도 일부 기능을 수행할 수 있다. 한 실시예에 따르면, 셀룰러 모듈(221)은 커뮤니케이션 프로세서(CP: communication processor)를 포함할 수 있다.

【0051】 WIFI 모듈(223), BT 모듈(225), GPS 모듈(227) 또는 NFC 모듈(228) 각각은, 예를 들면, 해당하는 모듈을 통해서 송수신되는 데이터를 처리하기 위한 프로세서를 포함할 수 있다. 어떤 실시예에 따르면, 셀룰러 모듈(221), WIFI 모듈(223), BT 모듈(225), GPS 모듈(227) 또는 NFC 모듈(228) 중 적어도 일부(예: 두 개 이상)는 하나의 integrated chip(IC) 또는 IC 패키지 내에 포함될 수 있다.

【0052】 RF 모듈(229)은, 예를 들면, 통신 신호(예: RF 신호)를 송수신할 수 있다. RF 모듈(229)은, 예를 들면, 트랜시버(transceiver), PAM(power amp module), 주파수 필터(frequency filter), LNA(low noise amplifier), 또는 안테나 등을 포함할 수 있다. 다른 실시예에 따르면, 셀룰러 모듈(221), WIFI 모듈(223), BT 모듈(225), GPS 모듈(227) 또는 NFC 모듈(228) 중 적어도 하나는 별개의 RF 모듈을 통하여 RF 신호를 송수신할 수 있다.

【0053】 SIM 카드(224)는, 예를 들면, 가입자 식별 모듈을 포함하는 카드 및 /또는 내장 SIM(embedded SIM)을 포함할 수 있으며, 고유한 식별 정보(예: ICCID(integrated circuit card identifier)) 또는 가입자 정보(예: IMSI(international mobile subscriber identity))를 포함할 수 있다.

【0054】 메모리(230)(예: 도 1의 메모리(130))는, 예를 들면, 내장 메모리(232) 또는 외장 메모리(234)를 포함할 수 있다. 내장 메모리(232)는, 예를 들면, 휘발성 메모리(예: DRAM(dynamic RAM), SRAM(static RAM), 또는 SDRAM(synchronous dynamic RAM) 등), 비휘발성 메모리(non-volatile Memory)(예: OTPROM(one time programmable ROM), PROM(programmable ROM), EPROM(erasable and programmable ROM), EEPROM(electrically erasable and programmable ROM), mask ROM, flash ROM, 플래시 메모리(예: NAND flash 또는 NOR flash 등), 하드 드라이브, 또는 솔리드 스테이트 드라이브(solid state drive(SSD)) 중 적어도 하나를 포함할 수 있다.

【0055】 외장 메모리(234)는 flash drive, 예를 들면, CF(compact flash), SD(secure digital), Micro-SD(micro secure digital), Mini-SD(mini secure digital), xD(extreme digital), 또는 메모리 스틱(memory stick) 등을 더 포함할 수 있다. 외장 메모리(234)는 다양한 인터페이스를 통하여 전자 장치(201)와 기능적으로 및/또는 물리적으로 연결될 수 있다.

【0056】 센서 모듈(240)은, 예를 들면, 물리량을 측정하거나 전자 장치(201)의 작동 상태를 감지하여, 측정 또는 감지된 정보를 전기 신호로 변환할 수 있다. 센서 모듈(240)은, 예를 들면, 제스처 센서(240A), 자이로 센서(240B), 기압 센서(240C), 마그네틱 센서(240D), 가속도 센서(240E), 그림 센서(240F), 근접 센서(240G), color 센서(240H(예: RGB(red, green, blue) 센서)), 생체 센서(240I), 온/습도 센서(240J), 조도 센서(240K), 또는 UV(ultra violet) 센서(240M) 등의 적

어도 하나를 포함할 수 있다. 추가적으로 또는 대체적으로, 센서 모듈(240)은, 예를 들면, 후각 센서(E-nose sensor), EMG 센서(electromyography sensor), EEG 센서(electroencephalogram sensor), ECG 센서(electrocardiogram sensor), IR(infrared) 센서, 홍채 센서 및/또는 지문 센서를 포함할 수 있다. 센서 모듈(240)은 그 안에 속한 적어도 하나 이상의 센서들을 제어하기 위한 제어 회로를 더 포함할 수 있다. 어떤 실시예에서는, 전자 장치(201)는 AP(210)의 일부로서 또는 별도로, 센서 모듈(240)을 제어하도록 구성된 프로세서를 더 포함하여, AP(210)가 슬립(sleep) 상태에 있는 동안, 센서 모듈(240)을 제어할 수 있다.

【0057】 입력 장치(250)는, 예를 들면, 터치 패널(touch panel(252)), (디지털) 펜 센서(pen sensor)(254), 키(key)(256), 또는 초음파(ultrasonic) 입력 장치(258)를 포함할 수 있다. 터치 패널(252)은, 예를 들면, 정전식, 감압식, 적외선 방식, 또는 초음파 방식 중 적어도 하나의 방식을 사용할 수 있다. 또한, 터치 패널(252)은 제어 회로를 더 포함할 수도 있다. 터치 패널(252)은 택타일 레이어(tactile layer)를 더 포함하여, 사용자에게 촉각 반응을 제공할 수 있다.

【0058】 (디지털) 펜 센서(254)는, 예를 들면, 터치 패널의 일부이거나, 별도의 인식용 쉬트(sheet)를 포함할 수 있다. 키(256)는, 예를 들면, 물리적인 버튼, 광학식 키, 또는 키패드를 포함할 수 있다. 초음파 입력 장치(258)는 초음파 신호를 발생하는 입력 도구를 통해, 전자 장치(201)에서 마이크(예: 마이크(288))로 음파를 감지하여 데이터를 확인할 수 있다.

【0059】 디스플레이(260)(예: 디스플레이(160))는 패널(262), 홀로그램 장치(264), 또는 프로젝터(266)를 포함할 수 있다. 패널(262)은, 도 1의 디스플레이(160)와 동일 또는 유사한 구성을 포함할 수 있다. 패널(262)은, 예를 들면, 유연하게(flexible), 투명하게(transparent), 또는 착용할 수 있게(wearable) 구현될 수 있다. 패널(262)은 터치 패널(252)과 하나의 모듈로 구성될 수도 있다. 홀로그램 장치(264)는 빛의 간섭을 이용하여 입체 영상을 허공에 보여줄 수 있다. 프로젝터(266)는 스크린에 빛을 투사하여 영상을 표시할 수 있다. 스크린은, 예를 들면, 전자 장치(201)의 내부 또는 외부에 위치할 수 있다. 한 실시예에 따르면, 디스플레이(260)는 패널(262), 홀로그램 장치(264), 또는 프로젝터(266)를 제어하기 위한 제어 회로를 더 포함할 수 있다.

【0060】 인터페이스(270)는, 예를 들면, HDMI(high-definition multimedia interface)(272), USB(universal serial bus)(274), 광 인터페이스(optical interface)(276), 또는 D-sub(D-subminiature)(278)을 포함할 수 있다. 인터페이스(270)는, 예를 들면, 도 1에 도시된 통신 인터페이스(170)에 포함될 수 있다. 추가적으로 또는 대체적으로, 인터페이스(270)는, 예를 들면, MHL(mobile high-definition link) 인터페이스, SD(secure digital) 카드/MMC(multi-media card) 인터페이스, 또는 IrDA(infrared data association) 규격 인터페이스를 포함할 수 있다.

【0061】 오디오 모듈(280)은, 예를 들면, 소리(sound)와 전기 신호를 쌍방향으로 변환시킬 수 있다. 오디오 모듈(280)의 적어도 일부 구성요소는, 예를 들면,

도 1 에 도시된 입출력 인터페이스(150)에 포함될 수 있다. 오디오 모듈(280)은, 예를 들면, 스피커(282), 리시버(284), 이어폰(286), 또는 마이크(288) 등을 통해 입력 또는 출력되는 소리 정보를 처리할 수 있다.

【0062】 카메라 모듈(291)은, 예를 들면, 정지 영상 및 동영상을 촬영할 수 있는 장치로서, 한 실시예에 따르면, 하나 이상의 이미지 센서(예: 전면 센서 또는 후면 센서), 렌즈, ISP(image signal processor), 또는 플래쉬(flash)(예: LED 또는 xenon lamp)를 포함할 수 있다.

【0063】 전력 관리 모듈(295)은, 예를 들면, 전자 장치(201)의 전력을 관리할 수 있다. 한 실시예에 따르면, 전력 관리 모듈(295)은 PMIC(power management integrated circuit), 충전 IC(charger integrated circuit), 또는 배터리 또는 연료 게이지(battery or fuel gauge)를 포함할 수 있다. PMIC는, 유선 및/또는 무선 충전 방식을 가질 수 있다. 무선 충전 방식은, 예를 들면, 자기공명 방식, 자기유도 방식 또는 전자기파 방식 등을 포함하며, 무선 충전을 위한 부가적인 회로, 예를 들면, 코일 루프, 공진 회로, 또는 정류기 등을 더 포함할 수 있다. 배터리 게이지는, 예를 들면, 배터리(296)의 잔량, 충전 중 전압, 전류, 또는 온도를 측정할 수 있다. 배터리(296)는, 예를 들면, 충전식 전지(rechargeable battery) 및/또는 태양 전지(solar battery)를 포함할 수 있다.

【0064】 인디케이터(297)는 전자 장치(201) 혹은 그 일부(예: AP(210))의 특정 상태, 예를 들면, 부팅 상태, 메시지 상태 또는 충전 상태 등을 표시할 수 있다. 모터(298)는 전기적 신호를 기계적 진동으로 변환할 수 있고, 진동

(vibration), 또는 햅틱(haptic) 효과 등을 발생시킬 수 있다. 도시되지는 않았으나, 전자 장치(201)는 모바일 TV 지원을 위한 처리 장치(예: GPU)를 포함할 수 있다. 모바일 TV 지원을 위한 처리 장치는, 예를 들면, DMB(digital multimedia broadcasting), DVB(digital video broadcasting), 또는 미디어 플로우(media flow) 등의 규격에 따른 미디어 데이터를 처리할 수 있다.

【0065】 전자 장치의 전술한 구성요소들 각각은 하나 또는 그 이상의 부품(component)으로 구성될 수 있으며, 해당 구성 요소의 명칭은 전자 장치의 종류에 따라서 달라질 수 있다. 다양한 실시예에서, 전자 장치는 전술한 구성요소 중 적어도 하나를 포함하여 구성될 수 있으며, 일부 구성요소가 생략되거나 또는 추가적인 다른 구성요소를 더 포함할 수 있다. 또한, 다양한 실시예에 따른 전자 장치의 구성 요소들 중 일부가 결합되어 하나의 개체(entity)로 구성됨으로써, 결합되기 이전의 해당 구성 요소들의 기능을 동일하게 수행할 수 있다.

【0067】 도 3은 다양한 실시예에 따른 프로그램 모듈(310)의 블록도(300)이다. 다양한 실시예에 따르면, 프로그램 모듈(310)(예: 도 1의 프로그램(140))은 전자 장치(예: 전자 장치(101))에 관련된 자원을 제어하는 운영 체제(operating system(OS)) 및/또는 운영 체제 상에서 구동되는 다양한 어플리케이션(예: 어플리케이션 프로그램(147))을 포함할 수 있다. 운영 체제는, 예를 들면, 안드로이드(android), iOS, 윈도우즈(windows), 심비안(symbian), 타이젠(tizen), 또는 바다(bada) 등이 될 수 있다.

【0068】 프로그램 모듈(310)은 커널(320), 미들웨어(330), 어플리케이션 프로그래밍 인터페이스(application programming interface (API))(360), 및/또는 어플리케이션(370)을 포함할 수 있다. 프로그램 모듈(310)의 적어도 일부는 전자 장치 상에 프리로드(preload) 되거나, 외부 전자 장치(예: 전자 장치(102), 전자 장치(104), 서버(106) 등)로부터 다운로드(download) 가능하다.

【0069】 커널(320)(예: 커널(141))은, 예를 들면, 시스템 리소스 매니저(321) 및/또는 디바이스 드라이버(323)를 포함할 수 있다. 시스템 리소스 매니저(321)는 시스템 리소스의 제어, 할당, 또는 회수 등을 수행할 수 있다. 다양한 실시예에 따르면, 시스템 리소스 매니저(321)는 프로세스 관리부, 메모리 관리부, 또는 파일 시스템 관리부 등을 포함할 수 있다. 디바이스 드라이버(323)는, 예를 들면, 디스플레이 드라이버, 카메라 드라이버, 블루투스 드라이버, 공유 메모리 드라이버, USB 드라이버, 키패드 드라이버, WiFi 드라이버, 오디오 드라이버, 또는 IPC(inter-process communication) 드라이버를 포함할 수 있다.

【0070】 미들웨어(330)는, 예를 들면, 어플리케이션(370)이 공통적으로 필요로 하는 기능을 제공하거나, 어플리케이션(370)이 전자 장치 내부의 제한된 시스템 자원을 효율적으로 사용할 수 있도록 API(360)를 통해 다양한 기능들을 어플리케이션(370)으로 제공할 수 있다. 다양한 실시예에 따르면, 미들웨어(330)(예: 미들웨어(143))는 런타임 라이브러리(335), 어플리케이션 매니저(application manager)(341), 윈도우 매니저(window manager)(342), 멀티미디어 매니저(multimedia manager)(343), 리소스 매니저(resource manager)(344), 파워 매니저

(power manager)(345), 데이터베이스 매니저(database manager)(346), 패키지 매니저(package manager)(347), 연결 매니저(connectivity manager)(348), 통지 매니저(notification manager)(349), 위치 매니저(location manager)(350), 그래픽 매니저(graphic manager)(351), 또는 보안 매니저(security manager)(352) 중 적어도 하나를 포함할 수 있다.

【0071】 런타임 라이브러리(335)는, 예를 들면, 어플리케이션(370)이 실행되는 동안에 프로그래밍 언어를 통해 새로운 기능을 추가하기 위해 컴파일러가 사용하는 라이브러리 모듈을 포함할 수 있다. 런타임 라이브러리(335)는 입출력 관리, 메모리 관리, 또는 산술 함수에 대한 기능 등을 수행할 수 있다.

【0072】 어플리케이션 매니저(341)는, 예를 들면, 어플리케이션(370) 중 적어도 하나의 어플리케이션의 생명 주기(life cycle)를 관리할 수 있다. 윈도우 매니저(342)는 화면에서 사용하는 GUI 자원을 관리할 수 있다. 멀티미디어 매니저(343)는 다양한 미디어 파일들의 재생에 필요한 포맷을 파악하고, 해당 포맷에 맞는 코덱(codec)을 이용하여 미디어 파일의 인코딩(encoding) 또는 디코딩(decoding)을 수행할 수 있다. 리소스 매니저(344)는 어플리케이션(370) 중 적어도 어느 하나의 어플리케이션의 소스 코드, 메모리 또는 저장 공간 등의 자원을 관리할 수 있다.

【0073】 파워 매니저(345)는, 예를 들면, 바이오스(BIOS: basic input/output system) 등과 함께 동작하여 배터리(battery) 또는 전원을 관리하고, 전자 장치의 동작에 필요한 전력 정보 등을 제공할 수 있다. 데이터베이스 매니저

(346)는 어플리케이션(370) 중 적어도 하나의 어플리케이션에서 사용할 데이터베이스를 생성, 검색, 또는 변경할 수 있다. 패키지 매니저(347)는 패키지 파일의 형태로 배포되는 어플리케이션의 설치 또는 업데이트를 관리할 수 있다.

【0074】 연결 매니저(348)는, 예를 들면, WiFi 또는 블루투스 등의 무선 연결을 관리할 수 있다. 통지 매니저(349)는 도착 메시지, 약속, 근접성 알림 등의 사건(event)을 사용자에게 방해되지 않는 방식으로 표시 또는 통지할 수 있다. 위치 매니저(350)는 전자 장치의 위치 정보를 관리할 수 있다. 그래픽 매니저(351)는 사용자에게 제공될 그래픽 효과 또는 이와 관련된 사용자 인터페이스를 관리할 수 있다. 보안 매니저(352)는 시스템 보안 또는 사용자 인증 등에 필요한 제반 보안 기능을 제공할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 전자 장치(101))가 전화 기능을 포함한 경우, 미들웨어(330)는 전자 장치의 음성 또는 영상 통화 기능을 관리하기 위한 통화 매니저(telephony manager)를 더 포함할 수 있다.

【0075】 미들웨어(330)는 전술한 구성요소들의 다양한 기능의 조합을 형성하는 미들웨어 모듈을 포함할 수 있다. 미들웨어(330)는 차별화된 기능을 제공하기 위해 운영 체제의 종류 별로 특화된 모듈을 제공할 수 있다. 또한, 미들웨어(330)는 동적으로 기존의 구성요소를 일부 삭제하거나 새로운 구성요소들을 추가할 수 있다.

【0076】 API(360)(예: API(145))는, 예를 들면, API 프로그래밍 함수들의 집합으로, 운영 체제에 따라 다른 구성으로 제공될 수 있다. 예를 들면, 안드로이드 또는 iOS의 경우, 플랫폼 별로 하나의 API 셋을 제공할 수 있으며, 타이젠(tizen)

의 경우, 플랫폼 별로 두 개 이상의 API 셋을 제공할 수 있다.

【0077】 어플리케이션(370)(예: 어플리케이션 프로그램(147))은, 예를 들면, 홈(371), 다이얼러(372), SMS/MMS(373), IM(instant message)(374), 브라우저(375), 카메라(376), 알람(377), 연락처(378), 음성 다이얼(379), 이메일(380), 달력(381), 미디어 플레이어(382), 앨범(383), 또는 시계(384), 건강 관리(health care)(예: 운동량 또는 혈당 등을 측정), 또는 환경 정보 제공(예: 기압, 습도, 또는 온도 정보 등을 제공) 등의 기능을 수행할 수 있는 하나 이상의 어플리케이션을 포함할 수 있다.

【0078】 다양한 실시예에 따르면, 어플리케이션(370)은 전자 장치(예: 전자 장치(101))와 외부 전자 장치(예: 전자 장치(102), 전자 장치(104)) 사이의 정보 교환을 지원하는 어플리케이션(이하, 설명의 편의 상, "정보 교환 어플리케이션")을 포함할 수 있다. 정보 교환 어플리케이션은, 예를 들면, 외부 전자 장치에 특정 정보를 전달하기 위한 알림 전달(notification relay) 어플리케이션, 또는 외부 전자 장치를 관리하기 위한 장치 관리(device management) 어플리케이션을 포함할 수 있다.

【0079】 예를 들면, 알림 전달 어플리케이션은 전자 장치의 다른 어플리케이션(예: SMS/MMS 어플리케이션, 이메일 어플리케이션, 건강 관리 어플리케이션, 또는 환경 정보 어플리케이션 등)에서 발생된 알림 정보를 외부 전자 장치(예: 전자 장치(102), 전자 장치(104))로 전달하는 기능을 포함할 수 있다. 또한, 알림 전달 어플리케이션은, 예를 들면, 외부 전자 장치로부터 알림 정보를 수신하여 사용자에게

게 제공할 수 있다.

【0080】장치 관리 어플리케이션은, 예를 들면, 전자 장치와 통신하는 외부 전자 장치(예: 전자 장치(102), 전자 장치(104))의 적어도 하나의 기능(예: 외부 전자 장치 자체(또는, 일부 구성 부품)의 턴-온/턴-오프 또는 디스플레이의 밝기(또는, 해상도) 조절), 외부 전자 장치에서 동작하는 어플리케이션 또는 외부 전자 장치에서 제공되는 서비스(예: 통화 서비스 또는 메시지 서비스 등)를 관리(예: 설치, 삭제, 또는 업데이트)할 수 있다.

【0081】다양한 실시예에 따르면, 어플리케이션(370)은 외부 전자 장치(예: 전자 장치(102), 전자 장치(104))의 속성에 따라 지정된 어플리케이션(예: 모바일 의료 기기의 건강 관리 어플리케이션 등)을 포함할 수 있다. 다양한 실시예에 따르면, 어플리케이션(370)은 외부 전자 장치(예: 서버(106) 또는 전자 장치(102), 전자 장치(104))로부터 수신된 어플리케이션을 포함할 수 있다. 다양한 실시예에 따르면, 어플리케이션(370)은 프리로드 어플리케이션(preloaded application) 또는 서버로부터 다운로드 가능한 제3자 어플리케이션(third party application)을 포함할 수 있다. 도시된 실시예에 따른 프로그램 모듈(310)의 구성요소들의 명칭은 운영 체제의 종류에 따라서 달라질 수 있다.

【0082】다양한 실시예에 따르면, 프로그램 모듈(310)의 적어도 일부는 소프트웨어, 펌웨어, 하드웨어, 또는 이들 중 적어도 둘 이상의 조합으로 구현될 수 있다. 프로그램 모듈(310)의 적어도 일부는, 예를 들면, 프로세서(예: 프로세서(210))에 의해 구현(implement)(예: 실행)될 수 있다. 프로그램 모듈(310)의 적어

도 일부는 하나 이상의 기능을 수행하기 위한, 예를 들면, 모듈, 프로그램, 루틴, 명령어 세트(sets of instructions) 또는 프로세스 등을 포함할 수 있다.

【0084】 도 4a 내지 도 4c는 다양한 실시예들에 따른 전자 장치를 도시한 도면이다.

【0085】 도 4a 내지 도 4c에는 3차원 X/Y/Z 직교 좌표계가 도시되었는데, 'Z축'은 전자 장치의 본체의 상하 방향(두께 방향)을, 'X축'은 본체의 가로 방향을, 'Y축'은 본체의 세로 방향을 각각 의미할 수 있다.

【0086】 도 4a는 본 발명의 다양한 실시예에 따른 전자 장치의 사시도를 도시하고 있다.

【0087】 도 4a를 참고하면, 전자 장치(400)(예: 도 1의 전자 장치(101))는 사용자의 손목에 착용할 수 있는 손목 착용형 웨어러블(wearable) 전자 장치일 수 있다. 하지만, 본 발명의 다양한 실시 예들은 이에 한정되지 않으며, 팔찌, 스트립, 밴드, 부착형(반창고형) 밴드, 벨트, 귀착용형 이어폰, 헤드폰, 의복형, 신발형, HMD(Head Mounted Display), 모자형, 장갑형, 끝무형(Finger-tip 착용형), 클립형, 암밴드(arm band)형, 콘택트렌즈 장치, 디지털의복, 또는 리모컨 중 하나일 수도 있다.

【0088】 다양한 실시예에 따르면, 전자 장치(400)는 본체(410)와 본체(410)에 장착되는 연결부(420)(예: 스트랩)를 포함할 수 있다.

【0089】 다양한 실시예들에 따르면, 전자 장치(400)는 내부에 전원 공급 수단으로 배터리(예: 충전용 배터리 등)를 포함할 수 있다. 전자 장치(400)는, 예를 들면, 휴대용 충전 크래들(cradle)에 선택적으로 장착되어 배터리를 충전시키도록 구현될 수도 있다.

【0090】 다양한 실시예들에 따르면, 본체(410)는 디스플레이(415), 적어도 하나의 키 버튼(430)(예를 들면, 사이드 키(K), 터치 센서, 또는 압력 센서) 또는/및 센서 모듈(예: 생체 센서 등) 등을 포함할 수 있다. 다양한 실시예에 따르면, 디스플레이(415)는 터치스크린을 포함하여, 터치 입력을 수신할 수 있다. 일 실시예에 따르면, 본체(410)는 전면(F)과, 작동 상태에서 사용자의 신체에 접촉하는 후면(R)을 포함할 수 있으며, 예를 들면, 디스플레이(415)는 본체(410)의 전면(F)에, 센서 모듈은 본체(410)의 후면(R)에 각각 배치될 수 있다. 하지만, 본 발명의 다양한 실시예들은 이에 한정되지 않으며, 디스플레이(415) 또는 센서 모듈은 전자 장치(400)의 다양한 영역에 배치될 수 있다. 예를 들면, 센서 모듈은 연결부(420)에 배치될 수도 있다.

【0091】 다양한 실시예에 따르면, 본체(410)는 바타입 형상이면서 적어도 부분적으로 사용자 신체에 대응하는 곡률(curvature)을 가질 수 있다. 예를 들어, 본체(410)는 세로 방향(Y축 방향)으로 연장된 직사각형 형상이면서 곡률을 가질 수 있다. 일 실시예에 따르면, 본체(410)의 측면에는 연결부(420)와 맞물리는 결속 홈이 형성될 수 있다.

【0092】 다양한 실시예에 따르면, 연결부(420)는 본체(410)와 결합하거나 분리 가능하게 구성될 수 있다. 다양한 실시예에 따르면, 연결부(420)는 탄성 재질로 구성될 수 있다. 이에 따라, 본체(410)를 사용자의 신체 피부에 밀착시킬 수 있다. 일 실시예에 따르면, 연결부(420)는 교환 가능한(changeable) 구조로 다양한 디자인이나 색상으로 구현될 수 있다. 이에 따라, 사용자는 사용자의 취향에 따라 연결부(420)를 교환하여 사용할 수 있다.

【0093】 다양한 실시예에 따르면, 연결부(420)는 본체(410)의 일부에 결합되고 사용자의 신체의 일부에 착탈가능하게 부착되는 결합구(422)를 포함할 수 있다. 일 실시예에 따르면, 전자 장치(400)는 속목에 올려놓은 상태에서 연결부(420)를 서로 엮어 체결하는 방식으로 착용될 수 있다. 예를 들어, 하나의 연결부(420)는 일정 간격으로 복수의 손목 조절용 개구를 포함하며 하나의 다른 연결부(420)는 손목 조절용 개구를 통해 체결될 수 있도록 결합구(422)를 포함할 수 있다.

【0094】 다양한 실시예들에 따르면, 본체(410)는 디스플레이(415)에 표시되는 다양한 사용자 인터페이스 환경을 조절할 수 있는 회전체(미도시)(예: 회전형 입력모듈)가 장착될 수 있다. 다양한 실시예에 따르면, 회전체는 용두(crown)와 같이 본체(410)의 측면에 돌출형으로 배치되어 회전될 수 있다. 다양한 실시예에 따르면, 본체(410)가 원형 타입 형상을 가질 경우, 회전체는 본체(410)의 하면, 상면 또는 측면에 회전 가능하게 설치될 수 있다. 예를 들어, 회전체는 본체(410)의 상면에 배치된 디스플레이(415)의 가장자리를 둘러싸는 베젤 부분에 배치될 수 있다.

【0095】 도 4b는 본 발명의 다양한 실시 예들 중 하나에 따른 전자 장치(400)의 본체를 나타내는 사시도이고, 도 4c는 본 발명의 다양한 실시 예들 중 하나에 따른 전자 장치(400)의 본체를 다른 방향에서 바라본 모습을 나타내는 사시도이다.

【0096】 도 4b 및 도 4c를 참조하면, 본체는 본체 하우징(440)과, 본체 하우징(440)에 장착된 디스플레이(450)를 포함할 수 있다. 다양한 실시예에 따르면, 본체 하우징(440)은 전면(front surface)(F)과, 후면(rear surface)(R)과, 전면(F)/후면(R)을 연결하는 측면(side surface)을 구비할 수 있다. 하지만, 본 발명의 다양한 실시예들은 이에 한정되지 않으며, 본체 하우징(440)은 전면(F)과 후면(R)으로 구성될 수 있다. 또한, 본체 하우징(440)의 전면(F), 후면(R), 및 측면은 일체형으로 형성될 수 있다. 일 실시예에 따르면, 전면(F) 및 후면(R)은 각각 곡률을 가지게 구성될 수 있다.

【0097】 다양한 실시예에 따르면, 본체 하우징(440)은 각종 정보를 입력하기 위한 키 버튼(예: 사이드 키(K), 터치 센서, 또는 압력 센서)을 더 포함할 수 있다. 일 실시예에 따르면, 전면(F)은 디스플레이(450)가 장착되는 면이고, 후면(R)은 사용자의 신체와 접촉하는 착용 면이 될 수 있다. 하지만, 본 발명의 다양한 실시예들은 이에 한정되지 않으며, 디스플레이(415) 또는 센서 모듈은 전자 장치(400)의 다양한 영역에 배치될 수 있다.

【0098】 다양한 실시예에 따르면, 하우징(440)의 전면(F)에 디스플레이(450)가 배치될 수 있다. 이에 따라, 사용자가 전자 장치를 통하여 디스플레이된 화면

보기가 편할 수 있다. 디스플레이(450)는 사용자의 신체 곡면을 반영한 형태를 예시하고 있으나, 평면(LCD, OLED) 디스플레이, 곡면 디스플레이 또는 플렉서블 디스플레이로 구성될 수 있다. 예컨대, 본 발명의 구체적인 실시 예들은 본체(410)가 곡면 디스플레이를 구비하는 것으로 예시되지만, 평면 디스플레이 또는 플렉서블 디스플레이로 구성될 수 있다.

【0099】도 4c를 참조하면, 일 실시예에 따르면, 후면(R)에는 센서 모듈(470)(예: 생체 센서)이 배치될 수 있다. 예를 들어, 센서 모듈은 사용자의 신체 손목에 밀착되는 형상으로 배치될 수 있다. 이에 따라, 사용자의 전자 장치에 대한 착용감을 향상시킬 수 있다.

【0100】하우징(440)에 구비되는 센서 모듈(470)(예: 센서 모듈(240) 또는 생체 센서)은 가속도 센서, 자이로 센서, 지자기 센서, 심박 센서, 근접 센서, 광 센서, GSR 센서, 심전도 센서, 근전도 센서, 혈당 센서, 혈압 센서, 압력 센서 또는 온도 센서 중 적어도 하나를 포함할 수 있다. 이 외에도 센서 모듈은 전자 장치(400)의 착용 여부를 판단하기 위한 센싱 값을 검출하는 다른 센서를 포함할 수도 있다. 다양한 실시예에 따르면, 센서 모듈(470)은 전자 장치(400)의 움직임을 측정하기 위한 센서(예: 가속도 센서, 지자기 센서, 또는 자이로 센서)와 신체 착용 여부를 판단하는 이용되는 센서(예: 근접 센서 또는 압력 센서) 및 생체 신호 측정을 위한 생체 센서(예: 심박 센서, 혈압 센서, 또는 체온 센서)를 포함하는 하나의 모듈 형태로서 하우징(440)의 후면(R)에 구비될 수 있다.

【0101】 다양한 실시예에 따르면, 하우징(440)의 후면(R)에는 접촉 부재들(460)(예: 충전 단자 또는 인터페이스(270) 등)이 배치될 수 있다.

【0103】 도 5a는 다양한 실시예들에 따른 프로세서(500)의 구성을 도시한 도면이다. 다양한 실시예에 따르면, 프로세서(500)는 전자 장치(예: 전자 장치(101) 또는 전자 장치(400))의 프로세서(예: 프로세서(120))일 수 있다.

【0104】 다양한 실시예에 따르면, 프로세서(500)는 장치상태 판단모듈(502), 파라미터 변경모듈(504), 사용자상태 판단모듈(506), 측정모듈(508)을 포함할 수 있다.

【0105】 다양한 실시예에 따르면, 장치상태 판단모듈(502)은 전자 장치의 상태(예: 작동 상태 또는 미작동 상태, 근접 상태 또는 근접하지 않은 상태, 또는 사용(또는, 소지) 상태 또는 사용(또는, 소지)하지 않는 상태)를 판단할 수 있다. 다양한 실시예에 따르면, 장치상태 판단모듈(502)은 적어도 하나의 센서에 기반하여 전자 장치의 상태를 판단할 수 있다. 일 실시예에 따르면, 장치상태 판단모듈(502)은 가속도 센서, 지자기 센서, 자이로 센서, 근접 센서, 기압 센서(또는, 압력 센서), 또는 온도 센서 중 적어도 하나의 센서를 통해 획득되는 정보에 기반하여 전자 장치의 상태를 판단할 수 있다.

【0106】 예를 들면, 장치상태 판단모듈(502)은, 예를 들면, 가속도 센서, 지자기 센서 또는 자이로 센서를 통하여, 전자 장치의 움직임이 감지된 경우, 전자

장치의 상태를 사용자에게 착용된 상태로 판단할 수 있다. 또한, 장치상태 판단 모듈(502)은, 예를 들면, 가속도 센서, 지자기 센서 또는 자이로 센서를 통하여, 전자 장치의 움직임이 감지되지 않은 경우, 전자 장치의 상태를 사용자에게 미착용된 상태로 판단할 수 있다.

【0107】 예를 들면, 장치상태 판단모듈(502)은, 예를 들면, 근접 센서를 통하여, 전자 장치로부터 선택된(또는, 지정된) 거리(예: 약 10cm) 내에 오브젝트(예: 사용자)가 감지된 경우, 전자 장치의 상태를 오브젝트로부터 근접한 상태(예: 사용자가 전자 장치를 사용 또는 소지하는 상태)로 판단할 수 있다. 또한, 장치상태 판단모듈(502)은, 예를 들면, 근접 센서를 통하여, 전자 장치로부터 선택된 거리 내에 오브젝트가 감지되지 않은 경우, 전자 장치의 상태를 오브젝트로부터 근접하지 않은 상태(예: 사용자가 전자 장치를 사용 또는 소지하지 않은 상태)로 판단할 수 있다.

【0108】 예를 들면, 장치상태 판단모듈(502)은, 예를 들면, 압력 센서 또는 온도 센서를 통하여, 선택된(또는, 지정된) 압력 또는 선택된(또는, 지정된) 온도(예: 약 37도)가 감지된 경우, 전자 장치의 상태를 사용자가 사용 또는 소지하는 상태로 판단할 수 있다. 또한, 장치상태 판단모듈(502)은, 예를 들면, 압력 센서 또는 온도 센서를 통하여, 선택된 압력 또는 선택된 온도가 감지되지 않은 경우, 전자 장치의 상태를 사용자가 사용 또는 소지하지 않는 상태로 판단할 수 있다.

【0109】 다양한 실시예에 따르면, 장치상태 판단모듈(502)은 전자 장치의 회로적 연결 또는 기구적 연결 여부에 기반하여 전자 장치의 상태를 판단할 수 있다.

예를 들면, 전자 장치가 스마트 안경인 경우, 스마트 안경이 사용자의 머리에 착용된 경우, 스마트 안경의 안경 다리에 포함된 스위치 회로가 연결될 수 있다. 이에 따라 장치상태 판단모듈(502)은 전자 장치의 상태를 착용 상태로 판단할 수 있다. 또한, 스마트 안경이 사용자의 머리에 착용되지 않은 경우, 스마트 안경의 안경 다리에 포함된 스위치 회로가 연결되지 않을 수 있다. 이에 따라, 장치상태 판단모듈(502)은 전자 장치의 상태를 미착용 상태로 판단할 수 있다.

【0110】 예를 들면, 전자 장치가 스마트 와치인 경우, 스마트 와치의 버클(buckle)이 연결된 경우, 장치상태 판단모듈(502)은 전자 장치의 상태를 착용 상태로 판단할 수 있다. 또한, 스마트 와치의 버클이 연결되지 않은 경우, 장치상태 판단모듈(502)은 전자 장치의 상태를 미착용 상태로 판단할 수 있다.

【0111】 다양한 실시예에 따르면, 사용자상태 판단모듈(506)은 사용자의 상태(예: 취침 상태)를 판단할 수 있다. 일 실시예에 따르면, 사용자상태 판단 모듈(506)은 사용자의 패턴에 기반하여 사용자의 상태를 판단할 수 있다. 예를 들어, 사용자의 패턴은 사용자의 움직임 패턴, 지정된 스케줄과 연관된 생활 패턴 등을 포함할 수 있다.

【0112】 다양한 실시예에 따르면, 사용자상태 판단모듈(506)은, 예를 들어, 생체 정보 측정 시, 사용자의 움직임, 또는 움직임 강도 등을 판단할 수 있다. 예를 들면, 사용자 상태 판단 모듈(506)은 사용자의 움직임 강도가 선택된(또는, 지정된) 강도 이하인 경우, 사용자의 상태를 취침 상태로 판단할 수 있다. 또한, 사용자 상태 판단 모듈(506)은 사용자의 움직임 강도가 선택된(또는, 지정된) 강도

이상인 경우, 사용자의 상태를 활동 상태(예: 취침하지 않는 상태)로 판단할 수 있다. 일 실시예에 따르면, 사용자상태 판단모듈(506)은 사용자의 움직임 정보를 측정모듈(508) 또는 파라미터 변경모듈(504)로 제공하여 생체 정보 측정 방식이 변경되도록 처리할 수 있다.

【0113】 다양한 실시예에 따르면, 사용자상태 판단모듈(506)은 전자 장치에 저장된 스케줄 정보에 기반하여 사용자의 상태를 판단할 수 있다. 예를 들면, 전자 장치에 저장된 스케줄에 제 1 시간(예: 약 오후 11시에서 약 오전 8시까지)은 ‘취침 시간’, 제 2 시간은 ‘활동 시간’ (예: 약 오전 8시에서 약 오후 11시)으로 지정될 수 있다. 이 경우, 사용자상태 판단모듈(506)은 스케줄에 기반하여 제 1 시간에 사용자의 상태를 ‘취침 상태’로 판단할 수 있다. 또한, 사용자상태 판단모듈(506)은 스케줄에 기반하여 제 2 시간에 사용자의 상태를 ‘활동 상태’로 판단할 수 있다.

【0114】 다양한 실시예에 따르면, 측정모듈(508)은 사용자의 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 측정모듈(508)은 적어도 하나의 센서에 기반하여 생체 정보를 측정할 수 있다. 일 실시예에 따르면, 측정모듈(508)은 신체의 적어도 일부와 접촉된 센서에 기반하여 혈압(blood pressure), 심박(heart rate), 심전도(electrocardiogram), 피부수분도(skin moisture), 산소포화도(saturation of peripheral oxygen), 체온(body temperature), 혈당(blood glucose), 뇌전도(electroencephalogram), 또는 근전도(electromyogram) 등 적어도 하나를 측정할 수 있다. 다양한 실시예에 따르면, 측정모듈(508)은 장치상태 판단모듈(502)을 통

하여 전자 장치의 상태가 선택된(또는, 지정된) 상태(예: 착용 상태)로 판단되는 경우에 활성화될 수 있다.

【0115】 다양한 실시예에 따르면, 측정모듈(508)은 생체 정보 측정 주기(예: 약 1분)가 도래하면 측정 동작을 수행할 수 있다. 다양한 실시예에 따르면, 측정모듈(508)은 생체 정보 측정 주기가 도래하면, 전자 장치 또는 사용자의 움직임에 기반하여 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 측정모듈(508)은 측정 주기가 도래하면, 주기가 도래하기 전에 측정된 사용자 또는 전자 장치의 움직임을 확인하고, 움직임이 선택된(또는, 지정된) 조건을 만족하는 경우(예: 움직임의 강도가 선택된(또는, 지정된) 강도보다 낮은 경우) 생체 정보를 측정할 수 있다. 다른 실시예에 따르면, 측정모듈(508)은 전자 장치 또는 사용자의 움직임이 선택된(또는, 지정된) 조건을 만족하지 못하는 경우(예: 움직임의 강도가 선택된(또는, 지정된) 강도보다 높은 경우) 주기가 도래한 시점을 기준으로 일정 시간(예: 약 10초) 동안 움직임 변화를 모니터링할 수 있다. 일 실시예에 따르면, 선택된(또는, 지정된) 시간 동안 움직임 변화가 선택된(또는, 지정된) 조건을 만족하는 경우(예: 움직임 데이터의 평균이 선택된(또는, 지정된) 값보다 낮은 경우), 측정모듈(508)은 생체 정보를 측정할 수 있다.

【0116】 다양한 실시예에 따르면, 파라미터 변경모듈(504)은 생체 정보 측정에 필요한 측정 파라미터를 변경할 수 있다. 일 실시예에 따르면, 파라미터 변경모듈(504)은 적어도 하나의 센서의 발광 시간, 생체 정보 측정 주기, 샘플링 비율, 생체 정보 측정 트리거, 또는 움직임 판단 조건 등을 변경할 수 있다. 이에 따라,

전자 장치에서 생체 정보 측정 시 발생하는 소모 전류가 줄어들거나, 생체 정보 측정의 정확도가 향상될 수 있다. 또한, 일 실시예에 따르면, 파라미터 변경모듈(504)은 생체 정보 측정 시, 생체 정보를 측정하는 동안의 전자 장치 또는 신체의 움직임 강도, 측정 대상의 신체 특성 또는 상태, 또는 측정 품질 또는 측정 결과 등과 연관된 상황 정보에 기반하여 측정 파라미터를 변경할 수 있다.

【0117】 다양한 실시예에 따르면, 장치상태 판단모듈(502), 파라미터 변경모듈(504), 사용자상태 판단모듈(506), 측정모듈(508)은 적어도 하나의 소프트웨어 구성일 수 있다.

【0118】 다양한 실시예에 따르면, 프로세서(500)는 다양한 실시예에 따라, 상술한 측정하는 동작을 수행하기 위해 장치상태 판단모듈(502), 파라미터 변경모듈(504), 사용자상태 판단모듈(506), 측정모듈(508) 중 적어도 하나의 모듈을 포함할 수 있다.

【0119】 도시되진 않았지만, 다양한 실시예에 따르면, 장치상태 판단모듈(502), 파라미터 변경모듈(504), 사용자상태 판단모듈(506), 및 측정모듈(508)은 복수의 프로세서(또는, 칩(chip))에 포함될 수 있다. 예를 들면, 장치상태 판단모듈(502) 및 사용자 상태 판단모듈(506)은 제 1 프로세서(또는, 제 1 칩)에 포함될 수 있다. 파라미터 변경모듈(504)은 제 2 프로세서(또는, 제 2 칩)에 포함될 수 있다. 또한, 측정 모듈(508)은 제 3 프로세서(또는, 제 3 칩)에 포함될 수 있다.

【0120】 다양한 실시예에 따르면, 전자 장치는 프로세서(500)와 다른 별도의 모듈로 상술한 생체 정보를 측정하는 동작을 수행할 수 있다.

【0122】 도 5b는 다양한 실시예들에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 전자 장치(400), 또는 프로세서(500))의 생체 정보 측정 동작을 도시한 도면이다.

【0123】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(500))는 선택된(또는, 지정된) 시간을 가지는 주기(예: 약 1분)를 간격으로 생체 정보의 측정을 시도할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(500))는 생체 정보의 측정을 시도하는 시기를 판단하기 위하여 선택된(또는, 지정된) 시간을 나타내는 제1 타이머를 동작시킬 수 있다. 예를 들어, 전자 장치(예: 프로세서(500))는 제1 타이머를 동작하여 1분 간격으로 생체 정보 측정을 시도할 수 있다.

【0124】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(500))는 제1 타이머의 동작이 만료(520)되기 전의 적어도 일부 시간 구간(522)(예: 제 1 타이머의 동작이 만료되기 전 약 10초 동안)에 전자 장치의 움직임을 측정할 수 있으며, 제1 타이머의 동작이 만료되면 측정된 움직임 중 적어도 일부의 움직임이 선택된(또는, 지정된) 기준을 만족하는지 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(500))는 제1 타이머가 만료되기 전 적어도 일부 구간(522)에 획득된 움직임의 평균을 측정하고, 측정된 움직임의 평균이 선택된(또는, 지정된) 임계값 이하인지 판

단할 수 있다.

【0125】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(500))는 측정된 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는 경우, 생체 정보를 측정하기 위한 적어도 하나의 센서(예: 생체 센서)를 동작시킬 수 있다.

【0126】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(500))는 측정된 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하지 못하는 경우, 제2 타이머가 동작하는 시간(524)(예: 약 30초) 동안 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는 상태로 변경되는지 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(500))는 이전 측정된 움직임(예: 움직임의 평균)이 임계값 이상으로 지속되면 생체 정보 측정 동작을 수행하지 않을 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(500))는 이전 측정된 움직임(예: 움직임의 평균)이 임계값 이하로 변경되면 생체 정보를 측정하기 위한 적어도 하나의 센서(예: 생체 센서)를 동작시킬 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(500))는 제2 타이머가 동작하는 시간(524)동안 추가적으로 전자 장치의 움직임을 측정할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(500))는 제2 타이머가 동작하는 시간 동안(524)에 선택된(또는, 지정된) 구간(예: 움직임의 평균이 측정된 구간(522)과 동일하거나 유사한 크기의 구간)의 움직임을 추가적으로 측정하고, 추가적으로 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하는지 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서)는 추가적으로 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하면 생체 정보를 측정하기 위한 적어도 하나의 센서

(예: 생체 센서)를 동작시킬 수 있다.

【0127】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(500))는 제2 타이머가 동작하는 시간(524)을 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(500))는 이전 측정된 움직임의 평균을 확인하는 시간(524)을 증가시키고, 증가된 시간 동안 생체 정보 측정이 가능한지를 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(500))는 전자 장치의 움직임(예: 움직임의 강도)에 기반하여 제2 타이머가 동작하는 시간(524)을 변경할 수 있다. 예를 들어, 전자 장치(예: 프로세서(500))는 이전 측정된 움직임(예: 움직임의 평균)에 기반하여 제2 타이머의 동작 시간(524)을 증가시키거나 또는 감소시킬 수 있다. 예를 들면, 전자 장치의 움직임(예: 움직임의 평균)이 감소하거나 임계값에 근접한 경우, 전자 장치(예: 프로세서(500))는 제 2 타이머의 동작 시간(524)을 증가시킬 수 있다. 예를 들면, 전자 장치의 움직임(예: 움직임의 평균)이 증가하거나 임계값에 근접하지 않은 경우, 전자 장치(예: 프로세서(500))는 제 2 타이머의 동작 시간(524)을 감소시킬 수 있다.

【0129】 다양한 실시예에 따른 전자 장치는 하우징, 상기 하우징의 일부에 결합되고, 사용자의 신체의 일부에 착탈가능하게 부착되는 결합구, 상기 전자 장치의 움직임을 감지할 수 있도록 구성된 적어도 하나의 제1 센서(예: 가속도 센서, 지자기 센서, 자이로 센서, 근접 센서 또는 이들의 조합), 사용자의 신체의 변화를 감지할 수 있도록 구성된 적어도 하나의 제2 센서(예: 심박 센서), 메모리 및 상기

적어도 하나의 제1 센서, 상기 적어도 하나의 제2 센서, 및 상기 메모리와 전기적으로 연결된 프로세서를 포함할 수 있다. 일 실시예에 따르면, 상기 메모리는 상기 프로세서가 예를 들면, 상기 적어도 하나의 제1 센서를 이용하여, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지(또는, 근접하였는지) 여부를 판단하고, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된(또는, 근접한) 시점으로부터 선택된(또는, 지정된) 제1 시간 주기를 경과하는 시점 바로 전 및/또는 바로 후의 제2 시간 주기 동안에 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하고, 상기 움직임에 관한 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 인스트럭션들을 저장할 수 있다.

【0130】 다양한 실시예에 따르면, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지 여부를 판단하는 동작은 신체의 살아있음(liveness)을 판단하는 동작을 포함할 수 있다. 일 실시예에 따르면, 신체의 살아있음을 판단하는 동작은 심박 센서, 혈압 센서, 체온 센서, 피부 수분도 센서, 산소 포화도 센서, 혈당 센서, 심전도 센서, 뇌전도 센서, 근전도 센서 중 적어도 하나의 센서를 이용하여 판단할 수 있다. 일 실시예에 따르면, 상기 적어도 하나의 제1 센서를 이용하여, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지(또는, 근접하였는지) 여부가 판단되더라도, 심박 센서로부터 수신되는 신호가 신체와 관련된 신호가 아니라고 판단되면 상기 사용자의 신체의 일부에 착용되지 않은 것으로 판단할 수 있다. 일 실시예에 따르면, 상기 적어도 하나의 제1 센서를 이용하여, 상기 전자 장

치가 상기 사용자의 신체의 일부에 착용되었는지(또는, 근접하였는지) 여부가 판단되더라도, 체온 센서로부터 수신되는 신호가 신체에 적합한 범위가 아니라고 판단되면 상기 사용자의 신체의 일부에 착용되지 않은 것으로 판단할 수 있다.

【0131】 다양한 실시예에 따르면, 상기 제2 시간 주기는 상기 제1 시간주기보다 짧을 수 있다.

【0132】 다양한 실시예에 따르면, 상기 적어도 하나의 제1 센서는, 근접센서, 자이로 센서, 지자기 센서 또는 가속도 센서를 포함할 수 있다.

【0133】 다양한 실시예에 따르면, 상기 적어도 하나의 제2 센서는 생체 센서를 포함할 수 있다. 일 실시예에 따르면, 상기 생체 센서는 심박 센서, 혈압 센서, 심전도 센서, 피부 수분도 센서, 또는 산소포화도 센서 중 적어도 하나를 포함할 수 있다.

【0134】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 움직임에 관한 정보가 선택된(또는, 지정된) 조건을 만족하지 않는 경우, 상기 제2 센서를 활성화(또는, 작동 또는 트리거)하지 않고, 상기 제2 시간 주기를 경과하는 시점 바로 후의, 상기 제1 시간 주기보다 짧은 제3 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고, 상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 설정될 수 있다.

【0135】 다양한 실시예에 따르면, 상기 제2 시간 주기 또는 상기 제3 시간 주기 중 적어도 하나는 상기 제1 시간 주기를 경과하는 시점을 포함하도록 설정될 수 있다.

【0136】 다양한 실시예에 따르면, 상기 제2 시간 주기는 상기 제1 시간 주기를 경과하는 시점을 포함하도록 설정될 수 있다.

【0137】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 활성화(또는, 작동 또는 트리거)된 제2 센서로부터 데이터를 선택된(또는, 지정된) 시간 내에 수신하지 않은 경우, 상기 선택된(또는, 지정된) 시간이 경과한 후에, 상기 제2 센서를 다시 활성화(또는, 작동 또는 트리거)하도록 설정될 수 있다.

【0138】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 선택된(또는, 지정된) 시간이 경과하는 시점 전의 상기 제2 시간 주기와 실질적으로 동일한 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고, 상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 다시 활성화(또는, 작동 또는 트리거)하도록 설정될 수 있다.

【0139】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고, 상기 획득된 데이터에 적어도 일부 기초하여 상기 제1 시간 주기 및/또는 제2 시간 주기 및 또는 제3 시간 주기를 변경하도록 설정될 수 있다.

【0140】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고, 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 적어도 하나의 조건을 변경하도록 설정될 수 있다.

【0141】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고, 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)시키는 시간 주기 또는 샘플링 비율 중 적어도 하나를 변경하도록 설정될 수 있다.

【0142】 다양한 실시예에 따르면, 전자 장치는 메모리 및 상기 메모리와 전기적으로 연결된 프로세서를 포함할 수 있다. 일 실시예에 따르면, 상기 메모리는, 상기 프로세서가 적어도 하나의 제 1 센서를 이용하여 상기 전자 장치가 사용자의 신체의 일부에 착용(또는, 근접)되었는지 여부를 판단하고, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된(또는, 근접한) 경우, 선택된(또는, 지정된) 시간 주기를 경과하는 시점에서 선택된(또는, 지정된) 시간 범위 내에 측정된 전자 장치의 움직임의 적어도 일부에 기반하여 적어도 하나의 제 2 센서를 활성화(또는, 작동 또는 트리거)하고, 상기 적어도 하나의 제 2 센서를 통하여 상기 사용자의 생체 정보를 측정하도록 하는 인스트럭션들을 저장할 수 있다.

【0143】 다양한 실시예에 따르면, 상기 적어도 하나의 제 1 센서는, 근접 센서, 자이로 센서, 지자기 센서 또는 가속도 센서 중 적어도 하나를 포함할 수 있다.

【0144】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 움직임의 적어도 일부가 선택된(또는, 지정된) 조건을 만족하지 않는 경우, 상기 제2 센서를 활성화(또는, 작동 또는 트리거)하지 않고, 상기 시간 주기를 경과하는 시점부터 선택된(또는, 지정된) 시간 동안 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고, 및 상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 설정될 수 있다.

【0145】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제 1 센서 또는 상기 적어도 하나의 제 2 센서 중 적어도 하나로 부터 획득된 데이터에 기반하여 상기 추가 정보를 획득하는 선택된(또는, 지정된) 시간을 변경하도록 설정될 수 있다.

【0146】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 생체 정보의 측정이 실패하면, 실패 횟수를 누적하고, 상기 누적된 실패횟수가 선택된(또는, 지정된) 조건을 만족하면 상기 제2 센서를 비활성화하도록 설정될 수 있다.

【0147】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고, 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 설정될 수 있다.

【0148】 다양한 실시예에 따르면, 상기 인스트럭션들은, 상기 프로세서가 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고, 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)시키는 시간 주기 또는 샘플링 비율 중 적어도 하나를 변경하도록 설정될 수 있다.

【0149】 다양한 실시예에 따르면, 전자 장치는 메모리 및 상기 메모리와 전기적으로 연결된 프로세서를 포함할 수 있다. 일 실시예에 따르면, 상기 메모리는, 상기 프로세서가 사용자의 신체에 대한 전자 장치의 근접 여부를 판단하고, 상기 전자 장치가 상기 사용자의 신체의 일부에 근접한 경우, 선택된(또는, 지정된) 주기를 경과하는 시점으로부터 선택된(또는, 지정된) 시간 범위 내에, 적어도 하나의 제 1 센서를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하고, 상기 움직임에 관한 정보에 적어도 일부 기초하여, 적어도 하나의 제 2 센서를 통하여 상기 사용자의 생체 정보를 획득하도록 설정될 수 있다.

【0151】 도 6은 본 발명의 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 생체 정보 측정 동작을 도시한 흐름도이다.

【0152】 동작 601과 같이, 전자 장치(예: 프로세서(120 또는 500) 또는 장치 상태 판단모듈(502))는 제1 센서에 기반하여 전자 장치의 상태(예: 착용 상태, 근접 상태, 사용 상태, 또는 소지 상태)를 판단할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 신체 착용(또는, 근접)과 연관된 전자 장치의 상태를 확인할 수 있다. 일 실시예에 따르면, 신체 착용과 연관된 전자 장치의 상태는 전자 장치가 신체의 적어도 일부에 접촉된 상태, 전자 장치가 신체의 적어도 일부로 접근된 상태 등이 될 수 있다. 다양한 실시예에 따르면, 제1 센서는 가속도 센서, 지자기 센서, 자이로 센서, 근접 센서, 기압 센서, 또는 온도 센서 중 적어도 하나의 센서로 구성될 수 있다.

【0153】 동작 603에서, 전자 장치(예: 프로세서(120 또는 500))는 상태 판단 동작을 통해서 신체 착용과 연관된 전자 장치의 상태(예: 착용 상태 또는 미착용 상태)를 확인할 수 있다.

【0154】 동작 603에서 신체 착용과 연관된 전자 장치의 상태가 선택된(또는, 지정된) 상태가 아닌 경우(예: 전자 장치의 상태가 미착용 상태인 경우), 동작 601과 같이 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 상태를 판단하는 동작을 수행할 수 있다.

【0155】 동작 603에서 신체 착용과 연관된 전자 장치의 상태가 선택된(또는,

지정된) 상태인 경우(예: 전자 장치의 상태가 착용 상태인 경우), 동작 605와 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서(예: 센서 모듈(470) 또는 생체 센서(240I))를 동작시킬 수 있다. 다양한 실시예에 따르면, 적어도 하나의 제2 센서는 생체 정보를 측정하기 위한 센서가 될 수 있으며, 혈압(blood pressure), 심박(heart rate), 심전도(electrocardiogram), 피부수분도(skin moisture), 산소포화도(saturation of peripheral oxygen), 체온(body temperature), 혈당(blood glucose), 뇌전도(electroencephalogram), 또는 근전도(electromyogram) 등 적어도 하나를 측정할 수 있다. 예를 들어, 제2 센서는 신체와 접촉될 수 있도록 전자 장치의 일면에 배치될 수 있다.

【0156】 다양한 실시예에 따르면, 적어도 하나의 제2 센서를 동작시키는 것은 선택된(또는, 지정된) 시간이 경과하는 경우, 적어도 하나의 제1 센서의 적어도 일부를 이용하여 획득한 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는 경우, 또는 선택된(또는, 지정된) 시각이 되는 경우 중 적어도 하나에 기반하여 동작시킬 수 있다. 일 실시예에 따르면, 전자 장치의 상태가 착용 상태로 판단되면, 선택된(또는, 지정된) 시간을 나타내는 제1 타이머를 동작시켜서 제1 타이머가 만료되면 적어도 하나의 제2 센서를 동작시킬 수 있다. 일 실시예에 따르면, 전자 장치의 상태가 착용 상태로 판단되면, 전자 장치의 움직임을 확인하여, 움직임이 선택된(또는, 지정된) 기준을 만족하면 적어도 하나의 제2 센서를 동작시킬 수 있다. 일 실시예에 따르면, 전자 장치의 상태가 착용 상태로 판단되면, 선택된(또는, 지정된) 시각(예: 착용 판단 시각이 약 6시 10분인 경우, 약 6시 11분)이 도래

하는지 확인하여 적어도 하나의 제2 센서를 동작시킬 수 있다.

【0157】 동작 607과 같이, 전자 장치(예: 프로세서(120 또는 500) 또는 측정 모듈(508))는 제2 센서에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서에 기반하여 측정된 생체 정보를 화면에 출력할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정된 생체 정보에 기반하여 헬스케어 기능(예: 심박수 제공 또는 운동량 제공)을 제공할 수도 있다.

【0159】 도 7은 본 발명의 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 상태 판단 방법의 수행 절차를 도시한 흐름도이다.

【0160】 다양한 실시예에 따르면, 상태 판단 방법의 수행 절차는 도 6에 기재된 동작 601에 대한 상세 동작일 수 있다.

【0161】 동작 701과 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서(예: 근접 센서)에 기반하여 외부 오브젝트(예: 사용자)의 근접이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 근접은 전자 장치가 신체로 근접된 상태, 전자 장치가 신체에 접촉된 상태 등과 연관될 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 제1 센서인 근접 센서를 통해 획득되는 정보에 기반하여 신체와 근접 여부를 판단할 수 있다.

【0162】 동작 701에서 사용자(예: 신체)와의 근접이 감지되면, 동작 703과 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서(예: 가속도 센서, 지자기 센서, 또는 자이로 센서)에 기반하여 전자 장치의 움직임을 측정할 수 있다. 다양한 실시예에 따르면, 움직임은 전자 장치 또는 사용자의 움직임과 연관될 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 근접 감지에 대응하여 가속도 센서, 지자기 센서, 자이로 센서 중 적어도 하나의 센서를 통해 획득되는 정보에 기반하여 움직임을 측정할 수 있다.

【0163】 동작 705와 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 조건을 만족하는 움직임이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 신체와 근접된 상태에서 선택된(또는, 지정된) 시간 동안 연속적인 움직임이 감지되는지 확인할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간 동안 연속적인 움직임이 감지되면 조건을 만족하는 움직임이 감지되었다고 판단할 수 있다.

【0164】 동작 705에서 선택된(또는, 지정된) 조건을 만족하는 움직임이 감지되면, 동작 707과 같이, 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관된 전자 장치의 상태로 판단할 수 있다.

【0165】 동작 705에서 선택된(또는, 지정된) 조건을 만족하지 움직임이 감지되지 않으면, 동작 709와 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서(예: 근접 센서)에 기반하여 신체와 근접이 감지되는지 확인할 수

있다. 예를 들어, 전자 장치는 제 1 센서인 근접 센서를 통해 획득되는 정보에 기반하여 신체와 근접 여부를 판단할 수 있다.

【0166】 동작 709에서 전자 장치와 신체의 근접이 감지되면, 동작 707과 같이, 전자 장치(예: 프로세서(120 또는 500))는 신체 작용과 연관된 전자 장치의 상태(예: 작용 상태)로 판단할 수 있다. 일 실시예로, 전자 장치 (예: 프로세서(120 또는 500))는 움직임이 없는 상황(예: 취침 상황)에서 전자 장치가 신체에 접촉 또는 근접되어 있는 경우 신체 작용과 연관된 상태로 판단할 수 있다.

【0167】 동작 709에서 전자 장치와 신체의 근접이 감지되지 않으면, 동작 713과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 상태를 신체 작용과 연관되지 않은 상태(예: 미작용 상태)로 판단할 수 있다.

【0168】 동작 701에서 근접이 감지되지 않으면, 동작 711과 같이 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 근접이 감지되지 않은 상태에서, 전자 장치의 움직임에 기반하여 근접 감지를 판단하는 동작의 재수행 여부를 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 근접이 감지되지 않았지만 전자 장치의 움직임이 감지되면 근접될 가능성이 있다고 판단할 수 있다.

【0169】 동작 711에서 움직임이 감지되면, 동작 701과 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 전자 장치가 신체와 근접하는지를 판단할 수 있다.

【0170】 동작 711에서 움직임이 감지되지 않으면, 동작 713과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 상태를 신체 착용과 연관되지 않은 상태(예: 미착용 상태)로 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 신체에 근접할 가능성이 없다고 판단하여 신체 착용과 연관되지 않은 상태로 판단할 수 있다.

【0171】 다양한 실시예에 따르면, 동작 701에서 동작 713 중 적어도 하나의 동작은 생략될 수 있다. 예를 들면, 동작 701에서 전자 장치(예: 프로세서(120 또는 500))는 사용자로부터 전자 장치의 근접이 감지되는지 확인할 수 있다. 전자 장치에서 사용자의 근접이 감지된 경우, 동작 707에서 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관된 전자 장치의 상태(예: 착용 상태)로 판단할 수 있다. 또한, 전자 장치에서 사용자의 근접이 감지되지 않은 경우, 동작 713에서 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관되지 않은 상태(예: 미착용 상태)로 판단할 수 있다.

【0172】 다양한 실시예에 따르면, 동작 703에서 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임을 측정할 수 있다. 동작 705에서 전자 장치(예: 프로세서(120 또는 500))는 지정된(또는, 선택된) 시간 동안 움직임(예: 연속적인 움직임)이 감지되는지 확인할 수 있다. 선택된(또는, 지정된) 시간 동안 움직임(예: 연속적인 움직임)이 감지되는 경우, 동작 707에서 전자 장치(예: 프로세서(120 또는 500))는 신체 착용(또는, 사용자가 소지 또는 사용하는 상태)과 연관된 전자 장치의 상태(예: 착용 상태)로 판단할 수 있다. 선택된(또는, 지정된)

시간 동안 연속적인 움직임이 감지되지 않는 경우, 동작 713에서 전자 장치(예: 프로세서(120 또는 500))는 신체 착용(또는, 사용자가 미소지 또는 사용하지 않는 상태)과 연관되지 않은 상태(예: 미착용 상태)로 판단할 수 있다.

【0174】 도 8은 본 발명의 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 제2 센서 동작 방법의 수행 절차를 도시한 흐름도이다.

【0175】 다양한 실시예에 따르면, 제2 센서 동작 방법의 수행 절차는 도 6에 기재된 동작 605에 대한 상세 동작일 수 있다.

【0176】 동작 801과 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간 동안 동작하는 제1 타이머를 동작시킬 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머의 동작에 기반하여 생체 정보의 측정을 시도할 수 있다. 일 실시예에 따르면, 제1 타이머는 생체 정보의 측정을 시도하는 시기를 판단하는 시간이 될 수 있다. 예를 들어, 제1 타이머는 약 1분 주기로 동작할 수 있다.

【0177】 동작 803과 같이, 전자 장치(예: 프로세서(120 또는 500))는 타이머의 동작에 대응하여 전자 장치의 움직임을 측정할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서(예: 가속도 센서 또는 자이로 센서)에 기반하여 전자 장치의 움직임을 감지할 수 있다. 다양한

실시예에 따르면, 제1 타이머의 구동에 기반하여, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머가 만료되기 전까지 전자 장치의 움직임 측정할 수 있다.

【0178】 동작 805와 같이, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머의 동작이 만료되는지 확인할 수 있다. 제1 타이머의 동작이 만료되는 시기는, 예를 들면, 생체 정보를 측정하는 동작의 수행 여부를 판단하는 시기가 될 수 있다. 생체 정보를 측정하는 동작은, 예를 들면, 생체 정보를 측정하기 위한 적어도 하나의 제2 센서를 구동시키는 것이 될 수 있다.

【0179】 동작 805에서 제1 타이머의 동작이 만료되지 않으면, 동작 803과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임을 측정할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머의 동작이 만료되기 전까지 전자 장치의 움직임을 측정할 수 있다.

【0180】 동작 805에서 제1 타이머의 동작이 만료되면, 동작 807과 같이, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머 동작 중 측정된 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는지 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 정확하게 측정할 수 있는 움직임이 발생되는지 판단할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머가 동작하는 동안 측정된 움직임 중 적어도 일부의 움직임이 선택된(또는, 지정된) 기준을 만족하는지 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머가 동작하는 동안 측정된 움직임의 평균을 측정하고, 측정된 움직임의 평균이

선택된(또는, 지정된) 조건을 만족하는지 판단할 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머가 만료되기 전 적어도 일부 구간에서 획득된 움직임의 평균을 측정하고, 측정된 움직임의 평균이 선택된(또는, 지정된) 조건을 만족하는지 판단할 수 있다. 다양한 실시예에 따르면, 선택된(또는, 지정된) 조건을 만족하는 상황은 선택된(또는, 지정된) 임계값 이하의 전자 장치의 움직임이 측정되는 상황이 될 수 있다. 예를 들어, 선택된(또는, 지정된) 임계값 이하의 전자 장치의 움직임이 측정되는 상황은, 전자 장치의 움직임이 발생되지 않거나 또는 움직임 미발생으로 인식될 수 있는 정도의 작은 움직임이 발생된 상태가 될 수 있다.

【0181】 동작 807에서 제1 타이머가 동작되는 동안 측정된 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는 경우, 동작 809와 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하기 위한 적어도 하나의 제2 센서를 동작시킬 수 있다.

【0182】 동작 807에서 제1 타이머 동작 중 측정된 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하지 못하는 경우, 동작 811과 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간 동안 동작하는 제2 타이머를 동작시킬 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 이미 측정된 기준을 만족하지 못하는 전자 장치의 움직임이 선택된(또는, 지정된) 기준을 만족하는 상태로 변경되는지를 판단하기 위하여 제2 타이머를 동작시킬 수 있다. 제2 타이머의 동작 시간은 제1 타이머의 동작 시간보다 짧을

수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정을 위한 다음 주기가 도래하기 전까지 전자 장치의 움직임이 변경되는지 확인할 수 있다.

【0183】 동작 813과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임 변화를 측정할 수 있다.

【0184】 동작 815와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 움직임의 변화가 선택된(또는, 지정된) 기준을 만족하는지 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정된 움직임(예: 기준을 만족하지 못하는 전자 장치의 움직임)이 생체 정보를 정확하게 측정할 수 있는 상태의 움직임으로 변경되는지 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머가 만료되는 시점에 측정된 움직임의 평균치가 선택된(또는, 지정된) 임계값 이하로 변경되는지 확인할 수 있다.

【0185】 동작 815에서 움직임 변화가 선택된(또는, 지정된) 기준을 만족하면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 예를 들어, 동작 809와 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하기 위한 적어도 하나의 제2 센서를 동작시킬 수 있다.

【0186】 동작 815에서 움직임 변화가 선택된(또는, 지정된) 기준을 만족하지 않으면, 동작 817과 같이, 전자 장치(예: 프로세서(120 또는 500))는 제2 타이머의 동작이 만료되는지 확인할 수 있다.

【0187】 동작 817에서 제2 타이머가 만료되지 않으면, 전자 장치(예: 프로세서(120 또는 500))는 제2 타이머가 동작하는 동안 전자 장치(예: 프로세서(120 또는 500))의 움직임 변화가 선택된(또는, 지정된) 기준을 만족하는지 재확인할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 동작 813과 같이 전자 장치의 움직임 변화를 측정할 수 있다.

【0188】 동작 817에서 제2 타이머가 만료되면, 동작 819와 같이, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머의 동작 시기를 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관된 전자 장치의 상태를 판단하는 동작을 수행할 수 있다. 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관된 전자 장치의 상태(예: 착용 상태)를 판단함에 따라 전자 장치의 상태를 판단하는 동작을 생략하고 제1 타이머를 구동시킬 수도 있다.

【0189】 다양한 실시예에 따르면, 동작 801에서 동작 819 중 적어도 하나의 동작은 생략될 수 있다. 동작 807에서 제1 타이머가 동작되는 동안 측정된 전자 장치(101)의 움직임이 선택된(또는, 지정된) 기준을 만족하는 경우, 동작 809와 같이, 전자 장치(예: 프로세서(120 또는 500))(101)는 생체 정보를 측정하기 위한 적어도 하나의 제2 센서를 동작시킬 수 있다. 동작 807에서 제1 타이머 동작 중 측정된 전자 장치(101)의 움직임이 선택된(또는, 지정된) 기준을 만족하지 못하는 경우, 동작 819와 같이, 전자 장치(예: 프로세서(120 또는 500))는 제1 타이머의 동작 시기를 판단할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는

신체 착용과 연관된 전자 장치의 상태를 판단하는 동작을 수행할 수 있다. 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 신체 착용과 연관된 전자 장치의 상태(예: 착용 상태)를 판단함에 따라 전자 장치의 상태를 판단하는 동작을 생략하고 제1 타이머를 구동시킬 수도 있다.

【0191】 도 9는 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 생체 정보 측정 방법의 수행 절차를 도시한 흐름도이다.

【0192】 다양한 실시예에 따르면, 생체 정보 측정 방법의 수행 절차는 도 6에 기재된 동작 607에 대한 상세 동작일 수 있다.

【0193】 동작 901과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정을 시도할 수 있다. 다양한 실시예에 따르면, 생체 정보 측정은 적어도 하나의 제2 센서를 동작시킴으로써 수행될 수 있다. 다양한 실시예에 따르면, 적어도 하나의 제2 센서는 빛을 발광시키는 발광부 및 발광부의 빛을 수신하는 수광부를 포함할 수 있으며, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서로부터 출력되는 신호에 기반하여 생체 정보를 측정할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 신체의 적어도 일부에 접촉된 적어도 하나의 제2 센서를 통해 출력되는 빛의 양에 기반하여 생체 정보(예: 심박수)를 측정할 수 있다. 예를 들어, 수축기에는 신체에 혈액이 많아져 약간 어두운 빛이 수신되고, 이완기에는 신체에 혈액이 빠지면서 밝은 빛이 수신될 수 있으며, 전자 장치(예:

프로세서(120 또는 500))는 이러한 빛의 양에 기반하여 심박 정보를 측정할 수 있다.

【0194】 동작 903에서, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보가 성공적으로 측정되는지 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 정상적인 측정 결과가 도출되는지 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정에 소요되는 시간 등에 기반하여 정상적인 측정 결과가 도출되는지 확인할 수 있다. 예를 들면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정이 시작되고 선택된(또는, 지정된) 시간 안에 측정 결과를 도출하지 못하는 경우 생체 정보 측정이 실패하였다고 판단할 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임에 기반하여, 정상적인 측정 결과가 도출되는지 확인할 수 있다. 예를 들면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정이 시작되고 선택된(또는, 지정된) 조건을 만족하지 못하는 움직임을 감지하는 경우 생체 정보 측정이 실패하였다고 판단할 수 있다.

【0195】 동작 903에서 생체 정보가 성공적으로 측정되면, 동작 905와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 생체 정보를 출력할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서에 기반하여 측정된 결과를 디스플레이(예: 화면)에 표시할 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정된 결과에 기반하여 헬스케어 기능(예: 심박수의 이상 여부 제공 기능)을 제공할 수도 있다.

【0196】 동작 903에서 생체 정보가 측정되지 않으면, 동작 907과 같이 전자 장치(예: 프로세서(120 또는 500))는 실패횟수를 누적시킬 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정을 시도한 후, 정상적인 결과가 도출되지 않을 때마다 실패횟수를 누적시킬 수 있다. 예를 들면, 생체 정보 측정의 정상적인 결과가 1회 도출되지 않았을 경우, 동작 907에서 전자 장치(예: 프로세서(120 또는 500))는 실패횟수를 1회로 결정할 수 있다. 또한, 생체 정보 측정의 정상적인 결과가 3회 도출되지 않았을 경우, 동작 907에서 전자 장치(예: 프로세서(120 또는 500))는 실패횟수를 3회로 결정할 수 있다.

【0197】 동작 909와 같이, 전자 장치(예: 프로세서(120 또는 500))는 누적된 실패횟수를 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 임계값(예: 약 3회) 이상의 실패횟수가 누적되었는지 확인할 수 있다. 다양한 실시예에 따르면, 임계값 이상의 실패횟수가 누적되면 지속적으로 측정 실패가 발생하여 생체 정보 측정이 불가능한 상황으로 판단 될 수 있다.

【0198】 동작 909에서 임계값 미만의 실패횟수가 누적되면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정을 재시도할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 동작 901과 연관된 동작을 재수행할 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임에 기반하여 생체 정보를 측정할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 동작 801에서 동작 819 중 적어도 하나의 동작을 수행할 수 있다.

【0199】 동작 909에서 임계값 이상의 실패횟수가 누적되면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 비활성화시킬 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정이 불가하다고 판단하여 적어도 하나의 제2 센서의 동작을 중단시킬 수 있다.

【0200】 다양한 실시예에 따르면, 동작 901에서 동작 911 중 적어도 하나의 동작은 생략될 수 있다. 예를 들면, 동작 907에서 동작 911은 생략될 수 있다. 이에 따라, 동작 903에서 생체 정보의 측정이 실패한 경우, 전자 장치(예: 프로세서(120 또는 500))는 동작 901을 다시 수행할 수 있다.

【0202】 도 10은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 생체 정보 측정 방법의 수행 절차를 도시한 흐름도이다.

【0203】 다양한 실시예에 따르면, 생체 정보 측정 방법의 수행 절차는 도 6에 기재된 동작 607에 대한 상세 동작일 수 있다.

【0204】 동작 1001과 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정과 연관된 상황 정보를 획득할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서 또는/및 적어도 하나의 제2 센서에 기반하여 상황 정보를 획득할 수 있다. 일 실시예에 따르면, 상황 정보는 생체 정보를 측정하는 동안의 전자 장치 또는 신체의 움직임, 측정 대상의 신체

특성, 측정 대상의 상태, 측정 품질, 또는 측정 결과 등을 포함할 수 있다.

【0205】 동작 1003과 같이, 전자 장치(예: 프로세서(120 또는 500))는 상황 정보에 기반하여 측정 파라미터를 변경할 수 있다. 예를 들어, 상황 정보에 기반하여 변경되는 측정 파라미터는 하기 <표1>과 같이 정의할 수 있다. 하지만, 본 발명의 다양한 실시예들은 <표1>에 한정되지 않는다. 예를 들면, 상황 정보가 ‘측정 품질’ 인 경우 변경 파라미터는 ‘타이머 주기’ 가 될 수 있다. 또한, 상황 정보가 신체 특성인 경우 변경 파라미터는 ‘샘플링 비율’ 이 될 수 있다.

【0206】 【표 1】

상황 정보	변경 파라미터
측정 품질	측정 시간(발광 시간)
움직임	샘플링 비율
측정 결과	측정 트리거
신체 조건	타이머 주기
사용자 상태	움직임 판단 조건

【0207】 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 상황 정보에 기반하여 측정 파라미터를 변경함으로써 생체 정보 측정 시 발생하는 전력 소모를 줄일 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 상황 정보에 기반하여 측정 파라미터를 변경함으로써 생체 정보 측정 시 측정 정확도를 향상시킬 수 있다.

【0208】 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 측정 품질이 좋지 않은 상태에서 생체 정보 측정은 무의미하다고 판단할 수 있다. 이에 따라, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서의 발광 시간을 줄여 소모 전류

를 줄일 수 있다. 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 중 정상적인 측정이 불가능한 전자 장치의 움직임 강도가 감지되면 측정 동작을 중단하여 소모 전류를 줄일 수 있다. 또 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 중 움직임에 의해 지속적으로 측정이 실패하면 움직임이 발생되지 않을 때까지 측정 동작을 생략하여 소모 전류를 줄일 수 있다. 또 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 중 움직임의 강도가 보정 가능할 정도라 판단되면 측정 동작 동안의 측정 샘플링 비율을 높임으로 측정 정확도를 향상시킬 수 있다. 또 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 기준값 이상의 제2 센서의 발광량을 요구하는 측정 대상의 신체 특성(예를 들어, 어두운 피부 색)을 감지하면 생체 정보 측정 횟수를 줄이도록 측정 주기를 변경하여 전력 소모를 줄일 수 있다. 또 다른 예로, 전자 장치(예: 프로세서(120 또는 500))는 움직임이 없는 사용자의 상황(예: 취침 상황)에 대해서는 전자 장치의 움직임을 판단하는 동작을 생략하여 전력 소모를 줄일 수도 있다.

【0209】 동작 1005와 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경된 파라미터에 기반하여 생체 정보를 측정하는 동작을 수행할 수 있다.

【0210】 동작 1007과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정이 완료되는지 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 결과가 성공적으로 도출되는지 확인할 수 있다.

【0211】 동작 1007에서 생체 정보 측정이 완료되지 않으면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 동작 1005와 같이 변경된 파라미터에 기반하여 생체 정보를 측정할 수 있다.

【0212】 동작 1007에서 생체 정보 측정이 완료되면, 동작 1009와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 생체 정보를 출력할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 디스플레이 또는 스피커를 통하여 측정된 생체 정보를 출력할 수 있다.

【0213】 다양한 실시예에 따르면, 동작 1001에서 동작 1009 중 적어도 하나의 동작은 생략될 수 있다. 예를 들면, 동작 1009는 생략될 수 있다.

【0215】 도 11은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0216】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0217】 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 품질이 좋지 않은 상태에서 생체 정보 측정은 무의미하다고 판단하여 제2 센서의 발광 시간을 줄여 소모 전류를 줄일 수 있다.

【0218】 동작 1101과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0219】 동작 1103과 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정 품질을 확인할 수 있다. 일 실시예에 따르면, 측정 품질은 적어도 하나의 제2 센서로부터 출력되는 신호의 품질이 될 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서로부터 출력되는 신호에 대한 신호대잡음비(signal to noise ratio)를 측정 품질로 획득할 수 있다.

【0220】 동작 1105와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정 품질이 선택된(또는, 지정된) 기준을 만족하는지 판단할 수 있다. 일 실시예에 따르면, 측정 품질이 선택된(또는, 지정된) 기준을 만족하지 못하는 것은 생체 정보 측정이 가능할 정도의 신호가 획득되지 않은 상태가 될 수 있다. 예를 들어, 측정 품질이 선택된(또는, 지정된) 기준을 만족하지 못하는 것은, 정상적인 측정 결과가 도출되지 않은 상태일 수 있다. 또 다른 예를 들어, 측정 품질이 선택된(또는, 지정된) 기준을 만족하지 못하는 것은, 선택된(또는, 지정된) 측정 시간 동안에 측정 결과가 도출되지 않은 상태일 수 있다.

【0221】 동작 1105에서 측정 품질이 선택된(또는, 지정된) 기준을 만족하면, 동작 1107과 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는,

지정된) 측정 시간에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 측정 시간은 적어도 하나의 제2 센서의 발광 시간과 연관될 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 품질이 선택된(또는, 지정된) 기준을 만족하면, 생체 정보 측정 주기(예: 약 1분 주기)가 도래할 때마다 선택된(또는, 지정된) 시간(예: 약 10초) 동안 적어도 하나의 제2 센서를 통해 빛을 발광하여 생체 정보를 획득할 수 있다.

【0222】 동작 1105에서 측정 품질이 선택된(또는, 지정된) 기준을 만족하지 않으면, 동작 1109와 같이, 전자 장치(예: 프로세서(120 또는 500))는 품질에 기반하여 선택된(또는, 지정된) 측정 시간을 변경할 수 있다.

【0223】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 소모 전류를 줄이기 위해 측정 시간을 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 측정 시간 동안 생체 정보의 측정 결과가 도출되지 않으면, 소모 전류를 줄이기 위해 측정 시간을 단축시킬 수 있다.

【0224】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과의 정확도를 향상시키기 위해 측정 시간을 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 측정 시간 동안 생체 정보의 측정 결과가 도출되지 않으면, 생체 정보 측정에 필요한 정보가 충분히 확보될 수 있도록 측정 시간을 증가시킬 수 있다.

【0225】 동작 1111과 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경

된 측정 시간에 기반하여 생체 정보를 측정할 수 있다.

【0226】 다양한 실시예에 따르면, 동작 1101에서 동작 1111 중 적어도 하나의 동작은 생략될 수 있다.

【0228】 도 12는 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0229】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0230】 동작 1201과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서(예: 생체 센서)를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0231】 동작 1203과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하는 동안 전자 장치의 움직임이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 전자 장치의 움직임을 측정할 수 있다.

【0232】 동작 1203에서 전자 장치의 움직임이 감지되지 않으면, 동작 1205와 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 샘플링 비율에 기반하여 적어도 하나의 제2 센서로부터 출력되는 정보를 획득할 수 있다.

【0233】 동작 1203에서 전자 장치의 움직임이 감지되면, 동작 1207과 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 움직임이 선택된(또는, 지정된) 기준값을 만족하는지 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과에 영향을 미치지 않는 정도의 작은 움직임 강도가 감지되는지 확인할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과에 영향을 미치지 않는 움직임 강도를 기준값을 만족하는 움직임으로 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과에 영향을 미칠 수도 있지만 보정 가능한 움직임 강도가 감지되는지 확인할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과에 영향을 미치지 않는 움직임 강도 보다 소정의 값 이상인 값을 기준값으로 정할 수 있다.

【0234】 동작 1207에서 기준값을 만족하는 움직임을 확인하면, 동작 1209와 같이, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서로부터 출력되는 신호의 샘플링 비율을 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 움직임에 따른 잡음을 보정하기 위해 샘플링 비율을 빠르게 하여 제2

센서로부터 출력되는 신호를 증가시킬 수 있다.

【0235】 동작 1207에서 기준값을 만족하는 움직임이 확인하지 못하면, 동작 1211과 같이, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서를 비활성화시킬 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 움직임에 따른 잡음 보정이 불가능하다고 판단하여 생체 정보 측정 동작을 중단할 수 있다.

【0236】 다양한 실시예에 따르면, 동작 1201에서 동작 1211 중 적어도 하나의 동작은 생략될 수 있다. 예를 들면, 동작 1207 및 동작 1211은 생략될 수 있다. 이 경우, 동작 1203에서 전자 장치의 움직임이 감지되는 경우, 동작 1209에서 샘플링 비율을 변경할 수 있다. 동작 1205에서 변경된 샘플링 비율에 따라서 생체 정보를 측정할 수 있다.

【0238】 도 13은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0239】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0240】 동작 1301과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간

동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0241】 동작 1303과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하는 동안 전자 장치의 움직임이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 전자 장치의 움직임을 측정할 수 있다.

【0242】 동작 1305와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하는지 확인할 수 있다. 일 실시예에 따르면, 선택된(또는, 지정된) 기준을 만족하는 것은 생체 정보 획득에 영향을 미치지 않는 정도의 움직임이 발생한 것이 될 수 있다.

【0243】 동작 1305에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하면, 동작 1307과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 측정 시간에 기반하여 적어도 하나의 제2 센서로부터 출력되는 정보를 획득할 수 있다.

【0244】 동작 1305에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하지 않으면, 동작 1309와 같이, 전자 장치(예: 프로세서(120 또는 500))는 결과물을 도출할 수 있는 정도의 정보가 수집된 상태인지 판단할 수 있다.

【0245】 동작 1309에서 결과를 도출할 수 있는 정도의 정보가 수집되면, 동작 1311과 같이, 전자 장치(예: 프로세서(120 또는 500))는 움직임 판단 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정이 거의 완료된 시점에서 측정 동작이 중단되는 것을 방지하기 위하여 움직임이 발생하더라도 제2 센서를 비활성화시키지 않고 움직임이 선택된(또는, 지정된) 기준을 만족할 때까지 움직임 판단 동작을 수행할 수 있다.

【0246】 동작 1309에서 결과를 도출할 수 있는 정도의 정보가 수집되지 않으면, 동작 1313과 같이, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서를 비활성화시킬 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정이 완료되기까지 소정의 시간이 남은 상태에서 선택된(또는, 지정된) 기준을 만족하지 않는 움직임이 발생했다고 판단하여 생체 정보 측정 동작을 중단할 수 있다.

【0247】 다양한 실시예에 따르면, 동작 1301에서 동작 1313 중 적어도 하나의 동작은 생략될 수 있다. 예를 들면, 동작 1309 및 동작 1311은 생략될 수 있다. 이 경우, 동작 1305에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하지 않는 경우, 동작 1313에서 전자 장치(예: 프로세서(120 또는 500))는 제 2 센서를 비활성화시킬 수 있다. 또한, 동작 1305에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하는 경우, 동작 1307에서 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정할 수 있다.

【0249】 도 14는 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0250】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0251】 동작 1401과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0252】 동작 1403과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하는 동안 전자 장치의 움직임이 감지되는지 확인할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 전자 장치의 움직임을 측정할 수 있다.

【0253】 동작 1405와 같이, 전자 장치(예: 프로세서(120 또는 500))는 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하는지 확인할 수 있다. 일 실시예에 따르면, 선택된(또는, 지정된) 기준을 만족하는 것은 생체 정보 획득에 영향을 미치지 않는 정도의 움직임이 발생한 것이 될 수 있다.

【0254】 동작 1405에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하면, 동작 1407과 같이, 전자 장치(예: 프로세서(120 또는 500))는 타이머 동작에 기반하여 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간을 가지는 주기 간격으로 생체 정보의 측정을 시도할 수 있다.

【0255】 동작 1405에서 측정된 움직임이 선택된(또는, 지정된) 기준을 만족하지 않으면, 동작 1409와 같이, 전자 장치(예: 프로세서(120 또는 500))는 움직임 기반으로 측정 트리거를 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간을 가지는 주기를 트리거로 하여 생체 정보를 측정하나, 선택된(또는, 지정된) 기준을 만족하는 움직임이 측정되지 않는 상황에서는 측정 주기가 도래하여도 측정 동작을 수행하지 않을 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행하지 않으면서 선택된(또는, 지정된) 기준을 만족하는 움직임이 측정되는지 확인하고, 선택된(또는, 지정된) 기준을 만족하는 움직임이 감지되면 생체 정보 측정 동작을 수행할 수 있다.

【0256】 동작 1411과 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경된 측정 트리거에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 동작 1401에서 동작 1411 중 적어도 하나의 동작은 생략될 수 있다.

【0258】 도 15는 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자

장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0259】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0260】 동작 1501과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0261】 동작 1503과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하는 동안 이전 측정 결과를 확인할 수 있다. 이전 측정 결과는 전자 장치(예: 프로세서(120 또는 500))의 적어도 하나의 제1 센서 또는 적어도 하나의 제2 센서 중 획득한 데이터에 기반한 결과일 수 있다.

【0262】 동작 1505와 같이, 전자 장치(예: 프로세서(120 또는 500))는 이전 측정 결과에 기반하여 측정 실패가 선택된(또는, 지정된) 횟수 이상 연속적으로 발생하는지 확인할 수 있다.

【0263】 동작 1505에서 측정 실패가 선택된(또는, 지정된) 횟수 이상 연속적으로 발생하지 않으면, 동작 1507과 같이, 전자 장치(예: 프로세서(120 또는 500))는 타이머 동작에 기반하여 생체 정보 측정 동작을 수행할 수 있다. 일 실시예

에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간을 가지는 주기 간격으로 생체 정보의 측정을 시도할 수 있다.

【0264】 동작 1505에서 측정 실패가 선택된(또는, 지정된) 횟수 이상 연속적으로 발생하면, 동작 1509와 같이, 전자 장치(예: 프로세서(120 또는 500))는 움직임 기반으로 측정 트리거를 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간을 가지는 주기를 트리거로 하여 생체 정보를 측정하던 것을, 선택된(또는, 지정된) 기준을 만족하는 움직임이 측정되는지 확인하고, 선택된(또는, 지정된) 기준을 만족하는 움직임이 감지되면 생체 정보를 측정하는 것으로 변경할 수 있다.

【0265】 동작 1511과 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경된 측정 트리거에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 동작 1501에서 동작 1511 중 적어도 하나의 동작은 생략될 수 있다.

【0267】 도 16은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 트리거 변경 방법의 수행 절차를 도시한 흐름도이다.

【0268】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 14에 기재된 동작 1409, 또는 도 15에 기재된 동작 1509에 대한 상세 동작일 수 있다.

【0269】 동작 1601과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임を確認할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 움직임 기반으로 측정 트리거가 변경되면 전자 장치의 움직임을 측정할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제1 센서에 기반하여 전자 장치의 움직임을 측정할 수 있다.

【0270】 동작 1603과 같이, 전자 장치(예: 프로세서(120 또는 500))는 전자 장치의 움직임이 조건을 만족하는지 확인할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정에 영향을 미치지 않는 정도의 움직임이 발생하였는지 판단할 수 있다.

【0271】 동작 1603에서 조건을 만족하는 움직임이 감지되지 않으면, 동작 1609와 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 측정하지 않고 대기하는 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정에 영향을 미치지 않는 움직임이 감지될 때까지 측정 동작을 수행하지 않고 대기할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 움직임 기반으로 측정 트리거를 변경하면 측정 주기가 도래해도 측정 동작을 수행하지 않을 수 있다.

【0272】 동작 1603에서 조건을 만족하는 움직임이 감지되면, 동작 1605와 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다.

【0273】 동작 1607과 같이, 전자 장치(예: 프로세서(120 또는 500))는 조건

을 만족하는 움직임에 기반하여 생체 정보를 측정 한 후, 타이머 기반으로 측정 트리거를 변경할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정에 영향을 미치지 않는 움직임이 감지된다고 판단하여 선택된(또는, 지정된) 시간을 가지는 주기 간격으로 생체 정보를 측정하도록 측정 트리거를 변경할 수 있다. 다양한 실시예에 따르면, 동작 1601에서 동작 1609 중 적어도 하나의 동작은 생략될 수 있다.

【0275】 도 17은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 트리거 변경 동작을 도시한 도면이다.

【0276】 도 17을 참조하면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 시간(예: 약 1분)을 가지는 주기를 간격으로 생체 정보의 측정을 시도할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 주기를 측정 트리거로 하여 생체 정보를 측정할 수 있다.

【0277】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 주기가 도래하면 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정할 수 있다.

【0278】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 주기가 도래한 시점(t_1 , t_2)에 전자 장치의 움직임이 선택된

(또는, 지정된) 조건을 만족하지 않으면(예: 생체 정보를 정확하게 측정할 수 없는 움직임이 감지되면), 측정 트리거를 변경할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보를 정확하게 측정할 수 있는 움직임이 감지될 때까지 생체 정보 측정 동작을 수행하지 않을 수 있다. 다시 말해서, 전자 장치(예: 프로세서(120 또는 500))는 조건을 만족하는 움직임(예: 생체 정보를 정확하게 측정할 수 있는 움직임)을 측정 트리거로 하여 생체 정보를 측정할 수 있다.

【0279】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 움직임에 따른 측정 실패가 선택된(또는, 지정된) 횟수(예를 들어, 약 2회) 이상 연속적으로 발생(1701)되면 측정 트리거를 변경할 수 있다.

【0280】 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 횟수 이상 연속적으로 측정에 실패한 이후에 움직임 감지로 측정 트리거를 변경하여 움직임을 모니터링하고 움직임이 감지되는 구간(1703)에는 타이머 기반으로 측정 동작을 수행하지 않고(1705), 움직임이 선택된(또는, 지정된) 조건을 만족하는 시점(t_3)에 측정 동작을 수행할 수 있다. 생체 정보의 획득에 성공하면 타이머 기반으로 측정 트리거를 변경할 수 있다(1707). 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 움직임 감지로 측정 트리거를 변경하여 생체 정보 획득에 성공하면 선택된(또는, 지정된) 시간을 가지는 주기 간격으로 생체 정보를 측정하도록 측정 트리거를 변경할 수 있다.

【0282】 도 18은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0283】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0284】 동작 1801과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0285】 동작 1803과 같이, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서가 접촉된 신체의 특성을 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 대상의 신체적 특성을 판단할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서가 접촉된 신체의 피부색을 신체의 특성으로 판단할 수 있다.

【0286】 동작 1805와 같이, 전자 장치(예: 프로세서(120 또는 500))는 신체 특성이 임계값 이상의 발광량을 요구하는지 판단할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 제2 센서의 수광부에 수신되는 반사광량에 기반하여 멜라닌 지수를 측정하고, 측정된 멜라닌 지수에 기반하여 피부색

및 측정시 요구되는 발광부의 발광량을 확인할 수 있다. 예를 들어, 멜라닌 지수에 따른 발광량은 하기 <표2>와 같이 정의할 수 있다.

【0287】 【표 2】

melanin Index	피부색 타입	전류 세기
5-10	Very light	31~40%
11-20	Light	41~50%
21-30	Intermediate	51~60%
31-40	Tan	61~70%
41-70	Brown	71~80%
71-99	black	81~100%

【0288】 예를 들어, 멜라닌 지수가 높을수록 어두운 색의 피부타입을 나타내며, 전자 장치(예: 프로세서(120 또는 500))는 밝은 색의 피부 타입보다 어두운 색의 피부 타입에 대하여 소모 전류가 높다고 판단할 수 있다.

【0289】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 사용자의 신체 특성을 저장하거나 또는 외부 장치로부터 제공받을 수도 있다.

【0290】 동작 1805에서 신체 특성이 임계값 이상의 발광량을 요구하지 않는다고 판단하면, 동작 1811과 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 측정 주기에 기반하여 생체 정보를 측정할 수 있다.

【0291】 동작 1805에서 신체 특성이 임계값 이상의 발광량을 요구한다고 판단하면, 동작 1807과 같이, 전자 장치(예: 프로세서(120 또는 500))는 신체 특성에 기반하여 측정 주기를 변경할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정시 임계값 이상의 발광량이 요구되면 측정 주기를 증가

시켜 측정 횟수를 감소시킬 수 있다.

【0292】 동작 1807에서 측정 주기를 변경하면, 동작 1809와 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경된 측정 주기에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 동작 1801에서 동작 1811 중 적어도 하나의 동작은 생략될 수 있다.

【0294】 도 19는 다양한 실시예에 따른 전자 장치(예: 전자 장치(101), 전자 장치(101)의 프로세서(120), 프로세서(500) 또는 전자 장치(400))의 측정 파라미터 변경 방법의 수행 절차를 도시한 흐름도이다.

【0295】 다양한 실시예에 따르면, 파라미터 변경 방법의 수행 절차는 도 10에 기재된 동작 1003에 대한 상세 동작일 수 있다.

【0296】 동작 1901과 같이, 전자 장치(예: 프로세서(120 또는 500))는 생체 정보 측정 동작을 수행할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 적어도 하나의 제2 센서를 동작시켜 선택된(또는, 지정된) 시간 동안 빛을 발광시키고, 적어도 하나의 제2 센서로부터 출력되는 신호를 획득할 수 있다.

【0297】 동작 1903과 같이, 전자 장치(예: 프로세서(120 또는 500))는 사용자의 상태를 확인할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 취침 상태 또는 활동 상태 등과 같은 사용자의 상태를 확인할 수 있다.

【0298】 동작 1905와 같이, 전자 장치(예: 프로세서(120 또는 500))는 움직임 조건 변경이 필요한 상태인지 판단할 수 있다. 다양한 실시예에 따르면, 사용자 상태가 취침 상태라면 움직임이 없거나 움직임의 강도가 낮을 것으로 판단할 수 있기 때문에 움직임 조건이 필요하지 않을 수 있다. 움직임 조건이 필요하지 않은 경우 움직임 조건 변경이 필요한 상태라고 판단될 수 있다.

【0299】 동작 1905에서 움직임 조건 변경이 필요하지 않는다고 판단되면, 동작 1911과 같이, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 움직임 조건에 기반하여 생체 정보를 측정할 수 있다.

【0300】 동작 1905에서 움직임 조건 변경이 필요하다고 판단되면, 동작 1907과 같이, 전자 장치(예: 프로세서(120 또는 500))는 사용자 상태에 기반하여 움직임 조건을 변경할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 움직임을 판단하는 조건을 생략하거나 만족하는 움직임 기준값을 감소시킬 수 있다.

【0301】 동작 1907에서 움직임 조건이 변경되면, 동작 1909와 같이, 전자 장치(예: 프로세서(120 또는 500))는 변경된 움직임 조건에 기반하여 생체 정보를 측정할 수 있다. 다양한 실시예에 따르면, 동작 1901에서 동작 1911 중 적어도 하나의 동작은 생략될 수 있다.

【0303】도 20은 다양한 실시예에 따른 전자 장치(예: 전자 장치(101 또는 400) 또는 전자 장치(101 또는 400)과 무선 또는 유선 통신을 통하여 연결된 외부 전자 장치(예: 스마트 폰))의 화면 구성을 도시한 도면이다.

【0304】다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 주기마다 측정된 생체 정보(예: 심박 정보)를 메모리 저장할 수 있으며, 입력에 기반하여 측정된 결과(2000)를 화면에 표시할 수 있다.

【0305】다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정된 심박 정보를 시간 단위(2001), 일 단위(2003), 월 단위(2005) 등으로 구분하여 표시할 수 있다. 일 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과를 시간 단위로 표시하는 경우, 매시간 동안에 측정된 심박 정보의 평균값을 시간 단위로 표시할 수 있다. 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과를 일 단위로 표시하는 경우, 매일 동안에 측정된 심박 정보의 평균값을 일 단위로 표시할 수 있다. 또 다른 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 측정 결과를 월 단위로 표시하는 경우, 매달 동안에 측정된 심박 정보의 평균값을 월 단위로 하여 표시할 수 있다. 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 심박 정보의 평균값뿐만 아니라, 시간 단위로 표시되는 심박 정보의 최고값 및 최소값(2011), (2013)을 표시할 수도 있다.

【0306】다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 심박 정보의 결과를 표시(2007)할 때, 평균값이 일정 기준보다 높거나 낮은 시점에

대하여 지시자를 함께 표시할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 선택된(또는, 지정된) 색상, 선택된(또는, 지정된) 문구, 선택된(또는, 지정된) 아이콘 등을 이용하여 지시자를 표시할 수 있다. 도시된 도면에서는 3시간 구간에 표시된 느낌표(2009)가 평균 값 이상의 심박 정보가 측정됨을 나타내는 지시자가 될 수 있다.

【0307】 다양한 실시예에 따르면, 전자 장치(예: 프로세서(120 또는 500))는 지시자에 대한 입력이 감지되면, 해당 구간과 관련된 상세 정보를 표시할 수 있다. 예를 들어, 전자 장치(예: 프로세서(120 또는 500))는 입력이 감지된 지시자에 대응하는 시간 구간에 해당하는 심박 정보(2021)와 해당 구간에 대한 전자 장치 또는 사용자의 상태 정보(예: 소모 칼로리 등)(2023)를 함께 표시할 수 있다. 도시된 도면에서는 사용자가 3시 15분부터 3시 20분까지 달리기를 수행하여 심박 정보가 증가하였음을 나타내고 있다.

【0309】 다양한 실시예에 따른 전자 장치의 운용 방법은, 적어도 하나의 제1 센서(예: 근접 센서, 가속도 센서, 자이로 센서 또는 지자기 센서 중 적어도 하나)를 이용하여 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지 여부를 판단하는 동작, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된(또는, 지정된) 제1 시간 주기 (a first period of time)를 경과하는 시점 바로 전(immediately before) 및/또는 바로 후(immediately after)의 제2 시간 주기 동안에 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자

장치의 움직임에 관한 정보를 획득하는 동작, 상기 움직임에 관한 정보에 적어도 일부 기초하여, 적어도 하나의 제2 센서(예: 심박 센서)를 활성화(또는, 작동 또는 트리거)하는 동작을 포함할 수 있다.

【0310】 다양한 실시예에 따르면, 상기 제2 시간 주기는, 상기 제1 시간 주기보다 짧을 수 있다.

【0311】 다양한 실시예에 따르면, 상기 적어도 하나의 제2 센서는, 생체 센서를 포함할 수 있다.

【0312】 다양한 실시예에 따르면, 상기 생체 센서는, 심박 센서, 혈압 센서, 심전도 센서, 피부 수분도 센서, 또는 산소포화도 센서 중 적어도 하나를 포함할 수 있다.

【0313】 다양한 실시예에 따르면, 상기 활성화(또는, 작동 또는 트리거) 동작은, 상기 움직임에 관한 정보가 선택된(또는, 지정된) 조건을 만족하지 않는 경우, 상기 제2 시간 주기를 경과하는 시점 바로 후의, 상기 제1 시간 주기 보다 짧은 제3 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하는 동작, 및 상기 추가 정보에 적어도 일부에 더 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하는 동작을 포함할 수 있다.

【0314】 다양한 실시예에 따르면, 상기 제2 시간 주기 또는 상기 제3 시간 주기 중 적어도 하나는, 상기 제1 시간 주기를 경과하는 시점을 포함할 수 있다.

【0315】 다양한 실시예에 따르면, 상기 제2 시간 주기는 상기 제1 시간 주기를 경과하는 시점을 포함할 수 있다.

【0316】 다양한 실시예에 따르면, 상기 활성화(또는, 작동 또는 트리거) 동작은, 상기 활성화(또는, 작동 또는 트리거)된 제2 센서로부터 데이터를 선택된(또는, 지정된) 시간 내에 수신하지 않은 경우, 상기 선택된(또는, 지정된) 시간이 경과한 후에, 상기 제2 센서를 다시 활성화(또는, 작동 또는 트리거)하는 동작을 포함할 수 있다.

【0317】 다양한 실시예에 따르면, 상기 활성화(또는, 작동 또는 트리거)하는 동작은, 상기 선택된(또는, 지정된) 시간이 경과하는 시점 전의 상기 제2 시간 주기와 실질적으로 동일한 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고, 상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 다시 활성화(또는, 작동 또는 트리거)하는 동작을 포함할 수 있다.

【0318】 다양한 실시예에 따르면, 상기 전자 장치의 운용 방법은, 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작, 및 상기 획득된 데이터에 적어도 일부 기초하여 상기 제1 시간 주기 및/또는 제2 시간 주기 및 또는 제3 시간 주기를 변경하는 동작을 더 포함할 수 있다.

【0319】 다양한 실시예에 따르면, 상기 전자 장치의 운용 방법은, 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작, 및 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 적어도 하나의 조건을 변경하는 동작을 더 포함할 수 있다.

【0320】 다양한 실시예에 따르면, 상기 전자 장치의 운용 방법은, 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작, 및 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)시키는 시간 주기 또는 샘플링 비율 중 적어도 하나를 변경하는 동작을 포함할 수 있다.

【0321】 다양한 실시예에 따른 전자 장치의 운용 방법은 적어도 하나의 제1 센서를 이용하여 전자 장치가 사용자의 신체의 일부에 착용되었는지 여부를 판단하는 동작, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된(또는, 지정된) 시간 주기를 경과하는 시점에서 선택된(또는, 지정된) 시간 범위 내에 측정된 전자 장치의 움직임의 적어도 일부에 기반하여 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하는 동작 및 상기 적어도 하나의 제2 센서를 통하여 생체 정보를 측정하는 동작을 포함할 수 있다.

【0322】 다양한 실시예에 따르면, 상기 적어도 하나의 제1 센서는, 근접 센서, 자이로 센서, 지자기 센서 또는 가속도 센서 중 적어도 하나를 포함할 수 있다.

【0323】 다양한 실시예에 따르면, 상기 움직임의 적어도 일부가 선택된(또는, 지정된) 조건을 만족하지 않는 경우, 상기 제2 센서를 활성화(또는, 작동 또는 트리거)하지 않고, 상기 시간 주기를 경과하는 시점부터 선택된(또는, 지정된) 시간 동안 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하는 동작 및 상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 동작을 포함할 수 있다.

【0324】 다양한 실시예에 따르면, 상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은 상기 생체 정보의 측정이 실패하면, 실패횟수를 누적하는 동작 및 상기 누적된 실패횟수가 선택된(또는, 지정된) 조건을 만족하면 상기 제2 센서를 비활성화하는 동작을 포함할 수 있다.

【0325】 다양한 실시예에 따르면, 상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작 및 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 적어도 하나의 조건을 변경하는 동작을 포함할 수 있다.

【0326】 다양한 실시예에 따르면, 상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작 및 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는

트리거)시키는 시간 주기 또는 샘플링 비율 중 적어도 하나를 변경하는 동작을 포함할 수 있다.

【0327】 다양한 실시예에 따르면, 상기 적어도 하나의 제 2 센서에 기반하여 생체 정보를 측정하는 동작은, 상기 사용자의 신체 특성(예: 피부 색) 또는 상태(예: 취침 상태 또는 활동 상태)에 대한 데이터를 획득하는 동작, 및 상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제 2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 적어도 하나의 조건을 변경하는 동작을 포함할 수 있다.

【0328】 다양한 실시예에 따르면, 상기 움직임에 관한 추가 정보를 획득하는 동작은 상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 획득된 데이터에 기반하여 상기 추가 정보를 획득하는 선택된(또는, 지정된) 시간을 변경하는 동작을 포함할 수 있다.

【0329】 다양한 실시예에 따르면, 전자 장치를 운용하는 방법은, 사용자의 신체에 대한 전자 장치의 근접 여부를 판단하는 동작, 상기 전자 장치가 상기 사용자의 신체의 일부에 근접한 경우, 선택된(또는, 지정된) 주기를 경과하는 시점으로부터 선택된(또는, 지정된) 시간 범위 내에, 적어도 하나의 제1 센서를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하는 동작, 및 상기 움직임에 관한 정보에 적어도 일부 기초하여, 적어도 하나의 제 2 센서를 통하여 상기 사용자의 생체 정보를 획득하는 동작을 포함할 수 있다.

【0331】 본 문서에서 사용된 용어 "모듈"은, 예를 들면, 하드웨어, 소프트웨어 또는 펌웨어(firmware) 중 하나 또는 둘 이상의 조합을 포함하는 단위(unit)를 의미할 수 있다. "모듈"은, 예를 들면, 유닛(unit), 로직(logic), 논리 블록(logical block), 부품(component), 또는 회로(circuit) 등의 용어와 바꾸어 사용(interchangeably use)될 수 있다. "모듈"은, 일체로 구성된 부품의 최소 단위 또는 그 일부가 될 수 있다. "모듈"은 하나 또는 그 이상의 기능을 수행하는 최소 단위 또는 그 일부가 될 수도 있다. "모듈"은 기계적으로 또는 전자적으로 구현될 수 있다. 예를 들면, "모듈"은, 알려졌거나 앞으로 개발될, 어떤 동작들을 수행하는 ASIC(application-specific integrated circuit) 칩, FPGAs(field-programmable gate arrays) 또는 프로그램 가능 논리 장치(programmable-logic device) 중 적어도 하나를 포함할 수 있다.

【0332】 다양한 실시예에 따른 장치(예: 모듈들 또는 그 기능들) 또는 방법(예: 동작들)의 적어도 일부는, 예컨대, 프로그램 모듈의 형태로 컴퓨터로 읽을 수 있는 저장매체(computer-readable storage media)에 저장된 명령어로 구현될 수 있다. 명령어가 프로세서(예: 프로세서(120))에 의해 실행될 경우, 하나 이상의 프로세서가 명령어에 해당하는 기능을 수행할 수 있다. 컴퓨터로 읽을 수 있는 저장매체는, 예를 들면, 메모리(130)가 될 수 있다.

【0333】 컴퓨터로 판독 가능한 기록 매체는, 하드디스크, 플로피디스크, 마그네틱 매체(magnetic media)(예: 자기테이프), 광기록 매체(optical media)(예: CD-ROM(compact disc read only memory), DVD(digital versatile disc), 자기-광

매체(magneto-optical media)(예: 플로티컬 디스크(floptical disk)), 하드웨어 장치(예: ROM(read only memory), RAM(random access memory), 또는 플래시 메모리 등) 등을 포함할 수 있다. 또한, 프로그램 명령에는 컴파일러에 의해 만들어지는 것과 같은 기계어 코드뿐만 아니라 인터프리터 등을 사용해서 컴퓨터에 의해서 실행될 수 있는 고급 언어 코드를 포함할 수 있다. 상술한 하드웨어 장치는 다양한 실시예의 동작을 수행하기 위해 하나 이상의 소프트웨어 모듈로서 작동하도록 구성될 수 있으며, 그 역도 마찬가지다.

【0334】 다양한 실시예에 따른 모듈 또는 프로그램 모듈은 전술한 구성요소들 중 적어도 하나 이상을 포함하거나, 일부가 생략되거나, 또는 추가적인 다른 구성요소를 더 포함할 수 있다. 다양한 실시예에 따른 모듈, 프로그램 모듈 또는 다른 구성요소에 의해 수행되는 동작들은 순차적, 병렬적, 반복적 또는 휴리스틱(heuristic)한 방법으로 실행될 수 있다. 또한, 일부 동작은 다른 순서로 실행되거나, 생략되거나, 또는 다른 동작이 추가될 수 있다.

【특허청구범위】

【청구항 1】

전자 장치에 있어서,

하우징;

상기 하우징의 일부에 결합되고, 사용자의 신체의 일부에 착탈가능하게 부착되는 결합구;

상기 전자 장치의 움직임을 감지할 수 있도록 구성된 적어도 하나의 제1 센서;

상기 사용자의 신체의 변화를 감지할 수 있도록 구성된 적어도 하나의 제2 센서;

메모리; 및

상기 적어도 하나의 제1 센서, 상기 적어도 하나의 제2 센서, 및 상기 메모리와 전기적으로 연결된 프로세서를 포함하고,

상기 메모리는, 실행 시에, 상기 프로세서가,

상기 적어도 하나의 제1 센서를 이용하여 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지 여부를 판단하고,

상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된 제1 시간 주기(a first period of time)를 경과하는 시점 바로 전(immediately before) 및/또는 바로 후(immediately after)의 제2 시간 주기 동안에 상기 적어도

하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하고,

상기 움직임에 관한 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화하도록 하는 인스트럭션들을 저장하는 것을 특징으로 하는 장치.

【청구항 2】

제1항에 있어서,

상기 제2 시간 주기는, 상기 제1 시간 주기보다 짧은 것을 특징으로 하는 장치.

【청구항 3】

제1항에 있어서,

상기 적어도 하나의 제2 센서는, 생체 센서를 포함하는 것을 특징으로 하는 장치.

【청구항 4】

제3항에 있어서,

상기 생체 센서는, 심박 센서, 혈압 센서, 심전도 센서, 피부 수분도 센서, 또는 산소포화도 센서 중 적어도 하나를 포함하는 것을 특징으로 하는 장치.

【청구항 5】

제1항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 움직임에 관한 정보가 선택된 조건을 만족하지 않는 경우,

상기 제2 시간 주기를 경과하는 시점 바로 후의, 상기 제1 시간 주기 보다 짧은 제3 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고,

상기 추가 정보의 적어도 일부에 더 기초하여, 상기 적어도 하나의 제2 센서를 활성화하도록 하는 것을 특징으로 하는 장치.

【청구항 6】

제5항에 있어서,

상기 제2 시간 주기 또는 상기 제3 시간 주기 중 적어도 하나는 상기 제1 시간 주기를 경과하는 시점을 포함하는 것을 특징으로 하는 장치.

【청구항 7】

제1항에 있어서,

상기 제2 시간 주기는 상기 제1 시간 주기를 경과하는 시점을 포함하는 것을

특징으로 하는 장치.

【청구항 8】

제1항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 활성화된 제2 센서로부터 데이터를 선택된 시간 내에 수신하지 않은 경우,

상기 선택된 시간이 경과한 후에, 상기 제2 센서를 다시 활성화하도록 하는 것을 특징으로 하는 장치.

【청구항 9】

제8항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 선택된 시간이 경과하는 시점 전의 상기 제2 시간 주기와 실질적으로 동일한 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하고,

상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 다시 활성화하도록 하는 것을 특징으로 하는 장치.

【청구항 10】

제1항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고,

상기 획득된 데이터에 적어도 일부 기초하여 상기 제1 시간 주기 및/또는 제2 시간 주기 및/또는 제3 시간 주기를 변경하도록 하는 것을 특징으로 하는 장치.

【청구항 11】

제1항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고,

상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화하도록 하는 적어도 하나의 조건을 변경하도록 하는 것을 특징으로 하는 장치.

【청구항 12】

제1항에 있어서,

상기 인스트럭션들은, 상기 프로세서가,

상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하고,

상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화시키는 시간 주기 또는 샘플링 비율 중 적어도 하나를 변경하도록 하는 것을 특징으로 하는 장치.

【청구항 13】

적어도 하나의 제1 센서를 이용하여 전자 장치가 사용자의 신체의 일부에 착용되었는지 여부를 판단하는 동작;

상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 경우, 선택된 시간 주기(a period of time)를 경과하는 시점에서 상기 선택된 시간 범위 내에 측정된 전자 장치의 움직임의 적어도 일부에 기반하여 적어도 하나의 제2 센서를 활성화하는 동작; 및

상기 적어도 하나의 제2 센서를 통하여 상기 사용자의 생체 정보를 측정하는 동작을 포함하는 방법.

【청구항 14】

제13항에 있어서,

상기 적어도 하나의 제1 센서는, 근접 센서, 자이로 센서, 지자기 센서 또는

가속도 센서 중 적어도 하나를 포함하는 것을 특징으로 하는 방법.

【청구항 15】

제13항에 있어서,

상기 움직임의 적어도 일부가 선택된 조건을 만족하지 않는 경우, 상기 제2 센서를 활성화하지 않고,

상기 시간 주기를 경과하는 시점부터 선택된 시간 동안 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 추가 정보를 획득하는 동작; 및

상기 추가 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화하도록 하는 동작을 포함하는 방법.

【청구항 16】

제15항에 있어서,

상기 움직임에 관한 추가 정보를 획득하는 동작은,

상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 획득된 데이터에 기반하여 상기 추가 정보를 획득하는 선택된 시간을 변경하는 동작을 포함하는 방법.

【청구항 17】

제13항에 있어서,
상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은,
상기 생체 정보의 측정이 실패하면, 실패횟수를 누적하는 동작; 및
상기 누적된 실패횟수가 선택된 조건을 만족하면 상기 제2 센서를 비활성화하는 동작을 포함하는 방법.

【청구항 18】

제13항에 있어서,
상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은,
상기 사용자의 신체 특성 또는 상태에 대한 데이터를 획득하는 동작; 및
상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화하도록 하는 적어도 하나의 조건을 변경하는 동작을 포함하는 방법.

【청구항 19】

제13항에 있어서,
상기 적어도 하나의 제2 센서에 기반하여 생체 정보를 측정하는 동작은,
상기 적어도 하나의 제1 센서 또는 상기 적어도 하나의 제2 센서 중 적어도 하나로부터 데이터를 획득하는 동작; 및

상기 데이터에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 작동 활성화하도록 하는 적어도 하나의 조건을 변경하는 동작을 포함하는 방법.

【청구항 20】

사용자의 신체에 대한 전자 장치의 근접 여부를 판단하는 동작;

상기 전자 장치가 상기 사용자의 신체의 일부에 근접한 경우, 선택된 주기를 경과하는 시점으로부터 선택된 시간 범위 내에, 적어도 하나의 제1 센서를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하는 동작; 및

상기 움직임에 관한 정보에 적어도 일부 기초하여, 적어도 하나의 제 2 센서를 통하여 상기 사용자의 생체 정보를 획득하는 동작을 실행시키기 위한 프로그램을 기록한 컴퓨터 판독 가능한 기록매체.

【요약서】

【요약】

본 발명은 배터리 소모를 줄이면서 생체 정보를 검출하기 위한 장치에 관한 것으로, 하우징, 상기 하우징의 일부에 결합되고, 사용자의 신체의 일부에 착탈가능하게 부착되는 결합구, 상기 전자 장치의 움직임을 감지할 수 있도록 구성된 적어도 하나의 제 1 센서, 상기 사용자의 신체의 변화를 감지할 수 있도록 구성된 적어도 하나의 제 2 센서, 메모리 및 상기 제1 센서, 상기 제2 센서, 및 상기 메모리와 전기적으로 연결된 프로세서를 포함할 수 있다. 상기 메모리는, 실행 시에, 상기 프로세서가, 상기 적어도 하나의 제1 센서를 이용하여 상기 전자 장치가 상기 사용자의 신체의 일부에 착용되었는지 여부를 판단하고, 상기 전자 장치가 상기 사용자의 신체의 일부에 착용된 시점으로부터 선택된 제1 시간 주기 (a first period of time)를 경과하는 시점 바로 전(immediately before) 및/또는 바로 후(immediately after)의, 상기 제1 시간 주기보다 짧은 제2 시간 주기 동안에, 상기 적어도 하나의 제1 센서의 적어도 일부를 이용하여 상기 전자 장치의 움직임에 관한 정보를 획득하고, 상기 움직임에 관한 정보에 적어도 일부 기초하여, 상기 적어도 하나의 제2 센서를 활성화(또는, 작동 또는 트리거)하도록 하는 인스트럭션들을 저장할 수 있다.

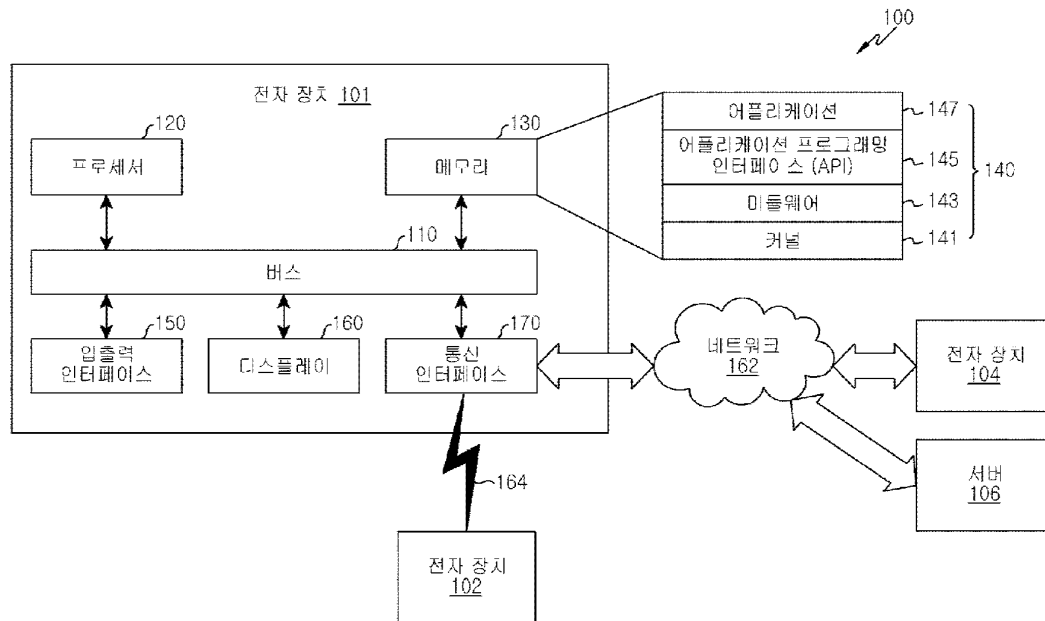
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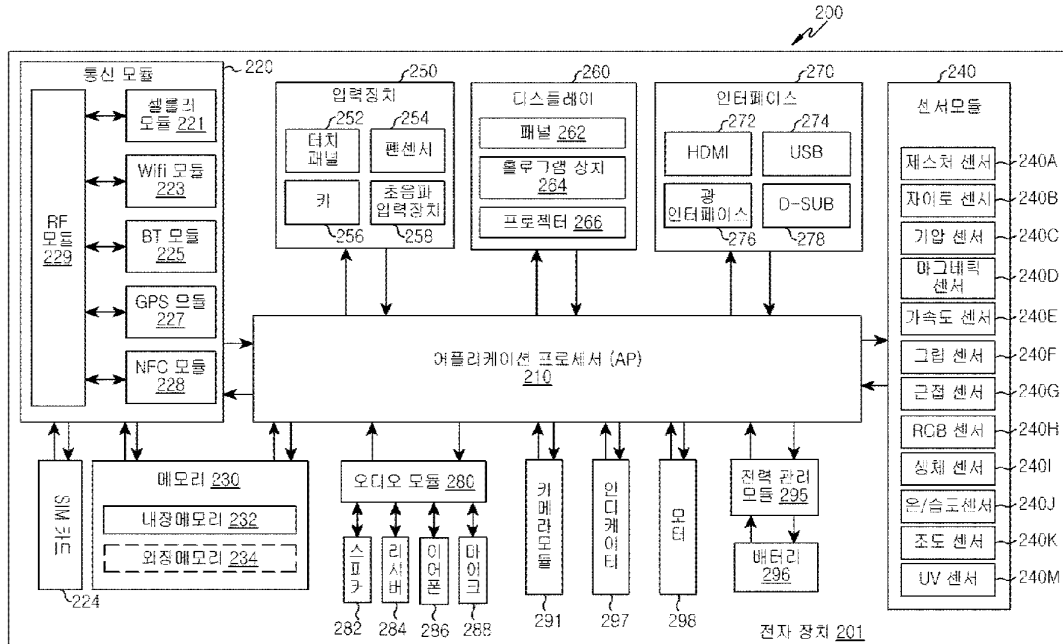
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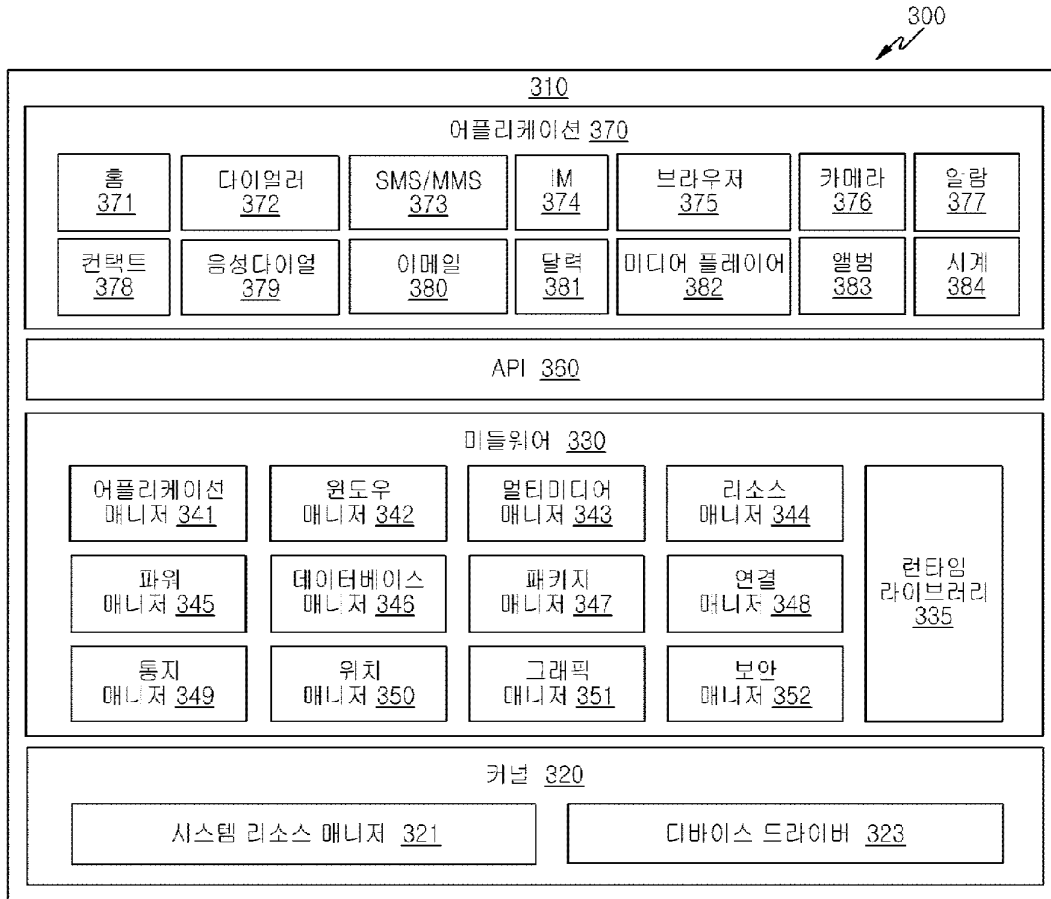
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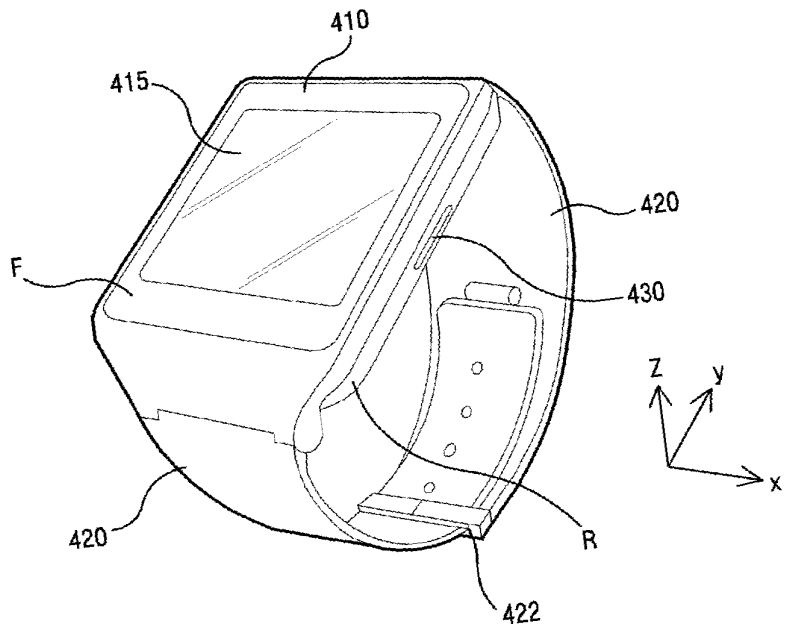
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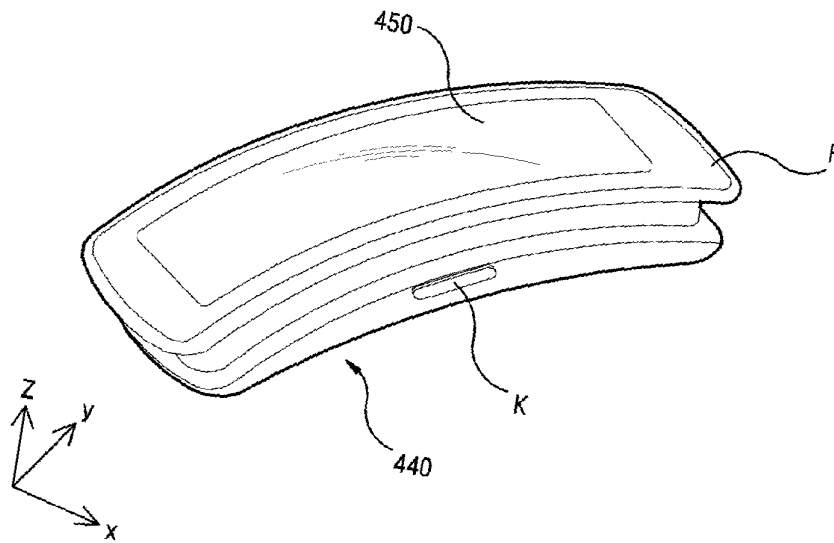


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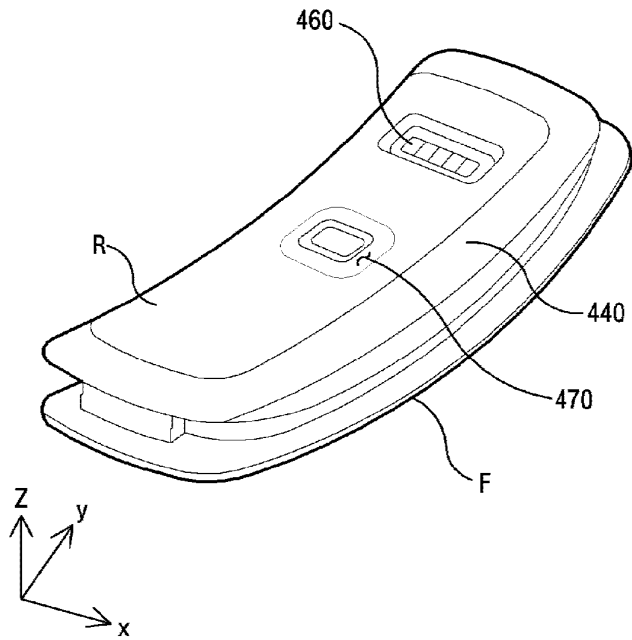


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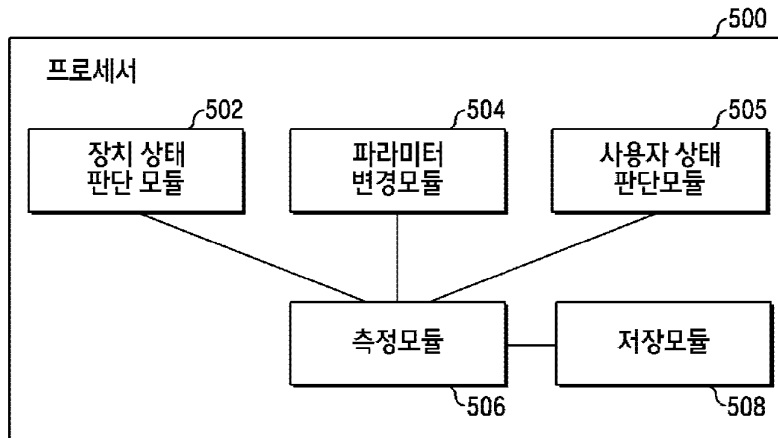
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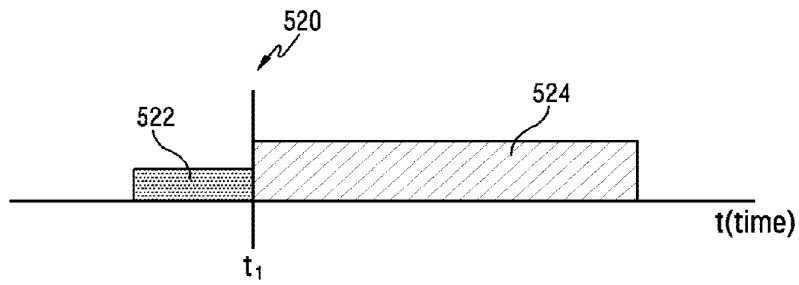
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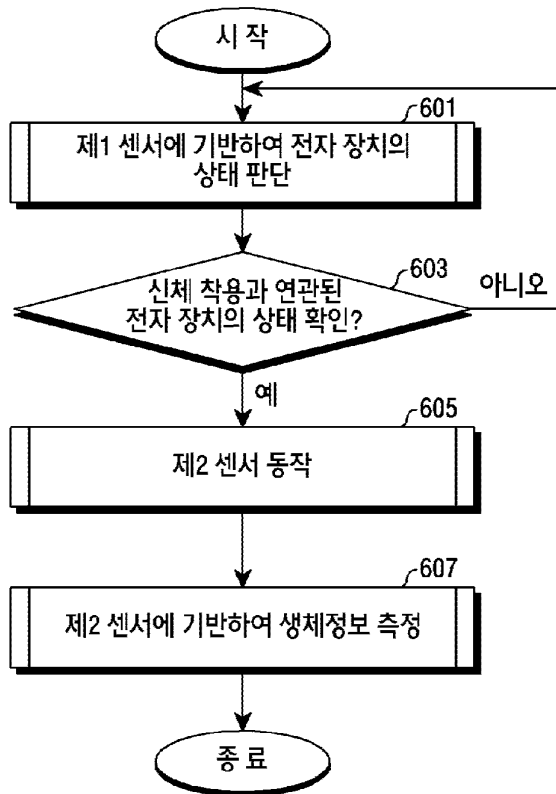
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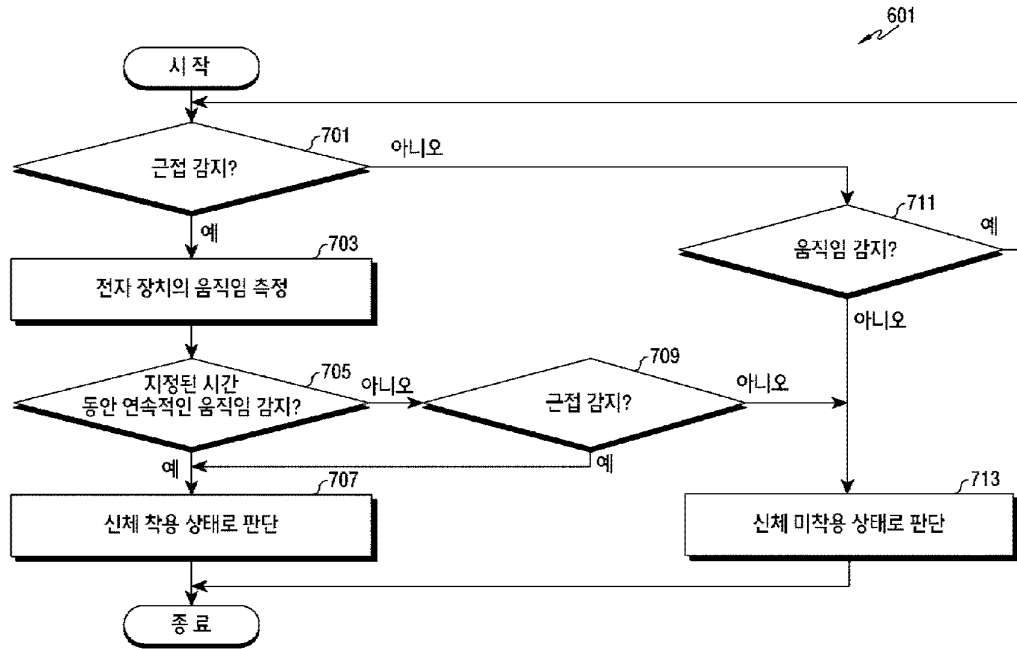
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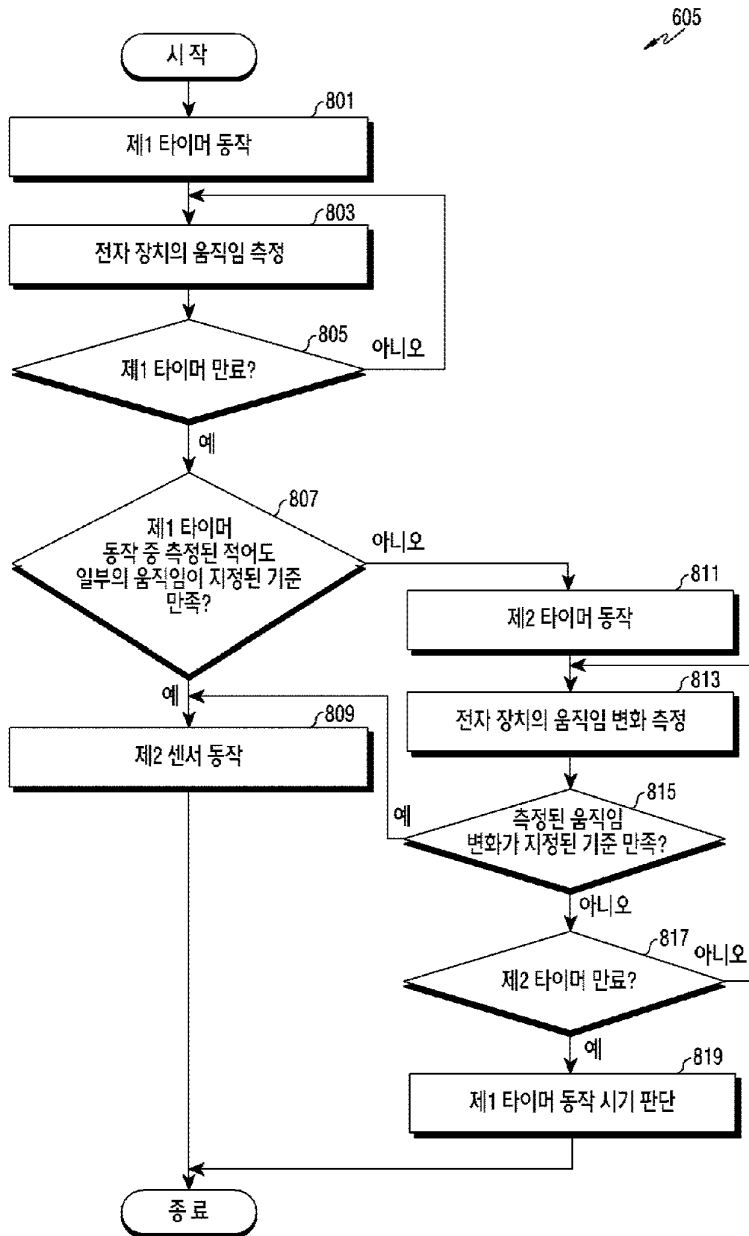
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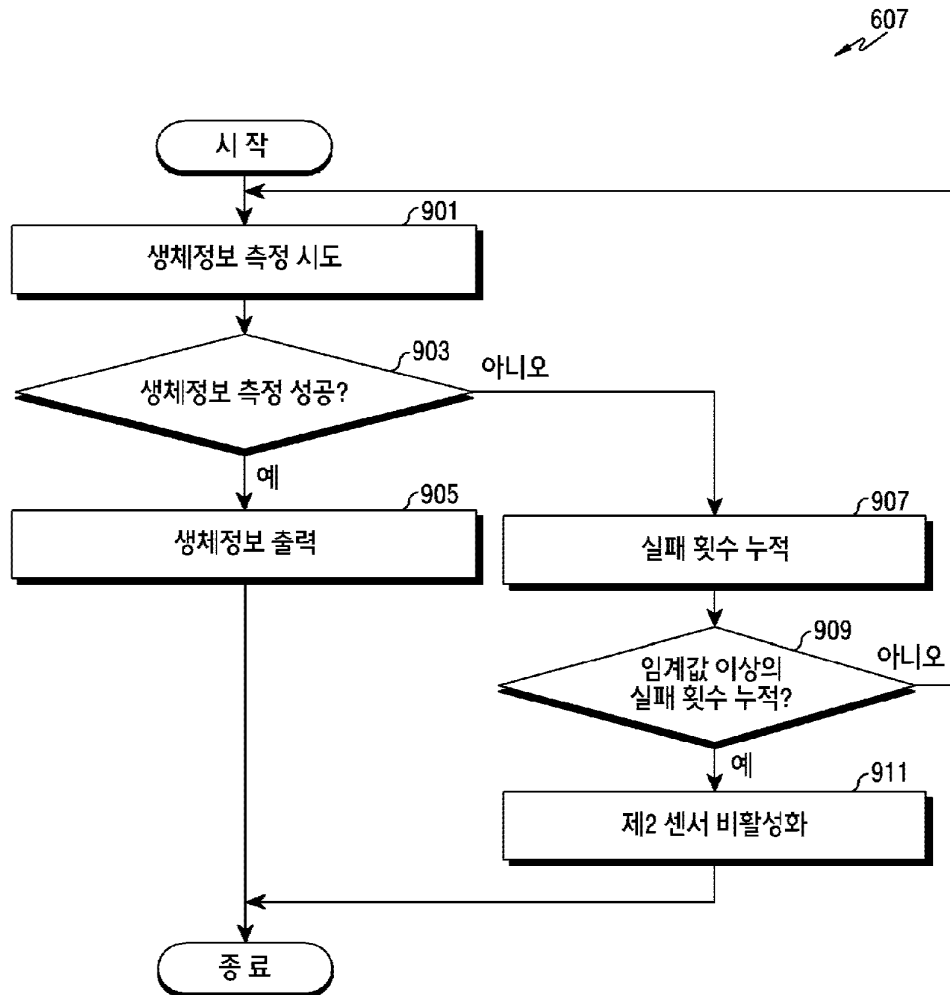
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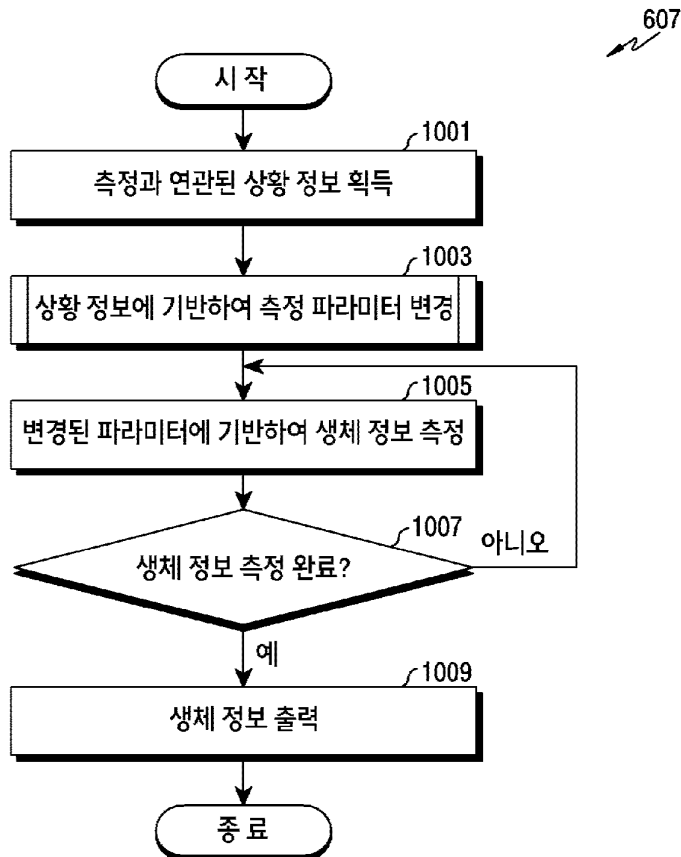
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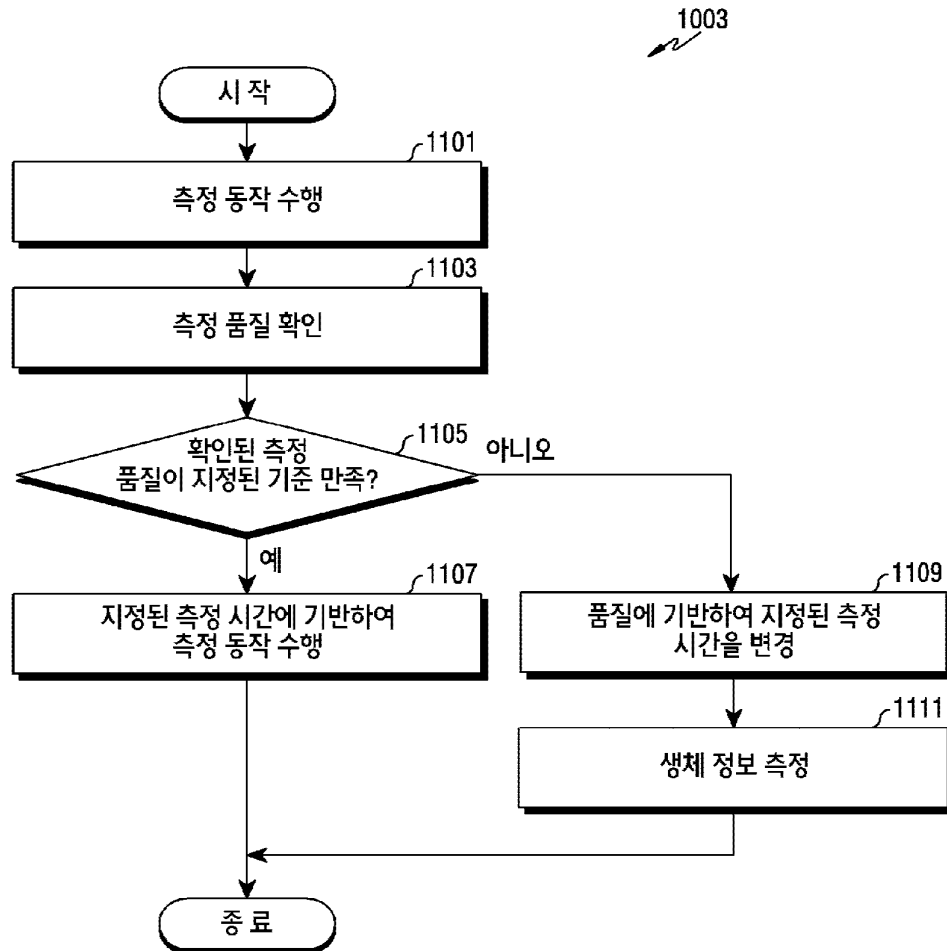
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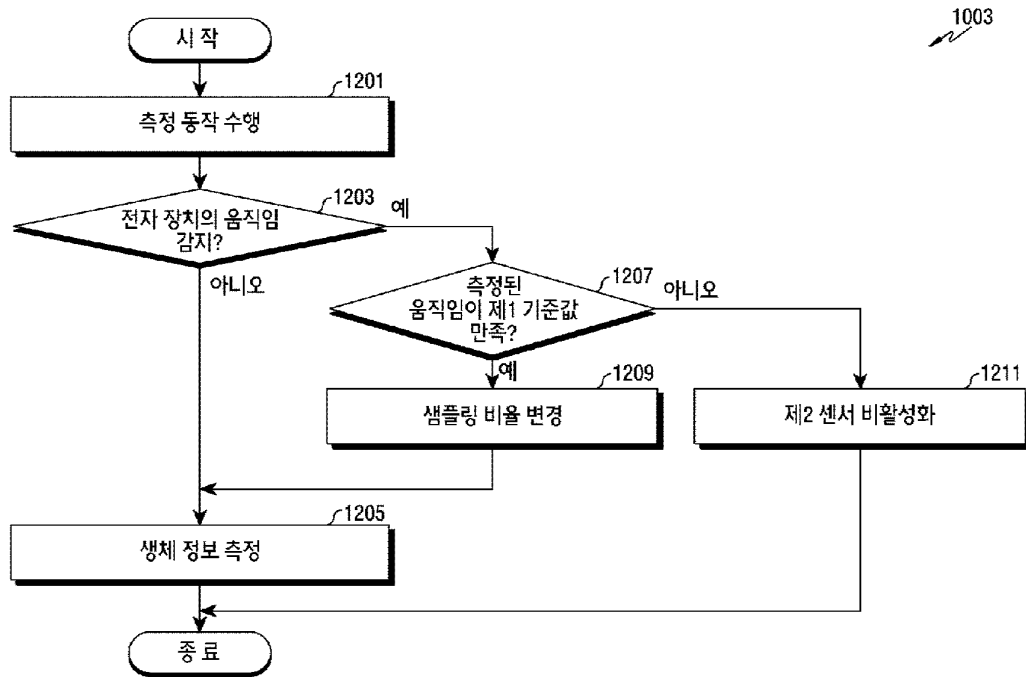
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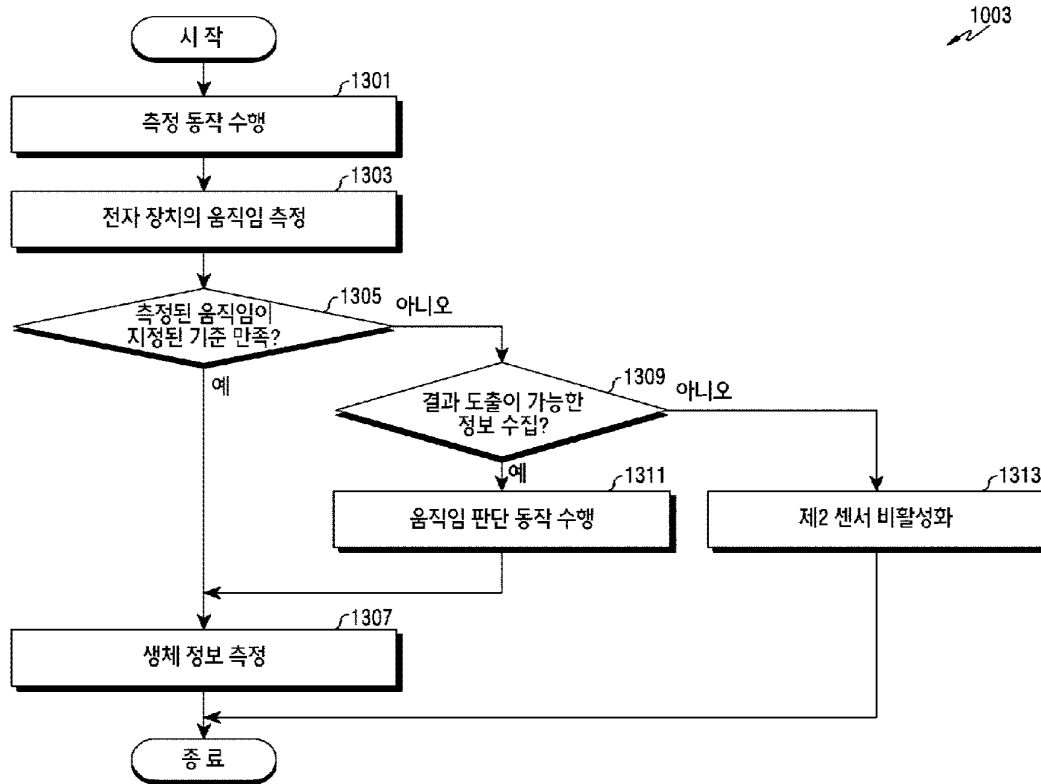
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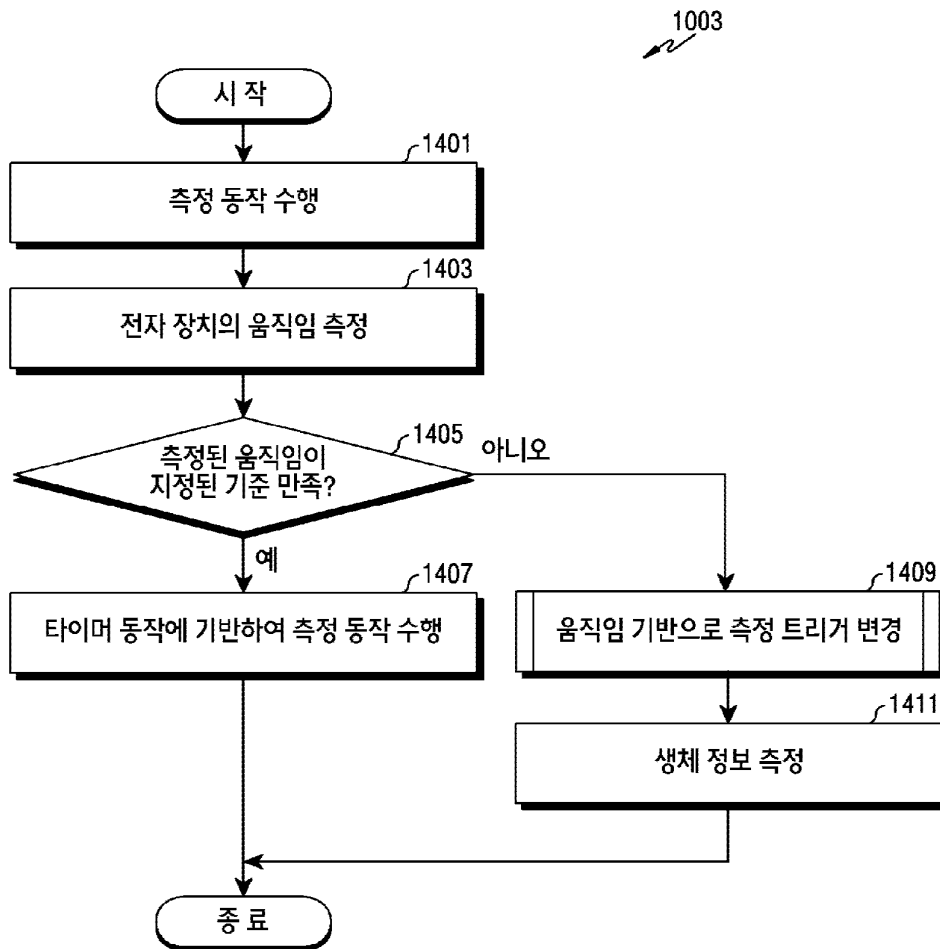
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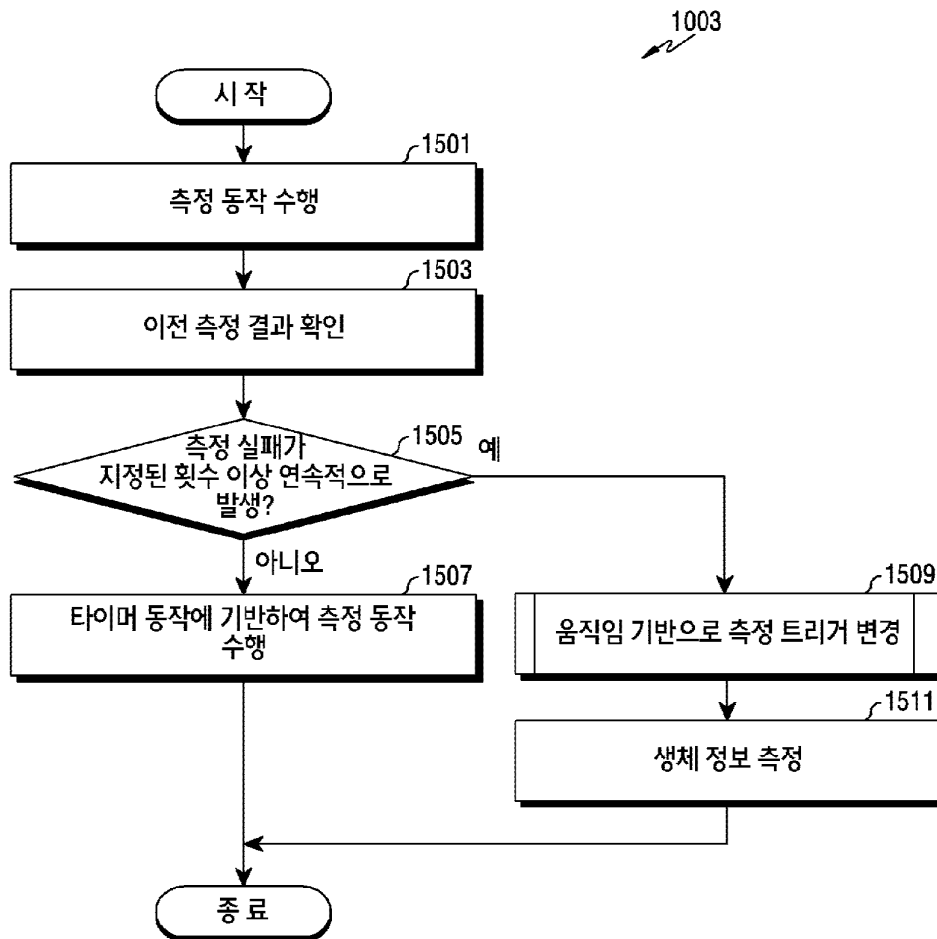
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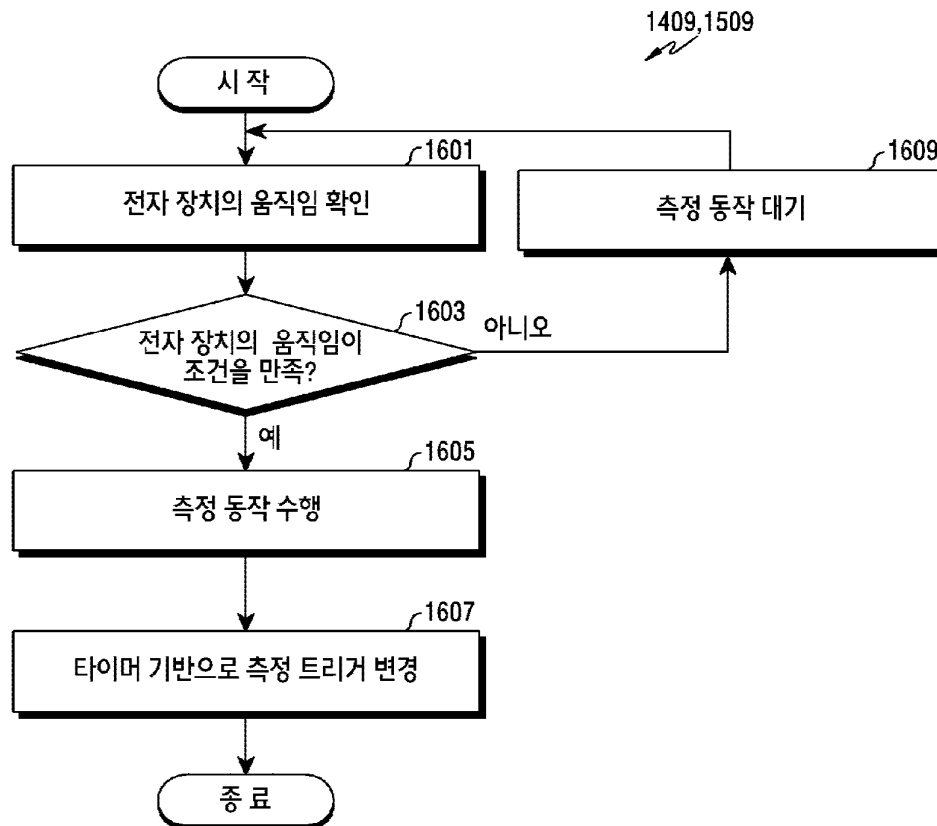
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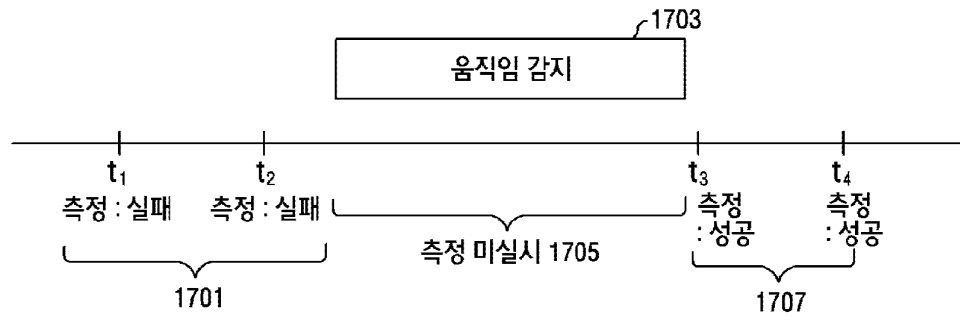
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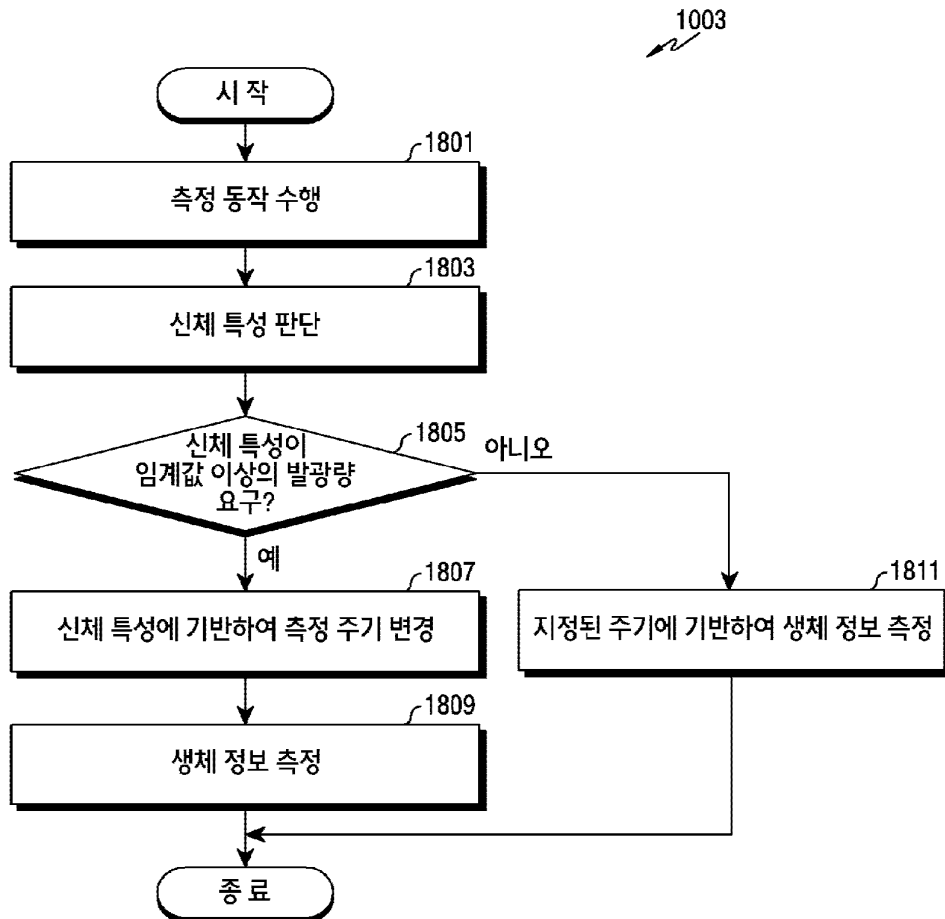
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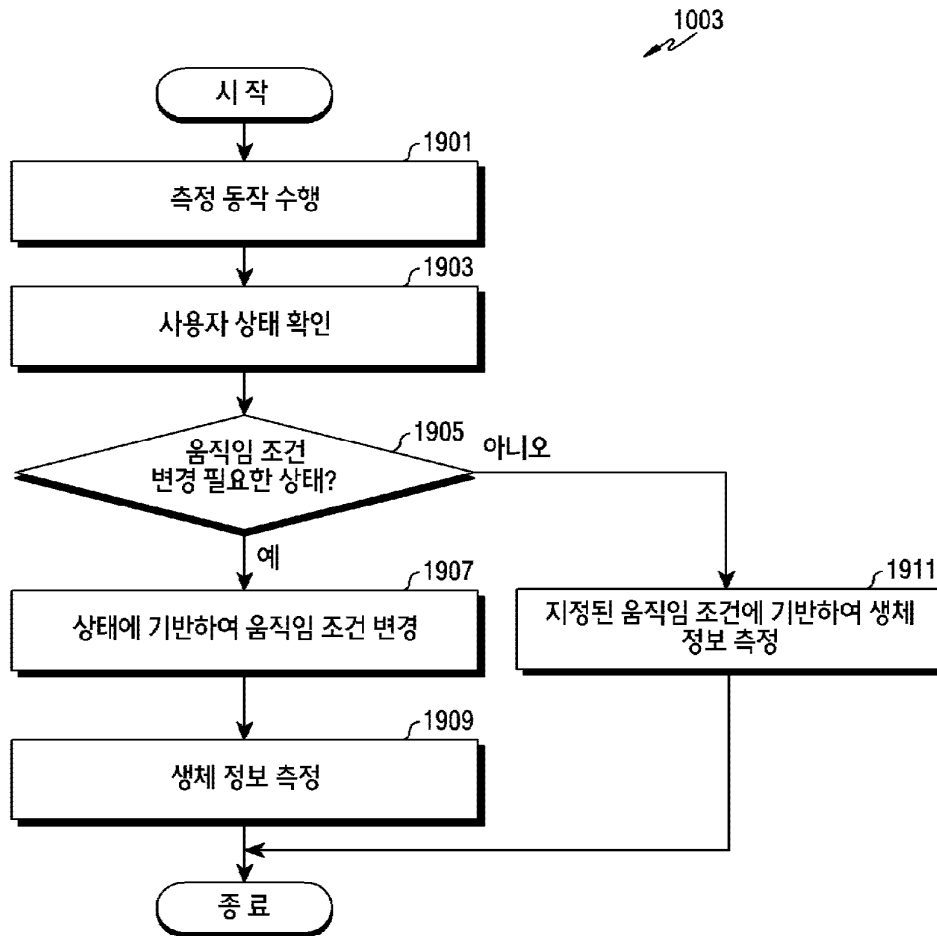
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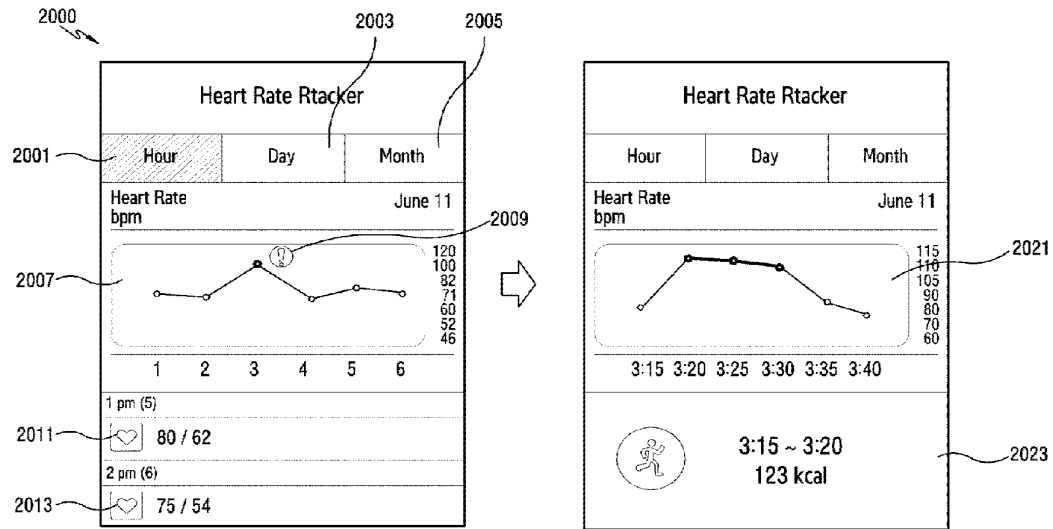
【도 18】



【도 19】



【도 20】



INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	16256497
	Filing Date	2019-01-24
	First Named Inventor	Gihung CHUNG
	Art Unit	3792
	Examiner Name	Niketa I PATEL
	Attorney Docket Number	0202-1231-1

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	1	2009140360	WO	A1	2009-11-19	KATERAAS et al.		

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		16256497
Filing Date		2019-01-24
First Named Inventor	Gihung CHUNG	
Art Unit	3792	
Examiner Name	Niketa I PATEL	
Attorney Docket Number	0202-1231-1	

1	Indian Office Action dated November 15, 2019; issued in a counterpart Indian Application No. 201747005402
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	3792
Examiner Name	Niketa I PATEL
Attorney Docket Number	0202-1231-1

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2020-02-04
Name/Print	Wonki Park	Registration Number	38991

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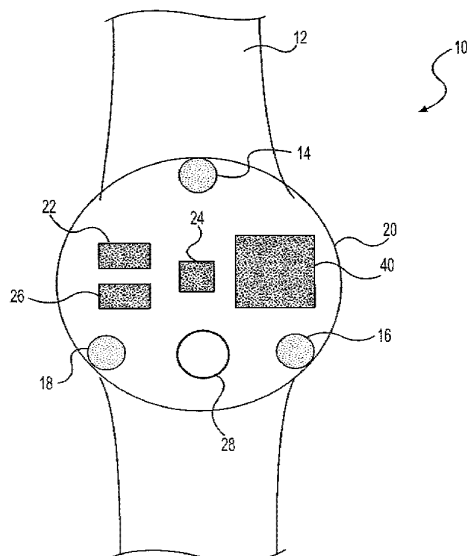


FIG. 1

(57) Abstract: A physical activity data collection unit includes one or more infrared sensors configured to provide an output indicative of a pulse rate of a user of the physical activity data collection unit, at least one temperature sensor configured to provide an output indicative of at least a body temperature of the user, and at least one accelerometer configured to provide an output indicative of movements of the user. The physical activity data collection unit can also include a microcontroller configured to determine a pulse rate, a body temperature, and movement characteristics of the user of the data collection unit based on outputs from the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer; determine a physical exertion level of the user based on one or more of the pulse rate, the body temperature, or the movement characteristics of the user; and store, in a memory, data indicative of the physical exertion level during a time period during which the physical exertion level exceeds a predetermined threshold.

DescriptionPHYSICAL ACTIVITY MONITOR AND DATA COLLECTION UNITTechnical Field

- [01] This application claims priority to U.S. Provisional Patent Application No. 61/071,701, filed on May 14, 2008, the contents of which are fully incorporated herein by reference.
- [02] The present disclosure relates to a sensor-based device configured to monitor the physical activity level of an individual, collect data during periods of physical exertion, and transmit the collected data to a data collection portal associated with a physical activity rewards allocation system and/or a physical activity tracking system.

Summary of the Invention

- [03] One aspect of the disclosure includes a physical activity data collection unit that includes one or more infrared sensors configured to provide an output indicative of a pulse rate of a user of the physical activity data collection unit, at least one temperature sensor configured to provide an output indicative of at least a body temperature of the user, and at least one accelerometer configured to provide an output indicative of movements of the user. The physical activity data collection unit can also include a microcontroller configured to determine a pulse rate, a body temperature, and movement characteristics of the user of the data collection unit based on outputs from the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer; determine a physical exertion level of the user based on one or more of the pulse rate, the body temperature, or the movement characteristics of the user; and store, in a memory, data indicative of the physical exertion level during a time period during which the physical exertion level exceeds a predetermined threshold.

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Another aspect of the disclosure includes a physical activity data collection unit including at least one physiological sensor configured to generate an output related to the physical activity level of a user and a microcontroller configured to: monitor the output of the at least one physiological sensor; determine a physical exertion level of the user based on the output of the at least one physiological sensor; and store, in a memory, data indicative of the physical exertion level during a time period during which the physical exertion level exceeds a predetermined threshold.

- [04] Yet another aspect of the disclosure includes a physical activity data collection unit that includes one or more infrared sensors configured to provide an output indicative of a pulse rate of a user of the physical activity data collection unit; at least one temperature sensor configured to provide an output indicative of at least a body temperature of the user; at least one accelerometer configured to provide an output indicative of movements of the user; and a microcontroller. The microcontroller may be configured to: sample the outputs of the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer; and store, in a memory, data derived from the sampled outputs of the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer.

Brief Description of the Drawings

- [05] Fig. 1 is a diagrammatic representation of a data collection unit according to an exemplary disclosed embodiment.
- [06] Fig. 2 is a functional block level diagram of a data collection unit according to an exemplary disclosed embodiment.

Detailed Description

- [07] Fig. 1 provides diagrammatic representation of a data collection unit according to an exemplary disclosed embodiment. As illustrated in Fig. 1, the disclosed data collection unit 10 may be configured as a wearable article. In

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certain embodiments, for example, the data collection unit may be incorporated into an article wearable on an individual's wrist. Such an article would offer the advantage of being minimally intrusive, as most people are accustomed to wearing articles fastened to the wrist. The wrist unit could be fashioned as a simple wrist band stylized in various colors and patterns. The band may be adjustable, shockproof, and secured to the wrist using a hook and loop closure, a buckle closure, an elastic material requiring no separate closure device, or with any other suitable fastening configuration. The band can be made from various materials including, for example, a waterproof material, neoprene, polymer, nylon, leather, metal, or any other wearable material.

[08] In one embodiment, data collection unit 10 may be embedded into a small, self-contained wrist band 12. In such a configuration, there may be little or no external indication of the presence of the hardware components of the data collection unit. In other embodiments, the data collection unit may be incorporated into a watch, bracelet, heart rate monitor or other wearable article to provide added functionality to those devices. In addition to the wrist, the disclosed data collection unit may be positioned over any portion of a user's body (e.g., the neck, chest, ankle, head, or thigh) that can provide suitable access to the biological markers needed for monitoring the user's level of physical exertion. For example, the data collection unit may be configured as or incorporated into shoe soles, ear clips, a necklace, ankle band, sock, belt, glove, ring, sunglasses, hat, and/or a headband.

[09] Data collection unit 10 includes a sensor array (including one or more sensors) configured to monitor biological markers that vary with the level of exertion of an individual. The monitored biological markers may include, for example, pulse rate, body temperature, blood oxygen content, or any other suitable marker. Within the sensor array, each sensor may be configured to monitor only a single biological marker. Alternatively, an individual sensor in the array may be configured to monitor multiple biological markers.

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- [10] In one embodiment, data collection unit 10 may include several sensors. These sensors may include any arrangement of one or more sensors capable of monitoring biological characteristics and/or movement associated with a user of data collection unit 10. In one exemplary embodiment, as shown in Fig. 1, data collection unit 10 may include at least one infrared sensor 14, a temperature sensor 22, and/or an accelerometer 24.
- [11] In the exemplary embodiment shown in Fig. 1, data collection unit 10 includes three infrared sensors 14, 16, 18. Suppliers of appropriate infrared transmitter/receivers include Vishay Semiconductors, among others.
- [12] Each infrared sensor may be configured as a transmitter/receiver capable of monitoring the oxygen content of blood passing through nearby blood vessels. Specifically, each infrared sensor can be configured to both emit infrared radiation into the body of the wearer of data collection unit 10 and detect the level of infrared radiation received at the sensor. The wavelength of the emitted radiation can be selected according to the requirements of a particular application. In one embodiment, infrared sensors 14, 16, and 18 can be configured to emit infrared radiation in a wavelength range of about 650 nm to about 950 nm.
- [13] The difference between the emitted radiation level and the detected radiation level is characteristic of the amount of infrared radiation absorbed by the body and, especially, by oxygen-carrying blood. This sensed absorption level can be used to determine the pulse rate of the wearer of data collection unit 10. Particularly, the infrared absorption level may be affected by the expansion and contraction of nearby blood vessels and the oxygen content of blood passing through nearby vessels, which are both physical characteristics that vary together with heart rate. Thus, the rate of observed changes in infrared absorption characteristics of the body can enable a calculation of the wearer's heart rate.
- [14] While only one infrared sensor may be needed depending on the functional requirements of a particular embodiment, including two or more

infrared sensors, or even three or more infrared sensors, can serve to increase the reliability of the data collected from these sensors. As illustrated in Fig. 1, infrared sensors 14, 16, and 18 may be spaced apart from one another. In certain embodiments, these sensors may be located along a perimeter of a central housing 20 of data collection unit 10. Spacing infrared sensors 14, 16, and 18 apart from one another can maximize the possibility that at least one sensor contacts the wearer's skin at all times, even during the movements associated with physical activities.

[15] A power management scheme may be employed to lower the power requirements of infrared sensors 14, 16, and 18. For example, the transmitter portion of each sensor may be pulsed at a predetermined duty cycle to conform to the power specifications of a particular configuration. In one exemplary embodiment, the infrared transmitters of sensors 14, 16, and 18 can be pulsed using a 1% duty cycle at a rate of about 8 pulses per second.

[16] Data collection unit 10 may also include a temperature sensor 22. Temperature sensor 22 may be configured to monitor the body temperature of the wearer of data collection unit 10 by measuring the temperature outside of housing 20 and, for example, against the skin of the wearer. Additionally, temperature sensor 22 may be configured to measure the temperature inside housing 20. Using the difference between the temperature measurements from inside and outside of housing 20, it can be determined whether an observed temperature change outside of the housing is likely attributable to atmospheric conditions or an actual change in body temperature of the wearer of data collection unit 10. While certain embodiments may include only one temperature sensor, other embodiments may include multiple temperature sensors in order to meet a desired set of operational characteristics (e.g., monitoring body temperature from multiple locations on data collection unit 10; separate temperature sensors to monitor the temperature inside and outside of housing 20; etc.).

[17] Temperature sensor 22 may include any suitable device for ascertaining the body temperature of an individual. For example, temperature sensor 22 may include a digital or analog device and may include thermocouples, diodes, resistance temperature detectors (RTDs), or infrared detectors. Suitable temperature sensors may be obtained from various suppliers, including Analog Devices Inc., Omega, or Texas Instruments. For certain types of temperature sensors, contact with the individual's skin may aid in obtaining accurate body temperature measurements. On the other hand, in certain instances where, for example, infrared sensors provide the primary mode of measuring body temperature, mere proximity to the individual's skin may be sufficient to accurately determine body temperature of the user.

[18] Additionally, data collection unit 10 may include an accelerometer 24 to monitor motion of data collection unit 10. In certain embodiments, accelerometer 24 includes only a single axis accelerometer configured to detect motion along one axis. Other embodiments, however, may include multiple accelerometers. In one exemplary embodiment, accelerometer 24 may include a three-axis accelerometer, which includes three accelerometers arranged orthogonally with respect to one another. With such an arrangement, accelerometer 24 may be able to detect or monitor movements along three separate axes.

[19] A three-axis accelerometer may be especially useful for the detection of movements associated with exercise and certain types of physical activity. Generally, most sports or types of physical activity produce a signature pattern of movements that can be detected using an accelerometer. In this way, accelerometer 24 can help confirm whether the wearer of data collection unit 10 is engaged in physical activity and, in certain cases, can help determine the type of sport or activity in which the wearer is engaged.

[20] Other embodiments of data collection unit 10 may include additional or different sensors. For example, data collection unit 10 may include

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a carbon dioxide detector, additional accelerometers, a breathing rate sensor, or any other type of sensor suitable for monitoring physical activity levels.

[21] In addition to the infrared sensors described above, the pulse of the wearer of data collection unit 10 may be ascertained using any other type of sensor suitable for monitoring the wearer's heart rate. In one embodiment, for example, electro-cardiogram based technology may be incorporated into data collection unit 10.

[22] Data collection unit 10 may also include a transceiver 26 for establishing communication with devices external to data collection unit 10. To address power requirements, data collection unit 10 may also include a battery 28.

[23] Fig. 2 provides a schematic, functional block level diagram of data collection unit 10, according to an exemplary disclosed embodiment. Within data collection unit 10, several sensed quantities can be provided to a microcontroller 40 for processing. For example, these sensed quantities may include outputs 30, 31, and 32 from infrared sensors 14, 16, and 18, respectively. Additionally, these sensed quantities may include temperature sensor outputs 33 and 34. Temperature output 33 may correspond to the temperature inside housing 20, for example, and temperature output 34 may correspond to the observed temperature outside of housing 20. The sensed quantities may also include accelerometer outputs 35, 36, and 37, each corresponding to a unique axis of movement.

[24] Microcontroller 40 can store the data associated with the sensed quantities in a memory 50 in raw form or, alternatively, after processing. Further, the data relating to the sensed quantities can be transmitted to a remote location by transceiver unit 26.

[25] Any suitable microcontroller 40 may be included in data collection unit 40. In one embodiment, microcontroller 40 includes a small microcontroller having dimensions of about 0.4 inches by 0.4 inches, or smaller. One suitable microcontroller includes the PIC18F series of microcontroller manufactured by

Microchip Inc. Preferably, microcontroller 40 would exhibit low power characteristics and would require from about 10 microamps to about 50 microamps during normal operation and between 5 milliamps to about 20 milliamps while transmitting data.

[26] Microcontroller 40 of data collection unit 10 has several responsibilities. Among these responsibilities, microcontroller 40 periodically collects data from the available sensors via an analog-to-digital converter 42. The frequency of data collection can be selected to meet the requirements of a particular application. In one embodiment, microcontroller 40 may sample the data from the sensors at least once per second. Higher or lower sampling frequencies, however, may also be possible.

[27] Microcontroller 40 may be configured with the ability for selecting from among multiple data sampling frequencies depending on sensed conditions. For example, microcontroller 40 may be programmed to sample the sensor outputs slower than once per second (e.g., once per every 10 seconds) when microcontroller 40 determines that the user of the device is at rest or at a normal level of physical exertion. Similarly, microcontroller 40 may be configured to sample the sensor outputs more frequently (e.g., at least once per second) when the user's physical exertion level exceeds a predetermined threshold. In certain embodiments, and during periods of physical exertion, microcontroller 40 may collect sensor data up to five times per second, ten times per second, or even more, to ensure that rapidly changing quantities such as pulse rate and blood oxygen, which may cycle on the order of 200 times per minute during periods of extreme physical exertion, can be accurately evaluated.

[28] When appropriate, microcontroller 40 may also enter a rest state to conserve power. For example, when infrared sensors 14, 16, or 18 provide no pulse readings or accelerometer 24 registers no movements over a certain period of time, microcontroller 40 may determine that data collection unit 10 is not being worn. Under such conditions, microcontroller 40 may slow the sensor

sampling period to once every thirty seconds, once every minute, or to another suitable sampling frequency. Additionally, microcontroller 40 may be configured to sample only a portion of the available sensors during times of physical inactivity or when data collection unit 10 is not being worn. In one embodiment, for example, once microcontroller 40 determines that the user is not wearing data collection unit 10, microcontroller 40 may begin sampling the output of temperature sensor 22 alone. In such a configuration, a perceived rapid change in temperature may indicate that data collection unit 10 is in use and may prompt the controller to “wake up” and restore full functioning data collection.

[29] Microcontroller 40 can be configured to analyze the data collected from the sensors onboard data collection unit 10. For example, data from infrared sensors 14, 16, 18 can be used to compare the transmitted infrared signal to the received infrared signal and calculate the blood oxygen saturation level via known algorithms. Microcontroller 40 may also be configured to calculate the pulse rate by monitoring the frequency of changes in the blood oxygen saturation level.

[30] As noted above, microcontroller 40 can be configured to store raw or processed data in memory 50 included in data collection unit 10. Memory 50 may include any suitable storage unit including, for example, a solid state non-volatile serial or parallel access memory. In certain embodiments, the memory may include a storage capacity of at least 32 MB. Suitable memory units include RAM, NVRAM, and Flash memory. It is also possible to use an internal microcontroller memory to store data, especially if microcontrollers are developed that include internal memory sizes greater than the currently available 64 kB sizes.

[31] In the case that microcontroller 40 is configured to store raw data, microcontroller 40 may sample the outputs of the sensors onboard data collection unit 10 and simply store those values in memory 50. Those stored values can

then later be downloaded from data collection unit 10 and processed using devices and/or systems external to data collection unit 10.

[32] While it is possible to store raw data collected from the sensor devices, microcontroller 40 may also be configured to process the data sampled from the sensors of data collection unit 10 prior to storage in memory 50. For example, microcontroller 40 may be configured to calculate pulse rate, temperature, acceleration and average each calculated value over periods of up to thirty seconds, sixty seconds, or more to remove noise and enhance accuracy of the readings. Microcontroller 40 can be further configured to store these time averaged, filtered pulse rate/temperature/acceleration readings at preselected intervals (e.g., once or twice per minute). Such a scheme may conserve memory and/or power resources yet still provide useful information. These processed or conditioned data signals stored in memory, in certain cases, can even be more useful, as they may exhibit less noise and rapidly fluctuating values, which can detract from the reliability of the data.

[33] Microcontroller 40 may be configured to condition the signals received from one or more of the sensors onboard data collection unit 10. During movement associated with physical activity, a significant amount of noise may be imparted to the signals generated by the onboard sensors. Such noise is especially prevalent in the data provided by the infrared sensors, which can be used to determine heart rate. Digital signal processing techniques may be employed to eliminate at least some of the noise from these signals and increase the accuracy of the heart rate calculation.

[34] Microcontroller 40 may also be configured to determine when the user is at rest and when the user is exercising. In addition to using this information to control the data collection and storage rates, this information can be used, for example, in conjunction with a physical activity rewards allocation system to provide rewards-based incentives to the user of data collection unit 10. That is, the user of data collection unit 10 may receive rewards in the form of

merchandise, merchandise discounts, currency, and/or free or discounted services based on the amount of time the user spends exercising and/or upon the level of physical exertion during exercise. The information may also be used to track physical activity levels for purposes of assessing the physical health of individuals. For example, the information may be tracked and used to determine the physical fitness, health, or well-being of private or public employees in order to provide worker incentives. Alternatively or additionally, this information could be used by the insurance industry to set rates/premiums tailored to an individual or discounted for a group of individuals participating in a physical activity tracking program.

[35] Microcontroller 40 can be configured to determine when the user's level of activity qualifies as exercise. For example, microcontroller 40 can assimilate one or more of the user's pulse rate, temperature, and acceleration levels into a exercise evaluation score. Comparing the exercise evaluation score with a predetermined threshold level, microcontroller 40 can determine that the user is exercising when the exercise evaluation score exceeds the threshold.

[36] The microcontroller's accuracy in determining the physical activity level or exertion level of a user can be refined according to any desired algorithm. In one embodiment, for example, microcontroller 40 may be configured to determine the relative reliability of the data provided by the sensors onboard data collection unit 10 and assign weighting factors (e.g., values between 0 and 1) to those outputs based on the perceived reliability of the data from each output. For example, if one of the infrared sensors is emitting a stable, oscillating output signal with a low noise level and another is emitting a noisy signal, then microcontroller 40 can assign a higher weighting factor to the higher quality signal and a lower weight to the noisy signal. In this way, microcontroller 40 can minimize the effects of extraneous noise and low quality data and maximize the measurement reliability when high quality data output signals are available.

[37] Microcontroller 40 can be programmed with a common baseline threshold for use with all users of the disclosed data collection unit 10. Alternatively, microcontroller 40 may be used to calculate and periodically update a unique threshold determined for a specific user of a particular data collection unit. For example, as the user wears and uses data collection unit 10 over a period of time, microcontroller 40 may “learn” about the user by monitoring and storing quantities (e.g., heart rate, acceleration levels, and temperature) associated with periods during which the user is at rest and exercising. Using a predefined exercise threshold algorithm, the microcontroller can use this information to tailor the exercise threshold and store a new, updated exercise threshold based on the current fitness level of the user. The predefined algorithm may be loaded into the microcontroller’s operating instruction set upon manufacture and may be updated via download from a central server system.

[38] Ultimately, microcontroller 40 can be configured to determine when the user’s level of physical activity surpasses the exercise threshold. Once the user exceeds the exercise threshold, the microcontroller may start a timer that monitors the amount of time the user spends above the exercise threshold. Further, via the sensed pulse rate, temperature, and acceleration levels measured, microcontroller 40 can determine and store a quantity that tracks the amount by which the user’s physical activity exceeds the exercise threshold. This information, together or separate from exercise time, may be used by microcontroller 40 or, more preferably, a remote rewards allocation system to determine a rewards quantity accrued by the user during each period of exercise. Alternatively or additionally, this information can be used by a physical activity tracking system to determine worker incentives or to set/adjust insurance rates/premiums.

[39] Data collection unit 10 may also include a feedback element, including, for example, a display, light, audible speaker, or other suitable sensory interface device. During periods when the user’s physical activity exceeds the

exercise threshold and qualifies for rewards accrual, microcontroller 40 may activate the feedback element to indicate to the user that the exercise threshold has been exceeded and rewards are being accrued. For example, an LED may be included that blinks during periods of qualifying exercise. In other embodiments, a speaker may emit an audible beep every few seconds during periods of qualifying exercise. In still other embodiments, a rewards indicator may be projected on a display during qualifying exercise sessions. Such an embodiment would be especially useful where data collection unit 10 was incorporated into a watch or other type of device including a display.

[40] Microcontroller 40 of data collection unit 10 may be configured to control transmission of data to one or more remote locations. In one embodiment, microcontroller 40 can activate transceiver 26, as illustrated in Fig. 2, with a low duty cycle of less than about 1% to detect the presence of suitable data collection portals. A data collection portal can include any intended recipient of the data acquired by data collection unit 10. In one embodiment, a data collection portal may be associated with a physical activity rewards allocation system and may forward the data received from data collection unit 10 to a central management facility that handles the operation of the rewards system. In another embodiment, the data collection portal may be associated with a threshold exercise tracking system for purposes of determining the physical fitness, health, or well-being of private and public employees for worker incentives. The data collection portal may also be associated with an insurance rate/premium setting system that tailors rates or adjusts premiums based on the physical activity level of individuals and/or groups.

[41] When data collection unit 10 detects a data collection portal (e.g., either through a wired or wireless data connection) and communication is established, download of the data will commence, for example, after proper identification of the user and of the portal has been achieved. This may prevent eavesdropping by unauthorized parties. Identification of the user may include

transmission of a unique code assigned to each data collection unit and/or user of the data collection unit. A user-selectable password can be used to allow data to be downloaded by the data collection portal. In other embodiments, passive identification of a user may displace the need for password protected downloads. For example, the microcontroller may be configured to determine and store a biological signature of an authorized user of the data collection unit. Such a signature may be determined using the same array of sensors used monitor temperature, pulse rate, and acceleration levels. Alternatively, one or more additional sensors (e.g., a skin pigment sensor, pH sensor, etc.) may be included to aid in user recognition.

[42] One or more other devices, including, e.g., an RFID tag may be employed to facilitate the transmission of data to a data collection portal. For example, in response to a radio frequency interrogation signal, an RFID tag located on data collection unit 10 may power on using an onboard power source, such as battery 28, or using energy provided by the interrogation signal. The RFID tag can respond to the interrogation signal by transmitting data to a location/receiver remotely located with respect to data collection unit 10. The information transmitted may include information about data collection unit 10. For example, the transmitted information may include a signature code associated with a particular data collection unit 10. Additionally, the transmitted information may include any other data that may aid in recognition of the particular data collection unit 10.

[43] Alternatively or additionally, an RFID tag or other similar device for transmitting data from data collection unit 10 (e.g., microcontroller 40 coupled with transceiver 26) may be used to transmit information about the user of data collection unit 10. This information can include, for example, medical emergency data, insurance information, name, home address, phone numbers, vital statistics, allergies, blood type, etc.

- [44] The transmitted information may also be used to recognize an individual wearer of data collection unit 10. For example, based on a particular piece of information (e.g., a signature code, name, address, etc.) an interrogating device or data portal may “recognize” the wearer of data collection unit 10. In response, the receiver of this information may take some action based on the recognition of the user of data collection unit 10. In certain embodiments, such information may be used to determine the location of a user of data collection unit 10; determine the frequency that the user visits a particular establishment, such as a health club, spa, pools; etc.
- [45] Data collection unit 10 may also be configured to detect potentially fraudulent use by a user. For example, because the user may receive rewards based on an indication by data collection unit 10 that the user had engaged in qualifying physical activity for a certain period of time, certain individuals may be motivated to simulate a state of physical activity, wear multiple data collection units, or engage in other types of fraudulent activity. With the robust sensor array included in data collection unit 10, the likelihood of data collection unit 10 being “fooled” by simulated physical activity is minimized.
- [46] Additionally, microcontroller 40 may be configured to generate and deliver a low power, low duty cycle pulse to metal contacts located, e.g., on the base of housing 20. These pulses may have a duration of less than about 100th of a millisecond per pulse and will be transmitted over short distances around data collection unit 10. The same metal contacts on the base of housing 20 can also serve as an antenna and can aid in detection of similar signals in close proximity. When such a signal is detected, it may indicate that a user is wearing more than one data collection unit devices. If the detected signal remains constant over a certain period of time, further suggesting that more than one data collection unit 10 is in use by a single user, then either the emitting or detecting data collection unit, or both, may be configured to shut down.

[47] Suitable data collection portals may include those located within a predetermined distance from data collection unit 10. In certain embodiments, data collection unit 10 may be configured to transmit data to portals located within about ten feet. In other embodiments, this transmission distance may be extended up to about 50 feet.

[48] Once transmission of data stored in data collection unit 10 commences, a handshaking process may be employed to validate the integrity of the data transmitted and to request retransmission of the data, if necessary. After data collection unit 10 establishes that the data has been successfully transmitted to the data collection portal, microcontroller 40 can delete the previously stored data.

[49] Transmission of data to a data collection portal may also be controlled based on the availability of stored data. For example, if no new data has been stored in memory 50 since the last successful download, then microcontroller 40 may determine that there is nothing to transmit. Under these conditions, microcontroller 40 may forego searching for a suitable data collection portal within range and will leave the data collection unit transceiver 26 powered down until data is subsequently stored in memory.

[50] Other schemes for data transmission initiation may be employed. For example, rather than the microcontroller periodically searching for a suitable data collection portal within range, microcontroller 40 may be configured to simply respond to an interrogation signal continuously or periodically emitted from a data collection portal. If microcontroller 40 receives such an interrogation and determines that the emitting data collection portal is within transmission range, then microcontroller 40 can activate transceiver 26 and commence data transmission.

[51] Data transmission may be accomplished via any suitable scheme for transmission of data. In one embodiment, the data stored in the data collection unit may be transferred via a wired connection including a cable and

cable interface. In one embodiment, data transmission can be accomplished via a USB data cable that enables charging of data collection unit 10 while data is downloaded. Data transmission may also be accomplished via a wireless connection including a radio frequency or optical transmission link. In certain embodiments, for example, data collection unit 10 can be Bluetooth or Zigbee enabled or may transmit data via an infrared optical link.

[52] In certain embodiments, data transmission can extend beyond the limits of the onboard transceiver. For example, using a Bluetooth enabled data collection unit coupled with an external device, such as a cell phone, PDA, personal computer, etc., data can be relayed from data collection unit 10 through the external device and on to a data collection portal or even directly to the management facility.

[53] Data collection unit 10 may include any suitable power source for meeting the power requirements of the unit. For example, data collection unit 10 may include a replaceable or rechargeable battery 28. In certain embodiments, three-volt lithium batteries contained within a 1.2 cm package may be included in data collection unit 10. Additionally, or alternatively, a solar cell may be included either alone or in combination with one or more batteries. In addition to serving as a stand alone power source, the solar cell may also function to recharge the batteries. In another embodiment, a motion activated regeneration device may be included for purposes of powering the data collection unit and/or recharging batteries.

[54] It will be apparent to those skilled in the art that various modifications and variations can be made in the disclosed sensor unit without departing from the scope of the disclosure. Other embodiments of the disclosed systems and methods will be apparent to those skilled in the art from consideration of the specification and practice of the disclosure disclosed herein.

Claims

What is claimed is:

1. A physical activity data collection unit, including:
 - one or more infrared sensors configured to provide an output indicative of a pulse rate of a user of the physical activity data collection unit;
 - at least one temperature sensor configured to provide an output indicative of at least a body temperature of the user;
 - at least one accelerometer configured to provide an output indicative of movements of the user; and
 - a microcontroller configured to:
 - determine a pulse rate, a body temperature, and movement characteristics of the user of the data collection unit based on outputs from the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer;
 - determine a physical exertion level of the user based on one or more of the pulse rate, the body temperature, or the movement characteristics of the user; and
 - store, in a memory, data indicative of the physical exertion level during a time period during which the physical exertion level exceeds a predetermined threshold.
2. The physical activity data collection unit of claim 1, further including a transceiver, and wherein the microcontroller is further configured to transmit data indicative of the exertion level of the user to an external data collection portal.

3. The physical activity data collection unit of claim 2, wherein the microcontroller determines the presence of stored physical exertion data and automatically initiates transmission of the stored data upon determining that the data collection has come within a predetermined range of a data collection portal.

4. The physical activity data collection unit of claim 2, wherein the microcontroller determines the presence of stored physical exertion data and initiates transmission of the stored data upon receiving an interrogation signal from a data collection portal.

5. The physical activity data collection unit of claim 2, wherein the transceiver is configured to transmit data via an optical or radio frequency link.

6. The physical activity data collection unit of claim 1, wherein the microcontroller is further configured to update the predetermined threshold based on the fitness of the user.

7. The physical activity data collection unit of claim 1, wherein the at least one temperature sensor includes at least one of a thermocouple, diode, resistive temperature device, or infrared device.

8. The physical activity data collection unit of claim 1, wherein the one or more infrared sensors each include an infrared transceiver unit.

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9. The physical activity data collection unit of claim 8, wherein the pulse rate is determined by monitoring changes in the blood oxygen saturation level.

10. The physical activity data collection unit of claim 1, wherein the microcontroller is configured to condition the output signals provided by the one or more infrared sensors, the at least one temperature sensor, or the at least one accelerometer and store conditioned signals in the memory.

11. The physical activity data collection unit of claim 1, wherein the microcontroller is configured to recognize potentially fraudulent use of the data collection unit and take at least one action in response to the recognized potentially fraudulent use.

12. The physical activity data collection unit of claim 11, wherein the microcontroller is configured to:

- generate a signal to be emitted by the data collection unit;
- detect the presence of similar signals generated in the vicinity of the data collection unit; and
- shut down the data collection unit if potentially fraudulent use is recognized based on the detection of the similar signals.

13. The physical activity data collection unit of claim 1, further including an RFID tag to aid in recognition of the data collection unit or the user of the data collection unit.

14. The physical activity data collection unit of claim 1, wherein the microcontroller is configured to communicate a unique code

associated with the data collection unit to aid in recognition of the data collection unit.

15. The physical activity data collection unit of claim 1, wherein the microcontroller is configured to apply variable weights to the outputs of the one or more infrared sensors, the at least one temperature sensor, or the at least one accelerometer when determining the physical exertion level of the user.

16. A physical activity data collection unit, including:
at least one physiological sensor configured to generate an output related to the physical activity level of a user; and
a microcontroller configured to:
monitor the output of the at least one physiological sensor;
determine a physical exertion level of the user based on the output of the at least one physiological sensor; and
store, in a memory, data indicative of the physical exertion level during a time period during which the physical exertion level exceeds a predetermined threshold.

17. A physical activity data collection unit, including:
one or more infrared sensors configured to provide an output indicative of a pulse rate of a user of the physical activity data collection unit;
at least one temperature sensor configured to provide an output indicative of at least a body temperature of the user;
at least one accelerometer configured to provide an output indicative of movements of the user; and
a microcontroller configured to:
sample the outputs of the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer; and

-22-

store, in a memory, data derived from the sampled outputs of the one or more infrared sensors, the at least one temperature sensor, and the at least one accelerometer.

18. The physical activity data collection unit of claim 17, further including a transceiver, and wherein the microcontroller is further configured to transmit data stored in the memory to an external data collection portal.

19. The physical activity data collection unit of claim 17, wherein the microcontroller is configured to condition the data prior to storing the data in the memory.

20. The physical activity data collection unit of claim 17, wherein the microcontroller is configured to recognize potentially fraudulent use of the data collection unit and take at least one action in response to the recognized potentially fraudulent use.

21. The physical activity data collection unit of claim 20, wherein the microcontroller is configured to:

- generate a signal to be emitted by the data collection unit;
- detect the presence of similar signals generated in the vicinity of the data collection unit; and
- shut down the data collection unit if potentially fraudulent use is recognized based on the detection of the similar signals.

22. The physical activity data collection unit of claim 17, further including an RFID tag to aid in recognition of the data collection unit or the user of the data collection unit.

23. The physical activity data collection unit of claim 17, wherein the microcontroller is configured to communicate a unique code associated with the data collection unit to aid in recognition of the data collection unit.

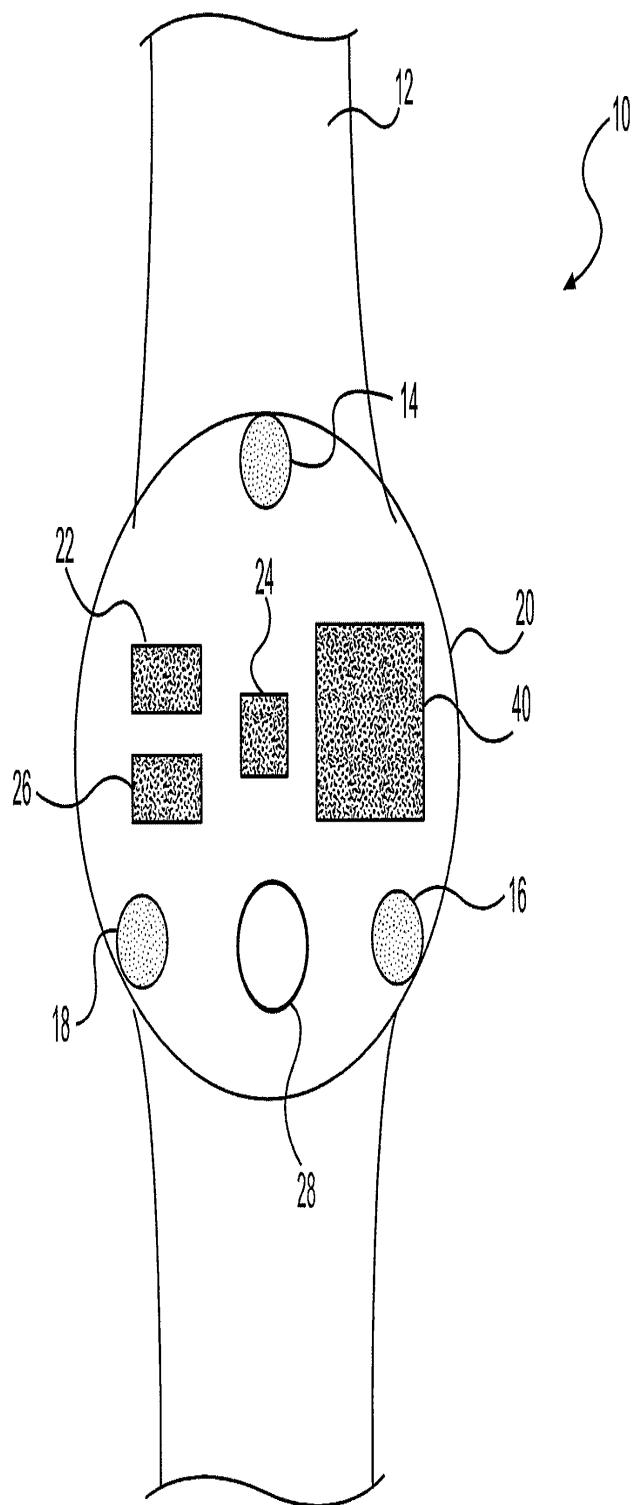


FIG. 1

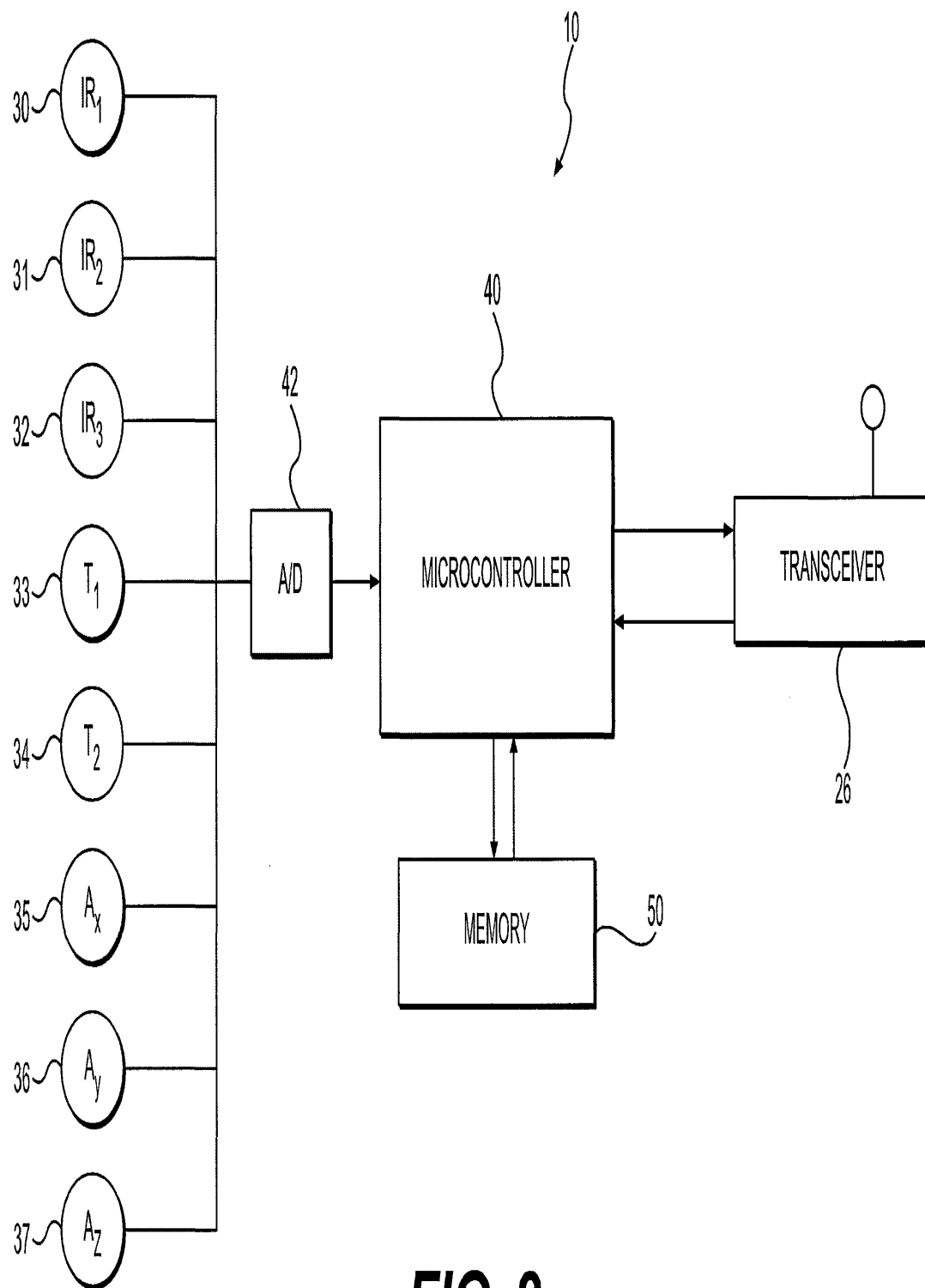


FIG. 2

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2009/043753

A. CLASSIFICATION OF SUBJECT MATTER
INV. A61B5/11

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
A61B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X Y	WO 2006/036911 A (IMPACT SPORTS TECHNOLOGIES INC [US]) 6 April 2006 (2006-04-06) page 6, lines 16-19 page 7, lines 18-30 page 9, lines 10-21 page 11, lines 6-9 page 13, lines 2-30 -----	1, 2, 5, 7-9, 15-19 10, 11, 13, 14, 19, 20, 22, 23
X	EP 1 702 560 A (BODYMEDIA INC [US]) 20 September 2006 (2006-09-20) paragraphs [0004], [0005], [0011] - [0019] ----- -/-	16

☒ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

- *A* document defining the general state of the art which is not considered to be of particular relevance
- *E* earlier document but published on or after the international filing date
- *L* document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
- *O* document referring to an oral disclosure, use, exhibition or other means
- *P* document published prior to the international filing date but later than the priority date claimed

- *T* later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
- *X* document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
- *Y* document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.
- *&* document member of the same patent family

Date of the actual completion of the international search

18 August 2009

Date of mailing of the international search report

27/08/2009

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INTERNATIONAL SEARCH REPORT

International application No

PCT/US2009/043753

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO 2006/044677 A (PULSETRACER TECHNOLOGIES INC [CA]; NAZARIAN RICHARD A [US]; LUCKE LORI) 27 April 2006 (2006-04-27) page 4, lines 14-29 page 5, line 25 - page 7, line 15 -----	10,13, 14,19, 22,23
Y	JP 2007 213196 A (OLYMPUS CORP) 23 August 2007 (2007-08-23) paragraphs [0001] - [0003], [0008] - [0013] -----	11,20
A	US 2006/025282 A1 (REDMANN WILLIAM G [US]) 2 February 2006 (2006-02-02) paragraphs [0001], [0058], [0080] - [0084], [0264] - [0266], [0291] -----	11,20

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/US2009/043753

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
WO 2006036911	A	06-04-2006	US 2006079794 A1	13-04-2006
			US 2006069319 A1	30-03-2006
EP 1702560	A	20-09-2006	AT 324066 T	15-05-2006
			AU 7009201 A	08-01-2002
			BR 0111918 A	13-05-2003
			CA 2413148 A1	03-01-2002
			DE 60119100 T2	31-08-2006
			EP 1292218 A1	19-03-2003
			ES 2260245 T3	01-11-2006
			JP 2004500949 T	15-01-2004
			MX PA02012643 A	10-09-2004
			PT 1292218 E	29-09-2006
			WO 0200111 A1	03-01-2002
			US 2006224051 A1	05-10-2006
			WO 2005016124 A2	24-02-2005
WO 2006044677	A	27-04-2006	CA 2582563 A1	27-04-2006
			EP 1809171 A1	25-07-2007
			JP 2008516691 T	22-05-2008
			US 2006084879 A1	20-04-2006
JP 2007213196	A	23-08-2007	NONE	
US 2006025282	A1	02-02-2006	NONE	



**INTELLECTUAL
PROPERTY INDIA**

एकसूत्र/PATENTS | अंकित/DESIGNS |
व्यापार चिह्न/TRADE MARKS | भौगोलिक
उपदेशिका/GEOGRAPHICAL INDICATIONS



सत्यमेव जयते

भारत सरकार

GOVERNMENT OF INDIA

एकसूत्र कार्यालय / THE PATENT OFFICE
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फैक्स / Fax: 044 22502066

Email/ई मेल : Patent@patent.gov.in
वेबसाइट / Website: <http://ipindia.nic.in>

अं.संख्या/Ref.No /आवेदन संख्या/Application No/ 201747005402

दिनांक/Date of Dispatch/Email: 15/11/2019

सेवा में, /To

SHARAD VADEHRA,

KAN AND KRISHME ADVOCATES, PATENT AND TRADEMARK ATTORNEYS KNK House, A-11, Shubham Enclave, Paschim Vihar, New Delhi-11 0063, India Telephone #: 91-11-43776666 (100 Lines) Facsimile #: 91-11-43776676, 43776677 E-mail: knk@kankrishme.com
Email : knk@kankrishme.com

विषय: एकसूत्र अधिनियम, 1970 की धारा 12 व 13 तथा एकसूत्र नियम, 2003 के अधीन परीक्षण रिपोर्ट

Subject: Examination report under sections 12 & 13 of the Patents Act, 1970 and the Patents Rules, 2003.

1. उपर्युक्त आवेदन के संदर्भ में परीक्षण रिपोर्ट (अर्थात, एकसूत्र नियम, 2003 (यथा संशोधित) के नियम 24-ख(3) में विनिर्दिष्ट आपत्तियों का प्रथम कथन) इसके साथ संलग्न है। यह रिपोर्ट परीक्षण हेतु अनुरोध दिनांक 15/02/2017 के अंतर में जारी की गयी है। परीक्षण रिपोर्ट का अंतर पोषित करने की अंतिम तिथि (अर्थात, इस रिपोर्ट में लगाई गयी सभी आवश्यकताओं के अनुपालन की अवधि) आवेदक को आपत्तियों का प्रथम कथन जारी होने की तिथि से छ माह है।

Please find enclosed herewith an Examination Report (i.e. a first statement of objections as specified in Rule 24-B(3) of The Patents Rules, 2003 (as amended)) in respect of above-mentioned application. This report is issued with reference to a request for examination dated 15/02/2017. The last date for filing a response to the Examination Report (i.e. a period to comply with all the requirements raised in this examination report) is six months from the date on which the first statement of objections is issued to the Applicant.

2. यदि रिपोर्ट के अंतर्गत लगाई गयी आवश्यकताओं का अनुपालन एकसूत्र नियम, 2003 (यथा संशोधित) के नियम 24 ख(5) में विनिर्दिष्ट अवधि के भीतर अंतर अनुपालन नहीं किया गया तो एकसूत्र अधिनियम 1970 की धारा 21(1) के अधीन वर्तमान आवेदन को परित्यक्त माना जाएगा।
The instant application shall be deemed to have been abandoned under Section 21(1) of The Patents Act, 1970, unless all the requirements raised in this report are complied with in the period as specified in Rule 24-B (5) of The Patents Rules, 2003 (as amended).
3. आपका ध्यान एकसूत्र नियम, 2003 के नियम 24 ख(6) के प्रावधानों की ओर भी आकर्षित किया जाता है।
Your attention is also invited to the provisions of Rule 24-B (6) of the Patents Rules 2003.
4. आपको सलाह दी जाती है कि शीघ्र निपटान हेतु अपना उत्तर शीघ्र प्रस्तुत करें।
You are advised to file the reply at the earliest for early disposal.

Pinkesh Jain
नियंत्रक पेटेंट/ Controller of Patents

संलग्न/Enclosed: अपरोक्त अनुसार/As above

टिप्पणी: यह इलेक्ट्रॉनिक रूप से उत्पन्न रिपोर्ट है।

NOTE: This is an electronically generated report.

सभी पत्राचार नियंत्रक एकसूत्र को उपरोक्त पते पर भेजा जाये।

All communications should be sent to the Controller of Patents at the above mentioned address.

THE PATENT OFFICE

परीक्षण रिपोर्ट / Examination Report

आवेदन संख्या / Application Number	201747005402
दाखिल करने की तिथि / Date of Filing	15/02/2017
पूर्वता दिनांक / Date of Priority	12/08/2015
पीसीटी अंतराष्ट्रीय आवेदन की संख्या व दिनांक / PCT International Application No. & Date	KR2016007727 – 15/07/2016
आवेदक / Applicant	SAMSUNG ELECTRONICS CO. LTD.
परीक्षण हेतु अनुरोध की संख्या व दिनांक / Request for Examination No. & Date	R20174005198 15/02/2017
प्रकाशन की तिथि / Date of Publication	28/04/2017

इस परीक्षण रिपोर्ट के चार भाग हैं, अर्थात् रिपोर्ट का सारांश, विस्तृत तकनीकी रिपोर्ट, औपचारिक आवश्यकताएँ तथा रिकॉर्ड में दस्तावेज़ / This examination report consists of four parts, namely summary of the report, detailed technical report, formal requirements and documents on record.

भाग -1: रिपोर्ट का सारांश

PART-I: SUMMARY OF THE REPORT

क्र. सं. /Sl. No.	अधिनियम के तहत आवश्यकताओं पर विस्तृत टिप्पणियाँ /Requirements under the Act	दावों की संख्या / Claim Numbers	टिप्पणी / Remarks
1.	धारा 2(1)(न) के तहत आविष्कार / Invention u/s 2(1)(j) नवीनता / Novelty आविष्कारी कदम / Inventive step औद्योगिक उपयोगिता / Industrial Applicability	दावे / Claims:	हाँ / Yes
		दावे / Claims: 1-15	नहीं / No
		दावे / Claims:	हाँ / Yes
		दावे / Claims: 1-15	नहीं / No
2.	धारा 3 के अधीन पेटेंट-अयोग्यता (यदि हाँ, संड 3(क-न) / Non-patentability u/s 3 (If yes, specify section 3(a-p))	दावे / Claims: 1-15	हाँ / Yes 3(k), 3(m), 3(i)
		दावे / Claims:	नहीं / No
3.	[धारा 10(5) व 10(4) (ग)] के अधीन दावे / Claims [u/s 10(5) & 10(4) (g)] परिभाषित / Definitive	दावे / Claims:	हाँ / Yes
		दावे / Claims: 1-15	नहीं / No
4.	अन्य आवश्यकता (एँ) / Other requirement(s): - Distinguishing features w.r.t prior art cited above should be furnished in the description. - The technical features of the claims should be referenced with numerals in parentheses to enhance intelligibility of claims.		

भाग -II विस्तृत तकनीकी रिपोर्ट

PART-II: DETAILED TECHNICAL REPORT

क. उद्धृत दस्तावेजों की सूची / A. List of documents cited:

(क) पेटेंट साहित्य / (a). Patent Literature :

THE PATENT OFFICE

क्र. सं. / Sl.no	दस्तावेजों का विवरण /Details of documents	प्रकाशन तिथि(दिन/मास/वर्ष) / Publication date	उद्धृत दस्तावेज का प्रासंगिक विवरण (पृष्ठ व अनुच्छेद संख्या) / Relevant description (page and paragraph no.) of cited document	उद्धृत दस्तावेज के प्रासंगिक दावे / Relevant claims of cited document	अधिकृत आविष्कार के दावे /Claims of alleged invention
1	D1: WO2009140360	19/11/2009	Refer para [07] - [54], fig. 1		1-15

(अ) गैर-पेटेंट साहित्य / (b) Non-patent literature

कोई दस्तावेज उद्धृत नहीं है / No Document Cited

ख. अधिनियम के तहत आवश्यकताओं पर विस्तृत टिप्पणियाँ / B. Detailed observations on the requirements under the Act:

(1).नवीनता / NOVELTY:

(i) ऊपर उद्धृत दस्तावेज के संदर्भ (1-15) में दिये गए फूटन के प्रकाशन को ध्यान में रखते हुए, निम्नलिखित कारणों से दावा(यों) (1-15) में नवीनता की कमी है /

Claim(s) (1-15) lack(s) novelty, being anticipated in view of disclosure in the document cited above under reference D1 for the following reasons:

Subject matter of claims 1-15 does not constitute an invention under section 2(1)(j) of the Patents Act, 1970 (as amended) because it lacks novelty in view of following documents:

D1: WO2009140360

D1 discloses an electronic device comprising: a housing (20); a fastener (12) coupled to a part of the housing and detachably attached to a part of a user's body; at least one first sensor (24) configured to detect a motion of the electronic device; at least one second sensor configured to detect a change in the user's body (14,22); a memory (50); and a processor (40) electrically connected to the at least one first sensor, the at least one second sensor, and the memory, wherein the memory stores instructions that configure the processor, when being executed, to: determine whether the electronic device is worn on a part of the user's body using the at least one first sensor (para. [27], [28]), acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/ or immediately after a first period of time, which is selected, passes from a time when the electronic device is worn on a part of the user's body, and activate the at least one second sensor based on at least a part of the information on the motion (para. [27], [28]). Therefore the subject-matter of independent claim 1 is not new.

Same reasoning as for the main claim above applies, mutatis mutandis, to the subject matter of the corresponding independent claim 10.

Dependent claims 2- 9 and 11- 15 is also disclosed in Document D1. (Refer para [07] - [54], fig. 1).

(2).आविष्कारी कदम / INVENTIVE STEP:

(i) ऊपर उद्धृत दस्तावेज(जों) के संदर्भ D1 में स्पष्ट अध्यापन(कों) को ध्यान में रखते हुए, निम्नलिखित कारणों से दावा(यों) (1-15) में आविष्कारी कदम की कमी है

Claim(s) (1-15) lack(s) inventive step, being obvious in view of teaching (s) of cited document(s) above under reference D1 for the following reasons:

Regarding claims 1-15, if the Applicant intended to argue novelty of said claims based on a slightly different interpretation of the features of said claims and those disclosed in document D1 (cited paragraphs as cited above in novelty objection), the subject-matter of said claims would not involve an inventive step, having regard to the disclosure of document D1.

(3).पेटेंट अयोग्यता /NON PATENTABILITY:

(I) निम्नलिखित कारणों से धारा 3 के खंड (3(k),3(m),3(i)) के प्रावधान के तहत दावा(दे) (1-15) आविष्कार रूप से पेटेंट योग्य नहीं हैं /
Claim(s) (1-15) are statutorily non-patentable under the provision of clause (3(k),3(m),3(i)) of Section 3 for the following reasons:

- Claims 1-15 do not define any structural features of the memory, processor rather they define computer instructions stored in memory and executable by processor. These instructions are nothing but computer program per se. Hence subject matter of said claims falls within scope of clause (k) of section (3) of the Patents Act, 1970 (as amended). Therefore invention claimed in said claims is not patentable.
- Claims 10-15 recite various method steps without disclosing what apparatus/structural component carried out said steps. In absence any structural limitations, subject matter of these claims is mere scheme and/or mental act and hence falls within scope of clause (m) of section (3) of the Patents Act, 1970 (as amended). Therefore invention claimed in said claim is not patentable.
- Without prejudice to objection U/S 2(1)(j), Subject matter of claim as filed in the instant application falls within scope of clause (k) of section (3) of the Patents Act, 1970 (as amended). Therefore invention claimed in said claims is not patentable. Method claims are a set of sequences used to implement an algorithm, without disclosing any constructional or structural aspects of the said features. Also system claims do not contribute in the constructional or structural aspects of the alleged invention and the alleged contribution lies only in algorithmic steps which make the system functional.
- Subject matter of claims 10-15 falls within scope of clause (i) of section (3) of the Patents Act, 1970 (as amended) being diagnostic method. Therefore invention claimed in said claims is not patentable.

(4).प्रकटन की दक्षता /SUFFICIENCY OF DISCLOSURE:

(I) सार /Abstract:

- The abstract should be prepared as the instructions given in rule 13(7)(a), 13(7)(b) and 13(7)(d) of the Patents Rules, 2003 (as amended).

(5).परिभाषिकता /DEFINITIVENESS:

(I) दावा(दे)1-15 निम्नलिखित कारणों से आविष्कार को पर्याप्त रूप से परिभाषित नहीं करता(ते) हैं
Claim(s) 1-15 do not sufficiently define the invention for the reasons as follows:

- i. The term "comprising/ at least" used in the claims shall be aptly reworded since it is not definitive.
- ii. Claims 1 and 10 are not clear as the purpose or object of the claimed apparatus is not mentioned in the preamble of the claims.

(6).अन्य आवश्यकताएँ /OTHERS REQUIREMENTS:

THE PATENT OFFICE

(I)

If any amendment is necessitated in the complete specification then it is required to clearly identify (submission of marked copy) the amendments carried out and to indicate the portion (page no and line no) of the complete specification as filed on which these amendments are based on. Further the pages wherever amendments are carried out need to be freshly typed on white pages and to be filed.

भाग - III: औपचारिक आवश्यकताएँ /PART-III: FORMAL REQUIREMENTS

आपत्तियाँ /Objections	टिप्पणी /Remarks
Statement & Under Taking (Form 3 Details)	Details as required under section 8(1) and 8(2) of the Act should be submitted.
Format of Specification (rule 13)	Irrelevant portion shall be deleted and blank spaces shall be scored out in the complete specification. Claims should be prefaced with the words "We Claim". Also the claims must be signed by the applicant/authorized patent agent.
Power of Attorney (Whether GPA, SPA, Stamped, requisite fee etc.)	Stamp duty should be paid in respect of Power of Attorney for the instant application.
Other Deficiencies	- Copy of Form 18 alongwith fee receipt details should be submitted. - All the submitted documents and forms have been presumed as originally signed by the authorized signatory under the provisions of the Patents Act, 1970. If not, submit the originally signed copy of the same failing to which the document may not be considered filed. - Fresh Form 1 , Form 5 should be submitted.
Format of Drawings	The Drawings referred to in the specification should be prepared in accordance with the instructions contained in the Rule 15 of the Patent Rules, 2003(as amended in 2006).
Translation of Pr.Document (Certified Copy etc.)	- Priority document or PCT/ IB/304 should be filed as per rule 21(1) of Patent Rules, 2003. - English translation of the priority document should be filed as per rule 21(2) of the Patents Rules, 2003.

भाग-IV: रिकॉर्ड में दस्तावेज़ /PART-IV: DOCUMENTS ON RECORD

निम्नलिखित दस्तावेज़ों के आधार पर यह परीक्षण रिपोर्ट तैयार की गयी है

The examination report has been prepared based on the following documents:

कार्यसूची तिथि / Docket Date	कार्यसूची संख्या /Docket Number	प्रविष्टि संख्या विवरण /Entry Number Description
15 Feb 2017	9444	1-New Application For Patent With Provisional /Complete Specification
15 Feb 2017	9444	28(i)-Request For Examination After 18 months Publication - Form 18
14 Mar 2017	17926	18(iii)-Changing Name/Address/Nationality/Address For Service -

THE PATENT OFFICE

DATE	NO.	FORM
03 Apr 2017	21999	Miscellaneous - Form 30
03 Apr 2017	21999	45-Form Of Authorisation Of Patent Agent - Form 26
13 Apr 2017	24152	OTHERS(NON CASH)
13 Apr 2017	24156	OTHERS(NON CASH)
19 Jul 2017	45608	3-Statement & Undertaking - Form 3
19 Jul 2017	45608	Section 8(2) information
18 Jun 2019	50820	3-Statement & Undertaking - Form 3

नियंत्रक का नाम /Name of the Controller: Pinkesh Jain

नियंत्रक स्थान /Controller Location: Mumbai

टिप्पणी: परीक्षण रिपोर्ट का उत्तर दाखिल करने की अंतिम तिथि / Note: Last date for filing response to the Examination Report:
15/05/2020

Electronic Acknowledgement Receipt

EFS ID:	38485755
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/Tizita Ayane
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	04-FEB-2020
Filing Date:	24-JAN-2019
Time Stamp:	12:41:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	0202-1231-1IDSAsFiled02-04-2020.pdf	1053288 f164ef6c8f442d3a8992e01bbbf24173c3c2d9b	no	4

Warnings:

Information:					
A U.S. Patent Number Citation or a U.S. Publication Number Citation is required in the Information Disclosure Statement (IDS) form for autoloading of data into USPTO systems. You may remove the form to add the required data in order to correct the Informational Message if you are citing U.S. References. If you chose not to include U.S. References, the image of the form will be processed and be made available within the Image File Wrapper (IFW) system. However, no data will be extracted from this form. Any additional data such as Foreign Patent Documents or Non Patent Literature will be manually reviewed and keyed into USPTO systems.					
2	Foreign Reference	0202-1231-1WO2009140360As Filed.pdf	1156061 0955a1e1644e539946381c3359f8eadf73d8e7c0	no	29
Warnings:					
Information:					
3	Non Patent Literature	0202-1231-1INOA_AsFiled.pdf	398520 39fd390b4c671bbfe8ba0d7a978af3075dd0cfb2	no	6
Warnings:					
Information:					
Total Files Size (in bytes):			2607869		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	01/24/2019	Gihsung CHUNG	0202-1231-1	6284
68103	7590	07/23/2020		
Jefferson IP Law, LLP 1130 Connecticut Ave., NW, Suite 420 Washington, DC 20036			EXAMINER JOHNSON, NICOLE F	
			ART UNIT	PAPER NUMBER
			3792	
			NOTIFICATION DATE	DELIVERY MODE
			07/23/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

usdocketing@jeffersonip.com

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Specification

The disclosure is objected to because of the following informalities: Any application numbers listed in the disclosure should be updated to reflect their patent status (e.g., pending, patented, abandoned, etc.) and publication numbers should be provided for an applications that have been published. Specifically, paragraph [e.g., 0001] of the disclosure requires correction. Furthermore, any references to attorney docket numbers appearing therein should be removed. Appropriate correction is required.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed.

Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP §§ 706.02(l)(1) - 706.02(l)(3) for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims 1 & 16 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1 & 20 of U.S. Patent No. 10/231675. Although the claims at issue are not identical, they are not patentably distinct from each other because both claim sets are directed towards an electronic device using a first and a second (motion) sensor, wherein the first signal corresponds to said device prior to a biological sing being obtained and the second signal corresponds to the device during the obtaining of a biological signal.

Allowable Subject Matter

Claims 2-15 & 17-20 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

The closest prior art, Fleck et al. (as cited in 15/173099), fails to disclose, suggest and/or teach the claimed invention since the reference is silent in regards to at least one processor configured to acquire first information on a motion of an electronic device using at least the part of at least one sensor...during detecting the biometric information, acquire second information on a motion of said device using said sensor and in response to identifying the motion of said device included in the second information is more than a threshold motion, inactive the at least one second sensor, in combination with the other claimed elements (e.g., see 'Arguments').

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Allowable Subject Matter

3. Claims 1-20 are allowed.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to NICOLE F JOHNSON whose telephone number is (571)270-5040. The examiner can normally be reached on **Mon-Thu 7:30 am-4:30 pm**.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **Jennifer McDonald** can be reached on **571-270-3061**. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/NICOLE F LAVERT/
Primary Examiner, Art Unit 3792

<i>Notice of References Cited</i>	Application/Control No. 16/256,497		Applicant(s)/Patent Under Reexamination CHUNG et al.	
	Examiner NICOLE F JOHNSON		Art Unit 3792	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20150223355-A1	08-2015	Fleck; Rod G.	H05K5/026	361/679.03
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.
	Examiner NICOLE F JOHNSON	Art Unit 3792

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner
A61B5/7285	07/20/2020	NFJ

US Classification - Searched*			
Class	Subclass	Date	Examiner
600	300-301	07/20/2020	NFJ

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search (PE2E)	07/20/2020	NFJ
EAST Search (see attachment)		
Consulted search notes for		

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16256497
	Filing Date		2019-01-24
	First Named Inventor	Gihung CHUNG	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		0202-1231-1

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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NON-PATENT LITERATURE DOCUMENTS			Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

1	European Office Action dated February 13, 2019; Application #: 16 183 812.3 - 1132; Ref. #P6061727EP
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If you wish to add additional non-patent literature document citation information please click the Add button

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EXAMINER SIGNATURE

Examiner Signature	/NICOLE F LAVERT/	Date Considered	07/20/2020
--------------------	-------------------	-----------------	------------

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /N.F.L/

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2019-05-07
Name/Print	Wonki Park	Registration Number	38991

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	16256497
	Filing Date	2019-01-24
	First Named Inventor	Gihsung CHUNG
	Art Unit	3792
	Examiner Name	Niketa I PATEL
	Attorney Docket Number	0202-1231-1

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	1	2009140360	WO	A1	2009-11-19	KATERAAS et al.		

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NON-PATENT LITERATURE DOCUMENTS			Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	3792
Examiner Name	Niketa I PATEL
Attorney Docket Number	0202-1231-1

1	Indian Office Action dated November 15, 2019; issued in a counterpart Indian Application No. 201747005402
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Examiner Signature	/NICOLE F LAVERT/	Date Considered	07/20/2020
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
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Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

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Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2020-02-04
Name/Print	Wonki Park	Registration Number	38991

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number

Filing Date

First Named Inventor

Art Unit

Examiner Name

Attorney Docket Number

2019-01-24

Gihung CHUNG

0202-1231-1

U.S.PATENTS

Remove

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4129125	A	1978-12-12	LESTER et al.	
	2	8942796	B2	2015-01-27	KASAMA	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20150074797	A1	2015-03-12	CHOI et al.	
	2	20110092780	A1	2011-04-21	ZHANG et al.	
	3	20150057967	A1	2015-02-26	ALBINALI	
	4	20150223355	A1	2015-08-06	FLECK et al.	

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Application Number		
Filing Date		2019-01-24
First Named Inventor	Gihung CHUNG	
Art Unit		
Examiner Name		
Attorney Docket Number	0202-1231-1	

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	2007/209430	JP	A	2007-08-23	TAKASHI et al.		
	2	10-2015-0046976	KR	A	2015-05-04	SONG		
	3	2989973	EP	A1	2016-03-02	CHOI et al.		
	4	2009140360	WO	A1	2009-11-19	KATERAAS et al.		
	5	2013043747	WO	A1	2013-03-28	KAHN et al.		
	6	2851001	EP	A2	2015-03-25	KOMAROMI et al.		
	7	2014120832	WO	A1	2014-08-07	LEE et al.		

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Filing Date	2019-01-24
First Named Inventor	Gihung CHUNG
Art Unit	
Examiner Name	
Attorney Docket Number	0202-1231-1

EXAMINER SIGNATURE

Examiner Signature	/NICOLE F LAVERT/	Date Considered	07/20/2020
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Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2019-01-24
Name/Print	Wonki Park	Registration Number	38,991

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Bibliographic Data

Application No: 16/256,497

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged: /NICOLE F LAVERT/

NFJ

Examiner's Signature

Initials

Title:

METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2019	600	3792	0202-1231-1
RULE			

APPLICANTS

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF

INVENTORS

Gihung CHUNG, Suwon-si, KOREA, REPUBLIC OF

Jooman HAN, Seongnam-si, KOREA, REPUBLIC OF

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Cheolho CHEONG, Seoul, KOREA, REPUBLIC OF

Jae-Woong CHUN, Suwon-si, KOREA, REPUBLIC OF

Won Suk CHOI, Seoul, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a CON of 15173099 06/03/2016 PAT 10231675

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF 10-2015-0113972 08/12/2015

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	7,619	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/07/20 09:15
L2	5,033	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/07/20 09:15
S1	10,582	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 19:59
S2	7,065	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 20:00
S3	3,935	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 20:02
S4	651,416	medical same device	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:25
S5	15,462	S4 and ((fastner or snap or button) same housing)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:26
S6	578	S5 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:26
S7	14	S6 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:27
S8	4,406	S4 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:30
S9	38	S8 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2018/03/02 07:32

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S10	143,580	(wearable same device)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:42
S11	707	S10 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:43
S12	68	S11 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:43
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S14	10,678	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S15	7,202	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S16	4,145	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S17	1	"20150223355" and (time or period)	USPAT	OR	OFF	2018/07/31 20:38
S18	10,753	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/10/08 07:09
S19	7,293	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/10/08 07:09
S20	4,237	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/10/08 07:09

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S21	2,972	600/300.ccls.	USPAT	OR	OFF	2018/10/08 07:20
S22	1,253	600/301.ccls.	USPAT	OR	OFF	2018/10/08 07:20
S23	548	a61b5/7285.cpc.	USPAT	OR	OFF	2018/10/08 07:20

7/20/2020 9:42:45 AM

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16256497	CTNF	07/23/2020	0202-1231-1
	892	07/23/2020	0202-1231-1
	1449	07/23/2020	0202-1231-1
	1449	07/23/2020	0202-1231-1
	1449	07/23/2020	0202-1231-1

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No.	: 16/256,497	Confirmation No.	: 6284
First Named Inventor	: Gihung CHUNG	TC/A.U.	: 3792
Filing or 371(c) Date	: January 24, 2019	Examiner	: Nicole F. JOHNSON
Title	: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME	Atty. Docket No.	: 0202-1231-1

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Sir:

In response to the Office Action dated July 23, 2020, please amend the above-identified application as follows:

Amendments to the Specification begin on page 2 of this paper.

Amendments to the Claims are reflected in the listing of claims which begins on page 3 of this paper.

Remarks begin on page 8 of this paper.

Electronic Acknowledgement Receipt

EFS ID:	40840163
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/Mariam Paulos
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	14-OCT-2020
Filing Date:	24-JAN-2019
Time Stamp:	12:37:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		0202-1231-1AmendmentAsFile d10-14-2020.pdf	104579 1c1045dd6cd2b2c236083dc2daa81c2e78d d9a32	yes	9

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Applicant Arguments/Remarks Made in an Amendment	8	9
	Claims	3	7
	Specification	2	2
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1

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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

REMARKS

I. Status of Claims

After the above amendments, claims 1-20 are pending, with claims 1, 9 and 16 being independent. By this Amendment, claim 20 is amended to improve antecedence. Applicant reserves the right to pursue the original claims and other claims in this and other applications. In view of at least the following, reconsideration and allowance are respectfully requested.

II. Indication of Allowable Subject Matter

Applicant thanks the Examiner for the indication that claims 2-15 and 17-20 would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims. Claims 2-15 and 17-20 depend from claims 1 and 16 which are rejected under non-statutory obviousness double patenting. A terminal disclaimer is submitted herewith which overcomes the rejection of claims 1 and 16. Accordingly, claims 1-20 are in condition for allowance.

III. Objection to the Specification

The disclosure is objected to over informalities. More specifically, the Office Action objects to the specification as not providing updated information (i.e., patent number of the parent application).

In order to overcome the objection, the specification is amended to include updated information in the "Cross-Reference to Related Application(s)" section by providing the U.S. patent number for the parent application, as requested by the Examiner. No new matter has been introduced.

Accordingly, withdrawal of the objection to the specification is respectfully requested.

IV. Rejection of Claims 1 and 16 Under Non-statutory Double Patenting

Claims 1 and 16 have been rejected under non-statutory obviousness double patenting as allegedly being unpatentable over U.S. Patent No. 10,231,675 to Chung et al.

A terminal disclaimer is submitted herewith that overcomes this rejection.

Accordingly, withdrawal of the rejection is respectfully requested.

V. Conclusion

In view of the foregoing, it is respectfully submitted that the present application is in condition for allowance, and an early indication of the same is courteously solicited. The Examiner is respectfully requested to contact the undersigned by telephone at the below listed telephone number, in order to expedite resolution of any issues and to expedite passage of the present application to issue, if any comments, questions, or suggestions arise in connection with the present application.

To the extent necessary, a petition for an extension of time under 37 CFR § 1.136 is hereby made.

It is believed that no additional fees are due in connection with the filing of this paper. However, please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account No. 50-4100, and please credit any excess fees to the same deposit account.

Respectfully submitted,

Date: October 14, 2020

/Wonki PARK/
Wonki PARK
Reg. No. 38,991
Representative for Applicant

Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, D.C. 20036
Main: (202)-293-0804

Amendments to the Claims:

This listing of claims will replace all prior versions and listing of the claims in the application.

Listing of Claims:

1. (Original) A wearable electronic device comprising:
a motion sensor;
one or more optical sensors includes a light emitting unit and a light receiving unit; and
at least one processor configured to:
obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,
emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,
obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,
obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and
refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.
2. (Original) The wearable electronic device of claim 1, wherein the at least one processor is further configured to:
determine whether the wearable electronic device is attached to the user using the one or more optical sensors; and
perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.

3. (Original) The wearable electronic device of claim 2, wherein the at least one processor is further configured to:

perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.

4. (Original) The wearable electronic device of claim 2, wherein the at least one processor is further configured to:

refrain from the emitting of the light further based at least in part on a determination that the wearable electronic device is not attached to the user.

5. (Original) The wearable electronic device of claim 1, wherein the at least one processor is further configured to:

perform the emitting of the light based at least in part on a specified time period.

6. (Original) The wearable electronic device of claim 5, wherein the at least one processor is further configured to:

adjust the specified time period based at least in part on context information corresponding to the biological signal.

7. (Original) The wearable electronic device of claim 1, wherein the at least one processor is further configured to:

generate a heart rate information corresponding to the user using the biological signal.

8. (Original) The wearable electronic device of claim 7, wherein the at least one processor is further configured to:

present a notification corresponding to the heart rate information based at least in part on a determination that the heart rate information is higher than a threshold.

9. (Original) A wearable electronic device comprising:

a first sensor;

a second sensor; and

at least one processor configured to:

obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,

activate the second sensor based at least in part of the movement,

obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and

refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.

10. (Original) The wearable electronic device of claim 9, wherein the at least one processor is further configured to:

control the second sensor to inactivate at least a portion thereof, as at least part of the refraining.

11. (Original) The wearable electronic device of claim 10, wherein the at least one processor is further configured to:

control the second sensor to activate the at least a portion thereof, based at least in part on a determination that the first information satisfies another specified range lower than the specified range after the refraining.

12. (Original) The wearable electronic device of claim 10, wherein the at least one processor is further configured to:

control the second sensor to activate the at least a portion thereof, further based at least in part on a specified time period.

13. (Original) The wearable electronic device of claim 12, wherein the at least one processor is further configured to:

adjust the specified time period based at least in part on context information generated using the first information and the second information.

14. (Original) The wearable electronic device of claim 9, wherein the at least one processor is further configured to:

generate a heart rate information corresponding to the user using the second information.

15. (Original) The wearable electronic device of claim 14, wherein the at least one processor is further configured to:

present a notification corresponding to the heart rate information based at least in part on a determination that the heart rate information is higher than a threshold.

16. (Original) A wearable electronic device comprising:

a first sensor;

a second sensor; and

at least one processor configured to:

obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,

obtain, using the second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and

refrain from the obtaining of the second information based at least in part on a determination that the first information satisfies a specified condition.

17. (Original) The wearable electronic device of claim 16, wherein the at least one processor is further configured to:

activate the second sensor prior to obtaining of the second information, in response to the movement of the wearable electronic device.

18. (Original) The wearable electronic device of claim 17, wherein the at least one processor is further configured to:

perform the activating based at least in part on a determination that the first information does not satisfy the specified condition.

19. (Original) The wearable electronic device of claim 16, wherein the at least one processor is further configured to:

control the second sensor to inactivate at least a portion thereof, as at least part of the refraining.

20. (Currently Amended) The wearable electronic device of claim 16, wherein the second sensor includes a light emitting unit, and wherein the processor is further configured to:

emit light, using the light emitting unit, as at least part of the obtaining of the second information, and

stop the emitting of the light, using the light emitting unit, as at least part of the refraining.

Amendments to the Specification:

Please replace paragraph [0001] of the specification with the following amended paragraph:

[0001] This application is a continuation application of prior Application number 15/173,099, filed on June 3, 2016 and issued as U.S. Patent No. 10,231,675 on March 19, 2019, which ~~claimed~~claims priority under 35 U.S.C § 119(a) of a Korean patent application number 10-2015-0113972, filed on August 12, 2015, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

Doc Code: DIST.E.FILE Document Description: Electronic Terminal Disclaimer - Filed		PTO/SB/26 U.S. Patent and Trademark Office Department of Commerce
Electronic Petition Request	TERMINAL DISCLAIMER TO OBIATE A DOUBLE PATENTING REJECTION OVER A "PRIOR" PATENT	
Application Number	16256497	
Filing Date	24-Jan-2019	
First Named Inventor	Gihung CHUNG	
Attorney Docket Number	0202-1231-1	
Title of Invention	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME	
☒ Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action ☒ This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.		
Owner	Percent Interest	
Samsung Electronics Co., Ltd.	100%	
The owner(s) with percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of prior patent number(s) 10231675 as the term of said prior patent is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the prior patent are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns. In making the above disclaimer, the owner does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the prior patent, "as the term of said prior patent is presently shortened by any terminal disclaimer," in the event that said prior patent later: - expires for failure to pay a maintenance fee; - is held unenforceable; - is found invalid by a court of competent jurisdiction; - is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321; - has all claims canceled by a reexamination certificate; - is reissued; or - is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer. ☒ Terminal disclaimer fee under 37 CFR 1.20(d) is included with Electronic Terminal Disclaimer request.		

☐ I certify, in accordance with 37 CFR 1.4(d)(4), that the terminal disclaimer fee under 37 CFR 1.20(d) required for this terminal disclaimer has already been paid in the above-identified application.

Applicant claims the following fee status:

- ☐ Small Entity
- ☐ Micro Entity
- ☒ Regular Undiscounted

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

- ☒ An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application
- Registration Number 38991
- ☐ A sole inventor
- ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application
- ☐ A joint inventor; all of whom are signing this request

Signature	/Wonki PARK/
Name	Wonki Park

*Statement under 37 CFR 3.73(b) is required if terminal disclaimer is signed by the assignee (owner).
Form PTO/SB/96 may be used for making this certification. See MPEP § 324.

Electronic Patent Application Fee Transmittal

Application Number:	16256497			
Filing Date:	24-Jan-2019			
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME			
First Named Inventor/Applicant Name:	Gihung CHUNG			
Filer:	Maeng-Ho Shin/Mariam Paulos			
Attorney Docket Number:	0202-1231-1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
STATUTORY OR TERMINAL DISCLAIMER	1814	1	170	170
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				170

Doc Code: DISQ.E.FILE

Document Description: Electronic Terminal Disclaimer – Approved

Application No.: 16256497

Filing Date: 24-Jan-2019

Applicant/Patent under Reexamination: CHUNG

Electronic Terminal Disclaimer filed on October 14, 2020

☒ APPROVED

This patent is subject to a terminal disclaimer

☐ DISAPPROVED

Approved/Disapproved by: Electronic Terminal Disclaimer automatically approved by EFS-Web

U.S. Patent and Trademark Office

Electronic Acknowledgement Receipt

EFS ID:	40837586
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/Mariam Paulos
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	14-OCT-2020
Filing Date:	24-JAN-2019
Time Stamp:	12:41:55
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 170
RAM confirmation Number	E20200DC41521155
Deposit Account	504100
Authorized User	Mariam Paulos

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Terminal Disclaimer-Filed (Electronic)	eTerminal-Disclaimer.pdf	33578	no	2
			71fab9b5d7b0c9166bf014d415345932a5babb74		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30462	no	2
			142127753974b816cbee5e4f78cbab36832e3370		

Warnings:

Information:

Total Files Size (in bytes):	64040
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

68103 7590 11/25/2020
Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036

EXAMINER

JOHNSON, NICOLE F

ART UNIT

PAPER NUMBER

3792

DATE MAILED: 11/25/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	01/24/2019	Gihung CHUNG	0202-1231-1	6284

TITLE OF INVENTION: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/25/2021

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

68103 7590 11/25/2020
Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	01/24/2019	Gihung CHUNG	0202-1231-1	6284

TITLE OF INVENTION: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/25/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
JOHNSON, NICOLE F	3792	600-301000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	01/24/2019	Gihung CHUNG	0202-1231-1	6284
68103	7590	11/25/2020	EXAMINER	
Jefferson IP Law, LLP 1130 Connecticut Ave., NW, Suite 420 Washington, DC 20036			JOHNSON, NICOLE F	
			ART UNIT	PAPER NUMBER
			3792	
DATE MAILED: 11/25/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

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The information provided by you in this form will be subject to the following routine uses:

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/256,497	Applicant(s) CHUNG et al.	
	Examiner NICOLE F JOHNSON	Art Unit 3792	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the arguments submitted on October 14, 2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some *c) ☐ None of the:

1. ☒ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date ____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material ____.	7. ☐ Other ____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. ____.	

/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792	
--	--

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

THE EXAMINER'S STATEMENT OF REASONS FOR ALLOWABILITY

The closest prior art, Fleck et al. (as cited in 15/173099), fails to disclose, suggest and/or teach the claimed invention since the reference is silent in regards to at least one processor configured to acquire first information on a motion of an electronic device using at least the part of at least one sensor...during detecting the biometric information, acquire second information on a motion of said device using said sensor and in response to identifying the motion of said device included in the second information is more than a threshold motion, inactive the at least one second sensor, in combination with the other claimed elements (e.g., see 'Arguments').

Allowable Subject Matter

Claims 1-15 are allowed.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to NICOLE F JOHNSON whose telephone number is (571)270-5040. The examiner can normally be reached on **Mon-thu 7:30am-4:30pm**.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **Jennifer McDonald** can be reached on **571-270-3061**. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

**/NICOLE F LAVERT/
Primary Examiner, Art Unit 3792**

Issue Classification 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.	
	Examiner NICOLE F JOHNSON	Art Unit 3792	

CPC						
Symbol					Type	Version
A61B	/	5	/	7285	F	2013-01-01
A61B	/	5	/	0205	I	2013-01-01
A61B	/	5	/	0402	I	2013-01-01
A61B	/	5	/	443	I	2013-01-01
A61B	/	5	/	681	I	2013-01-01
A61B	/	5	/	11	I	2013-01-01
A61B	/	5	/	6843	I	2013-01-01
A61B	/	5	/	1118	I	2013-01-01
A61B	/	5	/	6801	I	2013-01-01
G06F	/	1	/	3231	I	2013-01-01
G06F	/	1	/	163	I	2013-01-01
G06F	/	1	/	1684	I	2013-01-01
G06F	/	1	/	1694	I	2013-01-01
G06F	/	1	/	3215	I	2013-01-01
G06F	/	1	/	325	I	2013-01-01
G04G	/	21	/	025	I	2013-01-01
A61B	/	5	/	7475	I	2013-01-01
A61B	/	5	/	021	A	2013-01-01
A61B	/	5	/	02438	A	2013-01-01
A61B	/	5	/	14542	A	2013-01-01
A61B	/	2562	/	0219	A	2013-01-01
A61B	/	2562	/	0223	A	2013-01-01
A61B	/	2562	/	0257	A	2013-01-01
A61B	/	2560	/	0209	A	2013-01-01
Y02D	/	10	/	00	A	2018-01-01

CPC Combination Sets					
Symbol				Type	Set

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 20	
/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792 (Primary Examiner) _____ (Date) _____		19 November 2020	O.G. Print Claim(s) 1
			O.G. Print Figure 1

<i>Issue Classification</i> 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.
	Examiner NICOLE F JOHNSON	Art Unit 3792

INTERNATIONAL CLASSIFICATION			
CLAIMED			
A61B	/	5	/ 00
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
600	300

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
600	301					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	20	
/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792	19 November 2020	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

<i>Issue Classification</i> 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.
	Examiner NICOLE F JOHNSON	Art Unit 3792

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	19	19										
2	2	11	11	20	20										
3	3	12	12												
4	4	13	13												
5	5	14	14												
6	6	15	15												
7	7	16	16												
8	8	17	17												
9	9	18	18												

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 20	
/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792 (Primary Examiner) _____ (Date) _____		19 November 2020 O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Search Notes</i> 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.
	Examiner NICOLE F JOHNSON	Art Unit 3792

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner
A61B5/7285	07/20/2020	NFJ
ABOVE UPDATED	11/19/2020	NFJ

US Classification - Searched*			
Class	Subclass	Date	Examiner
600	300-301	07/20/2020	NFJ
ABOVE	UPDATED	11/19/2020	NFJ

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search (PE2E)	07/20/2020	NFJ
EAST Search (see attachment)		
Consulted search notes for		

/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792	
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<i>Search Notes</i> 	Application/Control No. 16/256,497	Applicant(s)/Patent Under Reexamination CHUNG et al.
	Examiner NICOLE F JOHNSON	Art Unit 3792

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
Searched all classes and subclasses from the Searched notes	Searched the USPAT, USOCR, USPGPU, EPO, JPO, DERWENT. etc databases	11/19/2020	NFJ

/NICOLE F LAVERT/ Primary Examiner, Art Unit 3792	
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Bibliographic Data

Application No: 16/256,497

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged: /NICOLE F LAVERT/

NFJ

Examiner's Signature

Initials

Title:

METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/24/2019	600	3792	0202-1231-1
RULE			

APPLICANTS

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INVENTORS

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CONTINUING DATA

This application is a CON of 15173099 06/03/2016 PAT 10231675

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF KR10-2015-0113972 08/12/2015

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02/11/2019

STATE OR COUNTRY

KOREA, REPUBLIC OF

ADDRESS

Jefferson IP Law, LLP

1130 Connecticut Ave., NW, Suite 420

Washington, DC 20036

UNITED STATES

FILING FEE RECEIVED

\$1,720

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	11,307	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/11/19 13:48
L2	7,597	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/11/19 13:49
L3	5,176	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/11/19 13:49
S1	10,582	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 19:59
S2	7,065	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 20:00
S3	3,935	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/02/28 20:02
S4	651,416	medical same device	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:25
S5	15,462	S4 and ((fastner or snap or button) same housing)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:26
S6	578	S5 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:26
S7	14	S6 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:27
S8	4,406	S4 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2018/03/02 07:30

			JPO; DERWENT; IBM_TDB			
S9	38	S8 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:32
S10	143,580	(wearable same device)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:42
S11	707	S10 and (detachably same attached)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:43
S12	68	S11 and (sensor\$2 same (motion and biometric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 07:43
S13	3	"20150223355" and memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/03/02 08:39
S14	10,678	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S15	7,202	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S16	4,145	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/07/31 18:46
S17	1	"20150223355" and (time or period)	USPAT	OR	OFF	2018/07/31 20:38
S18	10,753	600/300.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/10/08 07:09
S19	7,293	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2018/10/08 07:09
S20	4,237	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2018/10/08 07:09

			JPO; DERWENT; IBM_TDB			
S24	7,619	600/301.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/07/20 09:15
S25	5,033	a61b5/7285.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2020/07/20 09:15

11/19/2020 2:04:41 PM

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16256497	NOA	11/25/2020	0202-1231-1

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UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	16256497
	Filing Date	2019-01-24
	First Named Inventor	Gihung CHUNG
	Art Unit	3792
	Examiner Name	Nicole F JOHNSON
	Attorney Docket Number	0202-1231-1

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	3792
Examiner Name	Nicole F JOHNSON
Attorney Docket Number	0202-1231-1

1	Summons to attend oral proceeding dated January 20, 2021, issued in European Application No. 16 183 812.3
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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	3792
Examiner Name	Nicole F JOHNSON
Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2021-02-05
Name/Print	Wonki Park	Registration Number	38,991

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



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Application No. 16 183 812.3 - 1132	Ref. P6061727EP	Date 20.01.2021
Applicant Samsung Electronics Co., Ltd.		

Summons to attend oral proceedings pursuant to Rule 115(1) EPC

You are hereby summoned to attend oral proceedings arranged in connection with the above-mentioned European patent application.

The matters to be discussed are set out in the communication accompanying this summons (EPO Form 2906).

The oral proceedings, which will not be public, will take place before the Examining Division

on 12.10.2021 at 09:00 hrs, as a videoconference.

No changes to the date of the oral proceedings can be made, except on serious grounds (see OJ EPO 2009, 68). If you do not appear as summoned, the oral proceedings may continue without you (R. 115(2) EPC, see also OJ EPO 2008, 471).

Your attention is drawn to Rule 4 EPC, regarding the language of the oral proceedings, and to Special edition No. 3 OJ EPO 2007, L.1, concerning the filing of authorisations for company employees and lawyers acting as representatives before the EPO.

The final date for making written submissions and/or amendments (R. 116 EPC) is 10.09.2021.

You are informed that the EPO supports the following videoconferencing technologies:

- Microsoft Skype for Business Clients and SfB Room Systems
- Browser-based Web Meeting via Skype for Business WebApp
- SIP/H.323 videoconferencing systems.

The EPO provides the applicant or his representative with the opportunity to test the videoconferencing system in advance of the oral proceedings. A test call can be arranged by contacting the EPO Serviceline on Mondays to Fridays (except days on which the EPO filing offices are closed) from 7:30 to 18:00 hrs CET, by telephone (+31) (0) 70 340 44 44 or by email serviceline@epo.org. Further technical details can be found on the EPO website (www.epo.org).

Please supply the EPO as soon as possible with the following information concerning your videoconference studio:

- name of contact person with telephone number and email address to be used by the EPO for communicating the connection details
- type and name of videoconference equipment
- telephone and fax number (if any) of the studio including country and area code.

IMPORTANT:

Please inform the EPO without delay in the event that the detailed information about the videoconference connection has not been received at least ten days before the scheduled date of the oral proceedings to be held via videoconference.

Examining Division:

Chairman:
2nd Examiner:
1st Examiner:

Völlinger, Martin
Trachterna, Morten
Pohjamo, Terhi



Annexes:

4 page/s reasons (EPO Form 2906)

- 1 With his reply on 21.06.2019 the Applicant filed a new set of claims. However, the new claims are still not allowable for the reasons given below.

Since arguments of both sides have been exchanged and no agreement on an allowable set of claims seems to be possible, the examining division considers it to be expedient to finally discuss all matters raised so far in oral proceedings, as requested by the Applicant. The Applicant should be aware that a decision pursuant to Art. 97 and Rule 111(1) EPC will be given at Oral Proceedings so that no submissions or experimental data subsequent to the oral proceedings will be accepted. If it is intended to submit further evidence or further claims, pursuant to Rule 116 EPC, the **final date** for making any written submissions in preparation for the oral proceedings is fixed as indicated on the front page of this summons to oral proceedings.

The summons is based on the following application documents:

Description, Pages

1-66 as originally filed

Claims, Numbers

1-15 filed in electronic form on 21-06-2019

Drawings, Sheets

1/23-23/23 as originally filed

- 2 The pending claims and the applicants arguments are commented as follows:

2.1 **Article 123(2) EPC**

The amendment filed with the letter dated 21.06.2019 introduces subject-matter which extends beyond the content of the application as filed, contrary to Article 123(2) EPC. The amendments concerned are the following:

At least for claims 7 and 9 there cannot be basis found. Claim 7 is worded "user using the biological signal" which has no basis or meaning in the originally filed application. Claim 9 states that the second sensor is a motion sensor, which is also contradictory to the general teaching of the application, where the first sensor is a motion sensor and the second sensor for determining eg. heart rate.

Generally, the amendments and basis for them should be well-defined, see Guidelines H-III.2.1. With regard to the present application and the amended claims, these requirements have not been fulfilled (listing of plural claims and/or paragraphs without further explanations is not considered to be specific enough). Therefore the claims contravene also Rule 137(4) EPC.

In order to comply with the requirements of Rule 137(4) EPC, the applicant should identify clearly and precisely the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based. This applies to all claims, for which only very vague (if any) basis have been given so far. As an example, there has been no arguments or basis given for the deletion or certain features (fastener,...) in claim 1, when comparing to the originally filed claims.

If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed or in documents with tracked changes.

2.2 **Novelty and Inventive Step**

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC.

2.2.1 D2 discloses:

An electronic device comprising:

at least one first sensor (24) configured to detect a motion of the electronic device;

at least one second sensor (14, 22) configured to detect a change in the user's body;

a memory (50); and

a processor (40) electrically connected to the at least one first sensor, the at least one second sensor, and the memory,

wherein the memory stores instructions that configure the processor, when being executed, to:

determine whether the electronic device is worn on a part of a user's body using the at least one first sensor (par. 27, 28; *"For example, when infrared sensors 14, 16, or 18 provide no pulse readings or accelerometer 24 registers no movements over a certain period of time, microcontroller 40 may determine that data collection unit 10 is not being worn."*),

acquire information on the motion corresponding to the electronic device generated during the electronic device is worn on a part of the user's body

(accelerometer 24 is acquiring motion data), and activate the at least one second sensor based on at least a part of the information on the acquired motion (par. 27, 28; *"microcontroller 40 may be configured to sample only a portion of the available sensors during times of physical inactivity or when data collection unit 10 is not being worn"*; this implicitly states that certain sensor(s) are being activated / sampling performed when the device is detected as being worn).

Thus, the argumentation of the applicant on D2 is not considered convincing.

2.2.2 Also D5 discloses:

An electronic device (150, 200, 300) comprising:
at least one first sensor (153, 353) configured to detect a motion of the electronic device (par. 38, 39, 97);
at least one second sensor (153, 353) configured to detect a change in the user's body (par. 38, 39, 97);
a memory (328); and
a processor (321) electrically connected to the at least one first sensor, the at least one second sensor, and the memory,
wherein the memory stores instructions that configure the processor, when being executed, to:
determine whether the electronic device is worn on a part of a user's body using the at least one first sensor (par. 94, 97),
acquire information on the motion corresponding to the electronic device generated during the electronic device is worn on a part of the user's body, and
activate the at least one second sensor based on at least a part of the information on the acquired motion (par. 97, 106, 108-120; different power states are being chosen based on detected motion, which equals to activating / deactivating other sensors based on detected motion).

2.2.3 Dependent claims, as far as they can be interpreted (see objections above), do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC in respect of novelty and/or inventive step. In particular measuring during different time windows to determine if there is eg. enough / too much motion for performing other sensing, or to adjust power consumption do not seem anything but known / obvious features from which the skilled person would choose from.

3 It is considered that the applicant had had the opportunity to amend the claims in accordance with Rule 137(3), first sentence. Newly-filed claims that contain a shift in the subject-matter with respect to the current claims will probably not be

accepted under Rule 137(3) EPC, second sentence (see also G2/92, "Reasons for the Opinion", point 2, paragraph 2, second sentence "At the examination stage ...").

In a case in which amended claims filed by the applicant would not be admitted by the examining division by virtue of Rule 137(3) EPC, the application would probably be refused under Article 97(2) EPC on the ground that there would be no version approved by the applicant in the sense of Article 113(2) EPC, on which a patent could be granted.

The examining division draws the attention of the applicant to the Notice from the European Patent Office concerning non-attendance at oral proceedings published in the Official Journal 10/2008 p. 471. From that it is clear that a duly summoned party who chooses not to attend oral proceedings has to expect that a decision will be taken at the oral proceedings even if it has submitted amended claims on which no communication was issued. Also, according to the Guidelines E-III.8.3.3.3. the applicant must expect a decision to be taken, based on objections arising during the proceedings.

Electronic Patent Application Fee Transmittal

Application Number:	16256497			
Filing Date:	24-Jan-2019			
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME			
First Named Inventor/Applicant Name:	Gihsung CHUNG			
Filer:	Maeng-Ho Shin/Mariam Paulos			
Attorney Docket Number:	0202-1231-1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260

Electronic Acknowledgement Receipt

EFS ID:	41847523
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/Mariam Paulos
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	05-FEB-2021
Filing Date:	24-JAN-2019
Time Stamp:	12:13:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 260
RAM confirmation Number	E202125C13300469
Deposit Account	504100
Authorized User	Mariam Paulos

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)

37 CFR 1.20 (Post Issuance fees)

37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	0202-1231-1IDSAsFiled02-05-2021.pdf	612042	no	4
			82d013b08a98dde9306c1ba9baf33f55d48ebc45		
Warnings:					
Information:					
A U.S. Patent Number Citation or a U.S. Publication Number Citation is required in the Information Disclosure Statement (IDS) form for autoloading of data into USPTO systems. You may remove the form to add the required data in order to correct the Informational Message if you are citing U.S. References. If you chose not to include U.S. References, the image of the form will be processed and be made available within the Image File Wrapper (IFW) system. However, no data will be extracted from this form. Any additional data such as Foreign Patent Documents or Non Patent Literature will be manually reviewed and keyed into USPTO systems.					
2	Non Patent Literature	1_0202-1231-1EPNPLsFiled.pdf	169254	no	6
			640d475044d7fc56bb27e441679fc72123ef5826		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30551	no	2
			24646d2f97fa89304ef2aaddf04754687d14f1f8		
Warnings:					
Information:					
Total Files Size (in bytes):			811847		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

68103 7590 11/25/2020
Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	01/24/2019	Gihung CHUNG	0202-1231-1	6284

TITLE OF INVENTION: METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	02/25/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
JOHNSON, NICOLE F	3792	600-301000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Jefferson IP Law, LLP
2
3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Samsung Electronics Co., Ltd.

Suwon-si, Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Wonki PARK/

Date February 8, 2021

Typed or printed name Wonki Park

Registration No. 38,991

Electronic Patent Application Fee Transmittal

Application Number:	16256497			
Filing Date:	24-Jan-2019			
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME			
First Named Inventor/Applicant Name:	Gihsung CHUNG			
Filer:	Maeng-Ho Shin/Mariam Paulos			
Attorney Docket Number:	0202-1231-1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	41862240
Application Number:	16256497
International Application Number:	
Confirmation Number:	6284
Title of Invention:	METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME
First Named Inventor/Applicant Name:	Gihsung CHUNG
Customer Number:	68103
Filer:	Maeng-Ho Shin/Mariam Paulos
Filer Authorized By:	Maeng-Ho Shin
Attorney Docket Number:	0202-1231-1
Receipt Date:	08-FEB-2021
Filing Date:	24-JAN-2019
Time Stamp:	13:11:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E202128D11383930
Deposit Account	504100
Authorized User	Mariam Paulos

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37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	0202-1231-1IssueFeeTransmittalAsFiled-02-08-2021.pdf	95321	no	1
			649761f1649cc5f2c76a737f12a7f3fc21a6443c		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30352	no	2
			a7113d6416192a08abfdaa6f4a3851dfcb0479f7		

Warnings:

Information:

Total Files Size (in bytes):	125673
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PRINTER RUSH
(PTO ASSISTANCE)

Application: 16256497

Examiner: JOHNSON

GAU: 3792

From:

Location: RTFM

Creation Date: 02/08/2021

Week Date: 11/23/2020

<u>DOC CODE</u>	<u>DOC DATE</u>	<u>MISCELLANEOUS</u>
☐ 1449	<u>02/05/2021</u>	☐ Continuing Data
☒ IDS		☐ Foreign Priority
☐ CLM		☐ Document Legibility
☐ IIFW/FWCLM		☐ Fees
☐ SRFW		☐ Petition (TC)
☐ DRW		☐ Other
☐ OATH		
☐ 312		
☐ SPEC		

[RUSH] Message:

Please respond to the 02/05/2021 Ids

Thank you

[XRUSH] Response:

Initials: _____

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APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR/ PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
16/256,497	01/24/2019	CHUNG et al.	0202-1231-1

Jefferson IP Law, LLP 1130 Connecticut Ave., NW, Suite 420 Washington, DC 20036		EXAMINER	
		NICOLE F JOHNSON	
		ART UNIT	PAPER
		3792	20210213

DATE MAILED: _____

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

The IDS dated February 5, 2021 has been considered, accepted and annotated

/NICOLE F LAVERT/
Primary Examiner, Art Unit 3792

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	16256497
	Filing Date	2019-01-24
	First Named Inventor	Gihung CHUNG
	Art Unit	3792
	Examiner Name	Nicole F JOHNSON
	Attorney Docket Number	0202-1231-1

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /N.F.L./

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
Filing Date	2019-01-24
First Named Inventor	Gihsung CHUNG
Art Unit	3792
Examiner Name	Nicole F JOHNSON
Attorney Docket Number	0202-1231-1

1	Summons to attend oral proceeding dated January 20, 2021, issued in European Application No. 16 183 812.3
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EXAMINER SIGNATURE

Examiner Signature	/NICOLE F LAVERT/	Date Considered	02/13/2021
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /N.F.L./

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	16256497
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Art Unit	3792
Examiner Name	Nicole F JOHNSON
Attorney Docket Number	0202-1231-1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Wonki PARK/	Date (YYYY-MM-DD)	2021-02-05
Name/Print	Wonki Park	Registration Number	38,991

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PRINTER RUSH
(PTO ASSISTANCE)

Application: 16256497

Examiner: JOHNSON

GAU: 3792

From:

Location: RTFM

Creation Date: 02/08/2021

Week Date: 11/23/2020

<u>DOC CODE</u>	<u>DOC DATE</u>	<u>MISCELLANEOUS</u>
☐ 1449 ☒ IDS ☐ CLM ☐ IIFW/FWCLM ☐ SRFW ☐ DRW ☐ OATH ☐ 312 ☐ SPEC	<u>02/05/2021</u>	☐ Continuing Data ☐ Foreign Priority ☐ Document Legibility ☐ Fees ☐ Petition (TC) ☐ Other

[RUSH] Message:

Please respond to the 02/05/2021 Ids

Thank you

[XRUSH] Response:

The IDS has been considered, accepted and annotated

Initials: /N.F.L./



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/256,497	03/16/2021	10945677	0202-1231-1	6284

68103 7590 02/24/2021
Jefferson IP Law, LLP
1130 Connecticut Ave., NW, Suite 420
Washington, DC 20036

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 81 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

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Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF;
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Cc: PAIR_eOfficeAction@uspto.gov
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Feb 26, 2021 04:33:49 AM

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Washington, DC 20036
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Application	Document	Mailroom Date	Attorney Docket No.
16256497	ISSUE.NTF	02/24/2021	0202-1231-1

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